

NEBIANS

OPEN LEARNING COMMUNITY FOR NEPAL

Physics

Grade 11 · Science Stream

Complete study notes for the NEB / CDC secondary education curriculum. Every unit of the official syllabus, with derivations, worked examples, diagrams, tables, summaries, practice questions and full model question papers.

About this book

These notes cover the complete official curriculum for Physics, Grade 11, as published by the Curriculum Development Centre (CDC), Sanathimi, and examined by the National Examination Board (NEB).

How this book is organised

Each chapter maps to one unit of the CDC curriculum and opens with the exact syllabus points that unit must cover, so you can tick off the syllabus as you study. Inside a chapter you will find:

- **Concept sections** following the order of the official curriculum.
- **Derivations** set out step by step, the way a full-mark answer should be written.
- **Worked examples** with the complete solution method, not just the answer.
- **Figures, tables and graphs** drawn to be redrawn by hand in the exam hall.
- **Key points, exam tips and common mistakes** pulled out of the running text.
- **A chapter summary and practice questions with answers**, in NEB question style.

Syllabus coverage at a glance

Content area	Chapters	Teaching hours
Mechanics	8	45
Heat and Thermodynamics	5	26
Waves and Optics	5	15
Electricity and Magnetism	5	27
Modern Physics	3	15

Examination pattern

- 3hr, FM 75 written (+25 internal) | Group A MCQ 11x1=11 | Group B short 8x5=40 | Group C long 3x8=24

PLEASE READ THIS

This is a community study aid, not an official CDC or NEB publication, and it does not replace your prescribed textbook. It was prepared from the official CDC curriculum documents and past NEB question papers. Edition 1.0 is a first release: if you spot an error, report it in the NEBians app so the next edition can fix it for everybody.

Edition 1.0 · 2082 BS (2025–26 AD) · Prepared for the NEBians open learning community · Free to share and print for study purposes.

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MECHANICS

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UNIT 1

Physical Quantities

Mechanics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Precision and significant figures
- Dimensions and uses of dimensional analysis

Physics is an experimental science: every law in this book began as a number read off an instrument. That number is never exact. This unit is about how much of a measured number you are entitled to believe — the language of precision, error and significant figures — and about the hidden structure that every physical quantity carries with it, its **dimensions**. Dimensions let you check an unfamiliar equation in ten seconds, convert a constant between unit systems, and sometimes guess a formula you were never taught.

WHAT THE EXAMINER IS TESTING

Group A nearly always carries one mark on significant figures or on the dimensional formula of a named quantity. Group B regularly asks you to check an equation dimensionally, to derive one, or to combine percentage errors. These are the cheapest marks in the paper if the rules are automatic.

1.1 Precision and significant figures

Measurement, units and standards

A **physical quantity** is any property of matter or of a phenomenon that can be measured. Measuring it means comparing it with a standard, so every measurement is a numerical value multiplied by a unit:

$$Q = n \times u$$

For a given quantity the product nu is fixed, so a smaller unit always gives a larger number. This one relation is the whole of unit conversion:

$$n_1 u_1 = n_2 u_2$$

Quantities that are independent of one another are called **fundamental** (base) quantities; everything else is **derived** from them. The SI system takes seven base quantities.

Base quantity	SI unit	Symbol	Dimension
Length	metre	m	$[L]$
Mass	kilogram	kg	$[M]$
Time	second	s	$[T]$
Electric current	ampere	A	$[I]$
Thermodynamic temperature	kelvin	K	$[K]$
Amount of substance	mole	mol	$[N]$
Luminous intensity	candela	cd	$[J]$

Plane angle (radian) and solid angle (steradian) are **supplementary** units; both are ratios of two lengths or two areas and are therefore dimensionless.

Accuracy, precision and least count

ACCURACY AND PRECISION

Accuracy is how close a measured value lies to the true or accepted value. **Precision** is how close repeated measurements lie to one another, and how finely the instrument can resolve a reading. A set of readings can be precise without being accurate, and accurate on average without being precise.

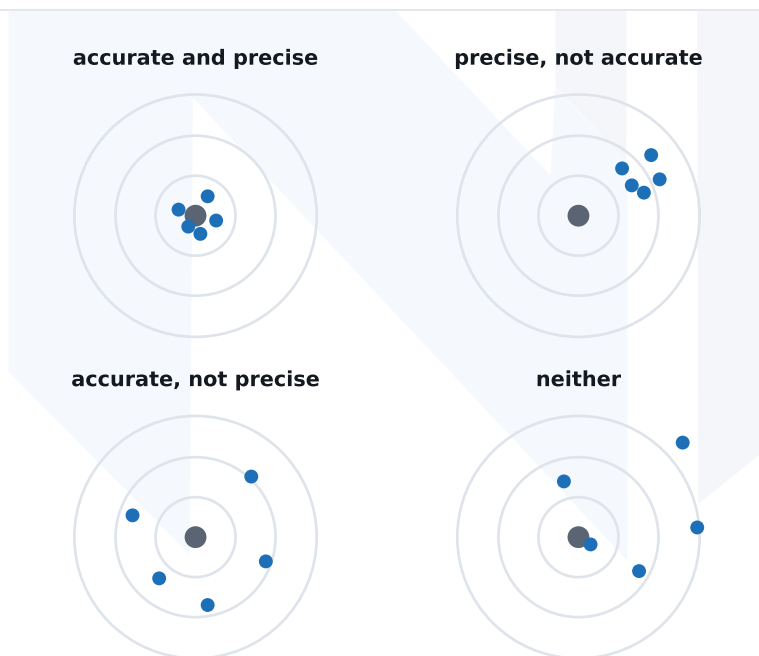


Figure 1. Accuracy is closeness to the true value (the bullseye); precision is closeness of the readings to each other. They are independent.

Precision is capped by the **least count** of the instrument — the smallest change it can display.

Instrument	Least count
Metre rule	1 mm = 0.1 cm
Vernier callipers (10 divisions)	0.1 mm = 0.01 cm
Screw gauge (pitch 1 mm, 100 divisions)	0.01 mm
Spherometer	0.01 mm
Mechanical stop-watch	0.1 s
Laboratory thermometer	1 °C

For a vernier scale, the least count is the difference between one main-scale division (MSD) and one vernier-scale division (VSD):

$$\text{L.C.} = 1 \text{ MSD} - 1 \text{ VSD} = \frac{\text{smallest main scale division}}{\text{number of vernier divisions}}$$

For a screw gauge, L.C. = pitch ÷ number of circular-scale divisions.

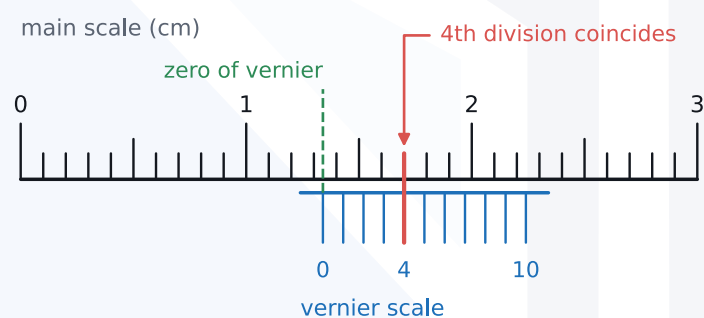


Figure 2. Vernier callipers. Ten vernier divisions span nine millimetres, so the least count is 0.1 mm. Here the vernier zero lies past 13 mm and its 4th division coincides, giving $13 + 4(0.1) = 13.4$ mm.

Errors in measurement

	Systematic error	Random error
Cause	faulty instrument, zero error, wrong technique, personal bias	unpredictable fluctuations: judgement, draughts, supply ripple
Sign	always in the same direction	equally likely either way
Effect	spoils accuracy	spoils precision
Remedy	calibrate, correct for zero error, change method	repeat and take the mean

If a quantity is measured n times giving $a_1, a_2, \dots, a_n$, the best estimate is the arithmetic mean $\bar{a}$. Then

$$\Delta a_i = |\bar{a} - a_i|, \quad \Delta a_{\text{mean}} = \frac{1}{n} \sum_{i=1}^n \Delta a_i$$

$$\text{relative error} = \frac{\Delta a_{\text{mean}}}{\bar{a}}, \quad \text{percentage error} = \frac{\Delta a_{\text{mean}}}{\bar{a}} \times 100$$

The result is quoted as $a = \bar{a} \pm \Delta a_{\text{mean}}$.

Significant figures

SIGNIFICANT FIGURES

The significant figures of a measured value are all the digits known with certainty together with the first digit that is uncertain. They record the precision of the measurement, not its size.

Rule	Example	Count
All non-zero digits count	2345	4
Zeros between non-zero digits count	2.005	4
Leading zeros never count	0.0025	2
Trailing zeros after a decimal point count	2.500	4
Trailing zeros in a whole number are ambiguous — use standard form	$4700 \rightarrow 4.70 \times 10^3$	3
Changing the unit never changes the count	$2.50 \text{ cm} = 25.0 \text{ mm} = 0.0250 \text{ m}$	3
Exact (counted) numbers and pure numbers such as π	12 eggs, 2π	infinite

ZEROS ARE THE TRAP

Leading zeros are only place-holders: 0.0025 has **two** significant figures, not four. But the zero in 2.50 was actually measured, so it counts. Never drop a trailing zero from a result — 2.5 cm and 2.50 cm are different claims.

Rounding off. If the digit to be dropped is greater than 5, raise the preceding digit by one; if it is less than 5, leave the preceding digit unchanged; if it is exactly 5 with nothing after it, round so that the preceding digit becomes even ($4.25 \rightarrow 4.2$, $4.35 \rightarrow 4.4$).

Arithmetic. Two different rules, and mixing them up is the commonest error:

- **Addition and subtraction** — the answer keeps as many decimal places as the term with the fewest decimal places.
- **Multiplication and division** — the answer keeps as many significant figures as the factor with the fewest significant figures.

WORKED EXAMPLE 1.1

Problem. (a) A rectangular plate measures 4.234 cm by 1.005 cm. Find its area to the correct number of significant figures. (b) Add 3.14 m, 2.7 m and 0.586 m.

Solution.

(a) $A = 4.234 \times 1.005 = 4.25517 \text{ cm}^2$. The factors have 4 and 4 significant figures, so the answer keeps **4**:

$$A = 4.255 \text{ cm}^2$$

(b) $3.14 + 2.7 + 0.586 = 6.426 \text{ m}$. The fewest decimal places is one (in 2.7), so the sum is rounded to one decimal place:

$$\text{sum} = 6.4 \text{ m}$$

Notice that (b) is not rounded to two significant figures — the decimal-place rule, not the significant-figure rule, governs addition.

Combining uncertainties

Let ΔA and ΔB be the absolute errors in A and B . For the **maximum possible** error in a result Z :

Operation	Rule
$Z = A + B$ or $Z = A - B$	$\Delta Z = \Delta A + \Delta B$ (absolute errors add)
$Z = AB$ or $Z = A/B$	$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$
$Z = A^p B^q / C^r$	$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$

WHERE THE ERROR-COMBINATION RULES COME FROM

We start from only one idea — a measured value may be anywhere between $A - \Delta A$ and $A + \Delta A$ — and we finish with every rule in the table above. The trick each time is the same: push both measurements to the worst possible end of their range.

(a) Sum. Let $Z = A + B$. The largest value Z can take is reached when both measurements are as large as they can be:

$$Z_{\max} = (A + \Delta A) + (B + \Delta B)$$

Group the true parts together and the error parts together. This is just rearranging the sum, so nothing has changed:

$$Z_{\max} = (A + B) + (\Delta A + \Delta B)$$

But $A + B$ is the answer Z itself, so whatever is left over is the error in Z :

$$\Delta Z = \Delta A + \Delta B$$

(b) Difference. Let $Z = A - B$. Now Z is largest when A is at its top and B at its **bottom**, because we are subtracting B :

$$Z_{\max} = (A + \Delta A) - (B - \Delta B)$$

Open the bracket. The two minus signs on ΔB make a plus:

$$Z_{\max} = (A - B) + (\Delta A + \Delta B)$$

So again

$$\Delta Z = \Delta A + \Delta B$$

Absolute errors **always add**, even for a subtraction. Errors never cancel, because we do not know which way each one went.

(c) Product. Let $Z = AB$. Take both measurements at their top end:

$$Z + \Delta Z = (A + \Delta A)(B + \Delta B)$$

Multiply out the two brackets, term by term:

$$Z + \Delta Z = AB + A \Delta B + B \Delta A + \Delta A \Delta B$$

Now drop the last term. ΔA and ΔB are both small, so their product is very small — for 1 % errors it is $0.01 \times 0.01 = 0.0001$ of the answer, which no instrument could ever show:

$$Z + \Delta Z \approx AB + A \Delta B + B \Delta A$$

Since $Z = AB$, subtract AB from both sides:

$$\Delta Z = A \Delta B + B \Delta A$$

Divide every term by $Z = AB$. Dividing both sides of an equation by the same quantity is always allowed:

$$\frac{\Delta Z}{AB} = \frac{A \Delta B}{AB} + \frac{B \Delta A}{AB}$$

Cancel A in the first term on the right and B in the second:

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

(d) Quotient. Let $Z = A/B$. Here Z is largest when the top is largest and the bottom smallest:

$$Z + \Delta Z = \frac{A + \Delta A}{B - \Delta B}$$

Pull A out of the top and B out of the bottom, so that only ratios are left inside the brackets:

$$Z + \Delta Z = \frac{A}{B} \cdot \frac{1 + \frac{\Delta A}{A}}{1 - \frac{\Delta B}{B}}$$

For any small number x , $\frac{1}{1-x} \approx 1+x$ (try $x = 0.01$: $1/0.99 = 1.0101$, and $1+x = 1.01$). Use this on the denominator:

$$Z + \Delta Z \approx \frac{A}{B} \left(1 + \frac{\Delta A}{A}\right) \left(1 + \frac{\Delta B}{B}\right)$$

Multiply the two brackets and again throw away the product of the two small terms:

$$Z + \Delta Z \approx \frac{A}{B} \left(1 + \frac{\Delta A}{A} + \frac{\Delta B}{B} \right)$$

Since $Z = A/B$, the bracket's "1" reproduces Z and the rest is ΔZ :

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

The same rule as for a product — division does not help you.

(e) Power. Let $Z = A^p$. Take A at its top end and pull out A^p :

$$Z + \Delta Z = (A + \Delta A)^p = A^p \left(1 + \frac{\Delta A}{A} \right)^p$$

For small x , $(1 + x)^p \approx 1 + px$ — this is the binomial expansion with only the first two terms kept, and the rest are negligible because x is small:

$$Z + \Delta Z \approx A^p \left(1 + p \frac{\Delta A}{A} \right)$$

Since $Z = A^p$,

$$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A}$$

Result. For $Z = A^p B^q / C^r$ the three contributions simply add:

$$\frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$$

What it means. Absolute errors add for $+$ and $-$; relative errors add for $\times$ and $\div$, each weighted by its power. So the measurement carrying the biggest power deserves the most care.

Condition used. Every step after (b) assumed the errors are **small** compared with the measurements ($\Delta A/A \ll 1$), so that products of two errors and higher binomial terms can be dropped. The rules give the maximum error, not the most likely one.

THE EXAMINER IS LOOKING FOR

For "derive the expression for the maximum error in a product": (i) write $Z + \Delta Z = (A + \Delta A)(B + \Delta B)$; (ii) expand fully; (iii) say in words that $\Delta A \Delta B$ is neglected because it is second order; (iv) divide through by $Z = AB$; (v) quote the final result with a box. Steps (iii) and (iv) are where marks are most often lost.

THE POWER RULE BITES HARDEST

A power multiplies the percentage error by that power, and it does so even for a quantity in the denominator. Measure the radius of a sphere to 1 % and the volume is only good to 3 %. So in the laboratory, measure most carefully the quantity that carries the highest power.

WORKED EXAMPLE 1.2

Problem. The radius of a sphere is measured as $r = 7.20 \pm 0.05$ cm. Find its volume and the percentage error in the volume.

Solution. $V = \frac{4}{3}\pi r^3 = \frac{4}{3}(3.1416)(7.20)^3$.

Since $(7.20)^3 = 373.248$ cm³,

$$V = 4.1888 \times 373.248 = 1563.5 \text{ cm}^3$$

Because $V \propto r^3$, the fractional errors combine as

$$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r} = 3 \times \frac{0.05}{7.20} = 0.0208$$

so the percentage error is **2.08 %**, and $\Delta V = 0.0208 \times 1563.5 = 32.6$ cm³.

The uncertainty is in the tens, so the answer is quoted as

$$V = (1.56 \pm 0.03) \times 10^3 \text{ cm}^3$$

1.2 Dimensions and uses of dimensional analysis

DIMENSIONS OF A PHYSICAL QUANTITY

The dimensions of a physical quantity are the powers to which the base quantities must be raised in order to represent it. The expression $[M^a L^b T^c]$ is its **dimensional formula**, and an equation stating that a quantity equals its dimensional formula is a **dimensional equation**.

For example, force = mass $\times$ acceleration, so

$$[F] = [M][LT^{-2}] = [MLT^{-2}]$$

Quantity	Defining relation	Dimensional formula	SI unit
Velocity	displacement/time	$[LT^{-1}]$	m s^{-1}
Acceleration	velocity/time	$[LT^{-2}]$	m s^{-2}
Force	mass $\times$ acceleration	$[MLT^{-2}]$	N
Momentum, impulse	mass $\times$ velocity	$[MLT^{-1}]$	kg m s^{-1}
Work, energy, torque	force $\times$ distance	$[ML^2T^{-2}]$	J
Power	work/time	$[ML^2T^{-3}]$	W
Pressure, stress, modulus	force/area	$[ML^{-1}T^{-2}]$	Pa
Density	mass/volume	$[ML^{-3}]$	kg m^{-3}
Frequency	1/period	$[T^{-1}]$	Hz
Gravitational constant G	Fr^2/m_1m_2	$[M^{-1}L^3T^{-2}]$	$\text{N m}^2 \text{kg}^{-2}$
Surface tension	force/length	$[MT^{-2}]$	N m^{-1}
Coefficient of viscosity	$F/(A \, dv/dx)$	$[ML^{-1}T^{-1}]$	Pa s
Strain, angle, refractive index	ratio	$[M^0L^0T^0]$	none

PRINCIPLE OF HOMOGENEITY OF DIMENSIONS

Only quantities of the same dimensions may be added, subtracted or equated. Therefore every term on both sides of a correct physical equation must have the same dimensional formula.

Use 1 — checking an equation

Test $v^2 = u^2 + 2as$. Left side $[LT^{-1}]^2 = [L^2T^{-2}]$. First term on the right $[L^2T^{-2}]$; second term $[LT^{-2}][L] = [L^2T^{-2}]$. All three agree, so the equation is dimensionally correct. Note that the pure number 2 is invisible to this test — dimensional analysis can prove an equation wrong but never fully right.

WORKED EXAMPLE 1.3

Problem. Find the dimensional formula of (a) Planck's constant h , defined by $E = h\nu$, and (b) the universal gas constant R , defined by $PV = nRT$.

Solution.

(a) $h = E/\nu$. Energy has $[ML^2T^{-2}]$ and frequency has $[T^{-1}]$, so

$$[h] = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

(This is the dimension of angular momentum — a useful check, since h is indeed measured in J s.)

(b) $R = PV/(nT)$. Pressure has $[ML^{-1}T^{-2}]$ and volume $[L^3]$, so $PV = [ML^2T^{-2}]$. Dividing by amount of substance $[N]$ and temperature $[K]$,

$$[R] = [ML^2T^{-2}K^{-1}N^{-1}]$$

with SI unit $J K^{-1} mol^{-1}$.

Use 2 — converting between systems of units

THE CONVERSION FORMULA FOR A QUANTITY OF DIMENSIONS $[M^aL^bT^c]$

We start from the fact that a physical quantity does not change when we change the units we describe it in, and we reach a formula that converts its number from one system to another.

Let the quantity be measured as n_1 units of size u_1 in the first system and n_2 units of size u_2 in the second. The quantity itself is the same object, so the two descriptions must be equal:

$$n_1 u_1 = n_2 u_2$$

Make n_2 the subject by dividing both sides by u_2 :

$$n_2 = n_1 \frac{u_1}{u_2}$$

Now we need u_1/u_2 . A unit of a quantity with dimensions $[M^aL^bT^c]$ is built out of that system's own mass, length and time units:

$$u_1 = M_1^a L_1^b T_1^c, \quad u_2 = M_2^a L_2^b T_2^c$$

Divide one by the other and collect the same base quantity together, using $x^a/y^a = (x/y)^a$:

$$\frac{u_1}{u_2} = \frac{M_1^a L_1^b T_1^c}{M_2^a L_2^b T_2^c} = \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$$

Put this back into $n_2 = n_1(u_1/u_2)$:

$$n_2 = n_1 \left(\frac{M_1}{M_2}\right)^a \left(\frac{L_1}{L_2}\right)^b \left(\frac{T_1}{T_2}\right)^c$$

What it means. Bigger units give smaller numbers. That is why the number of ergs in a joule is huge (10^7): the erg is a tiny unit.

Condition used. The quantity must have a pure power-law dimensional formula in M, L and T only. It fails for anything whose definition hides an extra base quantity (temperature, current, amount of substance) unless you include that base too.

WORKED EXAMPLE 1.4

Problem. Convert 1 joule into erg (the CGS unit of energy).

Solution. Energy has $[ML^2T^{-2}]$, so $a = 1$, $b = 2$, $c = -2$. Going from SI to CGS, $M_1/M_2 = 1 \text{ kg}/1 \text{ g} = 10^3$, $L_1/L_2 = 1 \text{ m}/1 \text{ cm} = 10^2$, and $T_1/T_2 = 1$.

$$n_2 = 1 \times (10^3)^1 (10^2)^2 (1)^{-2} = 10^3 \times 10^4 = 10^7$$

Hence $1 \text{ J} = 10^7 \text{ erg}$.

Use 3 — deriving a relation

If a quantity is known to depend on at most three others, assume a product of powers and match dimensions on both sides.

WORKED EXAMPLE 1.5

Problem. The period T of a simple pendulum may depend on its mass m , length l and the acceleration due to gravity g . Derive the relation.

Solution. Assume $T = k m^a l^b g^c$, where k is a dimensionless constant. Writing dimensions,

$$[T] = [M]^a [L]^b [LT^{-2}]^c = [M^a L^{b+c} T^{-2c}]$$

Comparing powers of M , L and T on the two sides:

$$a = 0, \quad b + c = 0, \quad -2c = 1$$

So $c = -\frac{1}{2}$, $b = +\frac{1}{2}$ and $a = 0$. Therefore

$$T = k l^{1/2} g^{-1/2} = k \sqrt{\frac{l}{g}}$$

The period does **not** depend on the mass of the bob — a real physical prediction from pure book-keeping. Experiment (or a full derivation) supplies $k = 2\pi$.

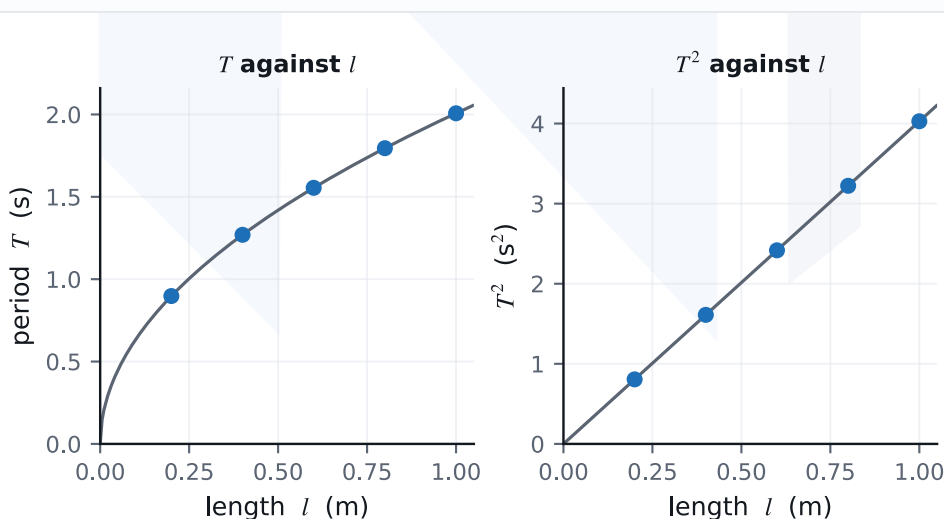


Figure 3. Testing the dimensional prediction $T \propto \sqrt{l}$ for a simple pendulum. The T^2 against l graph is a straight line through the origin of slope $4\pi^2/g$.

Limitations of dimensional analysis

1. It cannot supply **dimensionless constants**. The 2π in $T = 2\pi\sqrt{l/g}$, and the $\frac{1}{2}$ in $\frac{1}{2}mv^2$, must come from theory or experiment.
2. It fails if the quantity depends on **more than three** base quantities, because M, L and T give only three equations.
3. It cannot handle equations that are a **sum of terms**, such as $s = ut + \frac{1}{2}at^2$ — it can only check them, not build them.
4. It cannot deal with **trigonometric, exponential or logarithmic** functions, though it does tell you their arguments must be dimensionless.
5. It cannot distinguish quantities with the same dimensions (work and torque, both $[ML^2T^{-2}]$), and cannot tell a scalar from a vector.

✓ Chapter summary

- A measurement is $Q = nu$, and $n_1u_1 = n_2u_2$; SI has seven base quantities.
- **Accuracy** = closeness to the true value; **precision** = closeness of repeated readings, limited by the least count.
- Vernier L.C. = $1 \text{ MSD} - 1 \text{ VSD}$; screw-gauge L.C. = $\text{pitch} \div \text{number of circular divisions}$.
- Systematic errors spoil accuracy and are removed by calibration; random errors spoil precision and are reduced by averaging. Quote results as $\bar{a} \pm \Delta a_{\text{mean}}$.
- Significant figures = certain digits + first uncertain digit. Addition keeps the least **decimal places**; multiplication keeps the least **significant figures**.
- Errors combine as $\Delta Z = \Delta A + \Delta B$ for sums, and $\Delta Z/Z = p \Delta A/A + q \Delta B/B$ for products and powers.
- The **principle of homogeneity** says every term of a physical equation has the same dimensions. Dimensional analysis checks equations, converts units via $n_2 = n_1(M_1/M_2)^a(L_1/L_2)^b(T_1/T_2)^c$, and derives simple relations.
- It cannot give dimensionless constants, handle more than three variables, build equations with several terms, or separate work from torque.

Practice questions

Group A — Multiple choice (1 mark each)

1. The number of significant figures in 0.00250 m is [1]

a 2
b 3
c 5
d 6
2. Which pair has the same dimensional formula? [1]

a work and power
c work and torque

b force and momentum
d pressure and force
3. The dimensional formula of the universal gravitational constant is [1]

a $[M^{-1}L^3T^{-2}]$
b $[ML^3T^{-2}]$
c $[M^{-1}L^2T^{-2}]$
d $[ML^2T^{-1}]$

4. The radius of a sphere is measured with an error of 2 %. The error in its volume is [1]
- a 2 % b 4 % c 6 % d 8 %
5. A vernier callipers has 10 vernier divisions matching 9 main-scale divisions of 1 mm each. Its least count is [1]
- a 1 mm b 0.5 mm c 0.1 mm d 0.01 mm
6. Which of the following is dimensionless? [1]
- a stress b strain c surface tension d force constant

ANSWERS TO GROUP A

1. (b) — leading zeros are place-holders; the 2, 5 and the trailing 0 count.
2. (c) — both are force $\times$ distance, $[ML^2T^{-2}]$.
3. (a) — $G = Fr^2/m_1m_2 = [MLT^{-2}][L^2]/[M^2]$.
4. (c) — $V \propto r^3$, so the percentage error is tripled.
5. (c) — L.C. = $1 - 0.9 = 0.1$ mm, i.e. $1 \text{ mm} \div 10$.
6. (b) — strain is a ratio of two lengths.

Group B — Short answer (5 marks each)

1. Distinguish between accuracy and precision, giving one example of each. State what limits the precision of an instrument. [5]
2. State the rules for counting significant figures. Hence give the number of significant figures in (i) 0.00602, (ii) 6.023×10^{23} , (iii) 4.700 m. [5]
3. The length, breadth and thickness of a metal block are 5.12 cm, 2.34 cm and 1.05 cm. Calculate its volume to the correct number of significant figures. [5]
4. Check the correctness of the equation $T = 2\pi\sqrt{l/g}$ by the method of dimensions, and explain why the test cannot confirm the factor 2π . [5]
5. A physical quantity is given by $P = a^3b^2/(\sqrt{c}d)$. If the percentage errors in a , b , c and d are 1 %, 2 %, 3 % and 4 % respectively, find the maximum percentage error in P . [5]

ANSWERS TO GROUP B

1. Outline: accuracy = closeness to the true value (a zero-error instrument is inaccurate); precision = closeness of repeated readings, set by the least count.
2. (i) 3 — leading zeros do not count. (ii) 4 — only the mantissa 6.023 counts. (iii) 4 — the trailing zeros follow a decimal point.
3. $V = 5.12 \times 2.34 \times 1.05 = 12.57984 \text{ cm}^3$. The fewest significant figures among the factors is 3 (in 1.05), so $V = 12.6 \text{ cm}^3$.
4. LHS = $[T]$. RHS = $\sqrt{[L]/[LT^{-2}]} = \sqrt{[T^2]} = [T]$. The two sides agree, so the equation is dimensionally correct. 2π is a pure number with dimensions $[M^0L^0T^0]$, so replacing it by any other number leaves the check unchanged — dimensional analysis cannot detect it.

$$5. \frac{\Delta p}{p} \times 100 = 3(1) + 2(2) + \frac{1}{2}(3) + 1(4) = 3 + 4 + 1.5 + 4 = 12.5 \%$$

Group C — Long answer (8 marks each)

- (a)** Define absolute error, mean absolute error, relative error and percentage error. [3]

(b) In an experiment the mass of a body is found to be 20.00 ± 0.05 g and its volume 5.00 ± 0.02 cm³. Calculate its density, the percentage error and the absolute error, and quote the result properly. [5]
- (a)** State the principle of homogeneity of dimensions and describe three uses of dimensional analysis. [4]

(b) The value of the gravitational constant is $G = 6.67 \times 10^{-11}$ N m² kg⁻² in SI units. Convert it into the CGS system. [4]

ANSWERS TO GROUP C

1. (a) See the definitions in §1.1.

1. (b) $\rho = m/V = 20.00/5.00 = 4.00$ g cm⁻³.

$$\frac{\Delta \rho}{\rho} = \frac{0.05}{20.00} + \frac{0.02}{5.00} = 0.0025 + 0.0040 = 0.0065$$

so the percentage error is 0.65 % and $\Delta \rho = 0.0065 \times 4.00 = 0.026 \approx 0.03$ g cm⁻³. The result is $\rho = 4.00 \pm 0.03$ g cm⁻³.

2. (a) Outline: state homogeneity, then the three uses — checking equations, converting units, deriving relations.

2. (b) $[G] = [M^{-1}L^3T^{-2}]$, so $a = -1$, $b = 3$, $c = -2$ with $M_1/M_2 = 10^3$, $L_1/L_2 = 10^2$, $T_1/T_2 = 1$:

$$n_2 = 6.67 \times 10^{-11} (10^3)^{-1} (10^2)^3 = 6.67 \times 10^{-11} \times 10^{-3} \times 10^6$$

giving $G = 6.67 \times 10^{-8}$ dyne cm² g⁻².

UNIT 2

Vectors

Mechanics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Triangle, parallelogram and polygon laws of vectors
- Resolution of vectors; Unit vectors
- Scalar and vector products

Some physical quantities are fully described by a number and a unit — mass, time, temperature. Others are useless until you also say which way: a force of 10 N pushing north is not the force of 10 N pushing south. Quantities of the second kind are **vectors**, and they obey their own arithmetic. This unit builds that arithmetic: how vectors add, how they are broken into components, and the two ways of multiplying them. Every later unit — dynamics, circular motion, gravitation, fields — is written in this language.

THE ONE IDEA BEHIND THE WHOLE UNIT

Vectors never add arithmetically. $3 + 4$ is 7 only if the two vectors point the same way; it is 5 if they are perpendicular and 1 if they are opposite. Adding vectors means adding displacements drawn head to tail, and the fastest safe way to do that is always to resolve into components.

2.1 Triangle, parallelogram and polygon laws of vectors

A **scalar** has magnitude only and obeys ordinary algebra (mass, speed, work, energy, charge, temperature). A **vector** has magnitude and direction and obeys the laws of this section (displacement, velocity, acceleration, force, momentum, torque). A vector is drawn as an arrow whose length, to some scale, is the magnitude and whose arrowhead gives the direction. It is written $\vec{A}$, and its magnitude as A or $|\vec{A}|$.

Type of vector	Meaning
Equal vectors	same magnitude and same direction, wherever they are drawn
Negative vector	same magnitude, exactly opposite direction
Unit vector	magnitude exactly 1; only carries direction
Null (zero) vector	magnitude zero, direction indeterminate
Collinear vectors	act along the same line or parallel lines
Coplanar vectors	all lie in one plane
Position vector	drawn from a chosen origin to the point

Triangle law of vector addition

TRIANGLE LAW

If two vectors are represented in magnitude and direction by the two sides of a triangle taken in order, then their resultant is represented in magnitude and direction by the third side of the triangle taken in the opposite order.

So to add $\vec{P}$ and $\vec{Q}$ you draw $\vec{P}$, then start $\vec{Q}$ at the head of $\vec{P}$; the arrow from the tail of $\vec{P}$ to the head of $\vec{Q}$ is $\vec{R} = \vec{P} + \vec{Q}$. Because the same third side results whichever vector is drawn first, vector addition is **commutative**, $\vec{P} + \vec{Q} = \vec{Q} + \vec{P}$, and it is also **associative**, $(\vec{P} + \vec{Q}) + \vec{S} = \vec{P} + (\vec{Q} + \vec{S})$.

Parallelogram law of vector addition

PARALLELOGRAM LAW

If two vectors acting simultaneously at a point are represented in magnitude and direction by the two adjacent sides of a parallelogram drawn from that point, their resultant is represented in magnitude and direction by the diagonal of the parallelogram drawn from the same point.

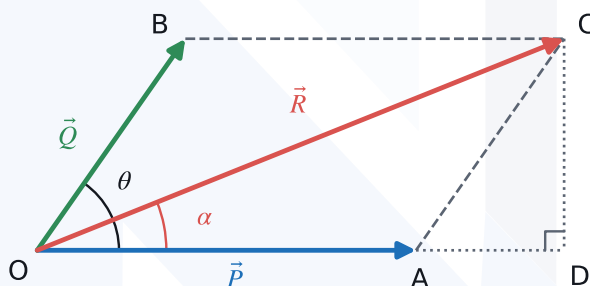


Figure 4. Parallelogram law. OC is the resultant of $\vec{P}$ and $\vec{Q}$; the perpendicular CD dropped on OA produced supplies the derivation.

MAGNITUDE AND DIRECTION OF THE RESULTANT OF TWO VECTORS

We start from the parallelogram of the figure — two vectors $\vec{P}$ and $\vec{Q}$ drawn from the same point O with angle θ between them — and we finish with a formula for how long the diagonal is and which way it points.

Setting up. Let $\vec{P} = \vec{OA}$ and $\vec{Q} = \vec{OB}$, with $\angle AOB = \theta$. Complete the parallelogram $OACB$, so the diagonal $\vec{OC} = \vec{R}$ is the resultant. Extend the line OA forwards and drop a perpendicular from C onto it, meeting it at D . We now have two right-angled triangles to work with, ADC and ODC .

Step 1 — the side AC equals Q . In a parallelogram opposite sides are equal and parallel, so AC is equal in length to OB and points the same way:

$$AC = OB = Q$$

Because AC is parallel to OB , the angle CAD that AC makes with the extended line OA is also θ (corresponding angles).

Step 2 — split AC into its two perpendicular parts. Triangle ADC has a right angle at D , with hypotenuse $AC = Q$ and angle θ at A . Ordinary trigonometry ($\cos = \text{adjacent/hypotenuse}$, $\sin = \text{opposite/hypotenuse}$) gives

$$AD = AC \cos \theta = Q \cos \theta, \quad DC = AC \sin \theta = Q \sin \theta$$

Step 3 — find the whole base OD . The point D lies beyond A , so the base is made of two pieces laid end to end:

$$OD = OA + AD$$

Put in $OA = P$ and $AD = Q \cos \theta$ from Step 2:

$$OD = P + Q \cos \theta$$

Step 4 — use Pythagoras in triangle ODC . This triangle is right-angled at D and its hypotenuse is exactly the resultant $OC = R$:

$$R^2 = OD^2 + DC^2$$

Substitute the two sides we found:

$$R^2 = (P + Q \cos \theta)^2 + (Q \sin \theta)^2$$

Step 5 — expand the first bracket. Using $(x + y)^2 = x^2 + 2xy + y^2$:

$$R^2 = P^2 + 2PQ \cos \theta + Q^2 \cos^2 \theta + Q^2 \sin^2 \theta$$

Step 6 — collect the two Q^2 terms. They share a factor Q^2 :

$$R^2 = P^2 + 2PQ \cos \theta + Q^2 (\cos^2 \theta + \sin^2 \theta)$$

Step 7 — use the identity. For any angle, $\sin^2 \theta + \cos^2 \theta = 1$, so the bracket is simply 1:

$$R^2 = P^2 + Q^2 + 2PQ \cos \theta$$

Step 8 — take the square root. A magnitude is never negative, so we keep the positive root:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

Step 9 — the direction. Let α be the angle between $\vec{R}$ and $\vec{P}$, i.e. $\angle COD$ in the right triangle ODC . Then $\tan \alpha = \text{opposite/adjacent}$:

$$\tan \alpha = \frac{DC}{OD}$$

Substitute $DC = Q \sin \theta$ and $OD = P + Q \cos \theta$:

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

Finally take the inverse tangent of both sides:

$$\alpha = \tan^{-1} \left(\frac{Q \sin \theta}{P + Q \cos \theta} \right)$$

Result.

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}, \quad \tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

What it means. The single vector $\vec{R}$ has exactly the same effect as $\vec{P}$ and $\vec{Q}$ acting together. The whole answer depends on the angle between them, not on their magnitudes alone — so "3 N and 4 N" has no single resultant until you are told θ .

Assumptions used. Both vectors act at the **same point** (they are concurrent) and they are the same kind of quantity (you may not add a force to a velocity). α is measured from $\vec{P}$; measuring from $\vec{Q}$ instead swaps P and Q in the direction formula.

WHEN IS THE RESULTANT LARGEST AND SMALLEST?

We start from the result just derived and reach the two extreme cases NEB likes to ask about.

In $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$, the only part that can change if P and Q are fixed is $\cos \theta$. So R is largest when $\cos \theta$ is largest and smallest when $\cos \theta$ is smallest.

For any angle, $\cos \theta$ can never be more than +1 and never less than -1.

Largest. Put $\cos \theta = +1$, which happens at $\theta = 0^\circ$ (the two vectors point the same way):

$$R_{\max} = \sqrt{P^2 + Q^2 + 2PQ}$$

The quantity under the root is a perfect square, $(P + Q)^2$:

$$R_{\max} = \sqrt{(P + Q)^2} = P + Q$$

Smallest. Put $\cos \theta = -1$, which happens at $\theta = 180^\circ$ (the vectors point in opposite directions):

$$R_{\min} = \sqrt{P^2 + Q^2 - 2PQ}$$

Again the inside is a perfect square, $(P - Q)^2$:

$$R_{\min} = \sqrt{(P - Q)^2} = |P - Q|$$

Result. R always lies between $|P - Q|$ and $P + Q$.

What it means. Two 3 N and 4 N pulls can give anything from 1 N to 7 N. The modulus sign matters: a magnitude cannot be negative, so if $Q > P$ the answer is $Q - P$.

THE EXAMINER IS LOOKING FOR

For the 5-mark parallelogram-law question: (i) statement of the law; (ii) a labelled parallelogram with the perpendicular CD drawn and O, A, B, C, D marked; (iii) $AD = Q \cos \theta$ and $DC = Q \sin \theta$ written down; (iv) Pythagoras in ODC ; (v) the identity $\sin^2 \theta + \cos^2 \theta = 1$ used visibly; (vi) both final boxed formulas — magnitude **and** direction. Leaving out the direction formula costs a full mark every time.

Angle θ between them	Resultant R	Note
0°	$P + Q$	maximum possible
90°	$\sqrt{P^2 + Q^2}$	$\tan \alpha = Q/P$
180°	$P - Q$ in magnitude	minimum possible
120° with $P = Q$	P	three equal vectors at 120° give zero

NEVER ADD MAGNITUDES

A body pulled by 3 N and 4 N does not feel 7 N unless the pulls are parallel. The resultant of 3 N and 4 N can be anything from 1 N to 7 N. Always ask for the angle first.

WORKED EXAMPLE 2.1

Problem. Two forces of 5 N and 3 N act at a point with an angle of 60° between them. Find the magnitude and direction of their resultant.

Solution. With $P = 5$ N, $Q = 3$ N, $\theta = 60^\circ$ and $\cos 60^\circ = 0.5$:

$$R = \sqrt{5^2 + 3^2 + 2(5)(3)(0.5)} = \sqrt{25 + 9 + 15} = \sqrt{49} = 7 \text{ N}$$

For the direction, with $\sin 60^\circ = 0.866$:

$$\tan \alpha = \frac{3(0.866)}{5 + 3(0.5)} = \frac{2.598}{6.5} = 0.3997$$

so $\alpha = 21.8^\circ$. The resultant is 7 N acting at 21.8° to the 5 N force.

Polygon law and subtraction**POLYGON LAW**

If a number of vectors are represented in magnitude and direction by the sides of an open polygon taken in order, their resultant is represented by the closing side of the polygon taken in the opposite order.

It follows that if the polygon **closes**, the resultant is zero — which is exactly the condition for a set of concurrent forces to be in equilibrium.

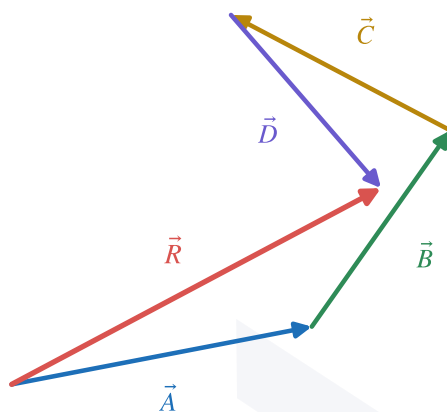


Figure 5. Polygon law: four vectors drawn head to tail, with the closing side (red) as the resultant. If the last head met the first tail, the resultant would be zero.

Subtraction is addition of the negative vector:

$$\vec{P} - \vec{Q} = \vec{P} + (-\vec{Q}), \quad |\vec{P} - \vec{Q}| = \sqrt{P^2 + Q^2 - 2PQ\cos\theta}$$

2.2 Resolution of vectors; Unit vectors

Resolution is the reverse of addition: replacing one vector by two (or three) vectors that together have the same effect. The useful choice is always two **mutually perpendicular** components, because perpendicular components never interfere with each other.

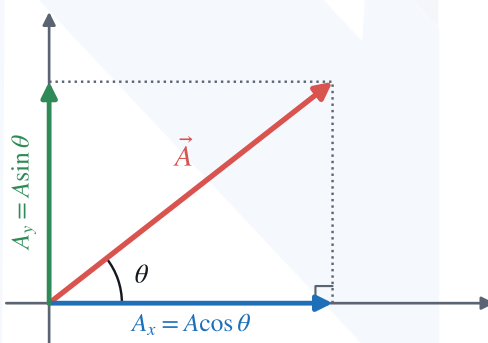


Figure 6. Rectangular resolution. $A_x = A\cos\theta$ and $A_y = A\sin\theta$ are the sides of the right triangle whose hypotenuse is $\vec{A}$.

If $\vec{A}$ makes an angle θ with the x -axis, the right triangle gives

$$A_x = A\cos\theta, \quad A_y = A\sin\theta$$

and going back the other way,

$$A = \sqrt{A_x^2 + A_y^2}, \quad \tan \theta = \frac{A_y}{A_x}$$

RESOLVE, ADD, RECOMBINE

To add any number of vectors: resolve each one into x and y components, add the components separately to get $R_x = \sum A_x$ and $R_y = \sum A_y$, then $R = \sqrt{R_x^2 + R_y^2}$ and $\tan \theta = R_y/R_x$. This method never fails, however many vectors there are.

Unit vectors**UNIT VECTOR**

A unit vector is a vector of magnitude one that specifies a direction only. The unit vector along $\vec{A}$ is

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}, \quad \text{so} \quad \vec{A} = A \hat{A}$$

The unit vectors along the x , y and z axes of a right-handed set are written $\hat{i}$, $\hat{j}$ and $\hat{k}$. Any vector in space is then

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}, \quad A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

The cosines of the angles α , β , γ that $\vec{A}$ makes with the three axes are its **direction cosines**:

$$l = \frac{A_x}{A}, \quad m = \frac{A_y}{A}, \quad n = \frac{A_z}{A}, \quad l^2 + m^2 + n^2 = 1$$

In component form addition is simply componentwise:

$$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$$

WORKED EXAMPLE 2.2

Problem. A vector is given by $\vec{A} = 2\hat{i} + 3\hat{j} - 6\hat{k}$. Find (a) its magnitude, (b) the unit vector along it, and (c) its direction cosines.

Solution.

$$(a) A = \sqrt{2^2 + 3^2 + (-6)^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$$

$$(b) \hat{A} = \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7} = \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} - \frac{6}{7}\hat{k}.$$

$$(c) l = 2/7 = 0.286, \quad m = 3/7 = 0.429, \quad n = -6/7 = -0.857.$$

Check: $l^2 + m^2 + n^2 = (4 + 9 + 36)/49 = 1$, as it must be.

A standard application: a block of weight mg on a plane inclined at θ has weight components $mg\sin\theta$ **down the slope** and $mg\cos\theta$ **perpendicular into the slope**. Almost every inclined-plane problem in the next four units starts from that one resolution.

2.3 Scalar and vector products

Scalar (dot) product

SCALAR PRODUCT

The scalar product of $\vec{A}$ and $\vec{B}$ is the scalar

$$\vec{A} \cdot \vec{B} = AB\cos\theta$$

where θ is the angle between them when they are drawn from a common point.

Geometrically, $\vec{A} \cdot \vec{B}$ is the magnitude of one vector times the **projection** of the other onto it. Its key properties:

- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$.
- Distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- If the vectors are perpendicular, $\vec{A} \cdot \vec{B} = 0$; if parallel, AB ; if antiparallel, $-AB$.
- $\vec{A} \cdot \vec{A} = A^2$.
- For the base unit vectors: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.

Hence in component form

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z, \quad \cos\theta = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

Physical examples: work $W = \vec{F} \cdot \vec{s}$, power $P = \vec{F} \cdot \vec{v}$, and magnetic flux $\Phi = \vec{B} \cdot \vec{A}$.

Vector (cross) product

VECTOR PRODUCT

The vector product of $\vec{A}$ and $\vec{B}$ is the vector

$$\vec{A} \times \vec{B} = AB\sin\theta \hat{n}$$

whose magnitude is $AB\sin\theta$ and whose direction $\hat{n}$ is perpendicular to the plane of $\vec{A}$ and $\vec{B}$, given by the right-hand screw rule: turn $\vec{A}$ into $\vec{B}$ and the screw advances along $\hat{n}$.

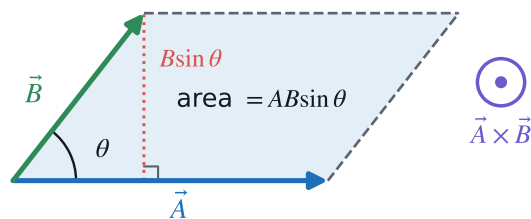


Figure 7. The magnitude of $\vec{A} \times \vec{B}$ equals the area of the parallelogram they span, base A times height $B \sin \theta$. The product vector points out of the page.

Properties:

- **Anticommutative:** $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$.
- $\vec{A} \times \vec{A} = \vec{0}$, and the product of parallel or antiparallel vectors is zero.
- It is maximum, AB , when the vectors are perpendicular.
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$, while $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (reverse the order and the sign flips).

In components,

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$

The magnitude $AB \sin \theta$ is the **area of the parallelogram** with the two vectors as adjacent sides; half of it is the area of the triangle. Physical examples: torque $\vec{\tau} = \vec{r} \times \vec{F}$, angular momentum $\vec{L} = \vec{r} \times \vec{p}$, and the magnetic force $\vec{F} = q \vec{v} \times \vec{B}$.

	Scalar product	Vector product
Result	scalar	vector
Formula	$AB \cos \theta$	$AB \sin \theta \hat{n}$
Zero when	vectors are perpendicular	vectors are parallel
Maximum when	vectors are parallel	vectors are perpendicular
Order matters?	no	yes — reversing changes the sign
Example	work, power	torque, angular momentum

SINE AND COSINE GET SWAPPED

The dot product uses $\cos \theta$, the cross product $\sin \theta$. So two perpendicular vectors have **zero dot product but maximum cross product** — which is why no work is done by a centripetal force, yet a force at right angles to the radius gives the largest torque.

WORKED EXAMPLE 2.3

Problem. For $\vec{A} = \hat{i} + \hat{j}$ and $\vec{B} = \hat{j} + \hat{k}$, find (a) $\vec{A} \cdot \vec{B}$ and the angle between the vectors, and (b) $\vec{A} \times \vec{B}$ and its magnitude.

Solution. Here $A = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$ and likewise $B = \sqrt{2}$.

(a) $\vec{A} \cdot \vec{B} = (1)(0) + (1)(1) + (0)(1) = 1$. So

$$\cos \theta = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

(b) With $\vec{A} = (1, 1, 0)$ and $\vec{B} = (0, 1, 1)$,

$$\vec{A} \times \vec{B} = (1 \cdot 1 - 0 \cdot 1)\hat{i} + (0 \cdot 0 - 1 \cdot 1)\hat{j} + (1 \cdot 1 - 1 \cdot 0)\hat{k} = \hat{i} - \hat{j} + \hat{k}$$

Its magnitude is $\sqrt{1 + 1 + 1} = \sqrt{3} = 1.73$. As a check, $AB \sin \theta = 2 \sin 60^\circ = 2(0.866) = 1.73$. The two answers agree.

WORKED EXAMPLE 2.4

Problem. (a) A force $\vec{F} = (3\hat{i} + 4\hat{j})$ N moves a body from the point $(1, 1, 0)$ m to $(4, 5, 0)$ m. Find the work done. (b) A force $\vec{F} = (3\hat{i} + 2\hat{j} - 4\hat{k})$ N acts at the point $\vec{r} = (2\hat{i} + \hat{j} - \hat{k})$ m from the origin. Find the torque about the origin.

Solution.

(a) Displacement $\vec{s} = (4 - 1)\hat{i} + (5 - 1)\hat{j} = 3\hat{i} + 4\hat{j}$ m.

$$W = \vec{F} \cdot \vec{s} = (3)(3) + (4)(4) = 9 + 16 = 25 \text{ J}$$

(b) $\vec{\tau} = \vec{r} \times \vec{F}$ with $\vec{r} = (2, 1, -1)$ and $\vec{F} = (3, 2, -4)$:

$$\tau_x = (1)(-4) - (-1)(2) = -2, \quad \tau_y = (-1)(3) - (2)(-4) = 5, \quad \tau_z = (2)(2) - (1)(3) = 1$$

$$\vec{\tau} = (-2\hat{i} + 5\hat{j} + \hat{k}) \text{ N m}$$

with magnitude $\sqrt{4 + 25 + 1} = \sqrt{30} = 5.48 \text{ N m}$.

✓ Chapter summary

- Scalars need a magnitude only; vectors need magnitude **and** direction and add by the triangle, parallelogram or polygon law.
- Parallelogram law: $R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$ and $\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$. Maximum $P + Q$ at 0° , minimum $P - Q$ at 180° .
- A closed polygon of vectors means zero resultant — the equilibrium condition.
- Resolution: $A_x = A\cos\theta$, $A_y = A\sin\theta$; recombine with $A = \sqrt{A_x^2 + A_y^2}$, $\tan\theta = A_y/A_x$.
- $\hat{A} = \vec{A}/A$; $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ with $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$ and $l^2 + m^2 + n^2 = 1$.
- Dot product $\vec{A} \cdot \vec{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$; it is a scalar and vanishes for perpendicular vectors.
- Cross product $\vec{A} \times \vec{B} = AB\sin\theta \hat{n}$; it is a vector, vanishes for parallel vectors, reverses sign on swapping, and its magnitude is the area of the parallelogram formed by the two vectors.

Practice questions

Group A — Multiple choice (1 mark each)

- Which of the following is a vector quantity? [1]

a work	b temperature	c torque	d electric charge
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- The resultant of two forces of 3 N and 4 N acting at right angles is [1]

a 1 N	b 5 N	c 7 N	d 12 N
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- If $\vec{A} \cdot \vec{B} = 0$ for two non-zero vectors, then the angle between them is [1]

a 0°	b 45°	c 90°	d 180°
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- The magnitude of $\hat{i} \times \hat{j}$ is [1]

a 0	b 1	c $\sqrt{2}$	d 2
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- Two equal vectors of magnitude P give a resultant of magnitude P . The angle between them is [1]

a 60°	b 90°	c 120°	d 180°
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- The area of the triangle formed by $\vec{A}$ and $\vec{B}$ as two of its sides is [1]

a $AB\sin\theta$	b $\frac{1}{2}AB\sin\theta$	c $AB\cos\theta$	d $\frac{1}{2}AB\cos\theta$
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ANSWERS TO GROUP A

- (c) — torque has a direction given by the right-hand rule; work and charge are scalars.
- (b) — $\sqrt{3^2 + 4^2} = 5$ N.
- (c) — $AB\cos\theta = 0$ requires $\cos\theta = 0$.
- (b) — $\sin 90^\circ = 1$ and both are unit vectors.
- (c) — $P^2 = 2P^2(1 + \cos\theta)$ gives $\cos\theta = -1/2$.

6. (b) — the cross product gives the parallelogram area; the triangle is half of it.

Group B — Short answer (5 marks each)

1. State the parallelogram law of vector addition and derive expressions for the magnitude and direction of the resultant of two vectors. [5]
2. Define a unit vector. If $\vec{A} = 3\hat{i} - 4\hat{j} + 12\hat{k}$, find its magnitude, the unit vector along it and its direction cosines. [5]
3. Two forces of 6 N and 8 N act on a body at right angles to each other. Find the magnitude and direction of the resultant force. [5]
4. Distinguish between the scalar and the vector product of two vectors, giving one physical example of each. [5]
5. Find the angle between the vectors $\vec{A} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{B} = 6\hat{i} - 3\hat{j} + 2\hat{k}$. [5]

ANSWERS TO GROUP B

1. Outline: state the law, then reproduce the derivation of $R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$ and $\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$ given in §2.1.
2. $A = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. $\hat{A} = (3\hat{i} - 4\hat{j} + 12\hat{k})/13$; direction cosines $l = 3/13 = 0.231$, $m = -4/13 = -0.308$, $n = 12/13 = 0.923$.
3. $R = \sqrt{6^2 + 8^2} = 10$ N; $\tan\alpha = 8/6 = 1.333$, so $\alpha = 53.1^\circ$ from the 6 N force.
4. Outline: use the comparison table in §2.3 — scalar versus vector result, $\cos\theta$ versus $\sin\theta$, order dependence; work and torque as examples.
5. $\vec{A} \cdot \vec{B} = 12 - 6 - 2 = 4$; $A = \sqrt{4 + 4 + 1} = 3$ and $B = \sqrt{36 + 9 + 4} = 7$. So $\cos\theta = 4/21 = 0.1905$ and $\theta = 79.0^\circ$.

Group C — Long answer (8 marks each)

1. (a) State and explain the triangle, parallelogram and polygon laws of vector addition. [4]
 (b) Two forces P and Q act at a point. Show that the resultant is maximum when they are parallel and minimum when they are antiparallel. Hence find the two forces whose maximum resultant is 14 N and whose minimum resultant is 2 N. [4]
2. (a) Define the scalar and vector products and state two properties of each. [4]
 (b) Given $\vec{A} = 4\hat{i} + 3\hat{j}$ and $\vec{B} = 2\hat{i} + 5\hat{j}$, find $\vec{A} \cdot \vec{B}$, $\vec{A} \times \vec{B}$, the angle between the vectors, and the area of the triangle having $\vec{A}$ and $\vec{B}$ as two of its sides. [4]

ANSWERS TO GROUP C

1. (b) $R^2 = P^2 + Q^2 + 2PQ\cos\theta$ increases with $\cos\theta$, which is greatest (+1) at $\theta = 0^\circ$, giving $R_{\max} = P + Q$, and least (-1) at $\theta = 180^\circ$, giving $R_{\min} = P - Q$. Then $P + Q = 14$ and $P - Q = 2$, so $P = 8$ N and $Q = 6$ N.

2. (b) $\vec{A} \cdot \vec{B} = (4)(2) + (3)(5) = 23$. $\vec{A} \times \vec{B} = (4 \times 5 - 3 \times 2)\hat{k} = 14\hat{k}$, of magnitude 14. $A = 5$, $B = \sqrt{29} = 5.385$, so $\cos \theta = 23/26.93 = 0.854$ and $\theta = 31.3^\circ$. Area of the triangle $= \frac{1}{2}(14) = 7$ square units.



UNIT 3

Kinematics

Mechanics • 5 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Instantaneous velocity and acceleration
- Relative velocity
- Equation of motion (graphical treatment)
- Motion of a freely falling body
- Projectile motion and its applications

Kinematics is the description of motion — where a body is, how fast it moves and how that speed changes — without asking why it moves. The "why" is dynamics, the next unit. Everything here rests on three quantities: displacement, velocity and acceleration, and on the two operations that connect them, differentiation and integration.

WHY THE GRAPHS MATTER

Almost every Group A and Group B kinematics question in recent NEB papers is either a graph-reading question or a projectile question. Learn to read a motion graph and half the unit is already done.

3.1 Instantaneous velocity and acceleration

If a particle moves along the x -axis and is at x_1 at time t_1 and at x_2 at t_2 , its **average velocity** over that interval is

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$

Average velocity tells you nothing about what happened inside the interval. To get the velocity at one instant we shrink the interval to zero. The limit is the **instantaneous velocity**:

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

INSTANTANEOUS VELOCITY

The instantaneous velocity of a particle is the first derivative of its position with respect to time. Its magnitude is the instantaneous speed, and its direction is the direction of motion at that instant — i.e. along the tangent to the path.

Geometrically, on a displacement–time graph the average velocity is the slope of the **chord** joining two points, and the instantaneous velocity is the slope of the **tangent** at a point. As the second point slides back towards the first, the chord rotates into the tangent.

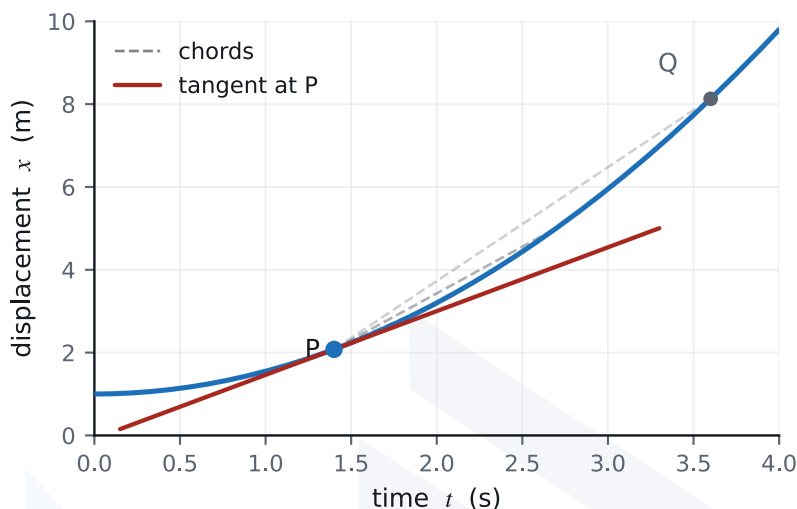


Figure 8. As Δt shrinks, the chord PQ rotates into the tangent at P. The slope of the tangent is the instantaneous velocity.

In the same way, **acceleration** is the rate of change of velocity:

$$a_{av} = \frac{\Delta v}{\Delta t}, \quad a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

THE THIRD FORM OF ACCELERATION, $a = v \frac{dv}{dx}$

We start from the definition $a = dv/dt$ and reach a form of acceleration with no time in it at all — useful whenever a question gives you velocity as a function of position.

Step 1 — the definition. Acceleration is the rate of change of velocity with time:

$$a = \frac{dv}{dt}$$

Step 2 — bring in x . The velocity v changes with time only because the particle has moved to a new position x , and x itself changes with time. So v depends on t through x . The chain rule of differentiation lets us insert dx on the top and bottom:

$$a = \frac{dv}{dx} \cdot \frac{dx}{dt}$$

(Think of it as $\frac{dv}{dt} = \frac{dv}{dx} \times \frac{dx}{dt}$ — the two dx factors formally cancel.)

Step 3 — recognise the second factor. By definition dx/dt is the velocity:

$$\frac{dx}{dt} = v$$

Step 4 — substitute. Replacing dx/dt by v :

$$a = v \frac{dv}{dx}$$

Result.

$$a = \frac{dv}{dt} = v \frac{dv}{dx}$$

What it means. Both forms describe the same acceleration. The first is "how fast does the speed change each second", the second is "how fast does the speed change each metre, multiplied by how many metres you cover per second".

Condition used. The particle must move along a single line (one dimension) and v must be a well-defined function of x . If the particle turns back and passes the same x at two different speeds, you must treat each part of the journey separately.

THE EXAMINER IS LOOKING FOR

For "show that $a = v \, dv/dx$ ": start from $a = dv/dt$, write the chain rule step explicitly, state that $dx/dt = v$, then substitute. Three lines, three marks — do not skip the chain-rule line.

WHEN TO USE EACH FORM

Use $a = dv/dt$ when the data is velocity against **time**; use $a = v \, dv/dx$ when it is velocity against **position**. Choosing the wrong one turns a two-line problem into a page of algebra.

Quantity	Defining relation	SI unit	Dimensional formula
Displacement	$\vec{s} = \vec{r}_2 - \vec{r}_1$	m	$[L]$
Velocity	$\vec{v} = d\vec{r}/dt$	m s^{-1}	$[LT^{-1}]$
Acceleration	$\vec{a} = d\vec{v}/dt$	m s^{-2}	$[LT^{-2}]$

WORKED EXAMPLE 3.1

Problem. A particle moves so that $x = 3t^3 - 4t^2 + 2t$ (metres, seconds). Find (a) its velocity and acceleration at $t = 2$ s, and (b) the time at which the acceleration is zero.

Solution.

(a) Differentiating once and twice,

$$v = \frac{dx}{dt} = 9t^2 - 8t + 2, \quad a = \frac{dv}{dt} = 18t - 8$$

At $t = 2$ s: $v = 9(4) - 8(2) + 2 = 22 \text{ m s}^{-1}$ and $a = 18(2) - 8 = 28 \text{ m s}^{-2}$.

(b) Setting $a = 0$ gives $18t - 8 = 0$, so $t = 4/9 \approx 0.44$ s.

3.2 Relative velocity

All velocity is measured with respect to some frame. If two bodies A and B have velocities $\vec{v}_A$ and $\vec{v}_B$ measured in the same frame, the velocity of A **relative to** B is

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

Read the subscripts as "A with respect to B". Reversing them reverses the vector: $\vec{v}_{BA} = -\vec{v}_{AB}$.

For motion along one line the vector subtraction is just a signed subtraction. In two dimensions, subtract the components, or use the parallelogram rule with $-\vec{v}_B$. If θ is the angle between $\vec{v}_A$ and $\vec{v}_B$, the magnitude is

$$v_{AB} = \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos \theta}$$

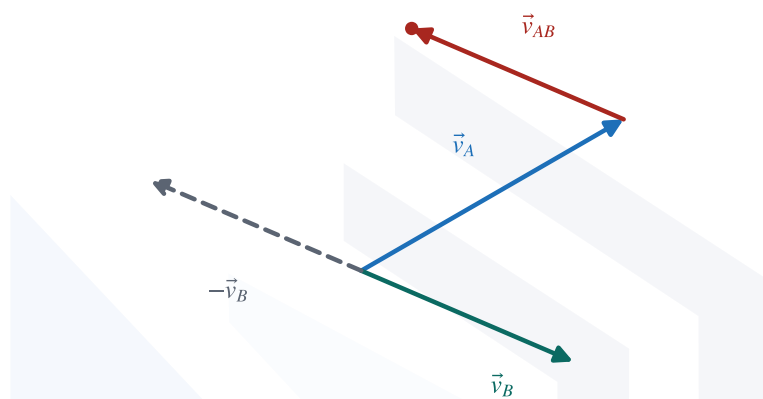


Figure 9. Relative velocity of A with respect to B is the vector $\vec{v}_A + (-\vec{v}_B)$.

WORKED EXAMPLE 3.2

Problem. Rain falls vertically at 4 m s^{-1} . A cyclist rides due east at 3 m s^{-1} . At what angle to the vertical must the cyclist tilt an umbrella?

Solution. The rain's velocity relative to the cyclist is $\vec{v}_{rc} = \vec{v}_r - \vec{v}_c$. Taking east as $+x$ and up as $+y$, $\vec{v}_r = (0, -4)$ and $\vec{v}_c = (3, 0)$, so $\vec{v}_{rc} = (-3, -4)$.

Its magnitude is $\sqrt{3^2 + 4^2} = 5 \text{ m s}^{-1}$ and it makes an angle θ with the vertical given by $\tan \theta = 3/4$, so $\theta = 36.87^\circ$. The umbrella must be tilted about 37° **forward**, towards the east.

THE COMMONEST RELATIVE-VELOCITY ERROR

Students subtract the speeds instead of the vectors. Velocity is a vector: always resolve into components, subtract componentwise, then recombine.

3.3 Equations of motion — graphical treatment

For **uniform acceleration** a , with initial velocity u and velocity v after time t , the velocity–time graph is a straight line of slope a and intercept u .

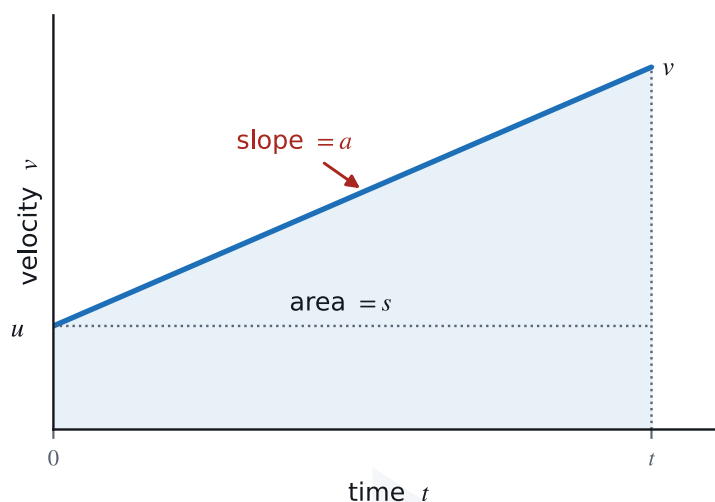


Figure 10. Velocity-time graph for uniform acceleration. The slope gives a ; the shaded area gives the displacement s .

THE THREE EQUATIONS OF MOTION FROM A VELOCITY-TIME GRAPH

We start from just two facts about a velocity-time graph — its **slope** is the acceleration and the **area** under it is the displacement — and we reach all three equations of uniformly accelerated motion.

Read the graph above: at $t=0$ the velocity is u (the intercept), at time t it is v , and because the acceleration is constant the line is straight.

First equation: $v = u + at$

Step 1. The acceleration is the slope of the straight line. Slope means "rise divided by run". The rise is the velocity gained, $v - u$; the run is the time taken, $t - 0 = t$:

$$a = \frac{v-u}{t}$$

Step 2. Multiply both sides by t to clear the fraction:

$$at = v - u$$

Step 3. Add u to both sides to make v the subject:

$$v = u + at$$

Second equation: $s = ut + \frac{1}{2}at^2$

Step 1. The displacement is the area between the line and the time axis. That shape is a **trapezium**. Cut it with a horizontal line at height u : below the cut is a rectangle, above it a triangle.

$$s = \text{area of rectangle} + \text{area of triangle}$$

Step 2. The rectangle has height u and width t , so its area is length $\times$ breadth:

$$\text{area of rectangle} = u \times t = ut$$

Step 3. The triangle has base t and height $(v - u)$, and the area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$:

$$\text{area of triangle} = \frac{1}{2} t (v - u)$$

Step 4. From the first equation, $v - u = at$. Substitute it:

$$\text{area of triangle} = \frac{1}{2} t (at) = \frac{1}{2} at^2$$

Step 5. Add the two areas:

$$s = ut + \frac{1}{2} at^2$$

Third equation: $v^2 = u^2 + 2as$

Step 1. Take the trapezium area in one go. The area of a trapezium is $\frac{1}{2} \times (\text{sum of the two parallel sides}) \times (\text{distance between them})$. The parallel sides are u and v , and the distance between them is t :

$$s = \frac{1}{2}(u + v) t$$

Step 2. We want no t in the answer, so get t from the first equation. Starting from $v = u + at$, subtract u and divide by a :

$$t = \frac{v - u}{a}$$

Step 3. Substitute this t into the expression for s :

$$s = \frac{1}{2}(u + v) \cdot \frac{v - u}{a}$$

Step 4. Multiply the two brackets. $(u + v)(v - u) = v^2 - u^2$, the difference-of-squares identity:

$$s = \frac{v^2 - u^2}{2a}$$

Step 5. Multiply both sides by $2a$:

$$2as = v^2 - u^2$$

Step 6. Add u^2 to both sides:

$$v^2 = u^2 + 2as$$

Result.

$$v = u + at, \quad s = ut + \frac{1}{2} at^2, \quad v^2 = u^2 + 2as$$

What it means. The first links velocity to time, the second links position to time, and the third links velocity to position with no time involved — so you pick the one that avoids the quantity you were not given.

Condition used. All three need the acceleration a to be **constant**; that is what makes the graph a straight line. They also need motion in a **straight line**, with one fixed positive direction chosen before you start. They do not apply to circular motion, or to a car whose acceleration is changing.

THE EXAMINER IS LOOKING FOR

A 5-mark "derive the three equations graphically" answer must contain: (i) a labelled $v-t$ graph with u , v , t marked; (ii) the sentence "slope = acceleration" and the sentence "area = displacement"; (iii) the algebra for each of the three, with the trapezium split into rectangle + triangle shown; (iv) the substitution $v - u = at$ in the second, and $t = (v - u)/a$ in the third. No graph means you cannot score the "graphical treatment" marks even if the algebra is perfect.

A fourth relation gives the distance travelled in the n^{th} second:

$$s_n = u + \frac{a}{2}(2n - 1)$$

s_n IS NOT A DISTANCE PER UNIT TIME

s_n is the distance covered **during** the n^{th} second, i.e. between $t = n - 1$ and $t = n$. Its unit is the metre, not m s^{-1} , even though the formula looks like a velocity.

Graph	Slope represents	Area under it represents
Displacement-time	velocity	— (no physical meaning)
Velocity-time	acceleration	displacement
Acceleration-time	jerk (not in syllabus)	change in velocity

3.4 Motion of a freely falling body

A body falling freely near the Earth's surface, with air resistance neglected, has constant acceleration $g \approx 9.8 \text{ m s}^{-2}$ directed downwards. The three equations apply unchanged with a replaced by g — you only have to be consistent about signs.

A SIGN CONVENTION THAT NEVER FAILS

Fix **upward as positive** before you write anything. Then $a = -g$ always, an upward throw has $u > 0$, and a downward throw has $u < 0$. Never change the convention halfway through a problem.

For a body projected vertically upwards with speed u :

$$t_{\text{up}} = \frac{u}{g}, \quad H_{\text{max}} = \frac{u^2}{2g}, \quad t_{\text{total}} = \frac{2u}{g}$$

and it returns to the point of projection with speed u — equal in magnitude, opposite in direction. Time of ascent equals time of descent.

WORKED EXAMPLE 3.3

Problem. A stone is dropped from a balloon rising at 5 m s^{-1} when the balloon is at 60 m . How long does the stone take to reach the ground? Take $g = 10 \text{ m s}^{-2}$.

Solution. The stone starts with the balloon's velocity: $u = +5 \text{ m s}^{-1}$ (upward positive). Displacement to the ground is $s = -60 \text{ m}$, and $a = -10 \text{ m s}^{-2}$.

$$s = ut + \frac{1}{2}at^2 \Rightarrow -60 = 5t - 5t^2$$

So $5t^2 - 5t - 60 = 0$, i.e. $t^2 - t - 12 = 0$, giving $(t - 4)(t + 3) = 0$. Taking the positive root, $t = 4 \text{ s}$.

Note the stone first rises for 0.5 s — forgetting its initial upward velocity is the standard trap in this question.

3.5 Projectile motion and its applications

A projectile is a body given an initial velocity and then left to move under gravity alone. The whole subject reduces to one idea:

THE INDEPENDENCE OF HORIZONTAL AND VERTICAL MOTION

The horizontal and vertical motions are completely independent. Horizontally there is no acceleration, so x -motion is uniform; vertically the acceleration is $-g$, so y -motion is uniformly accelerated. Solve them separately and link them only through the common time t .

THE PATH OF A PROJECTILE IS A PARABOLA

We start from the two separate equations of motion, one for the horizontal direction and one for the vertical, and we reach a single equation linking y to x with no time in it. Its shape is what we want to identify.

Step 1 — split the initial velocity. A speed u at angle θ to the horizontal has a horizontal part and a vertical part, found by resolving:

$$u_x = u \cos \theta, \quad u_y = u \sin \theta$$

Step 2 — horizontal motion. There is no horizontal force (we ignore air resistance), so there is no horizontal acceleration and the horizontal velocity stays $u \cos \theta$ for ever. Distance = speed $\times$ time:

$$x = (u \cos \theta) t$$

Step 3 — vertical motion. Vertically the acceleration is g downwards. Using $s = ut + \frac{1}{2}at^2$ with $u \rightarrow u \sin \theta$ and $a \rightarrow -g$ (taking up as positive):

$$y = (u \sin \theta) t - \frac{1}{2}gt^2$$

Step 4 — make t the subject of the horizontal equation. Divide both sides of Step 2 by $u \cos \theta$:

$$t = \frac{x}{u \cos \theta}$$

Step 5 — substitute this t into the vertical equation. This is legal because it is the same t in both — the two motions share one clock:

$$y = (u \sin \theta) \left(\frac{x}{u \cos \theta} \right) - \frac{1}{2} g \left(\frac{x}{u \cos \theta} \right)^2$$

Step 6 — tidy the first term. The u cancels, and $\sin \theta / \cos \theta = \tan \theta$:

$$y = x \tan \theta - \frac{1}{2} g \left(\frac{x}{u \cos \theta} \right)^2$$

Step 7 — square the bracket in the second term. Squaring top and bottom:

$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$$

Step 8 — read off the shape. For a given launch, u and θ are fixed numbers, so we may write $A = \tan \theta$ and $B = g/(2u^2 \cos^2 \theta)$, both constants:

$$y = Ax - Bx^2$$

Result. The trajectory equation is

$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$$

and since y is a quadratic in x with a negative x^2 coefficient, the path is a **parabola opening downwards**.

What it means. Gravity pulls the projectile below the straight line it would have followed, and the amount it falls behind grows as t^2 — that x^2 term is exactly the free fall the body would have had.

Assumptions used. Air resistance is neglected, g is constant in size and direction over the flight, and the Earth's curvature and rotation are ignored.

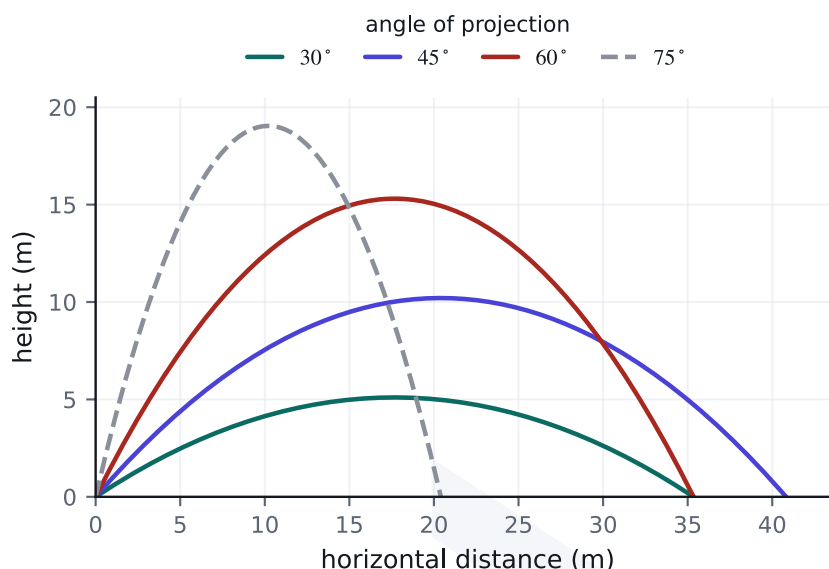


Figure 11. Trajectories for a fixed speed at several angles. Complementary angles (30° and 60°) give the same range; 45° gives the maximum.

TIME OF FLIGHT, MAXIMUM HEIGHT AND HORIZONTAL RANGE

We start from the same two motion equations, $x = (u \cos \theta)t$ and $y = (u \sin \theta)t - \frac{1}{2}gt^2$, and reach the three numbers a question always asks for: how long the flight lasts, how high it goes, and how far it lands.

Time of flight T

Step 1. The projectile is back on the ground when its height is zero again, so put $y = 0$ and $t = T$:

$$0 = (u \sin \theta)T - \frac{1}{2}gT^2$$

Step 2. Take out the common factor T :

$$0 = T(u \sin \theta - \frac{1}{2}gT)$$

Step 3. A product is zero when one factor is zero. So either $T = 0$ (that is the launch instant, which we do not want) or the bracket is zero:

$$u \sin \theta - \frac{1}{2}gT = 0$$

Step 4. Move the second term across:

$$\frac{1}{2}gT = u \sin \theta$$

Step 5. Multiply both sides by 2 and divide by g :

$$T = \frac{2u \sin \theta}{g}$$

Maximum height H

Step 1. At the highest point the projectile has stopped rising and has not yet started falling, so its **vertical** velocity is zero there. Apply $v^2 = u^2 + 2as$ to the vertical direction only, with initial vertical velocity $u \sin \theta$, final vertical velocity 0, acceleration $-g$ and displacement H :

$$0^2 = (u \sin \theta)^2 + 2(-g)H$$

Step 2. Write out the squares and the product:

$$0 = u^2 \sin^2 \theta - 2gH$$

Step 3. Move $2gH$ to the left:

$$2gH = u^2 \sin^2 \theta$$

Step 4. Divide both sides by $2g$:

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Horizontal range R

Step 1. Horizontally there is no acceleration, so the range is just the constant horizontal speed multiplied by the whole time of flight:

$$R = (u \cos \theta) T$$

Step 2. Put in $T = 2u \sin \theta / g$ from above:

$$R = (u \cos \theta) \cdot \frac{2u \sin \theta}{g}$$

Step 3. Multiply the two u factors together:

$$R = \frac{2u^2 \sin \theta \cos \theta}{g}$$

Step 4. Use the double-angle identity $2 \sin \theta \cos \theta = \sin 2\theta$:

$$R = \frac{u^2 \sin 2\theta}{g}$$

Maximum range

Step 1. In $R = u^2 \sin 2\theta / g$, only $\sin 2\theta$ can change if u is fixed. The largest value of a sine is 1:

$$R_{\max} = \frac{u^2 \times 1}{g} = \frac{u^2}{g}$$

Step 2. This happens when $\sin 2\theta = 1$, i.e. when $2\theta = 90^\circ$:

$$\theta = 45^\circ$$

Result.

$$T = \frac{2u \sin \theta}{g}, \quad H = \frac{u^2 \sin^2 \theta}{2g}, \quad R = \frac{u^2 \sin 2\theta}{g}, \quad R_{\max} = \frac{u^2}{g} \text{ at } 45^\circ$$

What it means. Time of flight and height depend only on the **vertical** component $u \sin \theta$; the range needs both components, which is why there is a best compromise angle of 45° . Note also $T = 2 \times$ (time to reach the top), so rise time equals fall time.

Assumptions used. No air resistance, constant g , and — important — the landing point is at the **same height** as the launch point. If the ground is lower (a ball thrown off a cliff), T and R must be recalculated from $y = -h$ instead of $y = 0$.

THE EXAMINER IS LOOKING FOR

(i) Resolve u into $u \cos \theta$ and $u \sin \theta$ and say the two motions are independent; (ii) for T , set $y = 0$ and reject the root $t = 0$ with a reason; (iii) for H , state that the vertical velocity is zero at the top; (iv) for R , use $R = u \cos \theta \times T$ and show the $2 \sin \theta \cos \theta = \sin 2\theta$ step; (v) state the same-level assumption. The rejected root and the double-angle step are the two places marks are usually dropped.

Three consequences follow at once and are examined constantly:

- R is maximum when $\sin 2\theta = 1$, i.e. $\theta = 45^\circ$, giving $R_{\max} = u^2/g$.
- Angles θ and $(90^\circ - \theta)$ give the **same range**, because $\sin 2\theta = \sin(180^\circ - 2\theta)$. This is why 30° and 60° land together.
- At the highest point the velocity is **not zero** — it is horizontal, of magnitude $u \cos \theta$.

VELOCITY AT THE TOP

"The velocity at the maximum height is zero" is wrong for a projectile. Only the vertical component vanishes. The speed there is $u \cos \theta$.

Horizontal projection. If a body is projected horizontally with speed u from height h , then $\theta = 0$ and

$$t = \sqrt{\frac{2h}{g}}, \quad R = u \sqrt{\frac{2h}{g}}, \quad v = \sqrt{u^2 + 2gh}$$

The time of fall does not depend on u : a bullet fired horizontally and one dropped from the same height hit the ground together.

WORKED EXAMPLE 3.4

Problem. A ball is thrown at 25 m s^{-1} at 53° to the horizontal. Taking $g = 10 \text{ m s}^{-2}$, $\sin 53^\circ = 0.8$ and $\cos 53^\circ = 0.6$, find the time of flight, maximum height, range, and the speed at the highest point.

Solution. Components: $u_x = 25(0.6) = 15 \text{ m s}^{-1}$, $u_y = 25(0.8) = 20 \text{ m s}^{-1}$.

$$T = \frac{2u_y}{g} = \frac{40}{10} = 4 \text{ s}$$

$$H = \frac{u_y^2}{2g} = \frac{400}{20} = 20 \text{ m}$$

$$R = u_x T = 15 \times 4 = 60 \text{ m}$$

At the highest point the velocity is purely horizontal, so the speed is $u_x = 15 \text{ m s}^{-1}$.

✓ Chapter summary

- Instantaneous velocity is $v = dx/dt$ (slope of the tangent to the $x-t$ graph); acceleration is $a = dv/dt = v \, dv/dx$.
- On a $v-t$ graph, slope is acceleration and area is displacement.
- For uniform acceleration: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$, $s_n = u + \frac{a}{2}(2n-1)$.
- Free fall is uniform acceleration with $a = -g$ (upward positive). Time up equals time down; the landing speed equals the launch speed.
- Projectile motion = uniform horizontal motion + free vertical fall. The path is a parabola, $T = 2u \sin \theta / g$, $H = u^2 \sin^2 \theta / 2g$, $R = u^2 \sin 2\theta / g$.
- Range is maximum at 45° ; complementary angles give equal ranges; the speed at the top is $u \cos \theta$, not zero.

Practice questions

Group A — Multiple choice (1 mark each)

- The slope of a displacement-time graph gives [1]
 - a** acceleration
 - b** velocity
 - c** distance
 - d** jerk
- A projectile has maximum range when the angle of projection is [1]
 - a** 30°
 - b** 45°
 - c** 60°
 - d** 90°
- At the highest point of its path a projectile's velocity is [1]
 - a** zero
 - b** vertical
 - c** horizontal
 - d** along the tangent to the vertical

ANSWERS TO GROUP A

- (b) — slope of $x-t$ is $dx/dt = v$.
- (b) — $R = u^2 \sin 2\theta / g$ is maximum when $2\theta = 90^\circ$.
- (c) — only the vertical component vanishes; the horizontal component $u \cos \theta$ survives.

Group B — Short answer (5 marks each)

- Define instantaneous velocity and acceleration. Show that $a = v \frac{dv}{dx}$. [5]
- Derive the three equations of uniformly accelerated motion from a velocity-time graph. [5]

3. A car accelerates uniformly from rest and covers 100 m in 5 s. Find its acceleration and the distance covered in the 5th second. [5]
4. Rain falls vertically at 6 m s^{-1} while a man walks east at 8 m s^{-1} . Find the magnitude and direction of the rain relative to the man. [5]

ANSWERS TO GROUP B

3. From $s = ut + \frac{1}{2}at^2$ with $u = 0$: $100 = \frac{1}{2}a(25)$, so $a = 8 \text{ m s}^{-2}$. Then $s_5 = u + \frac{a}{2}(2 \times 5 - 1) = 0 + 4(9) = 36 \text{ m}$.
4. $\vec{v}_{rm} = (0, -6) - (8, 0) = (-8, -6)$; magnitude $\sqrt{64 + 36} = 10 \text{ m s}^{-1}$, at $\tan^{-1}(8/6) = 53.1^\circ$ to the vertical, tilted towards the west (i.e. the man must tilt his umbrella 53° forward, to the east).

Group C — Long answer (8 marks each)

1. (a) Show that the trajectory of a projectile is a parabola. [4]
 (b) Derive expressions for its time of flight, maximum height and horizontal range, and hence show that the range is maximum at 45° . [4]
2. A body is projected at $u = 40 \text{ m s}^{-1}$ at 30° from the top of a tower 50 m high. Taking $g = 10 \text{ m s}^{-2}$, find
 (a) the time to reach the ground,
 (b) the horizontal distance from the foot of the tower, and
 (c) the speed on landing. [8]

ANSWER TO GROUP C QUESTION 2

$$u_x = 40 \cos 30^\circ = 34.64 \text{ m s}^{-1}, u_y = 40 \sin 30^\circ = 20 \text{ m s}^{-1}.$$

(a) Taking upward positive, $-50 = 20t - 5t^2$, so $5t^2 - 20t - 50 = 0$, i.e. $t^2 - 4t - 10 = 0$, giving $t = 2 + \sqrt{14} = 5.74 \text{ s}$.

(b) $x = u_x t = 34.64 \times 5.74 = 198.8 \text{ m}$.

(c) $v_y = u_y - gt = 20 - 57.4 = -37.4 \text{ m s}^{-1}$, so $v = \sqrt{34.64^2 + 37.4^2} = 51.0 \text{ m s}^{-1}$.

UNIT 4

Dynamics

Mechanics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Linear momentum, Impulse
- Conservation of linear momentum
- Application of Newton's laws
- Moment, torque and equilibrium
- Solid friction: Laws of solid friction and their verifications

Kinematics described motion; dynamics explains it. The explanation lies in Newton's three laws, and the quantity that makes them work is **linear momentum**. From it come force and impulse; from the third law comes the conservation of momentum; from careful bookkeeping of forces come the free-body diagrams that solve most mechanics problems. The unit closes with two effects present in every real machine — torque and friction.

HOW TO ATTACK ANY DYNAMICS PROBLEM

Draw the body alone. Mark every force acting **on** it. Choose axes. Write $\sum F = ma$ along each axis and, if the body can turn, $\sum \tau = 0$ about a convenient point. Most marks in this unit are earned by that sequence.

4.1 Linear momentum, Impulse

A body stays at rest or in uniform motion unless an external force acts — **Newton's first law**. It defines force as that which changes the state of motion, and names the resistance to that change **inertia**, measured by mass.

LINEAR MOMENTUM

The linear momentum of a body is the product of its mass and its velocity,

$$\vec{p} = m\vec{v}$$

It is a vector along the velocity. Its SI unit is kg m s^{-1} (equivalently N s) and its dimensional formula is $[MLT^{-1}]$.

Newton's second law states that the rate of change of linear momentum of a body is directly proportional to the applied force and takes place in the direction of the force:

$$\vec{F} \propto \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt}$$

$F = ma$ **FROM NEWTON'S SECOND LAW**

We start from the statement of the second law — force is proportional to the rate of change of momentum — and reach the familiar working formula $F = ma$.

Step 1 — write the law as an equation. "Proportional to" means "equal to, apart from a constant". Call that constant k :

$$F = k \frac{dp}{dt}$$

Step 2 — put in the definition of momentum, $p = mv$:

$$F = k \frac{d(mv)}{dt}$$

Step 3 — take the mass outside the differentiation. This step needs the mass to be **constant**; a constant multiplier can always be moved outside a derivative:

$$F = k m \frac{dv}{dt}$$

Step 4 — recognise the derivative. By definition dv/dt is the acceleration a :

$$F = k m a$$

Step 5 — fix the constant k by choosing the unit of force. We define one newton as the force that gives a mass of 1 kg an acceleration of 1 m s⁻². Putting $F = 1$, $m = 1$, $a = 1$ into Step 4:

$$1 = k \times 1 \times 1 \implies k = 1$$

Step 6 — substitute $k = 1$:

$$F = ma$$

Result. $\vec{F} = m\vec{a}$, with the acceleration in the same direction as the force.

What it means. The same force gives a small mass a big acceleration and a big mass a small one — mass is the measure of inertia.

Condition used. Step 3 assumed the **mass does not change**. For a rocket burning fuel or a rain-drop collecting water, go back to $F = dp/dt$ and differentiate the product properly. The law also holds only in an inertial (non-accelerating) frame.

Impulse

IMPULSE

Impulse is the product of a force and the time for which it acts. For a constant force $\vec{J} = \vec{F} \Delta t$; for a varying force,

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt$$

IMPULSE-MOMENTUM THEOREM: $J = mv - mu$

We start from Newton's second law in its momentum form and reach the statement that the impulse of a force equals the change in momentum it produces.

Step 1 — the second law. Force is the rate of change of momentum:

$$\vec{F} = \frac{d\vec{p}}{dt}$$

Step 2 — multiply both sides by dt . This just moves dt to the other side, so that we have "force times a tiny time" on the left:

$$\vec{F} dt = d\vec{p}$$

Step 3 — add up all the tiny bits. The contact lasts from t_1 to t_2 , during which the momentum goes from $\vec{p}_1$ to $\vec{p}_2$. Adding up tiny pieces is what integration does, so we integrate both sides between matching limits:

$$\int_{t_1}^{t_2} \vec{F} dt = \int_{\vec{p}_1}^{\vec{p}_2} d\vec{p}$$

Step 4 — do the right-hand integral. Integrating $d\vec{p}$ simply gives $\vec{p}$, so we evaluate it at the two limits:

$$\int_{t_1}^{t_2} \vec{F} dt = \vec{p}_2 - \vec{p}_1$$

Step 5 — name the left-hand side. By definition, that integral is the impulse $\vec{J}$:

$$\vec{J} = \vec{p}_2 - \vec{p}_1$$

Step 6 — put in $p = mv$. With initial velocity $\vec{u}$ and final velocity $\vec{v}$:

$$\vec{J} = m\vec{v} - m\vec{u}$$

Step 7 — the constant-force version. If we replace the real changing force by its average value F_{av} acting for the whole time $\Delta t = t_2 - t_1$, the integral becomes a simple product:

$$F_{av} \Delta t = mv - mu$$

Result.

$$\vec{J} = \int_{t_1}^{t_2} \vec{F} dt = m\vec{v} - m\vec{u} = F_{av} \Delta t$$

What it means. The area under a force–time graph is the momentum handed over. For a fixed change of momentum, a longer contact time means a smaller force — which is the whole idea behind airbags, crumple zones and a cricketer drawing his hands back while catching.

Condition used. The mass is constant, and $\vec{F}$ is the **net** force on the body. Impulse is a vector, so in a rebound problem you must give u and v opposite signs.

THE EXAMINER IS LOOKING FOR

(i) $F = dp/dt$ quoted as the starting point; (ii) the step $F dt = dp$; (iii) integration with **both** sets of limits written in; (iv) the final statement in words — "impulse = change in momentum"; (v) the unit $\text{N s} = \text{kg m s}^{-1}$. In numerical parts, signs: a rebound needs v and u with opposite signs.

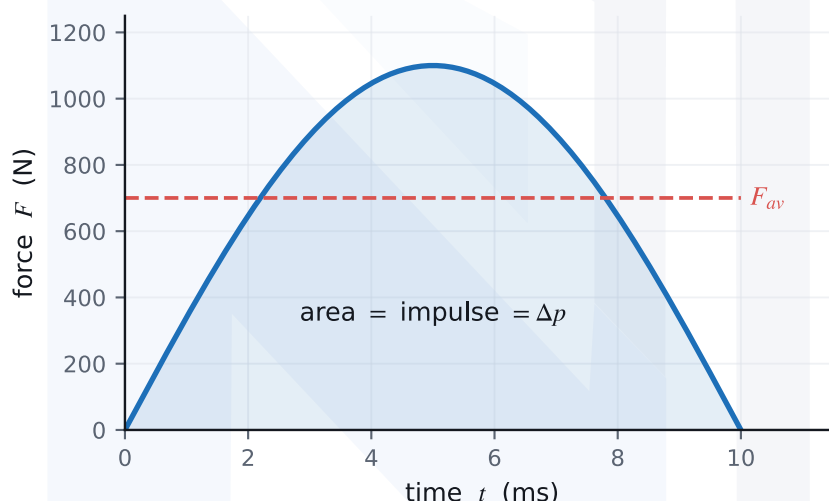


Figure 12. Force–time graph for a collision lasting 10 ms. The shaded area is the impulse; the dashed line is the constant average force that would deliver the same impulse.

WHY SAFETY DEVICES WORK

For a given collision Δp is fixed. Since $F_{av} = \Delta p / \Delta t$, the only way to cut the force is to **stretch the time** — the physics of airbags, crumple zones, helmet padding and the long-jump sand pit.

WORKED EXAMPLE 4.1

Problem. A ball of mass 0.2 kg moving at 20 m s^{-1} strikes a wall normally and rebounds at 15 m s^{-1} . If the contact lasts 0.01 s , find the impulse and the average force on the ball.

Solution. Take the direction away from the wall as positive, so $u = -20 \text{ m s}^{-1}$ and $v = +15 \text{ m s}^{-1}$.

$$J = mv - mu = 0.2 [15 - (-20)] = 0.2 \times 35 = 7 \text{ N s}$$

$$F_{av} = \frac{J}{\Delta t} = \frac{7}{0.01} = 700 \text{ N}$$

4.2 Conservation of linear momentum

Newton's third law: to every action there is an equal and opposite reaction. The two forces act on **different** bodies, along the same line, and appear and vanish together.

CONSERVATION OF LINEAR MOMENTUM FROM NEWTON'S THIRD LAW

We start from Newton's third law applied to two colliding bodies and reach the statement that their total momentum is the same before and after the collision.

Setting up. Two smooth spheres A and B of masses m_1 and m_2 move along the same straight line with velocities u_1 and u_2 , where $u_1 > u_2$ so that A catches up with B. They touch for a short time t and then separate with velocities v_1 and v_2 . No outside force (no friction, no push) acts on the pair.

Step 1 — name the two contact forces. While they touch, B pushes A with a force F_{12} and A pushes B with a force F_{21} .

Step 2 — apply the third law. Action and reaction are equal in size and opposite in direction:

$$F_{12} = -F_{21}$$

Step 3 — write each force as a rate of change of momentum. By the second law, the force on a body equals its change in momentum divided by the time. For A the momentum changes from $m_1 u_1$ to $m_1 v_1$:

$$F_{12} = \frac{m_1 v_1 - m_1 u_1}{t}$$

and for B, from $m_2 u_2$ to $m_2 v_2$:

$$F_{21} = \frac{m_2 v_2 - m_2 u_2}{t}$$

Step 4 — substitute both into Step 2:

$$\frac{m_1 v_1 - m_1 u_1}{t} = - \frac{m_2 v_2 - m_2 u_2}{t}$$

Step 5 — multiply both sides by t . The contact time is the same for both bodies — they touch and separate together — so it cancels:

$$m_1 v_1 - m_1 u_1 = -(m_2 v_2 - m_2 u_2)$$

Step 6 — open the bracket on the right:

$$m_1 v_1 - m_1 u_1 = -m_2 v_2 + m_2 u_2$$

Step 7 — collect the "before" terms on one side and the "after" terms on the other. Add m_1u_1 to both sides and add m_2v_2 to both sides:

$$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$$

Result.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

What it means. Whatever violence happens during the contact — however the bodies squash, heat up or break — the total momentum of the pair is unchanged. Momentum is only passed from one body to the other.

Conditions used. (i) No **net external force** acts on the pair during the collision; this is why the third-law pair could cancel. (ii) The contact time is the same for both bodies. (iii) Momentum is a vector: in one dimension you must give leftward velocities a minus sign, and in two dimensions you conserve the x and y components separately. Note that kinetic energy is not assumed to be conserved — this proof works for inelastic collisions too.

THE EXAMINER IS LOOKING FOR

(i) A "before and after" diagram with $m_1, m_2, u_1, u_2, v_1, v_2$ labelled; (ii) the third-law statement $F_{12} = -F_{21}$; (iii) each force written as change of momentum over time; (iv) the cancelling of t with the reason; (v) the final equation and the sentence "total momentum before = total momentum after, provided no external force acts". The condition in (v) carries a mark on its own.

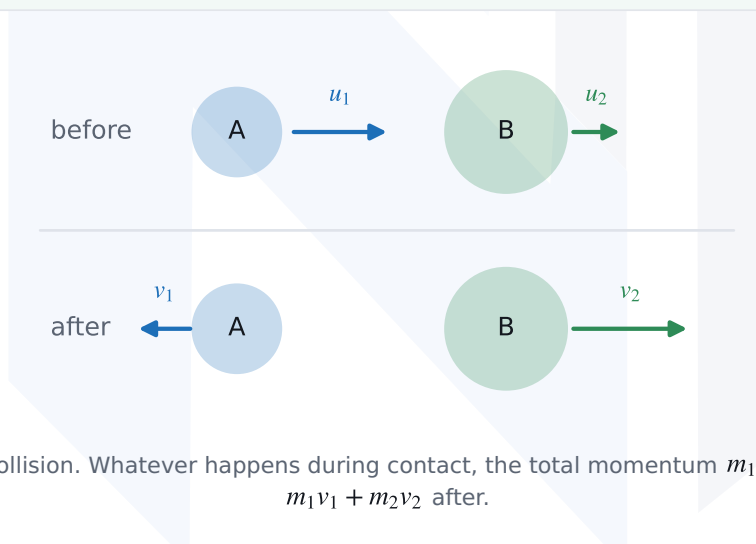


Figure 13. Head-on collision. Whatever happens during contact, the total momentum $m_1u_1 + m_2u_2$ before equals $m_1v_1 + m_2v_2$ after.

It also follows directly from $\vec{F} = d\vec{p}/dt$: zero net external force means $d\vec{p}/dt = 0$.

PRINCIPLE OF CONSERVATION OF LINEAR MOMENTUM

If no net external force acts on a system, its total linear momentum stays constant in magnitude and direction, however the bodies inside it interact.

Situation	How momentum is conserved
Recoil of a gun	forward momentum of bullet = backward momentum of gun
Explosion of a shell at rest	vector sum of the fragment momenta is zero
Rocket and jet propulsion	gases ejected backwards; rocket gains equal forward momentum

ACTION AND REACTION NEVER CANCEL

The two third-law forces act on **different bodies**, so they never add to zero on one body. A horse pulls a cart because the forward force on the cart and the backward force on the horse act on different objects.

WORKED EXAMPLE 4.2

Problem. A bullet of mass 20 g is fired from a gun of mass 4 kg with a muzzle velocity of 400 m s^{-1} . Find the recoil velocity of the gun.

Solution. Before firing the system is at rest, so the total momentum is zero. With V the recoil velocity,

$$0 = m_b v_b + m_g V = (0.020)(400) + 4V \Rightarrow V = -\frac{8}{4} = -2 \text{ m s}^{-1}$$

The gun recoils at 2 m s^{-1} , opposite to the bullet and far slower because it is 200 times more massive.

4.3 Application of Newton's laws

The routine is always the same: isolate one body, draw its **free-body diagram**, choose axes along and across the motion, and write $\sum F = ma$ for each axis.

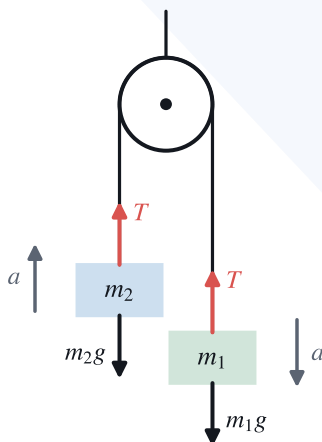


Figure 14. Atwood machine. Isolating each mass gives $m_1 g - T = m_1 a$ and $T - m_2 g = m_2 a$.

ACCELERATION AND TENSION IN AN ATWOOD MACHINE

We start from $F = ma$ written separately for each hanging mass, and reach formulas for the acceleration of the system and the tension in the string.

Setting up. Two masses m_1 and m_2 , with $m_1 > m_2$, hang from the two ends of a light inextensible string passing over a smooth pulley. Two facts follow from those words, and they are worth marks:

- Inextensible string: the string cannot stretch, so when m_1 goes down by 1 cm, m_2 goes up by 1 cm. Both masses therefore have the **same** acceleration a .
- Light string over a smooth pulley: no force is needed to accelerate the string itself and none is lost to friction, so the tension T is the **same** everywhere in the string.

Since $m_1 > m_2$, common sense says m_1 goes down and m_2 goes up.

Step 1 — free-body equation for m_1 . Two forces act on it: its weight m_1g downwards and the tension T upwards. It accelerates downwards, so take downwards as positive for this mass. Net force = mass $\times$ acceleration:

$$m_1g - T = m_1a$$

Step 2 — free-body equation for m_2 . The same two forces act, but this mass accelerates upwards, so take upwards as positive for it:

$$T - m_2g = m_2a$$

Step 3 — add the two equations. We add them because T appears with a minus sign in one and a plus sign in the other, so it will disappear:

$$(m_1g - T) + (T - m_2g) = m_1a + m_2a$$

Step 4 — cancel T and collect terms. On the left $-T + T = 0$; on the right take out the common factor a :

$$m_1g - m_2g = (m_1 + m_2)a$$

Step 5 — take out g on the left:

$$(m_1 - m_2)g = (m_1 + m_2)a$$

Step 6 — divide both sides by $(m_1 + m_2)$ to get a alone:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

Step 7 — now find T . Go back to Step 2 and make T the subject by adding m_2g to both sides:

$$T = m_2g + m_2a = m_2(g + a)$$

Step 8 — substitute the a we just found:

$$T = m_2 \left(g + \frac{(m_1 - m_2)g}{m_1 + m_2} \right)$$

Step 9 — put the bracket over one common denominator ($m_1 + m_2$):

$$T = m_2 \left(\frac{g(m_1 + m_2) + g(m_1 - m_2)}{m_1 + m_2} \right)$$

Step 10 — expand the top line:

$$T = m_2 \left(\frac{m_1 g + m_2 g + m_1 g - m_2 g}{m_1 + m_2} \right)$$

Step 11 — the two $m_2 g$ terms cancel, leaving $2m_1 g$:

$$T = m_2 \cdot \frac{2m_1 g}{m_1 + m_2}$$

Step 12 — write it tidily:

$$T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

Result.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}, \quad T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

What it means. The unbalanced force is only the difference in weights $(m_1 - m_2)g$, but it has to move the total mass $(m_1 + m_2)$ — so the machine slows gravity down, which is exactly why Atwood used it to measure g . If $m_1 = m_2$ then $a = 0$ and $T = m_1 g$, as expected. Also note $m_2 g < T < m_1 g$: the tension always lies between the two weights.

Assumptions used. Light inextensible string, frictionless and massless pulley, no air resistance. If the pulley had mass, part of the torque would go into spinning it and a would be smaller.

Apparent weight in a lift. A weighing machine reads the normal reaction R it exerts, not the true weight.

Motion of the lift	Equation	Apparent weight R
At rest or uniform velocity	$R - mg = 0$	mg
Accelerating upward with a	$R - mg = ma$	$m(g + a)$ — feels heavier
Accelerating downward with a	$mg - R = ma$	$m(g - a)$ — feels lighter
Cable breaks, free fall $a = g$	$mg - R = mg$	0 — weightlessness

Smooth inclined plane. Resolving the weight along and perpendicular to a plane inclined at θ gives

$$mg \sin \theta = ma \Rightarrow a = g \sin \theta, \quad R = mg \cos \theta$$

WORKED EXAMPLE 4.3

Problem. Masses of 5 kg and 3 kg hang from the ends of a light string over a frictionless pulley. Find the acceleration of the system and the tension in the string. Take $g = 10 \text{ m s}^{-2}$.

Solution.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{(5 - 3)(10)}{8} = 2.5 \text{ m s}^{-2}$$

$$T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2(5)(3)(10)}{8} = 37.5 \text{ N}$$

Check: $T - m_2g = 37.5 - 30 = 7.5 \text{ N} = (3)(2.5)$. ✓

4.4 Moment, torque and equilibrium

MOMENT OF A FORCE (TORQUE)

The moment, or torque, of a force about a point is the product of the force and the perpendicular distance of its line of action from that point:

$$\tau = F \times d_{\perp} = rF \sin \theta, \quad \vec{\tau} = \vec{r} \times \vec{F}$$

Its SI unit is the newton metre (N m) and its dimensional formula is $[ML^2T^{-2}]$. It is a vector directed along the axis of rotation.

A torque is positive when it turns the body anticlockwise. A force whose line of action passes through the point has zero moment, and a force at right angles to the arm has the largest moment — hence the door handle far from the hinge.

A **couple** is a pair of equal, opposite, parallel forces on different lines. Their resultant is zero, so a couple gives pure rotation. Its moment, $\tau = F \times$ (perpendicular distance between them), is the same about **every** point.

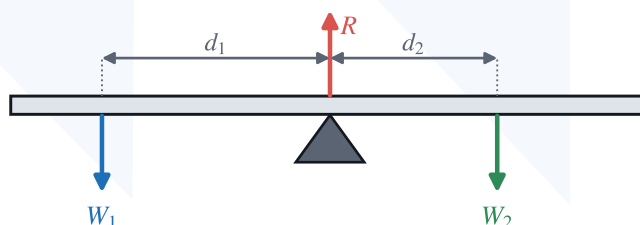


Figure 15. Principle of moments. The beam balances when $W_1d_1 = W_2d_2$; the pivot supplies the upward reaction R .

CONDITIONS FOR EQUILIBRIUM

A rigid body under coplanar forces is in equilibrium when **both**:

1. $\sum \vec{F} = 0$, i.e. $\sum F_x = 0$ and $\sum F_y = 0$ (no translation);
2. $\sum \tau = 0$ about **any** point (no rotation).

Condition 2 alone is the **principle of moments**: clockwise moments equal anticlockwise moments.

Type of equilibrium	On slight displacement	Centre of gravity	Example
Stable	returns to its original position	rises	cone on its base
Unstable	moves further away	falls	cone on its apex
Neutral	stays in the new position	height unchanged	cone on its side

WORKED EXAMPLE 4.4

Problem. A uniform plank 4 m long and of weight 200 N rests on supports at its two ends. A boy of weight 500 N stands 1 m from the left-hand support. Find the reaction at each support.

Solution. Let R_1 and R_2 be the left and right reactions. The weight of the uniform plank acts at its centre, 2 m from either end.

Vertical equilibrium: $R_1 + R_2 = 200 + 500 = 700$ N.

Taking moments about the left-hand support:

$$R_2 \times 4 = 200 \times 2 + 500 \times 1 = 900$$

$$R_2 = 225 \text{ N}, \quad R_1 = 700 - 225 = 475 \text{ N}$$

The support nearer the boy carries the larger share.

4.5 Solid friction: Laws of solid friction and their verifications**SOLID FRICTION**

Friction is the force acting along the surface of contact between two solid bodies that opposes their relative motion, or the tendency to it. It arises from interlocking and adhesion between the microscopic high points of the surfaces.

Push a resting block gently and it does not move: friction has grown to match the push exactly. This self-adjusting force is **static friction**. At some push the block just begins to slide, and the friction then is the **limiting friction**. Once sliding, the resistance settles at a smaller, nearly constant **kinetic (dynamic) friction**.

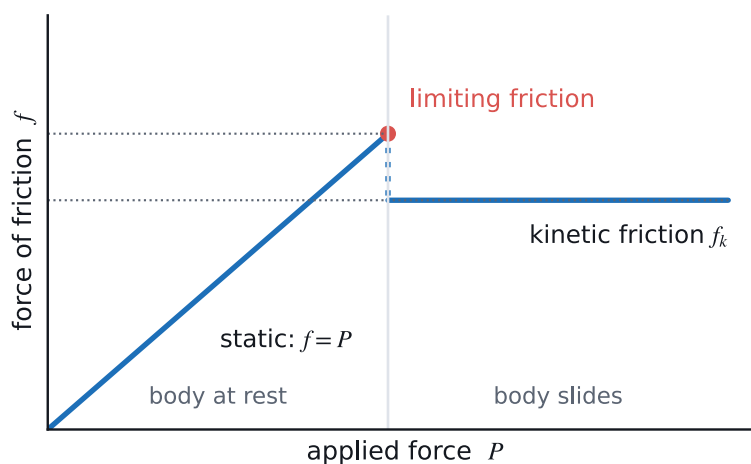


Figure 16. Friction against applied force. Static friction grows to match the applied force up to the limiting value, then drops to the smaller, roughly constant kinetic friction.

The **coefficients of friction** are ratios to the normal reaction N :

$$\mu_s = \frac{f_{\text{limiting}}}{N}, \quad \mu_k = \frac{f_k}{N}, \quad \mu_k < \mu_s$$

A ratio of two forces, μ has no units: about 0.6 for rubber on concrete, 0.3 for wood on wood, 0.04 for steel on ice.

Laws of solid friction

Static friction

1. Limiting friction is directly proportional to the normal reaction: $f_{\text{limiting}} = \mu_s N$.
2. It is independent of the apparent area of contact, provided the normal reaction is unchanged.
3. It depends on the nature, material and roughness of the two surfaces.

Kinetic friction

1. Kinetic friction is directly proportional to the normal reaction, $f_k = \mu_k N$.
2. It is independent of the apparent area of contact.
3. It is independent of the relative velocity at moderate speeds.
4. It depends on the nature of the surfaces and is always less than the limiting friction.

Angle of friction and angle of repose

The **angle of friction** λ is the angle between the normal reaction and the resultant of the normal reaction and the limiting friction, so

$$\tan \lambda = \frac{\mu_s N}{N} = \mu_s$$

The **angle of repose** θ is the smallest inclination of a plane at which a body placed on it just begins to slide.

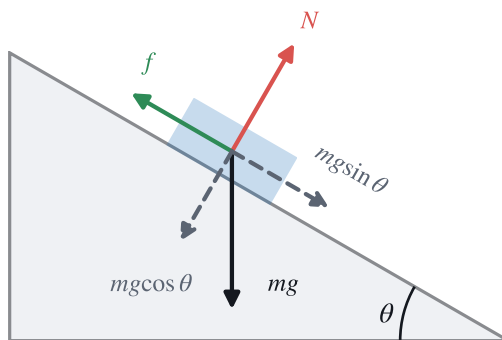


Figure 17. A block on the point of sliding down a rough plane. Along the slope $mg \sin \theta = f = \mu_s mg \cos \theta$, so $\mu_s = \tan \theta$.

THE ANGLE OF REPOSE EQUALS THE ANGLE OF FRICTION

We start from a block that is just about to slide down a rough slope, and reach the result $\mu_s = \tan \theta$ — which also proves that the angle of repose and the angle of friction are the same angle.

Setting up. A block of mass m rests on a rough plane whose tilt θ is slowly increased. At the **angle of repose** the block is on the very point of sliding: it is still in equilibrium, but the friction has reached its largest possible value, $f_{\text{limiting}} = \mu_s N$.

Choose axes along the slope and perpendicular to the slope — not horizontal and vertical. Then the only force needing to be resolved is the weight mg , which acts vertically downwards.

Step 1 — resolve the weight. The angle between the weight and the perpendicular to the slope is also θ (simple geometry of the tilted plane). So the weight splits into a part pressing into the slope and a part pulling down the slope:

$$\text{into the slope} = mg \cos \theta, \quad \text{down the slope} = mg \sin \theta$$

Step 2 — balance perpendicular to the slope. The block does not sink into the plane or lift off it, so the normal reaction N exactly balances the pressing component:

$$N = mg \cos \theta$$

Step 3 — balance along the slope. The block is not yet moving, so the friction up the slope exactly balances the component pulling it down:

$$f_{\text{limiting}} = mg \sin \theta$$

Step 4 — divide Step 3 by Step 2. Dividing one true equation by another is allowed as long as we are not dividing by zero, and $N \neq 0$ here:

$$\frac{f_{\text{limiting}}}{N} = \frac{mg \sin \theta}{mg \cos \theta}$$

Step 5 — cancel mg , which appears on top and bottom:

$$\frac{f_{\text{limiting}}}{N} = \frac{\sin \theta}{\cos \theta}$$

Step 6 — use $\sin \theta / \cos \theta = \tan \theta$:

$$\frac{f_{\text{limiting}}}{N} = \tan \theta$$

Step 7 — identify the left-hand side. By definition, limiting friction divided by normal reaction is the coefficient of static friction:

$$\mu_s = \tan \theta$$

Step 8 — compare with the angle of friction. The angle of friction λ is defined by $\mu_s = \tan \lambda$. Two tangents are equal, so for angles between 0° and 90° the angles themselves are equal:

$$\tan \theta = \tan \lambda \implies \theta = \lambda$$

Result.

$$\mu_s = \tan \theta, \quad \text{angle of repose} = \text{angle of friction}$$

What it means. mg cancelled in Step 5, so the angle of repose does **not** depend on the mass of the block — only on what the two surfaces are made of. This is why a heap of dry sand always settles to the same slope, whether the heap is small or huge, and it gives a one-line laboratory method for μ_s : tilt until it slides, then measure $\tan \theta = h/b$.

Conditions used. The block is on the **point** of sliding (so friction is at its limiting value — not before, not after), the plane is straight, and no other force such as a push or a string tension acts on the block.

THE EXAMINER IS LOOKING FOR

(i) A labelled diagram with mg , N , f and the slope angle θ marked; (ii) axes taken **along and perpendicular to** the plane; (iii) the two equilibrium equations $N = mg \cos \theta$ and $f = mg \sin \theta$; (iv) the division step; (v) the sentence "the mass cancels, so the angle of repose is independent of mass". Students who resolve horizontally and vertically instead usually get lost — resolve along the plane.

Verifying the laws

(a) Horizontal board, pulley and scale pan. A block on a horizontal board is joined by a light string passing over a smooth pulley at the edge to a scale pan. Weights are added slowly until the block just begins to move; the pan weight then equals the limiting friction.

- Add known weights **on top of the block** to change N . The graph of limiting friction against N is a straight line through the origin of slope μ_s — the first law.
- Rest the same block on a face of different area under the same load: the pan weight is unchanged — the second law.
- Change the material, or polish the board: the reading changes — the third law.

(b) Inclined-plane method. Raise one end of a hinged plane until the block just slides and measure the height h and base b : $\mu_s = \tan \theta = h/b$. Different masses give the same angle, so μ is independent of mass and area.

THREE THINGS STUDENTS GET WRONG ABOUT FRICTION

1. Friction is **not always** μN — that is only its limiting or kinetic value. A block at rest under a 5 N push feels 5 N of friction, even if $\mu N = 20$ N.
2. Friction opposes relative motion. The friction driving a walking person or a car forwards acts **forwards**.
3. μ can exceed 1, and it has no units.

Friction makes walking, gripping and braking possible, but wastes energy as heat and wears parts away; it is reduced by lubricants, bearings and polishing.

WORKED EXAMPLE 4.5

Problem. A block of mass 10 kg rests on a horizontal floor for which $\mu_s = 0.4$ and $\mu_k = 0.3$. Take $g = 10 \text{ m s}^{-2}$. (a) What least horizontal force starts it moving? (b) What is its acceleration if that force keeps acting?

Solution. The normal reaction is $N = mg = 100 \text{ N}$.

(a) $f_{\text{limiting}} = \mu_s N = 0.4 \times 100 = 40 \text{ N}$, so the least force is 40 N.

(b) Once sliding, friction falls to $f_k = \mu_k N = 0.3 \times 100 = 30 \text{ N}$:

$$ma = P - f_k = 40 - 30 = 10 \text{ N} \Rightarrow a = 1 \text{ m s}^{-2}$$

The block jerks into motion — the reason a heavy crate shoots forward the moment it starts to slide.

✓ Chapter summary

- $\vec{p} = m\vec{v}$ (kg m s^{-1} , $[MLT^{-1}]$); $\vec{F} = d\vec{p}/dt$, becoming $\vec{F} = m\vec{a}$ only for constant mass.
- Impulse $\vec{J} = \vec{F}\Delta t = \int \vec{F}dt = m\vec{v} - m\vec{u}$, the area under the $F-t$ graph; a longer Δt means a smaller F_{av} .
- $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$: momentum of an isolated system is conserved — recoil, explosions, rocket propulsion.
- Atwood: $a = (m_1 - m_2)g/(m_1 + m_2)$, $T = 2m_1m_2g/(m_1 + m_2)$; apparent weight in a lift is $m(g \pm a)$, zero in free fall.
- Torque $\tau = rF\sin\theta = \vec{r} \times \vec{F}$ (N m); a couple has moment $F \times d$ about every point.
- Equilibrium needs $\sum \vec{F} = 0$ **and** $\sum \tau = 0$ (principle of moments).
- Friction: $f_{\text{limiting}} = \mu_s N$, $f_k = \mu_k N$, $\mu_k < \mu_s$, independent of apparent area, and $\mu_s = \tan \theta = \tan \lambda$.

Practice questions

Group A — Multiple choice (1 mark each)

- The rate of change of linear momentum of a body is equal to [1]
 - impulse
 - the applied force
 - work done
 - power
- A rigid body is in complete equilibrium when [1]
 - $\sum F = 0$ only
 - $\sum \tau = 0$ only
 - both $\sum F = 0$ and $\sum \tau = 0$
 - its velocity is zero
- The coefficient of friction [1]
 - has the unit newton
 - has the unit N m^{-1}
 - is dimensionless
 - has dimension $[MLT^{-2}]$
- A body just slides down a plane inclined at 30° to the horizontal. The coefficient of static friction is [1]
 - 0.50
 - 0.58
 - 0.87
 - 1.73

ANSWERS TO GROUP A

- (b) — Newton's second law is $F = dp/dt$.
- (c) — translational and rotational equilibrium are both needed.
- (c) — it is a ratio of two forces.
- (b) — $\mu_s = \tan 30^\circ = 0.577$.

Group B — Short answer (5 marks each)

- Define linear momentum and impulse. Show that the impulse of a force equals the change of momentum it produces, and explain why a long-jump pit is filled with sand. [5]
- State Newton's third law and hence prove the principle of conservation of linear momentum for two bodies colliding head-on. [5]
- A bullet of mass 10 g is fired from a gun of mass 5 kg with a muzzle velocity of 300 m s^{-1} . Calculate the recoil velocity of the gun. [5]
- Two masses of 6 kg and 4 kg hang from the ends of a light string over a smooth pulley. Find the acceleration of the system and the tension in the string. Take $g = 10 \text{ m s}^{-2}$. [5]
- A uniform metre rule of weight 2 N balances on a knife edge at the 30 cm mark when a weight W hangs at the 10 cm mark. Find W and the reaction at the knife edge. [5]

ANSWERS TO GROUP B

- Outline: state both definitions, derive $J = mv - mu$ from $F = dp/dt$, and note that sand lengthens Δt so $F_{av} = \Delta p / \Delta t$ falls.
- Outline: as in the derivation in §4.2.
- Total momentum before firing is zero, so $0 = (0.010)(300) + 5V$, giving $V = -0.6 \text{ m s}^{-1}$: the gun recoils at 0.6 m s^{-1} .

4. $a = \frac{(6-4)(10)}{6+4} = 2 \text{ m s}^{-2}$ and $T = \frac{2(6)(4)(10)}{10} = 48 \text{ N}$.

5. The weight of the uniform rule acts at the 50 cm mark, 20 cm right of the knife edge, while W hangs 20 cm to the left. Moments about the knife edge give $W \times 20 = 2 \times 20$, so $W = 2 \text{ N}$, and the reaction is $R = 2 + 2 = 4 \text{ N}$ upwards.

Group C — Long answer (8 marks each)

- (a) State Newton's second law of motion and use it to derive $F = ma$, explaining why this form fails for a body of changing mass. [4]

(b) A machine gun fires 20 bullets per second, each of mass 30 g, with a muzzle speed of 400 m s^{-1} . Find the force required to hold the gun steady. [4]
- (a) State the laws of solid friction and describe an experiment to verify that limiting friction is directly proportional to the normal reaction. [5]

(b) A block of mass 5 kg rests on a horizontal surface with $\mu_s = 0.5$ and $\mu_k = 0.4$. Taking $g = 10 \text{ m s}^{-2}$, find (i) the least horizontal force that starts it moving and (ii) its acceleration if that force continues to act. [3]

ANSWERS TO GROUP C

1. (b) Each bullet carries momentum $mv = 0.030 \times 400 = 12 \text{ kg m s}^{-1}$, and 20 are fired per second, so

$$F = \frac{\Delta p}{\Delta t} = 20 \times 12 = 240 \text{ N}$$

2. (b) $N = mg = 50 \text{ N}$. (i) $f_{\text{limiting}} = 0.5 \times 50 = 25 \text{ N}$. (ii) $f_k = 0.4 \times 50 = 20 \text{ N}$, so $a = (25 - 20)/5 = 1 \text{ m s}^{-2}$.

UNIT 5

Work, energy and power

Mechanics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Work done by a constant force and a variable force
- Power
- Work-energy theorem; Kinetic and potential energy
- Conservation of Energy
- Conservative and non-conservative forces
- Elastic and inelastic collisions

Newton's laws tell you the force on a body at every instant. Often you do not care about every instant — you only want to know how fast the body ends up, or how much fuel the engine burned. Work and energy answer those questions in one line where dynamics would need a page. This unit builds the idea of work, turns it into kinetic and potential energy, and ends with the one law that survives every situation: energy is conserved.

WHAT THE EXAMINER ASKS HERE

Three things come up year after year: the work–energy theorem applied to a rough surface, the conservation of energy on a track or pendulum, and one-dimensional collisions. Learn to decide in one glance whether a problem is a force problem or an energy problem — energy is almost always faster.

5.1 Work done by a constant force and a variable force

In ordinary speech "work" means effort. In physics it means something much narrower: a force does work only when its point of application **moves**, and only the component of the force **along** the displacement counts.

WORK DONE BY A CONSTANT FORCE

If a constant force of magnitude F acts on a body while the body undergoes a displacement s , and θ is the angle between the force and the displacement, the work done by the force is

$$W = F s \cos \theta = \vec{F} \cdot \vec{s}$$

Work is a **scalar**. Its SI unit is the joule (J); $1 \text{ J} = 1 \text{ N m}$.

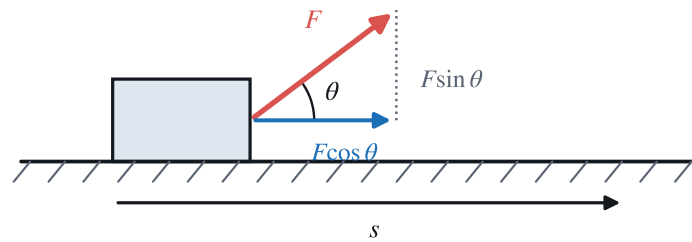


Figure 18. Only the component $F \cos \theta$ along the displacement does work. The vertical component $F \sin \theta$ reduces the normal reaction but does no work.

The cosine controls the sign, and the sign is physical:

Angle θ	$\cos \theta$	Work	Example
0°	+1	positive, maximum	a falling stone and its weight
acute	+	positive	pulling a cart with a slanted rope
90°	0	zero	normal reaction; centripetal force; a coolie carrying a load on level ground
obtuse	−	negative	brakes on a moving car
180°	−1	negative, maximum	friction on a sliding block

ZERO WORK IS NOT ZERO EFFORT

A porter on the Ghorepani trail carrying a 40 kg doko horizontally does **no** work on the load in the physics sense, because the supporting force is vertical and the displacement is horizontal. He tires because his muscles keep contracting and relaxing — that is physiological work, not mechanical work on the load.

Variable force. Most real forces change as the body moves — the pull of a spring, the drag on a bus, the thrust of a rocket. Split the displacement into strips so narrow that F is effectively constant across each strip, do $F \, dx$ for each, and add:

$$W = \lim_{\Delta x \rightarrow 0} \sum F \, \Delta x = \int_{x_1}^{x_2} F \, dx$$

WORK IS THE AREA UNDER THE FORCE-DISPLACEMENT GRAPH

This is true whatever the shape of the graph. Area above the axis is positive work, area below is negative work.

For a spring obeying Hooke's law the applied force needed is $F = kx$, a straight line through the origin. Stretching it from 0 to x sweeps out a triangle of base x and height kx :

$$W = \int_0^x kx \, dx = \frac{1}{2}kx^2$$

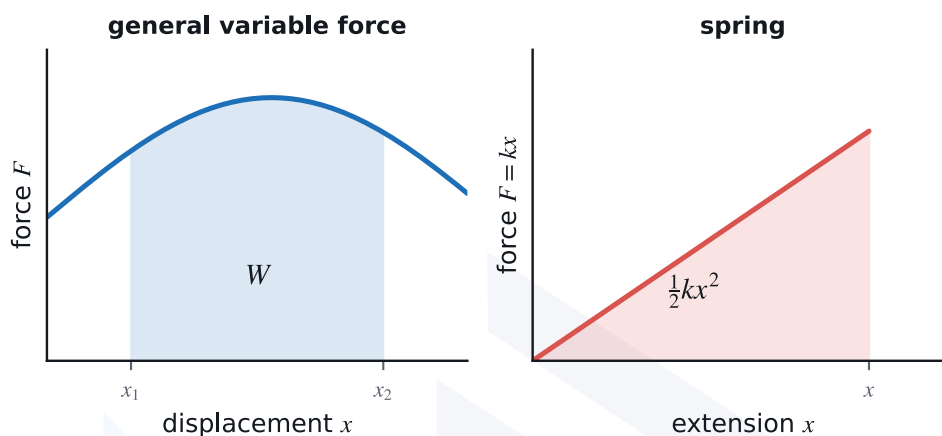


Figure 19. Work done is the area under the F - x graph. For a spring $F = kx$, so the area from 0 to x is the triangle $\frac{1}{2}kx^2$.

WORKED EXAMPLE 5.1

Problem. A crate of mass 10 kg is dragged 10 m across a horizontal floor by a rope inclined at 37° above the horizontal with a tension of 100 N. The coefficient of kinetic friction is 0.2. Take $g = 10 \text{ m s}^{-2}$, $\cos 37^\circ = 0.8$, $\sin 37^\circ = 0.6$. Find the work done by each force and the final speed if the crate starts at rest.

Solution. Work by the tension:

$$W_T = T \cos \theta = 100 \times 10 \times 0.8 = 800 \text{ J}$$

The vertical component of the tension lifts part of the weight, so the normal reaction is

$$N = mg - T \sin \theta = 100 - 60 = 40 \text{ N}$$

Friction is $f = \mu N = 0.2 \times 40 = 8 \text{ N}$, opposite the motion, so

$$W_f = -fs = -8 \times 10 = -80 \text{ J}$$

Weight and normal reaction are both perpendicular to the displacement, so each does **zero** work.

Net work = $800 - 80 = 720 \text{ J}$. By the work-energy theorem,

$$\frac{1}{2}mv^2 = 720 \implies 5v^2 = 720 \implies v = 12 \text{ m s}^{-1}$$

5.2 Power

Two labourers carry the same load of bricks to the same roof. One takes ten minutes, the other twenty. Both do the same work; the first is twice as **powerful**.

POWER

Power is the rate of doing work, or equivalently the rate of transfer of energy.

$$P_{av} = \frac{W}{t}, \quad P = \frac{dW}{dt}$$

The SI unit is the watt (W): $1 \text{ W} = 1 \text{ J s}^{-1}$. Commercial units are the kilowatt (10^3 W) and the horsepower ($1 \text{ hp} = 746 \text{ W}$).

If a constant force $\vec{F}$ moves a body at velocity $\vec{v}$, then in time dt the displacement is $\vec{v} dt$ and $dW = \vec{F} \cdot \vec{v} dt$, so

$$P = \vec{F} \cdot \vec{v} = Fv \cos \theta$$

This is the form to use for vehicles: at steady speed the driving force equals the total resisting force, so $P = Fv$ immediately gives the engine power.

KILOWATT-HOUR IS ENERGY, NOT POWER

The unit on a Nepal Electricity Authority bill, the kilowatt-hour, is a unit of **energy**: $1 \text{ kW h} = 1000 \times 3600 = 3.6 \times 10^6 \text{ J}$. "Units consumed" means joules used, not power.

WORKED EXAMPLE 5.2

Problem. A pump raises 600 litres of water per minute from a well 20 m deep in Chitwan. Taking $g = 10 \text{ m s}^{-2}$ and the density of water as 1000 kg m^{-3} , find (a) the useful output power and (b) the input power if the pump is 80% efficient.

Solution. 600 litres has mass 600 kg, delivered in 60 s, so the mass lifted per second is $\dot{m} = 10 \text{ kg s}^{-1}$.

(a) Useful power:

$$P_{out} = \frac{mgh}{t} = \dot{m}gh = 10 \times 10 \times 20 = 2000 \text{ W} = 2 \text{ kW}$$

(b) Efficiency $\eta = P_{out}/P_{in}$, so

$$P_{in} = \frac{2000}{0.80} = 2500 \text{ W} = 2.5 \text{ kW}$$

5.3 Work–energy theorem; kinetic and potential energy

Energy is the capacity to do work; it is measured in joules, like work. Mechanical energy comes in two forms.

THE WORK-ENERGY THEOREM, AND WHY $K = \frac{1}{2}mv^2$

We start from the definition of work and Newton's second law, and reach the result that the work done on a body equals the change in the quantity $\frac{1}{2}mv^2$ — which is why we give that quantity the special name kinetic energy.

Setting up. A constant net force F acts on a body of mass m in the direction it is moving. The body's speed rises from u to v while it moves a distance s .

Step 1 — write down the work. The force is along the displacement, so $\theta = 0$ and $\cos \theta = 1$:

$$W_{net} = Fs$$

Step 2 — replace F using Newton's second law. The net force on a constant mass is $F = ma$:

$$W_{net} = m a s$$

Step 3 — get as from the equations of motion. The third equation of motion for constant acceleration is

$$v^2 = u^2 + 2as$$

Step 4 — make as the subject. Subtract u^2 from both sides:

$$v^2 - u^2 = 2as$$

Step 5 — divide both sides by 2:

$$as = \frac{v^2 - u^2}{2}$$

Step 6 — substitute this into Step 2. This is the key move: it replaces the force-and-distance picture by a speed picture:

$$W_{net} = m \left(\frac{v^2 - u^2}{2} \right)$$

Step 7 — multiply the m into the bracket, term by term:

$$W_{net} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

Step 8 — name the two terms. The expression $\frac{1}{2}mv^2$ depends only on the mass and the speed at that instant, so it is a property of the state of the body. We call it the kinetic energy:

$$K = \frac{1}{2}mv^2$$

Step 9 — rewrite Step 7 in that language. With $K_i = \frac{1}{2}mu^2$ and $K_f = \frac{1}{2}mv^2$:

$$W_{net} = K_f - K_i = \Delta K$$

Result. The **work-energy theorem**:

$$W_{\text{net}} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \Delta K$$

The net work done by all the forces acting on a body equals the change in its kinetic energy. As a special case, a body at rest that is accelerated to speed v has had $\frac{1}{2}mv^2$ of work done on it — so that is the energy stored in its motion.

What it means. Positive net work speeds a body up; negative net work (friction, brakes) slows it down. Because K depends on v^2 , doubling the speed needs four times the work — the reason stopping distances grow so fast with speed.

Conditions used. Steps 1–5 assumed a **constant** force acting **along** the motion and a constant mass. The theorem itself is more general: for a varying force you replace Fs by $\int F dx$ and reach the same answer. W_{net} must be the work of the **net** force — friction work included, with its minus sign.

THE EXAMINER IS LOOKING FOR

(i) $W = Fs$ with the statement that the force is along the displacement; (ii) $F = ma$ substituted; (iii) $v^2 = u^2 + 2as$ quoted and rearranged for as ; (iv) the substitution, and (v) the final line named as the work-energy theorem in words. If the question says "for a variable force", you must start from $W = \int F dx$ instead.

Because $p = mv$, kinetic energy and momentum are linked by

$$K = \frac{p^2}{2m}, \quad p = \sqrt{2mK}$$

Potential energy is energy a body has because of its position or configuration. Lifting a mass m slowly through a height h needs an upward force mg through a distance h , so the work stored is

$$U = mgh$$

measured from whatever level you choose as zero. For a stretched or compressed spring the stored **elastic potential energy** is $U = \frac{1}{2}kx^2$, the area found in §5.1.

CHOOSE THE ZERO OF POTENTIAL ENERGY FIRST

Only changes in potential energy are physical. Write "taking the ground as $U = 0$ " as the first line of your answer — examiners give credit for it, and it stops sign errors later.

WORKED EXAMPLE 5.3

Problem. A block of mass 2 kg slides along a rough horizontal surface at 10 m s^{-1} and comes to rest. If $\mu = 0.25$ and $g = 10 \text{ m s}^{-2}$, how far does it travel? What is the retarding force?

Solution. The only force doing work is friction, $f = \mu mg = 0.25 \times 2 \times 10 = 5 \text{ N}$, acting backwards. By the work-energy theorem, $-fs = 0 - \frac{1}{2}mu^2$:

$$s = \frac{mu^2}{2f} = \frac{2 \times 100}{2 \times 5} = 20 \text{ m}$$

The retarding force is 5 N and the stopping distance is 20 m. Note that the stopping distance goes as u^2 : double the speed, four times the distance. That is the physics behind highway speed limits on the Mugling road.

5.4 Conservation of energy

PRINCIPLE OF CONSERVATION OF ENERGY

Energy can neither be created nor destroyed; it can only be converted from one form to another. The total energy of an isolated system is constant.

When only gravity and other conservative forces act, the mechanical form of the law applies:

$$K + U = \text{constant} \quad \Rightarrow \quad \frac{1}{2}mv^2 + mgh = \frac{1}{2}mu^2 + mgh_0$$

MECHANICAL ENERGY IS CONSERVED FOR A FREELY FALLING BODY

We start with a body of mass m released from rest at a height H , and show that at every point of its fall the sum $K + U$ has the same value mgH .

Setting up. Take the ground as the level where $U = 0$ (always write this line down). Let the body have fallen to a height h above the ground, so the distance already fallen is $H - h$.

Step 1 — the potential energy at height h . By definition $U = mgh$:

$$U = mgh$$

Step 2 — find the speed at height h . Apply $v^2 = u^2 + 2as$ to the fall so far. It started from rest, so $u = 0$; the acceleration is g downwards and the distance fallen is $(H - h)$:

$$v^2 = 0 + 2g(H - h)$$

so

$$v^2 = 2g(H - h)$$

Step 3 — the kinetic energy at height h . Put this v^2 into $K = \frac{1}{2}mv^2$:

$$K = \frac{1}{2}m \cdot 2g(H - h)$$

Step 4 — the 2 and the $\frac{1}{2}$ cancel:

$$K = mg(H - h)$$

Step 5 — add the two energies:

$$K + U = mg(H - h) + mgh$$

Step 6 — open the bracket:

$$K + U = mgH - mgh + mgh$$

Step 7 — the two mgh terms cancel:

$$K + U = mgH$$

Result. At every height h during the fall,

$$K + U = mgH = \text{constant}$$

Check the two ends. At the top, $h=H$: $U=mgH$ and $K=0$. At the ground, $h=0$: $U=0$ and $K=mgH$, so the landing speed is $v = \sqrt{2gH}$. The answer mgH is the same at both ends and everywhere in between.

What it means. Potential energy is not destroyed, it is converted: for every joule of U lost, exactly one joule of K appears. Note that h disappeared from the answer — that is the whole content of the conservation law.

Conditions used. Air resistance is neglected (otherwise some energy becomes heat and $K+U$ falls), g is constant over the height H , and the body is released from **rest**. If it were thrown down with speed u , the constant total would be $mgH + \frac{1}{2}mu^2$ instead.

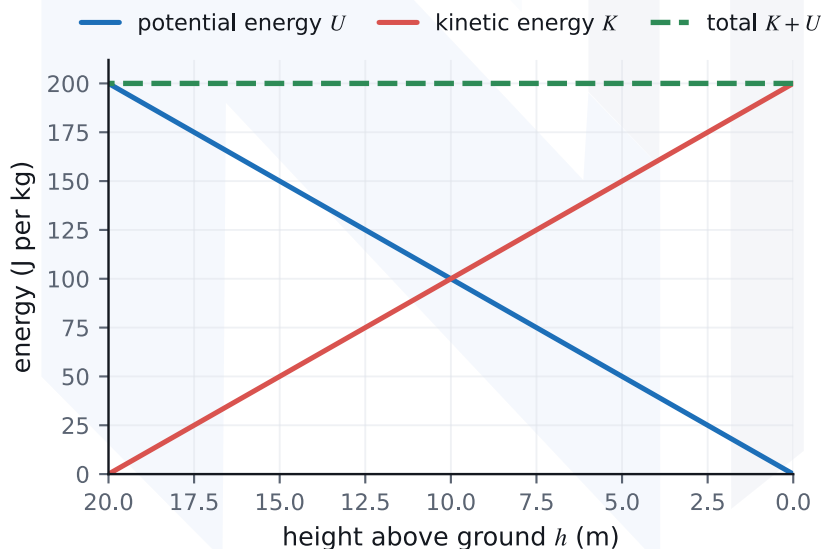


Figure 20. Energy of a body falling freely from height H . U falls linearly, K rises linearly, and the total $K + U = mgH$ stays constant.

CONSERVATION IS NOT "MECHANICAL ENERGY IS ALWAYS CONSTANT"

If friction, air resistance or a collision is involved, mechanical energy is **not** conserved — some is converted to heat and sound. Total energy is still conserved. In that case use the work-energy theorem with the friction work included, not $K + U = \text{constant}$.

5.5 Conservative and non-conservative forces

CONSERVATIVE FORCE

A force is **conservative** if the work it does in moving a body between two points is independent of the path taken, and consequently the work it does around any closed path is zero. A potential energy function can be defined for it, with $F = -dU/dx$. A force for which this fails is **non-conservative**.

Lift a 1 kg bag from the courtyard to the first floor of a house in Bhaktapur. Carry it straight up the stairs, or up to the second floor and back down — the work done against gravity is the same, mgh , because only the change of height matters. Gravity is conservative. Drag the same bag across the floor by a long route and a short route, and friction takes far more energy on the long route. Friction is non-conservative.

	Conservative	Non-conservative
Work depends on path?	No	Yes
Work in a closed loop	zero	non-zero (usually negative)
Potential energy defined?	Yes	No
Energy recoverable?	Fully	No — degraded to heat
Examples	gravity, spring force, electrostatic force	friction, air drag, viscous force, tension in an in-elastic string

When both kinds act, the work-energy theorem splits as

$$W_c + W_{nc} = \Delta K, \quad W_c = -\Delta U \implies W_{nc} = \Delta K + \Delta U = \Delta E$$

So the work done by non-conservative forces measures exactly how much mechanical energy is lost.

5.6 Elastic and inelastic collisions

In every collision between two isolated bodies the external forces are negligible during the short contact time, so **linear momentum is always conserved**. Whether kinetic energy is also conserved decides the type.

	Elastic	Inelastic	Perfectly inelastic
Momentum	conserved	conserved	conserved
Kinetic energy	conserved	not conserved	maximum loss
Bodies after impact	separate	separate	move together
Coefficient of restitution e	1	$0 < e < 1$	0
Example	collision of gas molecules, alpha particle and nucleus	ball bouncing to a lower height	bullet embedding in a block; mud thrown at a wall

FINAL VELOCITIES IN A ONE-DIMENSIONAL ELASTIC COLLISION

We start from the two conservation laws — momentum and kinetic energy — and reach formulas for the velocities of both bodies after the collision.

Setting up. Masses m_1 and m_2 move along the same straight line with velocities u_1 and u_2 before impact and v_1 and v_2 after it. All four are signed quantities: rightward positive, leftward negative.

Step 1 — momentum is conserved (no external force during the short contact):

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad (1)$$

Step 2 — kinetic energy is conserved (this is what "elastic" means):

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

Step 3 — clear the $\frac{1}{2}$ by multiplying every term by 2:

$$m_1 u_1^2 + m_2 u_2^2 = m_1 v_1^2 + m_2 v_2^2 \quad (2)$$

Step 4 — group body 1 on the left in equation (1). Take the m_1 terms to the left and the m_2 terms to the right:

$$m_1 u_1 - m_1 v_1 = m_2 v_2 - m_2 u_2$$

Take out the common factors:

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad (3)$$

Step 5 — do the same to equation (2):

$$m_1 u_1^2 - m_1 v_1^2 = m_2 v_2^2 - m_2 u_2^2$$

$$m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad (4)$$

Step 6 — factorise both differences of squares using $x^2 - y^2 = (x - y)(x + y)$:

$$m_1(u_1 - v_1)(u_1 + v_1) = m_2(v_2 - u_2)(v_2 + u_2)$$

Step 7 — divide this by equation (3). The left side of (3) is identical to the first factor on the left here, and the same for the right — so those factors cancel. (This needs $u_1 \neq v_1$, i.e. a real collision actually happened.)

$$u_1 + v_1 = v_2 + u_2 \quad (5)$$

Step 8 — rearrange (5) into its physical form. Subtract u_2 and v_1 from both sides:

$$u_1 - u_2 = v_2 - v_1$$

In words: **the relative velocity of separation equals the relative velocity of approach.** This is the most useful single line in the whole topic.

Step 9 — make v_2 the subject of (5):

$$v_2 = u_1 + v_1 - u_2$$

Step 10 — substitute this v_2 into the momentum equation (1):

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 (u_1 + v_1 - u_2)$$

Step 11 — open the bracket:

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 u_1 + m_2 v_1 - m_2 u_2$$

Step 12 — collect every v_1 term on the right and everything else on the left. Move $m_2 u_1$ across (it becomes negative) and $-m_2 u_2$ across (it becomes positive):

$$m_1 u_1 - m_2 u_1 + m_2 u_2 + m_2 u_2 = m_1 v_1 + m_2 v_1$$

Step 13 — factorise both sides:

$$(m_1 - m_2)u_1 + 2m_2 u_2 = (m_1 + m_2)v_1$$

Step 14 — divide both sides by $(m_1 + m_2)$:

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \frac{2m_2 u_2}{m_1 + m_2}$$

Step 15 — get v_2 the same way. From (5), $v_1 = v_2 + u_2 - u_1$. Put that into (1):

$$m_1 u_1 + m_2 u_2 = m_1 (v_2 + u_2 - u_1) + m_2 v_2$$

Step 16 — open the bracket:

$$m_1 u_1 + m_2 u_2 = m_1 v_2 + m_1 u_2 - m_1 u_1 + m_2 v_2$$

Step 17 — collect the v_2 terms on the right and the rest on the left:

$$2m_1u_1 + m_2u_2 - m_1u_2 = (m_1 + m_2)v_2$$

Step 18 — factorise the u_2 terms and divide by $(m_1 + m_2)$:

$$v_2 = \frac{2m_1u_1}{m_1 + m_2} + \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2$$

Result.

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_2u_2}{m_1 + m_2}, \quad v_2 = \left(\frac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \frac{2m_1u_1}{m_1 + m_2}$$

What it means. The outcome depends on the **mass ratio**, not on how hard the bodies hit. If $m_1 = m_2$ the first bracket is zero and the bodies simply swap velocities. If $m_1 \ll m_2$ the first bracket is nearly -1 , so the light body bounces straight back.

Conditions used. One-dimensional (head-on) collision; kinetic energy exactly conserved ($e = 1$); no external force during contact; signs kept consistently. Step 7 also assumed the bodies really did interact ($u_1 \neq v_1$).

THE EXAMINER IS LOOKING FOR

(i) Both conservation equations written down and labelled; (ii) the factorising of the energy equation into a difference of two squares; (iii) the division step with the words "dividing (4) by (3)"; (iv) the relative-velocity result $u_1 - u_2 = v_2 - v_1$ stated in words — it is often a mark on its own; (v) substitution back into momentum to get both final velocities. Check your answer by putting $m_1 = m_2$ and seeing the velocities swap.

Three special cases are worth memorising, all with $u_2 = 0$:

- **Equal masses** ($m_1 = m_2$): $v_1 = 0$, $v_2 = u_1$ — the bodies exchange velocities. This is the carrom-striker result.
- **Heavy hits light** ($m_1 \gg m_2$): $v_1 \approx u_1$, $v_2 \approx 2u_1$ — the light body is flung away at twice the speed.
- **Light hits heavy** ($m_1 \ll m_2$): $v_1 \approx -u_1$, $v_2 \approx 0$ — the light body bounces straight back with almost unchanged speed.

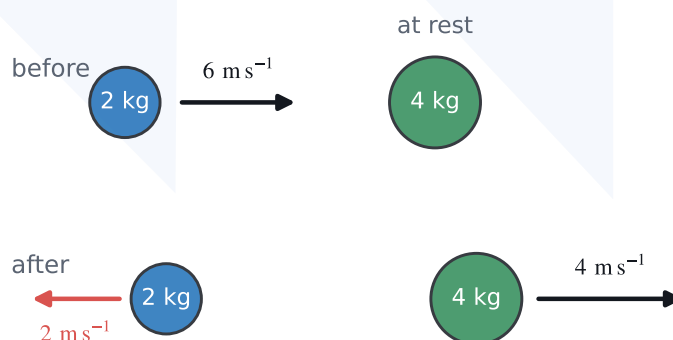


Figure 21. Head-on elastic collision of a 2 kg body at 6 m s⁻¹ with a stationary 4 kg body. The lighter body rebounds; momentum and kinetic energy both balance.

Perfectly inelastic collision. The bodies stick together and move with a common velocity v . Momentum conservation alone fixes it:

$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

and the kinetic energy lost, which appears as heat, sound and deformation, is

$$\Delta K = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) (u_1 - u_2)^2$$

WORKED EXAMPLE 5.4

Problem. A body of mass 2 kg moving at 6 m s^{-1} collides head-on with a stationary body of mass 4 kg. Find the velocities after impact if the collision is (a) perfectly elastic, (b) perfectly inelastic, and find the energy lost in (b).

Solution. (a) With $u_1 = 6 \text{ m s}^{-1}$, $u_2 = 0$:

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 = \frac{2-4}{2+4} \times 6 = -2 \text{ m s}^{-1}$$

$$v_2 = \frac{2m_1}{m_1 + m_2} u_1 = \frac{4}{6} \times 6 = 4 \text{ m s}^{-1}$$

The 2 kg body rebounds at 2 m s^{-1} . Check: momentum before = 12 kg m s^{-1} , after = $2(-2) + 4(4) = 12 \checkmark$; kinetic energy before = 36 J , after = $4 + 32 = 36 \text{ J} \checkmark$.

(b) Sticking together:

$$v = \frac{2 \times 6 + 0}{2 + 4} = 2 \text{ m s}^{-1}$$

Kinetic energy before = 36 J ; after = $\frac{1}{2}(6)(2)^2 = 12 \text{ J}$. Energy lost = 24 J , i.e. two-thirds of the original kinetic energy.

WORKED EXAMPLE 5.5

Problem. A bullet of mass 10 g travelling at 400 m s^{-1} embeds itself in a wooden block of mass 1.99 kg suspended by a long string. Find the speed of the block just after impact and the height it rises. Take $g = 10 \text{ m s}^{-2}$.

Solution. The collision is perfectly inelastic, so momentum is conserved during the impact but kinetic energy is not:

$$v = \frac{0.010 \times 400}{0.010 + 1.99} = \frac{4.0}{2.00} = 2 \text{ m s}^{-1}$$

After the impact no friction acts, so mechanical energy is conserved during the swing:

$$\frac{1}{2} M v^2 = M g h \implies h = \frac{v^2}{2g} = \frac{4}{20} = 0.2 \text{ m}$$

The block rises 20 cm. This arrangement is the **ballistic pendulum**, used to measure bullet speeds.

DO NOT USE ENERGY CONSERVATION DURING THE COLLISION

Momentum is conserved in every collision; kinetic energy only in an elastic one. Write the momentum equation for the impact and the energy equation for what happens before or after it — never the other way round.

✓ **Chapter summary**

- Work by a constant force: $W = F s \cos \theta = \vec{F} \cdot \vec{s}$, a scalar in joules. Zero when the force is perpendicular to the displacement.
- Work by a variable force: $W = \int F dx$ = area under the F - x graph. For a spring, $W = \frac{1}{2} k x^2$.
- Power $P = W/t = \vec{F} \cdot \vec{v}$; 1 hp = 746 W and 1 kW h = 3.6×10^6 J.
- Work-energy theorem: $W_{net} = \Delta K$, with $K = \frac{1}{2} m v^2 = p^2/2m$.
- Potential energy: $U = mgh$ (gravitational), $U = \frac{1}{2} k x^2$ (elastic).
- Conservation of energy: $K + U$ is constant when only conservative forces act; otherwise $W_{nc} = \Delta K + \Delta U$.
- Conservative forces do path-independent work and zero work round a closed loop; friction and drag do not.
- All collisions conserve momentum. Elastic ones also conserve kinetic energy and give $u_1 - u_2 = v_2 - v_1$; perfectly inelastic ones lose $\frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} (u_1 - u_2)^2$.

Practice questions**Group A — Multiple choice (1 mark each)**

1. A coolie carries a load horizontally along a level platform. The work done by him on the load is [1]
 - a mgh
 - b zero
 - c negative
 - d mgs
2. The area under a force-displacement graph represents [1]
 - a power
 - b impulse
 - c work done
 - d momentum
3. A body of mass m has momentum p . Its kinetic energy is [1]
 - a mp^2
 - b $p^2/2m$
 - c p^2/m
 - d $2mp^2$
4. Which of the following is a non-conservative force? [1]
 - a gravitational force
 - b spring force
 - c electrostatic force
 - d viscous force
5. In a perfectly inelastic collision, the quantity that is **not** conserved is [1]
 - a linear momentum
 - b total energy
 - c kinetic energy
 - d mass

ANSWERS TO GROUP A

1. (b) — the supporting force is vertical, the displacement horizontal, so $\theta = 90^\circ$.

2. (c) — $W = \int F dx$ is exactly the area under the curve.
3. (b) — $K = \frac{1}{2}mv^2$ and $p = mv$, so $K = p^2/2m$.
4. (d) — viscous drag depends on the path and dissipates energy as heat.
5. (c) — momentum and total energy are conserved; kinetic energy is lost to heat and deformation.

Group B — Short answer (5 marks each)

1. Define work, energy and power, and state their SI units. Show that the work done by a variable force is the area under the force–displacement graph. [5]
2. State and prove the work–energy theorem for a body moving under a constant force. [5]
3. Distinguish between conservative and non-conservative forces with two examples of each. [5]
4. A car of mass 1000 kg moves up an incline of 1 in 20 at a steady 20 m s^{-1} against a frictional resistance of 200 N. Calculate the power developed by the engine. Take $g = 10 \text{ m s}^{-2}$. [5]
5. A spring of force constant 200 N m^{-1} is compressed by 0.10 m and used to push a block of mass 0.50 kg along a frictionless table. Find the elastic potential energy stored and the speed of the block when the spring returns to its natural length. [5]

ANSWERS TO GROUP B

4. At steady speed the driving force balances gravity component plus friction: $F = mg \sin \theta + R = 1000 \times 10 \times \frac{1}{20} + 200 = 500 + 200 = 700 \text{ N}$. Then $P = Fv = 700 \times 20 = 14000 \text{ W} = 14 \text{ kW}$.
5. $U = \frac{1}{2}kx^2 = \frac{1}{2}(200)(0.10)^2 = 1.0 \text{ J}$. All of it becomes kinetic energy: $\frac{1}{2}(0.50)v^2 = 1.0$, so $v^2 = 4.0$ and $v = 2.0 \text{ m s}^{-1}$.
- 1., 2., 3. See §5.1, the derivation box in §5.3, and the comparison table in §5.5.

Group C — Long answer (8 marks each)

1. (a) State the principle of conservation of energy. [2]
- (b) Show that for a body falling freely from a height H the sum of its kinetic and potential energies is the same at every point of the fall. [4]
- (c) A stone of mass 0.5 kg is dropped from 45 m. Find its kinetic energy and its speed when it is 20 m above the ground ($g = 10 \text{ m s}^{-2}$). [2]
2. (a) Derive expressions for the final velocities of two bodies undergoing a perfectly elastic head-on collision, and deduce the result for equal masses. [6]
- (b) A 5 kg body moving at 4 m s^{-1} collides with a stationary 3 kg body and they move off together. Find their common velocity and the kinetic energy lost. [2]

ANSWERS TO GROUP C

1. (b) See §5.4: at height h , $v^2 = 2g(H - h)$ so $K = mg(H - h)$ while $U = mgh$, giving $K + U = mgH$ everywhere. (c) The stone has fallen $45 - 20 = 25 \text{ m}$, so $K = mg(H - h) = 0.5 \times 10 \times 25 = 125 \text{ J}$ and $v = \sqrt{2K/m} = \sqrt{500} = 22.4 \text{ m s}^{-1}$.

2. (a) See the derivation box in §5.6; for $m_1 = m_2$ the prefactor $(m_1 - m_2)/(m_1 + m_2)$ vanishes, so with $u_2 = 0$ we get $v_1 = 0$, $v_2 = u_1$ — the bodies exchange velocities. (b) $v = \frac{5 \times 4 + 0}{5 + 3} = 2.5 \text{ m s}^{-1}$. $K_i = \frac{1}{2}(5)(16) = 40 \text{ J}$, $K_f = \frac{1}{2}(8)(2.5)^2 = 25 \text{ J}$, so the loss is 15 J.



UNIT 6

Circular Motion

Mechanics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Angular displacement, velocity and acceleration
- Relation between angular and linear velocity and acceleration
- Centripetal acceleration
- Centripetal force
- Conical pendulum
- Motion in a vertical circle
- Applications of banking

A bus rounding a bend on the Prithvi Highway, a stone whirled on a string, the Moon going round the Earth and an electron in a magnetic field are all doing the same thing: moving on a curved path at a speed that may not change at all, yet accelerating every instant. This unit explains that apparent contradiction, builds the language of angular quantities, and applies it to three standard systems — the conical pendulum, the vertical circle and the banked road.

THE ONE IDEA BEHIND THE WHOLE UNIT

Velocity is a **vector**. A body going round a circle at constant speed is changing the direction of its velocity all the time, so it is accelerating. That acceleration points to the centre, and something real must supply the force that causes it.

6.1 Angular displacement, velocity and acceleration

For a particle moving on a circle of radius r , it is easier to describe the position by the angle θ swept out at the centre than by the arc length.

ANGULAR DISPLACEMENT

The **angular displacement** θ is the angle turned through by the radius joining the particle to the centre. It is measured in **radians**:

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r}$$

One complete revolution is $2\pi \text{ rad} = 360^\circ$, so $1 \text{ rad} = 57.3^\circ$. The radian is dimensionless.

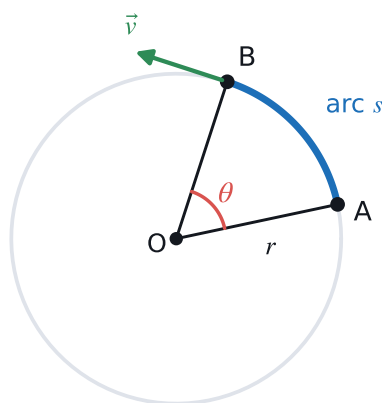


Figure 22. Angular displacement $\theta = s/r$. The direction of $\vec{\omega}$ is given by the right-hand rule, along the axis of rotation.

Angular velocity ω is the rate of change of angular displacement, and **angular acceleration** α the rate of change of ω :

$$\omega_{av} = \frac{\Delta\theta}{\Delta t}, \quad \omega = \frac{d\theta}{dt} \quad \alpha_{av} = \frac{\Delta\omega}{\Delta t}, \quad \alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$$

Both are vectors directed along the axis of rotation, their sense given by the right-hand rule. For one revolution in a time T (the **period**), with frequency $f = 1/T$,

$$\omega = \frac{2\pi}{T} = 2\pi f$$

Because the equations $\theta = \omega t$ and so on have exactly the same form as the linear ones, the kinematic equations carry over term by term:

Linear motion	Angular motion
$v = u + at$	$\omega = \omega_0 + \alpha t$
$s = ut + \frac{1}{2}at^2$	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$
units m, m s ⁻¹ , m s ⁻²	units rad, rad s ⁻¹ , rad s ⁻²

ALWAYS CONVERT RPM TO RAD S⁻¹

A speed quoted in revolutions per minute must be multiplied by $2\pi/60$ before it goes into any formula. $300 \text{ rpm} = 300 \times 2\pi/60 = 10\pi \approx 31.4 \text{ rad s}^{-1}$, not 300 rad s^{-1} .

6.2 Relation between angular and linear velocity and acceleration

THE RELATIONS $v = r\omega$ **AND** $a_t = r\alpha$

We start from the definition of angle in radians and reach the two formulas that connect the linear quantities of a point on a wheel to the angular quantities of the wheel itself.

Step 1 — the definition of the radian. If a point moves a distance s along a circle of radius r , the angle turned in radians is $\theta = s/r$. Multiply both sides by r :

$$s = r\theta$$

Step 2 — differentiate both sides with respect to time. The radius r is a fixed number for a given circle, so it is a constant multiplier and stays outside the derivative:

$$\frac{ds}{dt} = r \frac{d\theta}{dt}$$

Step 3 — name the two derivatives. ds/dt is the linear speed v along the circle, and $d\theta/dt$ is the angular velocity ω :

$$v = r\omega$$

Step 4 — differentiate once more with respect to time. Same reasoning: r stays outside:

$$\frac{dv}{dt} = r \frac{d\omega}{dt}$$

Step 5 — name these derivatives. dv/dt is the rate at which the speed changes — the **tangential** acceleration a_t — and $d\omega/dt$ is the angular acceleration α :

$$a_t = r\alpha$$

Result.

$$v = r\omega, \quad a_t = r\alpha$$

In vector form $\vec{v} = \vec{\omega} \times \vec{r}$ and $\vec{a}_t = \vec{\alpha} \times \vec{r}$.

What it means. Every point on a rigid rotating body shares the same ω and α , but a point twice as far from the axis moves twice as fast and gains speed twice as quickly.

Conditions used. θ must be in **radians** — in degrees, $s = r\theta$ is simply false. The radius r must be constant (the body is rigid, the point does not spiral in or out). Also, a_t is only the part of the acceleration along the tangent; there is always a second, radial part as well, which the next section works out.

ALL POINTS OF A RIGID BODY SHARE ω , NOT v

On a rotating wheel every point has the same ω and α , but $v = r\omega$ means the rim moves faster than a point near the hub. This is why the outer gear-teeth of a bicycle wheel travel further per turn than the inner ones.

WORKED EXAMPLE 6.1

Problem. A flywheel of radius 0.40 m rotates at 300 rpm. It is then brought uniformly to rest in 10 s. Find (a) the initial angular velocity, (b) the linear speed of a point on the rim, (c) the angular retardation, and (d) the number of revolutions made before stopping.

Solution.

$$(a) \omega_0 = \frac{2\pi N}{60} = \frac{2\pi \times 300}{60} = 10\pi = 31.4 \text{ rad s}^{-1}$$

$$(b) v = r\omega_0 = 0.40 \times 31.4 = 12.6 \text{ m s}^{-1}$$

$$(c) \alpha = \frac{\omega - \omega_0}{t} = \frac{0 - 31.4}{10} = -3.14 \text{ rad s}^{-2}$$

$$(d) \theta = \frac{\omega_0 + \omega}{2} t = \frac{31.4}{2} \times 10 = 157 \text{ rad, so the number of revolutions is } \frac{157}{2\pi} = 25.$$

6.3 Centripetal acceleration

CENTRIPETAL ACCELERATION $a = v^2/r$

We start from the fact that a body going round a circle at constant speed keeps changing the direction of its velocity, and we reach a formula for how big the resulting acceleration is and where it points.

Setting up. A particle moves on a circle of radius r at constant speed v . In a short time Δt it travels from point A to point B, and the radius to it turns through a small angle $\Delta\theta$.

Step 1 — the angle turned. By the definition of angular velocity, $\omega = \Delta\theta/\Delta t$, so

$$\Delta\theta = \omega \Delta t$$

Step 2 — the velocity turns through the same angle. The velocity is always along the tangent, and the tangent turns exactly as fast as the radius does. So $\vec{v}_A$ and $\vec{v}_B$ have the **same length** v but differ in direction by $\Delta\theta$.

Step 3 — find the change in velocity. Redraw $\vec{v}_A$ and $\vec{v}_B$ from one common point (right-hand figure). The change is the vector from the tip of $\vec{v}_A$ to the tip of $\vec{v}_B$:

$$\Delta\vec{v} = \vec{v}_B - \vec{v}_A$$

Step 4 — measure its length. The two vectors and $\Delta\vec{v}$ form an isosceles triangle: two equal sides of length v with $\Delta\theta$ between them. Splitting that triangle down the middle gives two right-angled triangles, each with angle $\Delta\theta/2$ and hypotenuse v , so half the base is $v\sin(\Delta\theta/2)$ and the whole base is twice that:

$$|\Delta\vec{v}| = 2v\sin \frac{\Delta\theta}{2}$$

Step 5 — use the small-angle approximation. When an angle in radians is very small, $\sin x \approx x$ (try $x = 0.01$: $\sin x = 0.0099998$). Since we are about to let $\Delta t \rightarrow 0$, the angle is as small as we like:

$$|\Delta\vec{v}| \approx 2v \cdot \frac{\Delta\theta}{2} = v \Delta\theta$$

Step 6 — divide by the time taken. Acceleration is change of velocity divided by time:

$$a = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{v \Delta \theta}{\Delta t}$$

Step 7 — take the limit as $\Delta t \rightarrow 0$ so that the answer is the instantaneous acceleration. The constant v comes outside the limit:

$$a = v \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t}$$

Step 8 — recognise the limit as ω :

$$a = v\omega$$

Step 9 — get the first standard form. Substitute $\omega = v/r$ (from $v = r\omega$):

$$a = v \cdot \frac{v}{r} = \frac{v^2}{r}$$

Step 10 — get the second standard form. Instead substitute $v = r\omega$ into $a = v\omega$:

$$a = (r\omega)\omega = r\omega^2$$

Step 11 — get the third form in terms of the period. Since $\omega = 2\pi/T$:

$$a = r\left(\frac{2\pi}{T}\right)^2 = \frac{4\pi^2 r}{T^2}$$

Step 12 — the direction. In Step 4's isosceles triangle, the two base angles are each $(180^\circ - \Delta\theta)/2$. As $\Delta\theta \rightarrow 0$ each base angle approaches 90° — so $\Delta\vec{v}$ becomes **perpendicular** to the velocity, pointing inwards along the radius, i.e. **towards the centre**.

Result.

$$a = \frac{v^2}{r} = r\omega^2 = \frac{4\pi^2 r}{T^2}, \quad \text{directed towards the centre}$$

That inward direction is why it is called the **centripetal** ("centre-seeking") acceleration.

What it means. A body can accelerate without speeding up. Here the size of the velocity never changes, only its direction — and changing direction is itself an acceleration, which needs a force pointing at the centre.

Conditions used. The speed v is constant (uniform circular motion) and the radius r is constant. Step 5 needs the angle in **radians**. If the speed also changes there is an extra tangential part $a_t = r\alpha$, and the total acceleration is $\sqrt{a_c^2 + a_t^2}$.

THE EXAMINER IS LOOKING FOR

(i) Two diagrams — the circle with $\vec{v}_A$, $\vec{v}_B$ and $\Delta\theta$, and the vector triangle; (ii) the statement that the magnitudes are equal but the directions differ by $\Delta\theta$; (iii) $|\Delta\vec{v}| = 2v\sin(\Delta\theta/2)$; (iv) the small-angle step, said out loud; (v) the limit, giving $a = v\omega$; (vi) all three final forms and, crucially, the sentence "directed towards the centre".

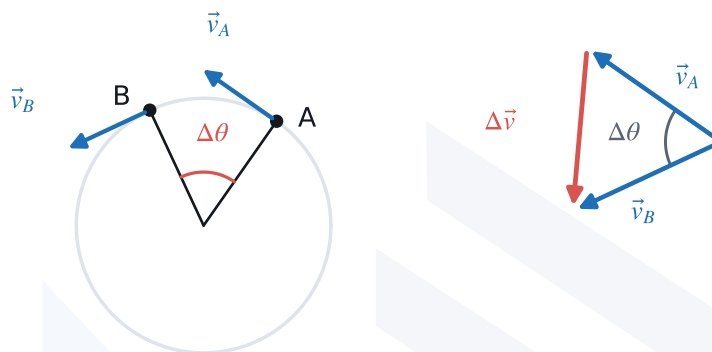


Figure 23. Left: velocities at A and B, each of magnitude v , turned through $\Delta\theta$. Right: the vector triangle gives $|\Delta\vec{v}| = 2v\sin(\Delta\theta/2)$, directed towards the centre as $\Delta\theta \rightarrow 0$.

If the speed also changes, the total acceleration has two perpendicular parts:

$$a = \sqrt{a_c^2 + a_t^2} = \sqrt{\left(\frac{v^2}{r}\right)^2 + (r\alpha)^2}$$

In **uniform** circular motion $a_t = 0$, so the acceleration is purely radial.

6.4 Centripetal force

By Newton's second law an acceleration needs a force. The force required to keep a body of mass m moving on a circle of radius r at speed v is

$$F_c = ma_c = \frac{mv^2}{r} = mr\omega^2 = \frac{4\pi^2 mr}{T^2}$$

CENTRIPETAL FORCE IS A ROLE, NOT A NEW FORCE

No new force of nature is involved. Some existing force plays the part of the centripetal force in each situation, and it always points to the centre.

Motion	Force acting as the centripetal force
Stone whirled on a string	tension in the string
Car on a level curve	friction between tyres and road
Car on a banked curve	horizontal component of the normal reaction
Moon round the Earth	gravitational attraction
Electron round a nucleus (Bohr model)	electrostatic attraction

CENTRIFUGAL FORCE IS NOT REAL

The outward "centrifugal force" a passenger feels on a bend is not a force at all; it is the passenger's inertia trying to continue in a straight line. It appears only in the rotating (non-inertial) frame. In an NEB answer, always resolve forces in the ground frame and put mv^2/r on the acceleration side.

WORKED EXAMPLE 6.2

Problem. A car of mass 1000 kg takes an unbanked curve of radius 50 m at 10 m s^{-1} . Find the centripetal force needed and the minimum coefficient of friction required. Take $g = 10 \text{ m s}^{-2}$.

Solution.

$$F_c = \frac{mv^2}{r} = \frac{1000 \times 100}{50} = 2000 \text{ N}$$

Friction must supply this: $\mu mg \geq \frac{mv^2}{r}$, so

$$\mu \geq \frac{v^2}{rg} = \frac{100}{50 \times 10} = 0.20$$

A wet road can drop below $\mu = 0.2$, which is why vehicles skid outwards on monsoon bends.

6.5 Conical pendulum

A conical pendulum is a bob on a string that moves in a **horizontal** circle while the string sweeps out a cone of semi-vertical angle θ .

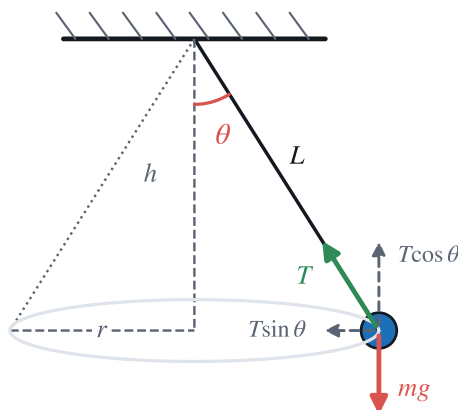


Figure 24. Conical pendulum. The vertical component $T \cos \theta$ balances the weight; the horizontal component $T \sin \theta$ provides the centripetal force mv^2/r .

PERIOD OF A CONICAL PENDULUM

We start from the two force equations for the bob — one vertical, one horizontal — and reach an expression for the time of one revolution.

Setting up. A bob of mass m hangs from a string of length L fixed at a point, and it moves in a horizontal circle so that the string sweeps out a cone at a constant angle θ to the vertical. From the geometry of that cone,

$$r = L \sin \theta \quad (\text{radius of the circle}), \quad h = L \cos \theta \quad (\text{depth of the circle below the support})$$

Only two forces act on the bob: its weight mg straight down, and the tension T along the string towards the support.

Step 1 — resolve the tension. The string makes angle θ with the vertical, so the tension has a vertical part $T \cos \theta$ (upwards) and a horizontal part $T \sin \theta$ (pointing at the centre of the circle).

Step 2 — vertical equation. The bob stays at the same height, so it has no vertical acceleration and the vertical forces balance:

$$T \cos \theta = mg$$

Step 3 — horizontal equation. Horizontally the bob is accelerating — it is going round a circle — and the only horizontal force is $T \sin \theta$. So this component must supply the whole centripetal force:

$$T \sin \theta = \frac{mv^2}{r} = mr\omega^2$$

Step 4 — divide Step 3 by Step 2. We divide because both T and m then cancel, which is exactly what we want:

$$\frac{T \sin \theta}{T \cos \theta} = \frac{mr\omega^2}{mg}$$

Step 5 — cancel and use $\sin/\cos = \tan$:

$$\tan \theta = \frac{r\omega^2}{g}$$

(The same division with mv^2/r instead gives the equally useful $\tan \theta = v^2/rg$.)

Step 6 — replace r by $L\sin \theta$:

$$\tan \theta = \frac{L\sin \theta \omega^2}{g}$$

Step 7 — write $\tan \theta$ as $\sin \theta / \cos \theta$:

$$\frac{\sin \theta}{\cos \theta} = \frac{L\sin \theta \omega^2}{g}$$

Step 8 — cancel $\sin \theta$ from both sides. This is allowed because the bob is actually circling, so $\theta \neq 0$ and $\sin \theta \neq 0$:

$$\frac{1}{\cos \theta} = \frac{L\omega^2}{g}$$

Step 9 — make ω^2 the subject. Multiply both sides by g and divide by L :

$$\omega^2 = \frac{g}{L\cos \theta}$$

Step 10 — take the square root:

$$\omega = \sqrt{\frac{g}{L\cos \theta}}$$

Step 11 — convert to a period. One revolution takes $T_p = 2\pi/\omega$:

$$T_p = \frac{2\pi}{\sqrt{\frac{g}{L\cos \theta}}}$$

Step 12 — turn the divided root upside down. Dividing by $\sqrt{g/x}$ is the same as multiplying by $\sqrt{x/g}$:

$$T_p = 2\pi\sqrt{\frac{L\cos \theta}{g}}$$

Step 13 — recognise $L\cos \theta$ as the depth h :

$$T_p = 2\pi\sqrt{\frac{h}{g}}$$

Step 14 — the tension. Go back to Step 2 and divide both sides by $\cos \theta$:

$$T = \frac{mg}{\cos \theta}$$

Result.

$$T_p = 2\pi\sqrt{\frac{L\cos \theta}{g}} = 2\pi\sqrt{\frac{h}{g}}, \quad \tan \theta = \frac{v^2}{rg}, \quad T = \frac{mg}{\cos \theta}$$

What it means. The mass cancelled, so a heavy bob and a light bob on the same string at the same angle take the same time to go round. The period depends only on the **depth** h of the circle below the support — so as the bob spins faster, θ grows, h shrinks and the period gets shorter. Because $\cos \theta < 1$, the tension $mg/\cos \theta$ is always **bigger** than the weight, and it would become infinite at $\theta = 90^\circ$ — which is why the string can never be truly horizontal.

Conditions used. The string is light and inextensible, the angle θ stays constant (steady motion, not a wobble), air resistance is neglected, and the circle is horizontal so there is no vertical acceleration.

THE EXAMINER IS LOOKING FOR

(i) A labelled diagram showing L , θ , r , h , mg and T resolved into $T\cos \theta$ and $T\sin \theta$; (ii) the vertical **balance** equation with the reason "no vertical acceleration"; (iii) the horizontal equation identified as the centripetal force; (iv) the division step; (v) $r = L\sin \theta$ substituted and $\sin \theta$ cancelled; (vi) the final period, plus the remark that it is independent of the mass.

The period depends on h , not on the mass of the bob — the same result as the simple pendulum, with h in place of L .

WORKED EXAMPLE 6.3

Problem. A bob of mass 0.50 kg hangs from a string of length 1.0 m and moves in a horizontal circle with the string making 60° with the vertical. Taking $g = 10 \text{ m s}^{-2}$, find the tension, the speed of the bob and the period.

Solution. $\cos 60^\circ = 0.5$, $\sin 60^\circ = 0.866$, $\tan 60^\circ = 1.732$.

$$T = \frac{mg}{\cos \theta} = \frac{0.50 \times 10}{0.5} = 10 \text{ N}$$

Radius: $r = L\sin \theta = 1.0 \times 0.866 = 0.866 \text{ m}$. From $\tan \theta = v^2/rg$:

$$v = \sqrt{rg \tan \theta} = \sqrt{0.866 \times 10 \times 1.732} = \sqrt{15.0} = 3.87 \text{ m s}^{-1}$$

$$T_p = 2\pi \sqrt{\frac{L \cos \theta}{g}} = 2\pi \sqrt{\frac{0.5}{10}} = 2\pi(0.2236) = 1.40 \text{ s}$$

6.6 Motion in a vertical circle

Whirl a bucket of water in a vertical circle and the water does not fall out at the top. Here gravity acts along the radius at the top and bottom, so the speed is **not** constant — the motion is non-uniform circular motion.

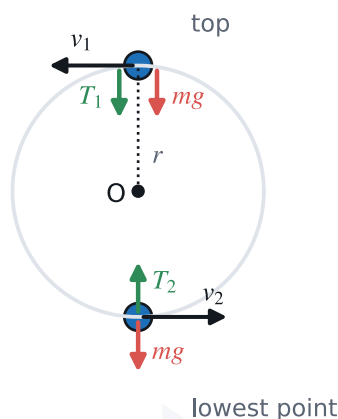


Figure 25. Motion in a vertical circle. At the top $T_1 + mg = mv_1^2/r$; at the bottom $T_2 - mg = mv_2^2/r$. Hence $T_2 - T_1 = 6mg$.

VERTICAL CIRCLE: MINIMUM SPEEDS AND THE $6mg$ TENSION DIFFERENCE

We start from $F = mv^2/r$ applied at the top and at the bottom of the circle, add conservation of energy between those two points, and reach three results: the least speed at the top, the least speed at the bottom, and the fact that the tension difference is always $6mg$.

Setting up. A mass m on a string of length r is whirled in a vertical circle. Let v_1 be its speed at the highest point and v_2 at the lowest point. At both of those two points the string is vertical, so the tension lies exactly along the radius — which is why these two points are the easy ones to analyse.

At the highest point

Step 1 — which way does each force point? At the top, the centre of the circle is **below** the mass. Both the weight mg and the tension T_1 (pulling towards the support, which is below) therefore point downwards, i.e. both point towards the centre.

Step 2 — write the centripetal equation. The sum of the inward forces equals mv^2/r :

$$T_1 + mg = \frac{mv_1^2}{r}$$

Step 3 — make T_1 the subject by subtracting mg from both sides:

$$T_1 = \frac{mv_1^2}{r} - mg$$

Step 4 — apply the physical limit. A string can pull but cannot push, so the tension can never be negative:

$$T_1 \geq 0$$

Step 5 — substitute Step 3 into that inequality:

$$\frac{mv_1^2}{r} - mg \geq 0$$

Step 6 — add mg to both sides:

$$\frac{mv_1^2}{r} \geq mg$$

Step 7 — cancel m and multiply both sides by r :

$$v_1^2 \geq gr$$

Step 8 — take the square root. The smallest allowed speed is the boundary case:

$$v_1(\text{min}) = \sqrt{gr}$$

At exactly this speed $T_1 = 0$: the string goes slack and **gravity alone** supplies the whole centripetal force.

At the lowest point

Step 9 — which way does each force point? At the bottom the centre is **above** the mass. The tension T_2 points up (towards the centre) and the weight points down (away from the centre), so the net inward force is $T_2 - mg$:

$$T_2 - mg = \frac{mv_2^2}{r}$$

Step 10 — make T_2 the subject:

$$T_2 = \frac{mv_2^2}{r} + mg$$

Linking top and bottom by energy

Step 11 — apply conservation of mechanical energy. The tension does no work (it is always perpendicular to the motion), so only gravity matters. Take the lowest point as the zero of potential energy; the highest point is a height $2r$ above it:

$$\frac{1}{2}mv_2^2 = \frac{1}{2}mv_1^2 + mg(2r)$$

Step 12 — multiply every term by 2:

$$mv_2^2 = mv_1^2 + 4mgr$$

Step 13 — cancel m throughout:

$$v_2^2 = v_1^2 + 4gr$$

Step 14 — put in the minimum top speed $v_1^2 = gr$:

$$v_2^2 = gr + 4gr = 5gr$$

Step 15 — take the square root:

$$v_2(\text{min}) = \sqrt{5gr}$$

The tension difference

Step 16 — subtract the top equation (Step 3) from the bottom one (Step 10):

$$T_2 - T_1 = \left(\frac{mv_2^2}{r} + mg \right) - \left(\frac{mv_1^2}{r} - mg \right)$$

Step 17 — remove the brackets carefully; the two mg terms now add:

$$T_2 - T_1 = \frac{mv_2^2}{r} - \frac{mv_1^2}{r} + 2mg$$

Step 18 — combine the first two terms over the common denominator r :

$$T_2 - T_1 = \frac{m(v_2^2 - v_1^2)}{r} + 2mg$$

Step 19 — use Step 13, which says $v_2^2 - v_1^2 = 4gr$:

$$T_2 - T_1 = \frac{m(4gr)}{r} + 2mg$$

Step 20 — cancel r :

$$T_2 - T_1 = 4mg + 2mg = 6mg$$

Result.

$$v_1(\text{min}) = \sqrt{gr}, \quad v_2(\text{min}) = \sqrt{5gr}, \quad T_2 - T_1 = 6mg$$

What it means. To keep the string taut all the way round you need a speed at the bottom of at least $\sqrt{5gr}$ — that is why you must swing a bucket of water briskly. The $6mg$ result is remarkable: the speed cancelled out, so the tension difference between bottom and top is the same no matter how fast you whirl it (as long as it does get round).

Conditions used. The string is light and inextensible, air resistance is neglected, and Step 11 assumed no energy is lost to friction. The $6mg$ result and $v_2^2 = v_1^2 + 4gr$ hold for **any** speed; only $v_1 = \sqrt{gr}$ and $v_2 = \sqrt{5gr}$ are the special minimum case. For a **rod** instead of a string the rod can push, so T_1 may be negative and the minimum top speed is zero.

THE EXAMINER IS LOOKING FOR

(i) A diagram with the forces drawn at the top and at the bottom; (ii) the correct signs — both forces inward at the top, opposing at the bottom; (iii) the argument "a string cannot push, so $T \geq 0$ " leading to $v_1 = \sqrt{gr}$; (iv) energy conservation over the height $2r$, with the remark that the tension does no work; (v) the subtraction giving $6mg$. Marks are most often lost on (ii) and on forgetting that the height difference is $2r$, not r .

WORKED EXAMPLE 6.4

Problem. A body of mass 1 kg tied to a string of length 1 m is rotated in a vertical circle. Its speed at the highest point is 5 m s^{-1} . Taking $g = 10 \text{ m s}^{-2}$, find the tension at the highest and lowest points, and the minimum speed needed at the top.

Solution. At the top:

$$T_1 = \frac{mv_1^2}{r} - mg = \frac{1 \times 25}{1} - 10 = 15 \text{ N}$$

Speed at the bottom: $v_2^2 = v_1^2 + 4gr = 25 + 40 = 65 \text{ m}^2\text{s}^{-2}$, so $v_2 = 8.06 \text{ m s}^{-1}$. Then

$$T_2 = \frac{mv_2^2}{r} + mg = 65 + 10 = 75 \text{ N}$$

Check: $T_2 - T_1 = 60 \text{ N} = 6mg$ ✓. The minimum speed at the top is $v_1(\text{min}) = \sqrt{gr} = \sqrt{10} = 3.16 \text{ m s}^{-1}$; the actual 5 m s^{-1} is comfortably above it.

6.7 Applications of banking

On a level road the centripetal force for a turn can only come from friction, which is unreliable when the surface is wet or dusty. So the outer edge of a curve is raised above the inner edge — the road is **banked** — and a component of the normal reaction does the job instead.

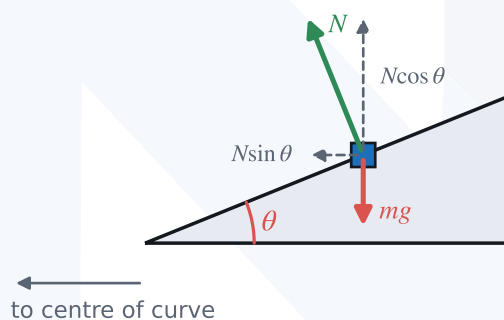


Figure 26. Cross-section of a banked road. The normal reaction N resolves into $N \cos \theta$ balancing mg and $N \sin \theta$ providing the centripetal force.

ANGLE OF BANKING OF A ROAD

We start from the two force equations for a car on a frictionless banked road and reach a formula for the correct banking angle.

Setting up. The road surface is tilted at an angle θ to the horizontal, with the outer edge raised. Take friction as negligible, so only two forces act on a car of mass m : its weight mg straight down, and the normal reaction N at right angles to the road surface. Because the surface is tilted by θ , the normal N is tilted by θ from the vertical.

Step 1 — resolve N into vertical and horizontal parts. Since N makes angle θ with the vertical:

vertical part = $N\cos\theta$, horizontal part = $N\sin\theta$ (pointing towards the centre)

Step 2 — the vertical equation. The car stays on the road and does not rise or sink, so it has no vertical acceleration; the vertical forces must balance:

$$N\cos\theta = mg$$

Step 3 — the horizontal equation. Horizontally the car is accelerating, because it is turning. The only horizontal force is $N\sin\theta$, so that component must be the whole centripetal force:

$$N\sin\theta = \frac{mv^2}{r}$$

Step 4 — divide Step 3 by Step 2. We divide because both N and m then cancel:

$$\frac{N\sin\theta}{N\cos\theta} = \frac{mv^2/r}{mg}$$

Step 5 — cancel N on the left and m on the right:

$$\frac{\sin\theta}{\cos\theta} = \frac{v^2}{rg}$$

Step 6 — use $\sin\theta/\cos\theta = \tan\theta$:

$$\tan\theta = \frac{v^2}{rg}$$

Step 7 — take the inverse tangent to get the angle:

$$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$$

Step 8 — or make v the subject instead. Multiply Step 6 by rg and take the square root:

$$v = \sqrt{rg\tan\theta}$$

Step 9 — the height of the outer edge. If the road is b wide, the raised edge is the "opposite" side of a right triangle whose hypotenuse is b :

$$h = b\sin\theta$$

Result.

$$\tan\theta = \frac{v^2}{rg}, \quad v = \sqrt{rg\tan\theta}, \quad h = b\sin\theta$$

What it means. The mass cancelled in Step 5, so one banking angle serves a motorcycle and a loaded truck equally — the engineer only has to choose the design speed v and the radius r . Sharper bends (small r) and faster roads (large v) need steeper banking. A cyclist leaning into a turn uses the identical relation.

Conditions used. Friction is neglected, so this gives the **one** speed at which no sideways friction is needed at all. The car is treated as a point mass (no toppling), the road is a circular arc of fixed ra-

dius, and θ is the same across the whole width. At speeds away from the design speed, friction must make up the difference — which is where the $v_{\max}$ formula below comes from.

THE EXAMINER IS LOOKING FOR

(i) A diagram of the tilted surface with mg , N , and N resolved into $N\cos\theta$ and $N\sin\theta$; (ii) the words "no vertical acceleration" beside the vertical equation; (iii) the horizontal component named as the centripetal force; (iv) the division step; (v) the final relation plus the statement "independent of the mass of the vehicle". A very common error is resolving mg along the slope instead of resolving N vertically — for banking, resolve N .

Including friction, the **maximum safe speed** on a banked road is

$$v_{\max} = \sqrt{\frac{rg(\mu + \tan\theta)}{1 - \mu\tan\theta}}$$

which reduces to $v_{\max} = \sqrt{\mu rg}$ for a level road ($\theta = 0$).

- **Roads.** Highway bends such as those on the Mugling–Naubise section are banked for a chosen design speed.
- **Railways.** The outer rail on a curve is laid higher than the inner rail ("superelevation") so that the flanges do not grind against the rails.
- **Aircraft.** A turning aeroplane banks its wings so that the horizontal component of the lift becomes the centripetal force.
- **Cyclists and bikers** lean inwards at an angle θ with $\tan\theta = v^2/rg$ — exactly the same relation.

WORKED EXAMPLE 6.5

Problem. A curve of radius 100 m on a highway is to be banked for a speed of 72 km h^{-1} . Find the angle of banking and, if the road is 10 m wide, the height of the outer edge above the inner edge. Take $g = 10 \text{ m s}^{-2}$.

Solution. First convert the speed: $72 \text{ km h}^{-1} = 72 \times \frac{5}{18} = 20 \text{ m s}^{-1}$.

$$\tan\theta = \frac{v^2}{rg} = \frac{400}{100 \times 10} = 0.40 \quad \Rightarrow \quad \theta = 21.8^\circ$$

The outer edge must be raised by

$$h = b\sin\theta = 10 \times \sin 21.8^\circ = 10 \times 0.371 = 3.71 \text{ m}$$

✓ Chapter summary

- $\theta = s/r$ in radians; $\omega = d\theta/dt = 2\pi/T = 2\pi f$; $\alpha = d\omega/dt$. Angular kinematic equations mirror the linear ones.
- $v = r\omega$ and $a_t = r\alpha$; all points of a rigid rotating body share ω but not v .
- Centripetal acceleration $a_c = v^2/r = r\omega^2 = 4\pi^2 r/T^2$, directed towards the centre. Total acceleration $= \sqrt{a_c^2 + a_t^2}$.
- Centripetal force $F_c = mv^2/r$ is supplied by tension, friction, gravity or a component of the normal reaction. Centrifugal force is a fictitious force.
- Conical pendulum: $\tan \theta = v^2/rg$, $T = mg/\cos \theta$, period $T_p = 2\pi\sqrt{L\cos \theta/g} = 2\pi\sqrt{h/g}$.
- Vertical circle: $v_{\text{top}}(\text{min}) = \sqrt{gr}$, $v_{\text{bottom}}(\text{min}) = \sqrt{5gr}$, $v_2^2 = v_1^2 + 4gr$, and $T_2 - T_1 = 6mg$ always.
- Banking: $\tan \theta = v^2/rg$, independent of mass; $v_{\text{max}} = \sqrt{rg(\mu + \tan \theta)/(1 - \mu \tan \theta)}$; on a level road $v_{\text{max}} = \sqrt{\mu rg}$.

Practice questions

Group A — Multiple choice (1 mark each)

1. A body moves in a circle at constant speed. Which quantity remains constant? [1]

a velocity	b acceleration	c kinetic energy	d momentum
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2. The angular velocity of a body making 300 revolutions per minute is [1]

a 5 rad s^{-1}	b $10\pi \text{ rad s}^{-1}$	c 300 rad s^{-1}	d $60\pi \text{ rad s}^{-1}$
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3. The angle of banking of a road is independent of [1]

a speed of the vehicle	b radius of the curve
c mass of the vehicle	d acceleration due to gravity
4. The minimum speed at the highest point of a vertical circle of radius r is [1]

a $\sqrt{5gr}$	b $\sqrt{2gr}$	c $\sqrt{gr}$	d zero
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5. The work done by the centripetal force in one complete revolution is [1]

a mv^2/r	b $2\pi r \cdot mv^2/r$	c zero	d $\frac{1}{2}mv^2$
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ANSWERS TO GROUP A

1. (c) — the speed is constant, so $\frac{1}{2}mv^2$ is constant; velocity, acceleration and momentum all change direction.
2. (b) — $\omega = 2\pi N/60 = 2\pi(300)/60 = 10\pi \text{ rad s}^{-1}$.
3. (c) — $\tan \theta = v^2/rg$ contains no mass term.
4. (c) — at the limit the tension is zero and gravity alone gives $mg = mv^2/r$.
5. (c) — the force is always perpendicular to the displacement, so $W = 0$.

Group B — Short answer (5 marks each)

1. Define angular velocity and angular acceleration. Derive the relations $v = r\omega$ and $a_t = r\alpha$. [5]
2. Derive an expression for the centripetal acceleration of a body moving uniformly in a circle of radius r with speed v . [5]
3. What is a conical pendulum? Derive an expression for its period. [5]
4. Show that for a body moving in a vertical circle the difference between the tensions at the lowest and the highest points is $6mg$. [5]
5. A stone of mass 0.20 kg is whirled in a horizontal circle of radius 0.50 m at 4 rev s^{-1} . Calculate its linear speed and the tension in the string. [5]

ANSWERS TO GROUP B

5. $\omega = 2\pi f = 2\pi(4) = 8\pi = 25.13 \text{ rad s}^{-1}$. Linear speed $v = r\omega = 0.50 \times 25.13 = 12.6 \text{ m s}^{-1}$. Tension $= mr\omega^2 = 0.20 \times 0.50 \times (25.13)^2 = 0.10 \times 631.7 = 63.2 \text{ N}$.

1., 2., 3., 4. See §6.2, the derivation box in §6.3, §6.5 and §6.6 respectively.

Group C — Long answer (8 marks each)

1. (a) What is meant by banking of a road? Derive an expression for the angle of banking of a curved road for a vehicle moving with speed v on a curve of radius r . [5]
 (b) A cyclist rides round a circular track of radius 20 m at 10 m s^{-1} . Find the angle through which he must lean from the vertical. Take $g = 10 \text{ m s}^{-2}$. [3]
2. (a) Derive expressions for the minimum speeds at the highest and lowest points for a body to just complete a vertical circle of radius r . [5]
 (b) A bucket of water of mass 2 kg is whirled in a vertical circle of radius 1 m . Find the minimum speed at the top for the water not to spill, and the tension in the rope at the lowest point in that case ($g = 10 \text{ m s}^{-2}$). [3]

ANSWERS TO GROUP C

1. (b) A leaning cyclist obeys the same relation, $\tan \theta = v^2/rg = 100/(20 \times 10) = 0.50$, so $\theta = 26.6^\circ$ from the vertical.

2. (b) Minimum speed at the top: $v_1 = \sqrt{gr} = \sqrt{10 \times 1} = 3.16 \text{ m s}^{-1}$. Then $v_2^2 = v_1^2 + 4gr = 10 + 40 = 50$, so $T_2 = \frac{mv_2^2}{r} + mg = \frac{2 \times 50}{1} + 20 = 120 \text{ N}$ (which is $6mg$, as expected when $T_1 = 0$).

UNIT 7

Gravitation

Mechanics • 10 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Newton's law of gravitation
- Gravitational field strength
- Gravitational potential; Gravitational potential energy
- Variation in value of 'g' due to altitude and depth
- Centre of mass and centre of gravity
- Motion of a satellite: orbital velocity and time period
- Escape velocity
- Potential and kinetic energy of the satellite
- Geostationary satellite
- GPS

Gravitation is the weakest of the four fundamental interactions, yet it shapes the largest structures in the universe, because it is always attractive and never cancels out. One equation written by Newton in 1687 explains a falling apple in Lumbini, the orbit of the Moon, the tides, the period of a communication satellite over the equator and the reading on a GPS receiver in Kathmandu. This unit builds that equation, measures the Earth with it, and then flies satellites.

WHAT THIS UNIT IS REALLY ABOUT

Two families of formulas, and knowing which family a question belongs to. The **field** family (F , g) is about forces and varies as $1/r^2$. The **energy** family (V , U) is about work and varies as $1/r$, and is always negative. Escape velocity, orbital energy and binding energy all come from the second family.

7.1 Newton's law of gravitation

NEWTON'S LAW OF UNIVERSAL GRAVITATION

Every particle in the universe attracts every other particle with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The force acts along the line joining them.

$$F = \frac{Gm_1m_2}{r^2}$$

G is the **universal gravitational constant**. Its value, first measured by Henry Cavendish in 1798 with a torsion balance, is

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

with dimensions $[M^{-1}L^3T^{-2}]$. It is "universal" because it is the same everywhere, for every pair of masses, at every temperature.

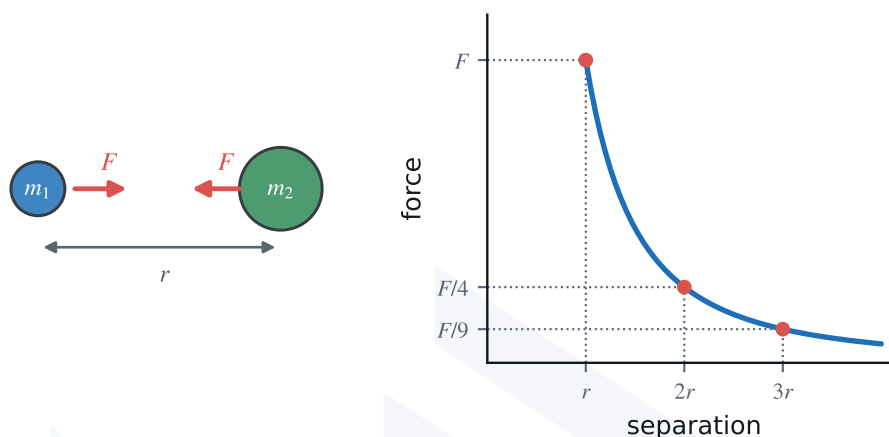


Figure 27. Left: the two forces are equal, opposite and along the line of centres. Right: the inverse-square fall-off — doubling r quarters the force.

Three points that carry marks:

1. The forces form a **Newton's third law pair** — equal in magnitude, opposite in direction — even when the masses are wildly different. The Earth pulls an apple with the same force with which the apple pulls the Earth.
2. The law is stated for **point masses**. For a uniform solid sphere it still holds exactly, with r measured from the **centre** — Newton proved this.
3. Gravitational force is **independent of the medium** between the bodies and of the presence of other masses. Forces from several masses simply add as vectors (the principle of superposition).

WORKED EXAMPLE 7.1

Problem. Taking $g = 9.8 \text{ m s}^{-2}$ at the Earth's surface and $R = 6.4 \times 10^6 \text{ m}$, calculate the mass and the mean density of the Earth. ($G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$.)

Solution. For a body of mass m on the surface, its weight is the gravitational pull:

$$mg = \frac{GMm}{R^2} \quad \Rightarrow \quad M = \frac{gR^2}{G}$$

$$M = \frac{9.8 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}} = \frac{4.01 \times 10^{14}}{6.67 \times 10^{-11}} = 6.0 \times 10^{24} \text{ kg}$$

Mean density:

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{6.0 \times 10^{24}}{\frac{4}{3}\pi (6.4 \times 10^6)^3} = \frac{6.0 \times 10^{24}}{1.10 \times 10^{21}} = 5.5 \times 10^3 \text{ kg m}^{-3}$$

This is about twice the density of surface rock, which is the first clue that the Earth has a dense iron core. Cavendish called this "weighing the Earth".

7.2 Gravitational field strength

A mass changes the space around it so that any other mass placed there feels a force. That region is the **gravitational field**.

GRAVITATIONAL FIELD STRENGTH

The gravitational field strength (or gravitational intensity) at a point is the force per unit mass experienced by a small test mass placed at that point:

$$\vec{E} = \frac{\vec{F}}{m}$$

It is a **vector**, measured in N kg^{-1} (or m s^{-2} , the same thing), directed towards the attracting mass.

For a point mass (or outside a uniform sphere) of mass M , at distance r ,

$$E = \frac{GM/r^2 \cdot m}{m} = \frac{GM}{r^2}$$

At the surface of the Earth $r = R$ and the field strength is exactly what we call g :

$$g = \frac{GM}{R^2}$$

So g **is not a property of the falling body — it is a property of the Earth and of where you stand**. Field lines point radially inwards; their crowding near the surface shows where the field is strong.

WEIGHT AND MASS ARE DIFFERENT

Mass (kg) is the amount of matter and is the same everywhere. Weight (N) is the gravitational force mg and changes with location. A 60 kg student has mass 60 kg on the Moon too, but weighs only about $60 \times 1.62 = 97$ N there instead of 588 N.

7.3 Gravitational potential; gravitational potential energy

Gravity is a conservative force, so a potential energy can be defined. The natural zero is at **infinite separation**, where the force vanishes.

GRAVITATIONAL POTENTIAL ENERGY $U = -GMm/r$ AND POTENTIAL $V = -GM/r$

We start from the definition of potential energy as the work needed to bring a mass in from infinity, and reach the $-GMm/r$ formula and the potential that follows from it.

Setting up. A mass M sits at the origin. We bring a small mass m from infinity (where we agree $U = 0$) slowly in to a distance r . "Slowly" means the body has no kinetic energy at either end, so all the work done shows up as stored energy.

Step 1 — the force at a general distance x . By Newton's law of gravitation the attraction on m when it is a distance x from M has magnitude

$$F_{\text{grav}} = \frac{GMm}{x^2}$$

and it points inwards, towards M .

Step 2 — fix a sign convention. Measure x outwards from M , so "outwards" is positive. Gravity pulls **inwards**, i.e. in the direction of decreasing x , so as a signed quantity the gravitational force is

$$F = -\frac{GMm}{x^2}$$

Step 3 — work done by gravity over a tiny displacement dx . Work is force times displacement:

$$dW = F \, dx = -\frac{GMm}{x^2} \, dx$$

Step 4 — add up all the tiny bits from $x = \infty$ to $x = r$:

$$W = \int_{\infty}^r -\frac{GMm}{x^2} \, dx$$

Step 5 — take the constants outside the integral. G , M and m do not change during the journey:

$$W = -GMm \int_{\infty}^r \frac{dx}{x^2}$$

Step 6 — integrate x^{-2} . Using $\int x^{-2} dx = -x^{-1}$:

$$W = -GMm \left[-\frac{1}{x} \right]_{\infty}^r$$

Step 7 — the two minus signs cancel:

$$W = GMm \left[\frac{1}{x} \right]_{\infty}^r$$

Step 8 — put in the limits (value at the top limit minus value at the bottom limit):

$$W = GMm \left(\frac{1}{r} - \frac{1}{\infty} \right)$$

Step 9 — use $1/\infty = 0$:

$$W = \frac{GMm}{r}$$

This is the work done **by gravity** as the mass falls in from infinity, and it is positive — gravity helped, as you would expect.

Step 10 — convert that into potential energy. Whenever a conservative force does work W , the potential energy falls by the same amount:

$$U(r) - U(\infty) = -W$$

Step 11 — put in $U(\infty) = 0$ (our chosen zero) and $W = GMm/r$:

$$U(r) - 0 = -\frac{GMm}{r}$$

so

$$U = -\frac{GMm}{r}$$

Step 12 — get the potential. The gravitational potential V is defined as the potential energy **per unit mass**, so divide by m :

$$V = \frac{U}{m} = -\frac{GM}{r}$$

Result.

$$U = -\frac{GMm}{r} \text{ (joule)}, \quad V = -\frac{GM}{r} \text{ (J kg}^{-1}\text{)}$$

Both are **scalars** — there is no direction to worry about, which is what makes potential so much easier to work with than field.

What it means. The sign is not a mistake. We chose $U=0$ at infinity, and gravity is always attractive, so every real point is a place you would have to do work to climb out of. Hence U is negative everywhere, deepest (most negative) close in and rising towards zero far away — a "potential well".

Conditions used. M is treated as a point mass (or a uniform sphere, for which this is exactly true outside it), the mass is moved **slowly** so no kinetic energy is involved, and the zero of potential is taken at infinity. U belongs to the pair of masses, not to either one alone.

THE EXAMINER IS LOOKING FOR

(i) The statement that $U=0$ at infinity — write it down; (ii) $F = GMm/x^2$ at a general distance x ; (iii) the integral with limits ∞ to r ; (iv) the integration of $1/x^2$ shown properly; (v) the negative sign explained in words; (vi) $V = U/m$ with the unit J kg^{-1} . The explanation of the minus sign is worth a mark on its own and is almost always missing.

WHY THE POTENTIAL IS NEGATIVE

Zero potential is at infinity. Since gravity is always attractive, work must be done against it to take a mass out to infinity, so every finite point is at a lower energy than infinity, i.e. negative. The potential is most negative at the centre of the well and rises towards zero as $r \rightarrow \infty$.

Field and potential are linked by

$$E = -\frac{dV}{dr}$$

— the field is the negative gradient (steepness) of the potential.

Near the surface, over heights small compared with R , the difference $U_2 - U_1$ reduces to the familiar mgh :

$$\Delta U = GMm\left(\frac{1}{R} - \frac{1}{R+h}\right) = \frac{GMmh}{R(R+h)} \approx \frac{GMmh}{R^2} = mgh \quad (h \ll R)$$

So $U = mgh$ is not a different formula — it is the $1/r$ formula in disguise, valid only close to the ground.

7.4 Variation in the value of g due to altitude and depth

VARIATION OF g WITH ALTITUDE (HEIGHT ABOVE THE SURFACE)

We start from $g = GM/R^2$ at the surface and reach a formula for g at a height h above it.

Step 1 — g at the surface. A mass m on the surface is a distance R from the centre of the Earth, so its weight is GMm/R^2 . But weight is also mg , so:

$$mg = \frac{GMm}{R^2} \implies g = \frac{GM}{R^2}$$

Step 2 — g at a height h . At height h the body is a distance $R+h$ from the centre. The only change is the distance, so

$$g_h = \frac{GM}{(R+h)^2}$$

Step 3 — form the ratio g_h/g . Divide Step 2 by Step 1. The GM cancels:

$$\frac{g_h}{g} = \frac{1/(R+h)^2}{1/R^2} = \frac{R^2}{(R+h)^2}$$

Step 4 — write it as a single squared bracket:

$$g_h = g\left(\frac{R}{R+h}\right)^2$$

Step 5 — divide the top and bottom inside the bracket by R . Dividing top and bottom by the same thing does not change a fraction:

$$g_h = g\left(\frac{1}{1+\frac{h}{R}}\right)^2$$

Step 6 — write the reciprocal as a negative power:

$$g_h = g\left(1 + \frac{h}{R}\right)^{-2}$$

This formula is **exact**. The next two steps only simplify it for small heights.

Step 7 — use the binomial expansion. For small x , $(1+x)^n \approx 1+nx$. Here $x = h/R$ and $n = -2$:

$$g_h \approx g\left(1 + (-2)\frac{h}{R}\right)$$

Step 8 — tidy up:

$$g_h \approx g\left(1 - \frac{2h}{R}\right)$$

Result.

$$g_h = g\left(1 + \frac{h}{R}\right)^{-2} \quad (\text{exact}), \quad g_h \approx g\left(1 - \frac{2h}{R}\right) \quad (h \ll R)$$

What it means. g gets smaller as you climb, and for small heights the fractional drop is $2h/R$ — twice the fraction of the Earth's radius you have climbed. At the top of Sagarmatha ($h \approx 8.8$ km) this is about 0.28%, so g falls from 9.800 to about 9.77 m s⁻². g only becomes zero at infinity.

Conditions used. The Earth is treated as a uniform sphere so that it acts like a point mass at its centre; rotation is ignored. Step 7 needs $h \ll R$ — for a satellite at $h = R$ you must use the exact form, which gives $g/4$, not $g(1 - 2) = -g$.

VARIATION OF g WITH DEPTH (BELOW THE SURFACE)

We start from the shell theorem and reach a formula for g at a depth d below the surface — and show that g becomes zero at the centre.

Setting up. Treat the Earth as a uniform sphere of radius R and density ρ . A body at depth d is at a distance $(R - d)$ from the centre.

Step 1 — the shell theorem. A uniform spherical shell exerts **no net gravitational force** on anything inside it: the pulls from all sides cancel exactly. So the outer shell, of thickness d , can be ignored, and only the inner sphere of radius $(R - d)$ attracts the body.

Step 2 — the mass of that inner sphere. Mass = density $\times$ volume, and the volume of a sphere is $\frac{4}{3}\pi(\text{radius})^3$:

$$M' = \frac{4}{3}\pi(R - d)^3\rho$$

Step 3 — the field at the body's position. The inner sphere behaves like a point mass M' at the centre, a distance $(R - d)$ away:

$$g_d = \frac{GM'}{(R - d)^2}$$

Step 4 — substitute M' from Step 2:

$$g_d = \frac{G}{(R - d)^2} \cdot \frac{4}{3}\pi(R - d)^3\rho$$

Step 5 — cancel $(R - d)^2$ against two of the three factors in $(R - d)^3$, leaving one:

$$g_d = \frac{4}{3}\pi G\rho (R - d)$$

Step 6 — do the same thing at the surface. Putting $d = 0$ in Step 5 gives the surface value:

$$g = \frac{4}{3}\pi G\rho R$$

Step 7 — divide Step 5 by Step 6. All the constants $\frac{4}{3}\pi G\rho$ cancel:

$$\frac{g_d}{g} = \frac{R-d}{R}$$

Step 8 — split the fraction on the right:

$$\frac{g_d}{g} = \frac{R}{R} - \frac{d}{R} = 1 - \frac{d}{R}$$

Step 9 — multiply both sides by g :

$$g_d = g\left(1 - \frac{d}{R}\right)$$

Step 10 — put $d=R$ (the centre of the Earth):

$$g_{\text{centre}} = g(1 - 1) = 0$$

Result.

$$g_d = g\left(1 - \frac{d}{R}\right), \quad g = 0 \text{ at the centre}$$

What it means. Unlike the altitude case, g falls **linearly** with depth — and it reaches exactly zero at the centre, where a body would be weightless because the pulls from all directions cancel. Notice also that going up a small distance h costs $2h/R$ of g , while going down the same distance costs only d/R — twice as much is lost by climbing as by digging.

Conditions used. The Earth is assumed to be of **uniform density**, which is the only way Step 2 works. (In reality the core is far denser, so g actually rises slightly for the first 3000 km before falling.) Rotation and the non-spherical shape are ignored.

THE EXAMINER IS LOOKING FOR

For the depth derivation: (i) the shell-theorem statement — "the shell above exerts no force" — in words; (ii) $M' = \frac{4}{3}\pi(R-d)^3\rho$; (iii) the cancellation in Step 5; (iv) the surface value with $d=0$; (v) the ratio, giving $g_d = g(1 - d/R)$; (vi) the substitution $d=R$ to show $g=0$ at the centre. State the uniform-density assumption — it is frequently a listed mark.

Three consequences follow:

- g decreases **linearly** with depth and reaches **zero at the centre** ($d=R$). A body at the centre of the Earth is weightless.
- g decreases with altitude as $1/(R+h)^2$, reaching zero only at infinity.
- Near the surface a given small distance costs twice as much g going up as going down ($2h/R$ against d/R).

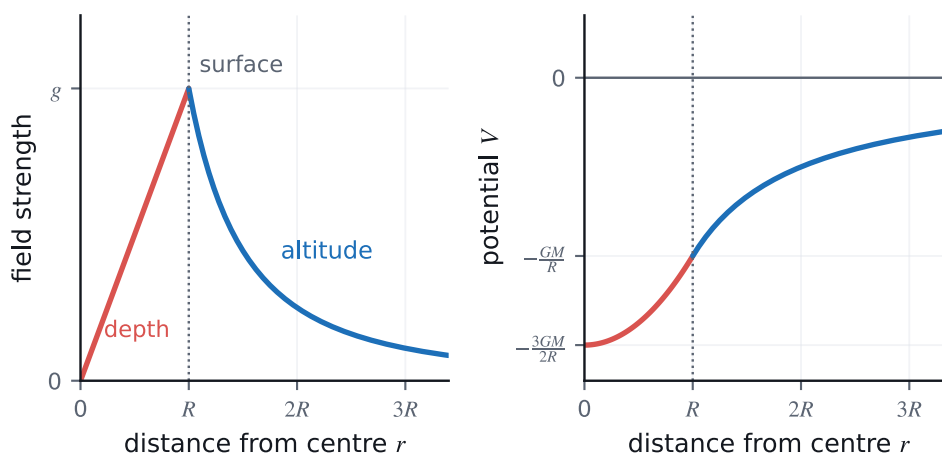


Figure 28. Left: g inside the Earth rises linearly from zero at the centre, peaks at the surface, then falls as $1/r^2$. Right: the potential well $V = -GM/r$, flat-bottomed inside the Earth and rising towards zero at large r .

Two further effects change g over the Earth's surface, both worth a mark:

Cause	Effect on g
Shape: the Earth bulges at the equator ($R_{eq} > R_{pole}$)	g is greatest at the poles (9.83), least at the equator (9.78)
Rotation: $g' = g - R\omega^2 \cos^2 \lambda$ at latitude λ	reduces g most at the equator, no effect at the poles

WORKED EXAMPLE 7.2

Problem. Take $g = 9.8 \text{ m s}^{-2}$ and $R = 6.4 \times 10^6 \text{ m}$. Find g (a) at the summit of Sagarmatha (Mt. Everest), height 8849 m, (b) at a height equal to the Earth's radius, and (c) at a depth of $R/2$.

Solution.

(a) Here $h = 8849 \text{ m} \ll R$, so use the approximation:

$$g_h = g \left(1 - \frac{2h}{R} \right) = 9.8 \left(1 - \frac{2 \times 8849}{6.4 \times 10^6} \right) = 9.8(1 - 0.002765) = 9.77 \text{ m s}^{-2}$$

Climbing the highest mountain on Earth reduces your weight by only about 0.28%.

(b) Here $h = R$ is not small, so use the exact form:

$$g_h = g \left(\frac{R}{R+h} \right)^2 = \frac{g}{4} = 2.45 \text{ m s}^{-2}$$

(c) $g_d = g \left(1 - \frac{d}{R} \right) = 9.8 \left(1 - \frac{1}{2} \right) = 4.9 \text{ m s}^{-2}$.

DO NOT USE THE BINOMIAL FORM WHEN h IS LARGE

$g_h = g(1 - 2h/R)$ is a first-order approximation. At $h = R$ it would give $g_h = -9.8 \text{ m s}^{-2}$, which is nonsense. Use it only when h is a few per cent of R at most; otherwise use $g(R/(R+h))^2$.

7.5 Centre of mass and centre of gravity

CENTRE OF MASS AND CENTRE OF GRAVITY

The **centre of mass** of a system is the point at which the whole mass may be considered concentrated for the purpose of describing translational motion:

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}, \quad y_{cm} = \frac{\sum m_i y_i}{\sum m_i}$$

The **centre of gravity** is the point through which the resultant weight of the body acts.

In a **uniform** gravitational field the two points coincide, because each mass element is weighted by the same g . They separate only when the body is so tall that g differs measurably across it — for a mountain or a long satellite, the centre of gravity lies slightly below the centre of mass, because the lower parts are pulled harder. For anything of laboratory size the distinction is academic, but NEB asks for it.

	Centre of mass	Centre of gravity
Depends on	mass distribution only	mass distribution and the field
Exists in zero gravity?	Yes	No
Position for a uniform field	same point	same point
Position for a very large body	fixed	slightly towards the stronger field

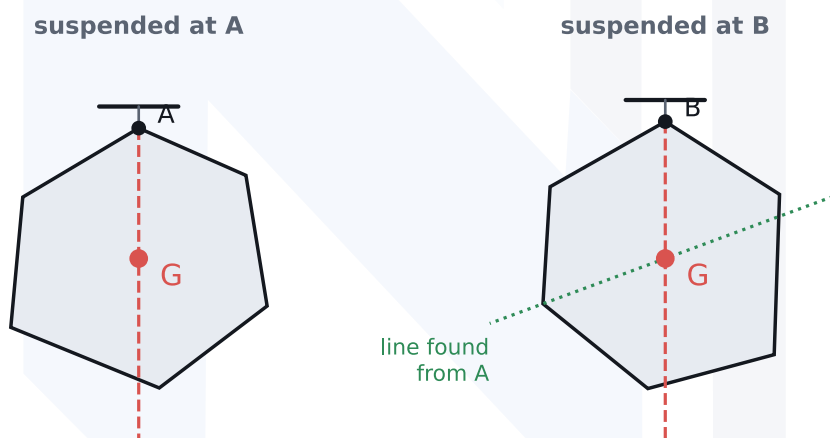


Figure 29. Locating the centre of gravity of an irregular lamina. When the body hangs freely, G lies on the vertical through the point of suspension; repeating with a second point fixes G at the intersection.

WORKED EXAMPLE 7.3

Problem. Three particles of masses 1 kg, 2 kg and 3 kg are placed at the points (0, 0), (6, 0) and (0, 6) metres. Locate their centre of mass.

Solution. Total mass $M = 1 + 2 + 3 = 6$ kg.

$$x_{cm} = \frac{1(0) + 2(6) + 3(0)}{6} = \frac{12}{6} = 2 \text{ m}$$

$$y_{cm} = \frac{1(0) + 2(0) + 3(6)}{6} = \frac{18}{6} = 3 \text{ m}$$

The centre of mass is at (2, 3) m. Note it lies nearer the heavier masses, and it need not be at the position of any actual particle.

7.6 Motion of a satellite: orbital velocity and time period

A satellite in a circular orbit of radius r around a planet of mass M is held there because the gravitational attraction plays the part of the centripetal force.

ORBITAL VELOCITY AND TIME PERIOD OF A SATELLITE

We start from the single physical idea that gravity is the centripetal force for a satellite, and reach expressions for its orbital speed and its period — and Kepler's third law as a bonus.

Setting up. A satellite of mass m moves in a circle of radius $r = R + h$ about a planet of mass M and radius R , at a height h above the surface, with orbital speed v_o .

Step 1 — the gravitational force on the satellite. By Newton's law of gravitation, at a distance r from the centre:

$$F_{grav} = \frac{GMm}{r^2}$$

Step 2 — the centripetal force it needs. To move in a circle of radius r at speed v_o , a mass m needs an inward force

$$F_{cent} = \frac{mv_o^2}{r}$$

Step 3 — equate them. Gravity is the only force acting, so it must be exactly the force required — no more and no less:

$$\frac{GMm}{r^2} = \frac{mv_o^2}{r}$$

Step 4 — cancel m from both sides. It appears once on each side, so the answer will not depend on the satellite's mass:

$$\frac{GM}{r^2} = \frac{v_o^2}{r}$$

Step 5 — multiply both sides by r :

$$\frac{GM}{r} = v_o^2$$

Step 6 — take the square root:

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{R+h}}$$

Step 7 — rewrite using surface quantities. At the surface $g = GM/R^2$, so $GM = gR^2$. Substitute:

$$v_o = \sqrt{\frac{gR^2}{R+h}}$$

Step 8 — take R^2 out of the root as R :

$$v_o = R\sqrt{\frac{g}{R+h}}$$

Step 9 — the period. The satellite travels one circumference $2\pi r$ in one period T at steady speed v_o , so time = distance $\div$ speed:

$$T = \frac{2\pi r}{v_o}$$

Step 10 — substitute v_o from Step 6:

$$T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

Step 11 — dividing by $\sqrt{GM/r}$ is the same as multiplying by $\sqrt{r/GM}$:

$$T = 2\pi r \sqrt{\frac{r}{GM}}$$

Step 12 — bring the r outside the root inside it, where it becomes r^2 :

$$T = 2\pi \sqrt{\frac{r^2 \cdot r}{GM}} = 2\pi \sqrt{\frac{r^3}{GM}}$$

Step 13 — square both sides:

$$T^2 = 4\pi^2 \frac{r^3}{GM}$$

Step 14 — group the constants. $4\pi^2$, G and M are all fixed for a given planet:

$$T^2 = \left(\frac{4\pi^2}{GM}\right)r^3 \implies T^2 \propto r^3$$

Result.

$$v_o = \sqrt{\frac{GM}{R+h}} = R\sqrt{\frac{g}{R+h}}, \quad T = 2\pi\sqrt{\frac{r^3}{GM}}, \quad T^2 \propto r^3$$

That last line is **Kepler's third law** — here derived from Newton's laws rather than assumed from observation.

What it means. The satellite's mass cancelled in Step 4, so a tiny cubesat and a space station at the same height orbit at the same speed and with the same period. A higher orbit is a slower orbit ($v_o \propto 1/\sqrt{r}$) with a longer period — which is why a geostationary satellite has to sit 36 000 km up. For a satellite skimming the surface, $h \approx 0$ and $v_o = \sqrt{gR} \approx 7.9 \text{ km s}^{-1}$.

Conditions used. The orbit is **circular** (for an ellipse, r becomes the mean distance), there is no air resistance, the planet is a uniform sphere, and the satellite's mass is far smaller than the planet's so the planet does not move.

THE EXAMINER IS LOOKING FOR

(i) The sentence "the gravitational force provides the centripetal force"; (ii) both force expressions written out; (iii) the cancellation of m , and a comment that the speed is therefore independent of the satellite's mass; (iv) the substitution $GM = gR^2$ if the question gives g and R rather than G and M ; (v) $T = 2\pi r/v_o$ and the final $T^2 \propto r^3$ named as Kepler's third law.

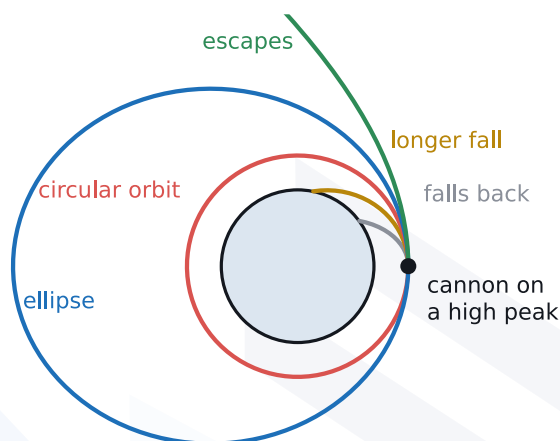


Figure 30. Newton's thought experiment. Fired slowly, the projectile falls back; at $v_o = \sqrt{GM/r}$ the fall exactly matches the curvature of the Earth and it orbits; beyond $v_e = \sqrt{2GM/r}$ it never returns.

WORKED EXAMPLE 7.4

Problem. A satellite orbits the Earth at a height of 300 km. Taking $g = 9.8 \text{ m s}^{-2}$ and $R = 6.4 \times 10^6 \text{ m}$, find its orbital speed and period.

Solution. Orbital radius $r = R + h = 6.4 \times 10^6 + 0.3 \times 10^6 = 6.7 \times 10^6 \text{ m}$, and $GM = gR^2 = 9.8 \times 4.096 \times 10^{13} = 4.01 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$.

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{4.01 \times 10^{14}}{6.7 \times 10^6}} = \sqrt{5.99 \times 10^7} = 7.74 \times 10^3 \text{ m s}^{-1}$$

$$T = \frac{2\pi r}{v_o} = \frac{2\pi \times 6.7 \times 10^6}{7.74 \times 10^3} = 5.44 \times 10^3 \text{ s} = 90.6 \text{ minutes}$$

A low-Earth-orbit satellite therefore circles the planet about 16 times a day.

7.7 Escape velocity

ESCAPE VELOCITY

The escape velocity is the minimum speed with which a body must be projected from the surface of a planet so that it escapes the planet's gravitational field and never returns — that is, it just reaches infinity with zero speed.

ESCAPE VELOCITY $v_e = \sqrt{2gR}$

We start from conservation of energy between the planet's surface and infinity, and reach the minimum launch speed needed to get away and never come back.

Setting up. A body of mass m is fired straight up from the surface of a planet of mass M and radius R with speed v_e . Take the gravitational potential energy to be zero at infinity, as usual.

Step 1 — energy at the surface. The body has kinetic energy $\frac{1}{2}mv_e^2$ and potential energy $-GMm/R$ (using $U = -GMm/r$ with $r = R$). The total is

$$E_{\text{surface}} = \frac{1}{2}mv_e^2 - \frac{GMm}{R}$$

Step 2 — energy at infinity. "Just escapes" means it arrives at infinity with no speed left over, so its kinetic energy there is zero. Its potential energy there is also zero, by our choice of zero level. So:

$$E_{\infty} = 0 + 0 = 0$$

Step 3 — apply conservation of energy. Gravity is a conservative force and we neglect air resistance, so the total energy does not change:

$$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0$$

Step 4 — move the second term to the right:

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R}$$

Step 5 — cancel m from both sides. It appears once on each side, so the answer does not depend on the body's mass:

$$\frac{1}{2}v_e^2 = \frac{GM}{R}$$

Step 6 — multiply both sides by 2:

$$v_e^2 = \frac{2GM}{R}$$

Step 7 — take the square root:

$$v_e = \sqrt{\frac{2GM}{R}}$$

Step 8 — rewrite in terms of g . At the surface $g = GM/R^2$, so $GM = gR^2$:

$$v_e = \sqrt{\frac{2gR^2}{R}}$$

Step 9 — cancel one R :

$$v_e = \sqrt{2gR}$$

Step 10 — a third form, in terms of density. The planet's mass is $M = \frac{4}{3}\pi R^3 \rho$. Putting this into Step 7:

$$v_e = \sqrt{\frac{2G}{R} \cdot \frac{4}{3}\pi R^3 \rho} = R\sqrt{\frac{8}{3}\pi G\rho}$$

Result.

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} = R\sqrt{\frac{8}{3}\pi G\rho}$$

For the Earth, $v_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = 11.2 \text{ km s}^{-1}$.

What it means. m cancelled, so a pebble and a rocket need the same escape speed. There is also no angle in the answer — energy depends only on distance, so the direction of projection does not matter either. Escape speed depends only on the planet: its mass (or density) and its radius.

Conditions used. Air resistance is neglected (a real rocket needs more), the body is given all its speed at once at the surface (not continuously pushed), the planet does not rotate, and there are no other bodies pulling on it.

ESCAPE VELOCITY IS $\sqrt{2}$ TIMES THE ORBITAL VELOCITY

We start from the two results just obtained and reach the ratio between them.

Step 1 — orbital speed for an orbit close to the surface. Put $h=0$, i.e. $r=R$, into $v_o = \sqrt{GM/r}$:

$$v_o = \sqrt{\frac{GM}{R}} = \sqrt{gR}$$

Step 2 — write down the escape speed:

$$v_e = \sqrt{2gR}$$

Step 3 — split the root. Since $\sqrt{ab} = \sqrt{a}\sqrt{b}$:

$$v_e = \sqrt{2} \cdot \sqrt{gR}$$

Step 4 — recognise the second factor as v_o from Step 1:

$$v_e = \sqrt{2} v_o$$

Result. $v_e = \sqrt{2} v_o \approx 1.41 v_o$.

What it means. A satellite in low orbit is already travelling at 7.9 km s^{-1} ; only 41 % more speed would let it leave the Earth altogether. In energy terms, escaping needs exactly **twice** the kinetic energy of orbiting, because $v_e^2 = 2v_o^2$.

Condition used. The orbit must be close to the surface ($h \ll R$), so that the same R appears in both formulas.

For the Earth, $v_e = \sqrt{2 \times 9.8 \times 6.4 \times 10^6} = \sqrt{1.254 \times 10^8} = 11.2 \text{ km s}^{-1}$.

Body	$g \text{ (m s}^{-2}\text{)}$	$R \text{ (km)}$	$v_e \text{ (km s}^{-1}\text{)}$
Moon	1.62	1738	2.4
Earth	9.8	6400	11.2
Mars	3.7	3390	5.0
Jupiter	24.8	71 500	59.5
Sun	274	696 000	618

WHY THE MOON HAS NO ATMOSPHERE

The escape velocity of the Moon (2.4 km s^{-1}) is comparable to the root-mean-square speed of ordinary gas molecules at its daytime temperature, so over geological time the gas molecules have all escaped. The Earth's 11.2 km s^{-1} is far above molecular speeds, so our atmosphere stays — except for hydrogen and helium, the lightest and fastest, which do leak away.

ESCAPE VELOCITY DOES NOT DEPEND ON THE DIRECTION OF PROJECTION

v_e contains no angle, because the derivation uses only energy, and potential energy depends only on r . Nor does it depend on the mass of the projectile. It does depend on the mass and radius of the planet.

7.8 Potential and kinetic energy of the satellite

For a satellite of mass m in a circular orbit of radius r , the orbital condition $GMm/r^2 = mv_o^2/r$ gives $mv_o^2 = GMm/r$, so

$$K = \frac{1}{2}mv_o^2 = +\frac{GMm}{2r}, \quad U = -\frac{GMm}{r}, \quad E = K + U = -\frac{GMm}{2r}$$

Three facts to remember:

- $U = -2K$, and $E = -K = U/2$. The total energy is **negative**, which is the signature of a **bound** orbit. If E were zero or positive the satellite would escape.
- The **binding energy** — the extra energy needed to send the satellite from its orbit to infinity — is $|E| = GMm/2r$.
- A satellite in a lower orbit has a more negative total energy but a higher speed. Losing energy to atmospheric drag makes a satellite spiral down and **speed up**, which is why decaying satellites burn up.

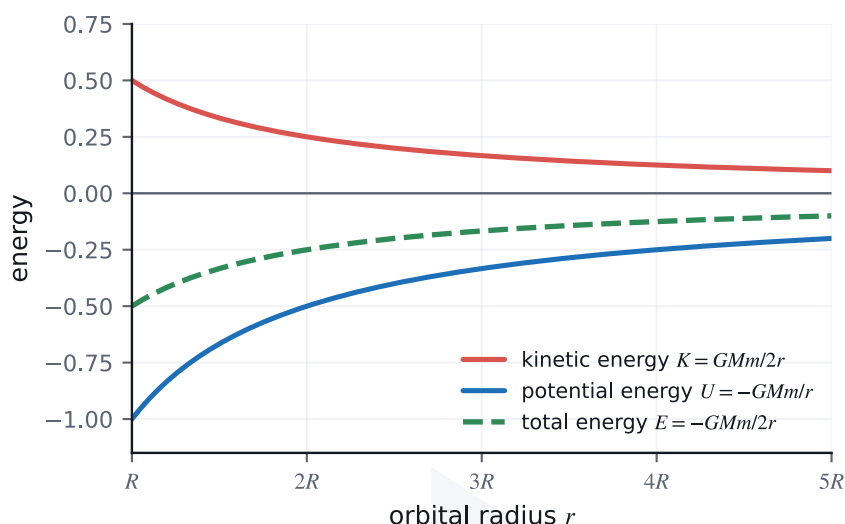


Figure 31. Energies of a satellite against orbital radius, in units of GMm/R . K is positive, U negative, and the total $E = -K$ is always negative for a bound orbit.

WORKED EXAMPLE 7.5

Problem. A satellite of mass 200 kg orbits the Earth at a height equal to the Earth's radius. Find its orbital speed, kinetic energy, potential energy, total energy and binding energy. ($g = 9.8 \text{ m s}^{-2}$, $R = 6.4 \times 10^6 \text{ m}$.)

Solution. Orbital radius $r = 2R = 1.28 \times 10^7 \text{ m}$ and $GM = gR^2 = 4.01 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$.

$$v_o = \sqrt{\frac{GM}{r}} = \sqrt{\frac{4.01 \times 10^{14}}{1.28 \times 10^7}} = \sqrt{3.14 \times 10^7} = 5.6 \times 10^3 \text{ m s}^{-1}$$

$$K = \frac{GMm}{2r} = \frac{4.01 \times 10^{14} \times 200}{2 \times 1.28 \times 10^7} = 3.14 \times 10^9 \text{ J}$$

$$U = -\frac{GMm}{r} = -6.27 \times 10^9 \text{ J}, \quad E = K + U = -3.14 \times 10^9 \text{ J}$$

The binding energy is $|E| = 3.14 \times 10^9 \text{ J}$ — the energy that must be supplied to free the satellite completely.

7.9 Geostationary satellite

GEOSTATIONARY SATELLITE

A geostationary satellite is one that appears permanently fixed above one point on the Earth's equator, because it revolves in the same sense as the Earth with a period exactly equal to the Earth's period of rotation, 24 hours.

Four conditions must all be met:

1. Its period must be 24 h (86400 s).
2. Its orbit must lie in the **equatorial plane**.
3. It must revolve **west to east**, the same sense as the Earth's spin.

4. The orbit must be **circular**, so that the angular speed is constant.

The height follows from the period. From $T^2 = 4\pi^2 r^3 / GM$,

$$r = \left(\frac{GMT^2}{4\pi^2} \right)^{1/3} = \left(\frac{4.01 \times 10^{14} \times (86400)^2}{4\pi^2} \right)^{1/3} = 4.23 \times 10^7 \text{ m}$$

so the height above the surface is

$$h = r - R = 4.23 \times 10^7 - 0.64 \times 10^7 = 3.59 \times 10^7 \text{ m} \approx 36\,000 \text{ km}$$

and the orbital speed is $v = 2\pi r / T = 3.08 \text{ km s}^{-1}$.

Geostationary satellites carry television broadcasts, telephone relays and weather imaging. Three of them, 120° apart, cover almost the whole globe except the polar caps. Nepal's own communications satellite slot, at 123.3°E , is a geostationary slot of exactly this kind; the INSAT and FengYun weather images used by the Department of Hydrology and Meteorology come from the same ring.

A GEOSTATIONARY SATELLITE IS NOT STATIONARY

It moves at over 3 km s^{-1} . It only appears stationary because the ground beneath it moves with the same angular velocity. Also, it cannot be placed over Kathmandu — the centre of any orbit must be the centre of the Earth, so a fixed-overhead satellite is only possible above the equator.

7.10 GPS

The **Global Positioning System** is a constellation of at least 24 satellites in six orbital planes at a height of about $20\,200 \text{ km}$, each with a period of $11 \text{ h } 58 \text{ min}$ — half a sidereal day, so each satellite traces the same ground track twice daily. From anywhere on Earth at least four are above the horizon.

Each satellite continuously broadcasts its own position and the exact time of transmission, kept by an atomic clock. The receiver measures the delay Δt and computes the distance

$$d = c \Delta t$$

where $c = 3 \times 10^8 \text{ m s}^{-1}$. One distance places the receiver on a sphere around that satellite; two spheres intersect in a circle; three fix it to two points, one of which is absurd (far out in space). This is **trilateration**. A **fourth** satellite is needed because the receiver's own cheap quartz clock has an unknown offset — four measurements solve for four unknowns (x , y , z and the clock error).

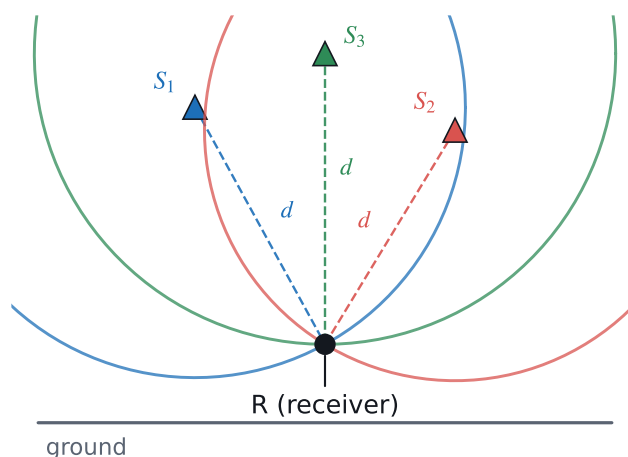


Figure 32. Trilateration. Each measured range puts the receiver on a sphere; three ranges intersect at the receiver R. A fourth satellite is needed to correct the receiver clock.

Two corrections matter. Because the satellites move fast, special relativity makes their clocks run slow by about $7 \mu\text{s}$ per day; because they are high in a weaker gravitational field, general relativity makes them run fast by about $45 \mu\text{s}$ per day. The net $38 \mu\text{s}$ per day is corrected in the satellite clocks — without it, positions would drift by about 11 km every day.

GPS is used in Nepal for land surveying and cadastral mapping, for tracking aircraft and trekking routes, for monitoring the slow northward creep of the Indian plate (about 2 cm per year) that builds the Himalaya and causes earthquakes, and it was used in the 2019–2020 survey in which Nepal and China jointly re-measured the height of Sagarmatha as 8848.86 m .

✓ Chapter summary

- Newton's law: $F = Gm_1m_2/r^2$ with $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$; forces are an action–reaction pair, independent of the medium.
- Field strength $E = GM/r^2$ (a vector, N kg^{-1}); at the surface $g = GM/R^2$, from which $M = gR^2/G$.
- Potential $V = -GM/r$ (a scalar, J kg^{-1}) and potential energy $U = -GMm/r$; $E = -dV/dr$, and $\Delta U \approx mgh$ near the surface.
- Altitude: $g_h = g(R/(R+h))^2 \approx g(1 - 2h/R)$. Depth: $g_d = g(1 - d/R)$, zero at the centre.
- Centre of mass $x_{cm} = \sum m_i x_i / \sum m_i$; it coincides with the centre of gravity only in a uniform field.
- Orbit: $v_o = \sqrt{GM/r}$, $T = 2\pi\sqrt{r^3/GM}$, hence $T^2 \propto r^3$ (Kepler's third law).
- Escape velocity $v_e = \sqrt{2GM/R} = \sqrt{2gR} = \sqrt{2} v_o = 11.2 \text{ km s}^{-1}$ for the Earth; independent of mass and direction.
- Satellite energies: $K = GMm/2r$, $U = -GMm/r$, $E = -GMm/2r = -K$; binding energy $= GMm/2r$.
- Geostationary orbit: $T = 24 \text{ h}$, equatorial, west to east, circular, at $r = 4.23 \times 10^7 \text{ m}$, i.e. about $36\,000 \text{ km}$ up.
- GPS: 24+ satellites at $20\,200 \text{ km}$, period $11 \text{ h } 58 \text{ min}$; range $d = c\Delta t$; four satellites fix position and receiver clock error.

Practice questions

Group A — Multiple choice (1 mark each)

- The dimensional formula of the universal gravitational constant G is [1]
 - $[ML^2T^{-2}]$
 - $[M^{-1}L^3T^{-2}]$
 - $[ML^{-3}T^2]$
 - $[M^{-1}L^2T^{-1}]$
- The value of g at the centre of the Earth is [1]
 - maximum
 - equal to the surface value
 - zero
 - infinite
- The gravitational potential at a distance r from a mass M is [1]
 - GM/r^2
 - $-GM/r$
 - $-GM/r^2$
 - GMm/r
- The escape velocity from the Earth's surface does **not** depend on [1]
 - the mass of the Earth
 - the radius of the Earth
 - the mass of the body
 - the value of g
- The total energy of a satellite in a circular orbit of radius r is [1]
 - GMm/r
 - $-GMm/r$
 - $-GMm/2r$
 - $+GMm/2r$
- A GPS receiver needs a minimum of how many satellites to fix its position and time? [1]
 - 2
 - 3
 - 4
 - 6

ANSWERS TO GROUP A

- (b) — from $G = Fr^2/m_1m_2$, $[MLT^{-2}][L^2]/[M^2] = [M^{-1}L^3T^{-2}]$.
- (c) — $g_d = g(1 - d/R)$ gives zero at $d = R$; there is no mass below you.
- (b) — potential is work per unit mass to bring the mass from infinity, and is negative.
- (c) — $v_e = \sqrt{2gR}$ contains no term for the mass of the escaping body.
- (c) — $K = GMm/2r$ and $U = -GMm/r$, so $E = -GMm/2r$.
- (c) — three for the three spatial coordinates, a fourth for the receiver clock error.

Group B — Short answer (5 marks each)

- State Newton's law of gravitation. Define G and give its SI unit and dimensional formula. [5]
- Define gravitational potential and gravitational potential energy. Show that the gravitational potential at a distance r from a mass M is $V = -GM/r$. [5]
- Derive an expression for the variation of g with depth below the Earth's surface, and hence show that g is zero at the centre. [5]
- Distinguish between centre of mass and centre of gravity. Under what circumstances do they not coincide? [5]
- Calculate the height and orbital speed of a geostationary satellite. ($g = 9.8 \text{ m s}^{-2}$, $R = 6.4 \times 10^6 \text{ m}$.) [5]
- Two spheres, each of mass 100 kg, have their centres 1.0 m apart. Find the gravitational force between them and comment on its size. [5]

ANSWERS TO GROUP B

5. $GM = gR^2 = 4.01 \times 10^{14}$. With $T = 86400$ s, $r = (GMT^2/4\pi^2)^{1/3} = (7.59 \times 10^{22})^{1/3} = 4.23 \times 10^7$ m. Height $h = r - R = 3.59 \times 10^7$ m $\approx 36\,000$ km; speed $v = 2\pi r/T = 3.08 \times 10^3$ m s⁻¹.

6. $F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 100 \times 100}{1.0^2} = 6.67 \times 10^{-7}$ N — about the weight of a grain of dust. Gravity is negligible between everyday objects and matters only when one mass is astronomically large.

1., 2., 3., 4. See §7.1, §7.3, the depth derivation in §7.4 and §7.5.

Group C — Long answer (8 marks each)

- (a) Define escape velocity and derive the expression $v_e = \sqrt{2gR}$. [4]

(b) Show that the escape velocity is $\sqrt{2}$ times the orbital velocity of a satellite close to the planet's surface. [2]

(c) Calculate the escape velocity from the Moon, for which $g_m = 1.62$ m s⁻² and $R_m = 1.74 \times 10^6$ m, and use your answer to explain why the Moon has no atmosphere. [2]
- (a) Derive expressions for the orbital velocity and the time period of a satellite revolving at a height h above the Earth's surface, and hence obtain Kepler's third law. [5]

(b) What is a geostationary satellite? State the conditions it must satisfy. [3]
- (a) Obtain expressions for the kinetic energy, potential energy and total energy of a satellite in a circular orbit of radius r , and explain the significance of the negative sign of the total energy. [5]

(b) A satellite of mass 500 kg orbits at a height of 3.6×10^6 m. Find its binding energy. ($g = 9.8$ m s⁻², $R = 6.4 \times 10^6$ m.) [3]

ANSWERS TO GROUP C

1. (c) $v_e = \sqrt{2g_m R_m} = \sqrt{2 \times 1.62 \times 1.74 \times 10^6} = \sqrt{5.64 \times 10^6} = 2.37 \times 10^3$ m s⁻¹ = 2.37 km s⁻¹. This is of the same order as the speeds of gas molecules at lunar surface temperatures, so over time all the gas has escaped.

2. See the derivation box in §7.6 and the four conditions listed in §7.9.

3. (b) $r = R + h = 6.4 \times 10^6 + 3.6 \times 10^6 = 1.0 \times 10^7$ m, $GM = gR^2 = 4.01 \times 10^{14}$. Binding energy $= \frac{GMm}{2r} = \frac{4.01 \times 10^{14} \times 500}{2 \times 1.0 \times 10^7} = 1.0 \times 10^{10}$ J.

UNIT 8

Elasticity

Mechanics • 5 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Hooke's law: Force constant
- Stress, Strain, Elasticity and plasticity
- Elastic modulus: Young modulus, bulk modulus, shear modulus
- Poisson's ratio
- Elastic potential energy

Every solid changes shape a little when you push or pull it. A bridge deck sags under a loaded truck on the Prithvi Highway, a steel cable in a Trishuli suspension bridge stretches, a rubber band grows longer. Elasticity asks two questions: how much does a body deform under a given force, and does it come back when the force is removed? The whole unit is built on two quantities — **stress** (force per unit area) and **strain** (fractional change in size) — and on the constants that link them.

WHAT THE EXAMINER WANTS

Group B nearly always carries one definition set (stress, strain, Hooke's law, modulus) plus one numerical on Young modulus or stored energy. Group C favours "define the three moduli and derive the energy stored in a stretched wire". Master the four boxed formulas in the summary and you cover most of the marks.

8.1 Hooke's law: force constant

When an external force acts on a body, its molecules are pulled apart from (or pushed towards) their equilibrium separations. The inter-molecular forces then act to restore the original spacing. The external force is called the **deforming force**; the internal force that opposes it is the **restoring force**. As long as the deformation is small, the two are equal in magnitude.

Robert Hooke stated the experimental law in 1678: for small deformations the extension is proportional to the load.

HOOKE'S LAW

Within the elastic limit, the extension produced in a body is directly proportional to the deforming force applied, the temperature remaining constant.

$$F \propto x \quad \Rightarrow \quad F = kx$$

The constant k is the **force constant** (also called stiffness, or spring constant for a spring). It is the force needed to produce unit extension:

$$k = \frac{F}{x}, \quad \text{unit: N m}^{-1}, \quad \text{dimensions } [MT^{-2}]$$

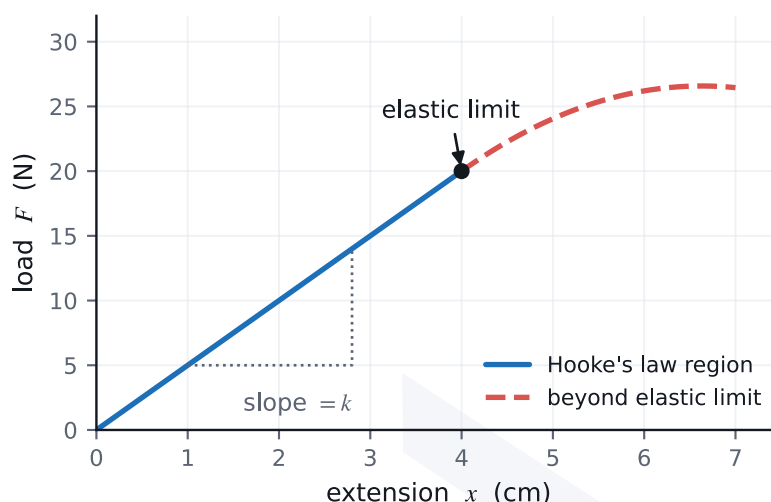


Figure 33. Load-extension graph. The straight part obeys Hooke's law and its slope is the force constant k . Past the elastic limit the graph curves and the body no longer returns to its original length.

The force constant is **not** a property of the material alone. For a wire of length L and area of cross-section A made of a material of Young modulus Y (section 8.3),

$$k = \frac{YA}{L}$$

so a short thick wire is stiffer than a long thin one of the same metal. When springs of constants k_1 and k_2 are joined **in series** the combination is softer, $1/k = 1/k_1 + 1/k_2$; joined side by side **in parallel** it is stiffer, $k = k_1 + k_2$.

WORKED EXAMPLE 8.1

Problem. A spring of natural length 20 cm stretches to 24 cm when a load of 20 N hangs from it. Find its force constant and the length of the spring when the load is 35 N.

Solution. Extension $x = 24 - 20 = 4 \text{ cm} = 0.04 \text{ m}$.

$$k = \frac{F}{x} = \frac{20}{0.04} = 500 \text{ N m}^{-1}$$

For $F = 35 \text{ N}$, $x = F/k = 35/500 = 0.07 \text{ m} = 7 \text{ cm}$, so the spring is $20 + 7 = 27 \text{ cm}$ long (assuming the elastic limit is not exceeded).

8.2 Stress, Strain, Elasticity and plasticity

STRESS AND STRAIN

Stress is the restoring force set up per unit area of cross-section of a deformed body: $\text{stress} = F/A$. SI unit N m^{-2} or pascal (Pa), dimensional formula $[ML^{-1}T^{-2}]$.

Strain is the ratio of the change in a dimension to the original value of that dimension. It is a pure number with no unit and no dimensions.

Stress has the same dimensions as pressure, but the two are not the same thing: pressure in a fluid always acts normal to a surface, while stress may act along a surface (shear). There are three kinds of stress and a matching strain for each.

Stress	How the force acts	Strain produced	Modulus
Tensile / compressive (longitudinal)	normal to the end faces, along the length	longitudinal strain = e/L	Young modulus Y
Hydraulic (volume)	normal and equal on every face	volume strain = $\Delta V/V$	Bulk modulus B
Tangential (shear)	parallel to one face, the opposite face fixed	shear strain = $\theta = x/L$	Shear modulus η

Elasticity is the property by which a body regains its original shape and size after the deforming force is removed. **Plasticity** is the opposite property: the body keeps the new shape. No real material is perfectly elastic or perfectly plastic; a quartz fibre is the nearest to perfectly elastic and wet clay or putty the nearest to perfectly plastic.

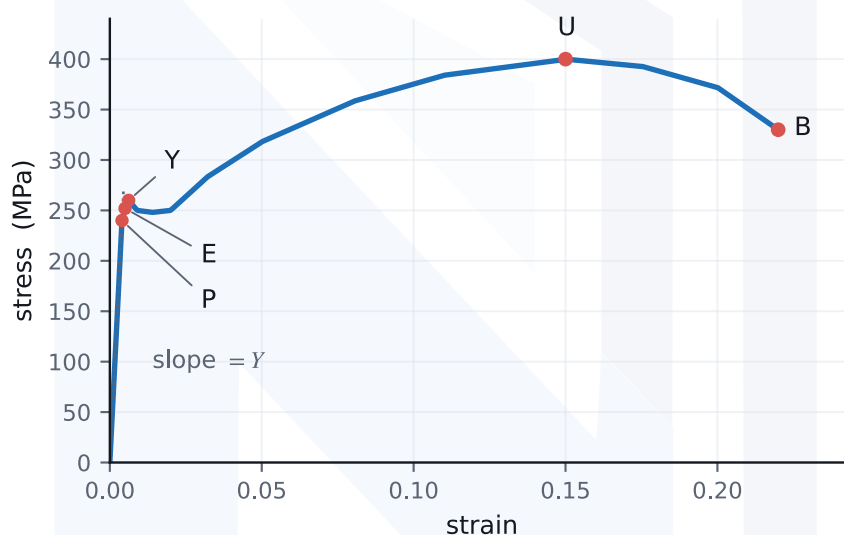


Figure 34. Stress-strain curve for a ductile metal such as mild steel. P is the proportional limit, E the elastic limit, Y the yield point, U the ultimate tensile stress and B the breaking point.

Reading the curve from the origin:

- **O to P — proportional limit.** Stress is proportional to strain; Hooke's law holds and the slope of this line is the Young modulus.
- **P to E — elastic limit.** The graph bends, so Hooke's law fails, but the wire still returns exactly to its original length when unloaded.
- **Beyond E — permanent set.** Unloading now leaves the wire longer than it started. At the **yield point** Y the wire begins to extend with almost no extra load.
- **U — ultimate tensile stress,** the largest stress the material can take. After U the wire develops a local neck, so the true area falls and the curve drops until the wire snaps at the **breaking point** B .

A material that stretches a great deal between Y and B (copper, mild steel, gold) is **ductile**; one that breaks very soon after the elastic limit (glass, cast iron, ceramic tile) is **brittle**.

BREAKING STRESS AND BREAKING FORCE ARE DIFFERENT

Breaking stress depends only on the material. **Breaking force** = breaking stress $\times$ area, so it depends on the thickness of the wire but **not** on its length. A 10 m wire and a 1 m wire of the same metal and diameter snap under exactly the same load.

Two related effects are worth naming. **Elastic after-effect** is the delay some materials (glass, rubber) show in returning to their original size. **Elastic fatigue** is the loss of elastic strength when a body is subjected to repeated cycles of stress — the reason a wire finally breaks after being bent back and forth many times, and why bridges and aircraft parts are replaced long before they look worn.

8.3 Elastic modulus: Young modulus, bulk modulus, shear modulus

MODULUS OF ELASTICITY

Within the elastic limit, the modulus of elasticity of a material is the ratio of the stress applied to the strain produced: $\text{modulus} = \text{stress}/\text{strain}$. Because strain has no unit, a modulus has the unit of stress, N m^{-2} (Pa). A large modulus means a stiff material.

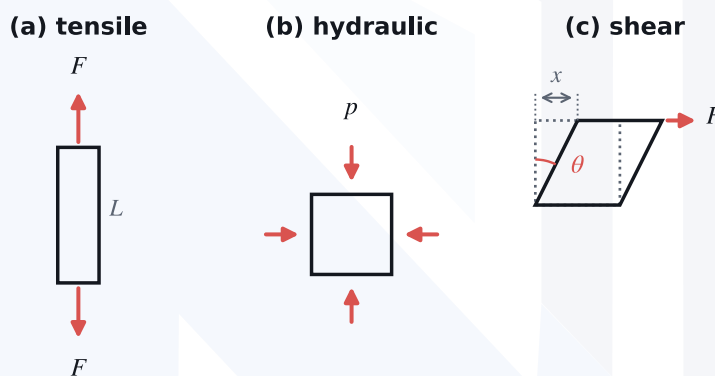


Figure 35. The three deformations. (a) tensile stress changes length, (b) hydraulic stress changes volume, (c) tangential stress changes shape through the angle θ .

Young modulus

$$Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{e/L} = \frac{FL}{Ae}$$

where e is the extension of a wire of original length L and area A under a load F . Rearranging gives $F = (YA/L)e$, which compared with $F = kx$ proves the result $k = YA/L$ quoted in section 8.1. Only solids have a Young modulus, because only a solid can be pulled into a wire.

WORKED EXAMPLE 8.2

Problem. A steel wire of length 2.0 m and diameter 1.0 mm carries a load of 50 N. Taking $Y = 2.0 \times 10^{11}$ Pa, find the stress, the strain and the extension.

Solution. Area of cross-section:

$$A = \frac{\pi d^2}{4} = \frac{3.1416 \times (1.0 \times 10^{-3})^2}{4} = 7.854 \times 10^{-7} \text{ m}^2$$

$$\text{stress} = \frac{F}{A} = \frac{50}{7.854 \times 10^{-7}} = 6.37 \times 10^7 \text{ Pa}$$

$$\text{strain} = \frac{\text{stress}}{Y} = \frac{6.37 \times 10^7}{2.0 \times 10^{11}} = 3.18 \times 10^{-4}$$

$$e = \text{strain} \times L = 3.18 \times 10^{-4} \times 2.0 = 6.37 \times 10^{-4} \text{ m} = 0.64 \text{ mm}$$

Bulk modulus

If a uniform increase of pressure p on all sides reduces a volume V by ΔV , then

$$B = -\frac{p}{\Delta V/V} = -\frac{pV}{\Delta V}$$

The minus sign is there because ΔV is negative when p increases, so that B comes out positive. Solids, liquids **and** gases all have a bulk modulus. Its reciprocal is the **compressibility**, $K = 1/B$, measured in Pa^{-1} .

WORKED EXAMPLE 8.3

Problem. The pressure on 1.0 m^3 of water is increased by $2.2 \times 10^7 \text{ Pa}$. If the bulk modulus of water is $2.2 \times 10^9 \text{ Pa}$, find the change in volume and the compressibility of water.

Solution.

$$\Delta V = -\frac{pV}{B} = -\frac{2.2 \times 10^7 \times 1.0}{2.2 \times 10^9} = -1.0 \times 10^{-2} \text{ m}^3$$

The volume falls by 0.010 m^3 , i.e. by 1.0 % — water is very nearly incompressible. The compressibility is

$$K = \frac{1}{B} = \frac{1}{2.2 \times 10^9} = 4.55 \times 10^{-10} \text{ Pa}^{-1}$$

Shear modulus (modulus of rigidity)

A tangential force F on the top face of area A of a block of height L slides that face sideways by x , turning the vertical edges through a small angle $\theta = x/L$ radians. Then

$$\eta = \frac{\text{tangential stress}}{\text{shear strain}} = \frac{F/A}{\theta} = \frac{FL}{Ax}$$

Only solids have a shear modulus: a liquid or a gas cannot sustain a tangential stress — it simply flows. For most metals η is roughly $Y/3$.

WORKED EXAMPLE 8.4

Problem. A rubber block of square face $10\text{ cm} \times 10\text{ cm}$ and height 5.0 cm is fixed at the bottom. A tangential force of 100 N is applied to the top face. If $\eta = 2.0 \times 10^6\text{ Pa}$, find the shear strain and the sideways displacement of the top face.

Solution. $A = 0.10 \times 0.10 = 1.0 \times 10^{-2}\text{ m}^2$.

$$\text{tangential stress} = \frac{100}{1.0 \times 10^{-2}} = 1.0 \times 10^4\text{ Pa}$$

$$\theta = \frac{\text{stress}}{\eta} = \frac{1.0 \times 10^4}{2.0 \times 10^6} = 5.0 \times 10^{-3}\text{ rad}$$

$$x = \theta L = 5.0 \times 10^{-3} \times 0.050 = 2.5 \times 10^{-4}\text{ m} = 0.25\text{ mm}$$

Approximate room-temperature values ($1\text{ GPa} = 10^9\text{ Pa}$):

Material	Y (GPa)	B (GPa)	η (GPa)	Poisson ratio σ
Steel	200	160	84	0.29
Copper	110	140	44	0.34
Brass	91	61	35	0.34
Aluminium	70	70	25	0.33
Glass	55	37	23	0.25
Water	—	2.2	0	—
Rubber	0.0005 – 0.05	≈ 2	≈ 0.0007	≈ 0.50

8.4 Poisson's ratio

A stretched wire does not only get longer — it also gets thinner. If a wire of length L and diameter d stretches by e and its diameter shrinks by Δd , then

$$\text{longitudinal strain} = \frac{e}{L}, \quad \text{lateral strain} = -\frac{\Delta d}{d}$$

POISSON'S RATIO

Poisson's ratio σ is the ratio of the lateral strain to the longitudinal strain, within the elastic limit:

$$\sigma = -\frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\Delta d/d}{e/L}$$

It is a pure number with **no unit and no dimensions**, and it is a constant of the material.

Theory allows σ to lie between -1 and $+0.5$. For all common materials it is positive and lies between about 0.2 and 0.4 ; rubber is the extreme case at nearly 0.5 . The upper bound has a simple physical meaning. For a wire of volume $V = \pi r^2 L$, small changes give

$$\frac{\Delta V}{V} = \frac{2\Delta r}{r} + \frac{\Delta L}{L} = (-2\sigma + 1)\frac{\Delta L}{L} = (1 - 2\sigma) \frac{\epsilon}{L}$$

If $\sigma = 0.5$ the volume does not change at all when the wire is stretched; σ larger than 0.5 would mean the volume **shrinks** on stretching, which no material does.

WORKED EXAMPLE 8.5

Problem. A wire of length 3.0 m and diameter 2.0 mm is stretched by 1.5 mm. If Poisson's ratio for the material is 0.30 , find the decrease in diameter and the fractional change in volume.

Solution.

$$\text{longitudinal strain} = \frac{\epsilon}{L} = \frac{1.5 \times 10^{-3}}{3.0} = 5.0 \times 10^{-4}$$

$$\text{lateral strain} = \sigma \times 5.0 \times 10^{-4} = 0.30 \times 5.0 \times 10^{-4} = 1.5 \times 10^{-4}$$

$$\Delta d = 1.5 \times 10^{-4} \times 2.0 \text{ mm} = 3.0 \times 10^{-4} \text{ mm}$$

So the diameter falls by $0.30 \mu\text{m}$. The volume change is

$$\frac{\Delta V}{V} = (1 - 2\sigma)\frac{\epsilon}{L} = (1 - 0.60) \times 5.0 \times 10^{-4} = 2.0 \times 10^{-4}$$

an increase of 0.02% .

8.5 Elastic potential energy

Stretching a wire takes work, and that work is stored in the wire as **elastic potential energy**. It is recovered when the wire is released — this is how a catapult, a bow and a clock spring work.

ENERGY STORED IN A STRETCHED WIRE, $U = \frac{1}{2}Fe$

We start from the work done by the stretching force and reach the energy stored in the wire, and then the energy stored in each cubic metre of it.

Setting up. A wire of original length L and cross-section area A is slowly stretched. Let x be the extension so far, and let e be the final extension when the full load F is on. The wire obeys Hooke's law throughout.

Step 1 — the force needed at extension x . The definition of Young modulus, $Y = \frac{F_x L}{A x}$, rearranged for the force, says that the force which holds the wire at extension x is

$$F_x = \frac{YA}{L} x$$

This is the key point: the force is **not** constant. It grows as the wire stretches, so we cannot just write work = force $\times$ distance.

Step 2 — take a stretch so small the force cannot change. Let the wire stretch a further tiny amount dx . Over such a small distance F_x is effectively constant, so the small work done is

$$dW = F_x dx = \frac{YA}{L} x dx$$

Step 3 — add up all the small bits. Adding up infinitely many small pieces means integrating, from no extension ($x=0$) to the final extension ($x=e$):

$$W = \int_0^e \frac{YA}{L} x dx$$

Step 4 — take the constants outside. Y , A and L do not change during the stretch:

$$W = \frac{YA}{L} \int_0^e x dx$$

Step 5 — do the integral. Using $\int x dx = x^2/2$:

$$W = \frac{YA}{L} \left[\frac{x^2}{2} \right]_0^e$$

Step 6 — put in the limits. At $x=e$ we get $e^2/2$; at $x=0$ we get 0:

$$W = \frac{YA}{L} \cdot \frac{e^2}{2}$$

Step 7 — split one factor of e off, ready for the next step:

$$W = \frac{1}{2} \left(\frac{Y Ae}{L} \right) e$$

Step 8 — recognise the bracket. Putting $x=e$ in Step 1 gives the final load, $F = Y Ae/L$. So the bracket is F :

$$W = \frac{1}{2} Fe$$

Step 9 — the work is stored, not lost. The stretching is slow and the wire stays elastic, so all this work is stored as elastic potential energy U :

$$U = \frac{1}{2} Fe = \frac{1}{2} \times \text{load} \times \text{extension}$$

Now the energy per unit volume.

Step 10 — divide by the volume. The wire's volume is $A \times L$:

$$u = \frac{U}{AL} = \frac{\frac{1}{2} Fe}{AL}$$

Step 11 — split the fraction so that F pairs with A and e pairs with L :

$$u = \frac{1}{2} \cdot \frac{F}{A} \cdot \frac{e}{L}$$

Step 12 — name the two factors. F/A is the stress and e/L is the strain:

$$u = \frac{1}{2} \times \text{stress} \times \text{strain}$$

Step 13 — two other forms. Since $\text{stress} = Y \times \text{strain}$, we may replace either factor. Writing ϵ for the strain:

$$u = \frac{1}{2} Y \epsilon^2 = \frac{1}{2} \cdot \frac{(\text{stress})^2}{Y}$$

Result.

$$U = \frac{1}{2} Fe, \quad u = \frac{1}{2} \times \text{stress} \times \text{strain}$$

What it means. The $\frac{1}{2}$ is the whole physical story: the force starts at zero and ends at F , so on average only $F/2$ acted over the distance e . Graphically, U is the **area under the load-extension line** — a triangle of base e and height F . The second result says the energy stored in each cubic metre depends only on how hard the material is squeezed and how much it gives, not on the size of the wire.

Conditions used. Hooke's law holds (we are below the elastic limit, so the graph is a straight line), the load is applied slowly so nothing vibrates, and no energy is lost as heat.

THE EXAMINER IS LOOKING FOR

1. The statement that the force is variable, $F_x = (YA/L)x$ — full marks are rarely given without it.
2. The element of work $dW = F_x dx$.
3. The integral $\int_0^e (YA/L)x dx$ **with correct limits**.
4. The step that turns $Y Ae/L$ back into F , giving $U = \frac{1}{2} Fe$.
5. For the energy-density part: dividing by volume AL and regrouping into $\frac{1}{2}$ stress $\times$ strain.

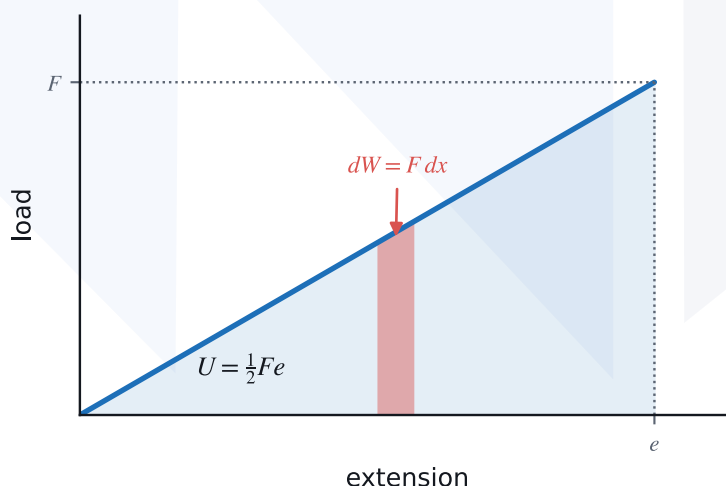


Figure 36. Work done in stretching is the area under the load-extension line, $U = \frac{1}{2} Fe$. The shaded strip is $dW = F dx$.

THE FACTOR OF ONE HALF

The energy is $\frac{1}{2}Fe$, **not** Fe . The load is built up gradually from zero to F , so the average force during the stretch is $F/2$. If instead the full load F is hung on suddenly, it does work Fe — twice the stored energy — and the extra $\frac{1}{2}Fe$ appears as vibration and heat.

WORKED EXAMPLE 8.6

Problem. A steel wire of length 4.0 m and cross-sectional area 2.0 mm^2 is stretched by 2.0 mm. Taking $Y = 2.0 \times 10^{11} \text{ Pa}$, find the load applied, the energy stored and the energy stored per unit volume.

Solution. With $A = 2.0 \times 10^{-6} \text{ m}^2$ and $e = 2.0 \times 10^{-3} \text{ m}$:

$$F = \frac{Y A e}{L} = \frac{2.0 \times 10^{11} \times 2.0 \times 10^{-6} \times 2.0 \times 10^{-3}}{4.0} = 200 \text{ N}$$

$$U = \frac{1}{2} F e = \frac{1}{2} \times 200 \times 2.0 \times 10^{-3} = 0.20 \text{ J}$$

Volume $= AL = 2.0 \times 10^{-6} \times 4.0 = 8.0 \times 10^{-6} \text{ m}^3$, so

$$u = \frac{0.20}{8.0 \times 10^{-6}} = 2.5 \times 10^4 \text{ J m}^{-3}$$

✓ Chapter summary

- Hooke's law: within the elastic limit $F = kx$, where the force constant $k = F/x$ (N m^{-1}). For a wire $k = YA/L$, so k depends on shape as well as material.
- Stress $= F/A$ (Pa, $[ML^{-1}T^{-2}]$); strain $= \text{change} \div \text{original value}$ (no unit). Modulus $= \text{stress} \div \text{strain}$.
- The stress-strain curve runs: proportional limit P , elastic limit E , yield point Y , ultimate stress U , breaking point B . Ductile materials have a long $Y-B$ stretch; brittle ones break just after E .
- Young modulus $Y = FL/Ae$; bulk modulus $B = -pV/\Delta V$ with compressibility $K = 1/B$; shear modulus $\eta = F/(A\theta)$ with $\theta = x/L$. Only B applies to liquids and gases.
- Poisson's ratio $\sigma = (\Delta d/d) \div (e/L)$; no unit, theoretical range -1 to 0.5 , real materials $0.2-0.4$. Also $\Delta V/V = (1 - 2\sigma)e/L$.
- Elastic potential energy $U = \frac{1}{2}Fe = \frac{1}{2} \frac{Y A e^2}{L}$, and energy per unit volume $u = \frac{1}{2} \times \text{stress} \times \text{strain} = \frac{1}{2} Y e^2$.
- Breaking force depends on the area of cross-section, never on the length.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of stress is [1]
 - N m
 - N m⁻²
 - N m⁻¹
 - it has no unit
- Strain is [1]
 - measured in N m⁻²
 - measured in m
 - a pure number
 - measured in N
- For a perfectly rigid body the Young modulus is [1]
 - zero
 - infinite
 - unity
 - negative
- The theoretical limits of Poisson's ratio are [1]
 - 0 to 1
 - 1 to 0.5
 - 0.5 to 1
 - 0 to 0.25
- The energy stored per unit volume of a stretched wire equals [1]
 - stress × strain
 - $\frac{1}{2} \times \text{stress} \times \text{strain}$
 - stress ÷ strain
 - $\frac{1}{2} \times \text{stress} \div \text{strain}$

ANSWERS TO GROUP A

- (b) — stress is force per unit area, N m⁻² = Pa.
- (c) — it is a ratio of two lengths (or two volumes), so it has no unit.
- (b) — a rigid body has zero strain, and $Y = \text{stress}/\text{strain} \rightarrow \infty$.
- (b) — σ larger than 0.5 would make the volume fall on stretching.
- (b) — the area under the stress-strain line is $\frac{1}{2} \times \text{stress} \times \text{strain}$.

Group B — Short answer (5 marks each)

- Define stress and strain and state their units. State Hooke's law and hence define modulus of elasticity. [5]
- Draw a stress-strain curve for a ductile material and mark the proportional limit, elastic limit, yield point and breaking point. Distinguish between elasticity and plasticity. [5]
- Define Young modulus. A steel wire 3.0 m long with cross-sectional area 2.0 mm² stretches by 1.5 mm under a load W . Find W , given $Y = 2.0 \times 10^{11}$ Pa. [5]
- Define bulk modulus and compressibility. The pressure on a sample of water is increased by 4.4×10^7 Pa. Find the percentage decrease in its volume if $B = 2.2 \times 10^9$ Pa. [5]
- Show that the energy stored per unit volume of a stretched wire is $\frac{1}{2} \times \text{stress} \times \text{strain}$. A wire of volume 1.0×10^{-6} m³ is stretched so that its strain is 1.0×10^{-3} . Find the energy stored if $Y = 2.0 \times 10^{11}$ Pa. [5]

ANSWERS TO GROUP B

- $F = \frac{Y A e}{L} = \frac{2.0 \times 10^{11} \times 2.0 \times 10^{-6} \times 1.5 \times 10^{-3}}{3.0} = 200 \text{ N}.$
- $\frac{\Delta V}{V} = \frac{p}{B} = \frac{4.4 \times 10^7}{2.2 \times 10^9} = 0.020$, i.e. a decrease of 2.0 %.

5. Outline: work done $= \int_0^e (YA/L)x \, dx = \frac{1}{2}Fe$; divide by volume AL to get $u = \frac{1}{2}(F/A)(e/L)$. Numerically $U = \frac{1}{2}Ye^2V = \frac{1}{2} \times 2.0 \times 10^{11} \times (1.0 \times 10^{-3})^2 \times 1.0 \times 10^{-6} = 0.10 \text{ J}$.

Group C — Long answer (8 marks each)

1. **(a)** Define the three elastic moduli and write the expression for each. [3]
(b) Derive an expression for the elastic potential energy stored in a stretched wire and hence for the energy per unit volume. [3]
(c) A wire of length 2.0 m and area 1.0 mm^2 is stretched by 1.0 mm with $Y = 1.1 \times 10^{11} \text{ Pa}$. Find the energy stored. [2]
2. A copper wire and a steel wire, each 2.0 m long and 1.0 mm in diameter, are joined end to end and a force of 100 N is applied to the free ends. Given $Y_{Cu} = 1.1 \times 10^{11} \text{ Pa}$ and $Y_{St} = 2.0 \times 10^{11} \text{ Pa}$, find the stress in each wire, the extension of each wire and the total extension. [8]

ANSWERS TO GROUP C

1. (c) $F = Y Ae/L = (1.1 \times 10^{11} \times 1.0 \times 10^{-6} \times 1.0 \times 10^{-3})/2.0 = 55 \text{ N}$, so
 $U = \frac{1}{2}Fe = \frac{1}{2} \times 55 \times 1.0 \times 10^{-3} = 2.75 \times 10^{-2} \text{ J}$.

2. The two wires are in series, so each carries the same force 100 N and, having the same diameter, the same stress.

$A = \pi d^2/4 = 7.854 \times 10^{-7} \text{ m}^2$, hence stress $= 100/7.854 \times 10^{-7} = 1.273 \times 10^8 \text{ Pa}$ in both.

Copper: $e = \text{stress} \times L/Y = (1.273 \times 10^8 \times 2.0)/1.1 \times 10^{11} = 2.32 \times 10^{-3} \text{ m} = 2.32 \text{ mm}$.

Steel: $e = (1.273 \times 10^8 \times 2.0)/2.0 \times 10^{11} = 1.27 \times 10^{-3} \text{ m} = 1.27 \text{ mm}$.

Total extension $= 2.32 + 1.27 = 3.59 \text{ mm}$. The copper wire stretches nearly twice as much because its Young modulus is about half that of steel.

UNIT 9

Heat and Temperature

Heat and Thermodynamics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Molecular concept of thermal energy, heat and temperature; cause and direction of heat flow
- Thermal equilibrium and Zeroth law of thermodynamics
- Thermal equilibrium as working principle of mercury thermometer

Put a hot glass of tea on a cold marble table and, after a while, both are lukewarm. Something flowed from one to the other, and it stopped flowing when some property of the two became equal. That "something" is **heat**; that property is **temperature**. This short unit separates two words that students use as if they meant the same thing, and then shows that the rule allowing temperature to be measured at all is a law of physics in its own right — the zeroth law.

THREE IDEAS, THREE MARKS EACH

(1) Heat is energy in transit, temperature is the average molecular kinetic energy. (2) Heat flows from high to low temperature until thermal equilibrium is reached. (3) The zeroth law is what lets a thermometer work at all. Almost every question from this unit is one of these three dressed up differently.

9.1 Molecular concept of thermal energy, heat and temperature; cause and direction of heat flow

Every substance is made of molecules that are never at rest. In a solid they vibrate about fixed lattice positions; in a liquid they vibrate and slide past one another; in a gas they fly freely and collide. Each molecule therefore has kinetic energy, and because the molecules attract one another they also have potential energy.

INTERNAL ENERGY AND THERMAL ENERGY

The **internal energy** of a body is the total energy of all its molecules — the sum of their kinetic energies (of translation, rotation and vibration) and the potential energy stored in the forces between them. The part of the internal energy that depends on temperature, i.e. the total molecular kinetic energy, is called the **thermal energy** of the body.

Temperature is not the total energy but the average energy per molecule. For an ideal gas the kinetic theory gives the average translational kinetic energy of one molecule as

$$\bar{E}_k = \frac{3}{2}k_B T$$

where $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$ is the Boltzmann constant and T is the absolute temperature in kelvin. Turn this around and it says something important: **temperature is a measure of the average kinetic energy of the molecules of a body.**

A BUCKET OF WATER HAS MORE THERMAL ENERGY THAN A MATCHSTICK FLAME

A flame at 800°C has far higher temperature than a bucket of water at 30°C , but the bucket holds far more thermal energy because it contains enormously more molecules. Temperature is per molecule; thermal energy is total. Never say "this body contains more heat".

HEAT

Heat is energy that flows from one body to another **because of a temperature difference between them**. It exists only while it is being transferred: once it has arrived it is no longer heat, it is internal energy of the receiving body. Its SI unit is the joule (J).

Cause and direction of heat flow. When two bodies at different temperatures touch, their surface molecules collide. On average the faster molecules of the hotter body lose kinetic energy in these collisions and the slower molecules of the colder body gain it. The net result is a transfer of energy from the body at higher temperature to the body at lower temperature. So:

- The **cause** of heat flow is the temperature difference, not a difference in the amount of heat, mass or size.
- The **direction** of heat flow is always from higher to lower temperature.
- Flow stops when the temperatures become equal — not when the energies become equal.

	Heat	Temperature
What it is	energy in transit	measure of average molecular KE
SI unit	joule (J)	kelvin (K)
Depends on mass?	yes	no
Measured with	calorimeter	thermometer
Symbol	Q	T or θ
Decides	how much energy moved	which way the energy moves

WORKED EXAMPLE 9.1

Problem. Find the average translational kinetic energy of a molecule of a gas at 27°C , and state the temperature at which this energy would be doubled. Take $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$.

Solution. $T = 27 + 273 = 300 \text{ K}$.

$$\bar{E}_k = \frac{3}{2}k_B T = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21} \text{ J}$$

Since $\bar{E}_k \propto T$, doubling the energy needs double the **absolute** temperature: $T = 600 \text{ K} = 327^\circ\text{C}$ — not 54°C .

9.2 Thermal equilibrium and Zeroth law of thermodynamics

Two bodies placed in thermal contact exchange energy until the net flow becomes zero. They are then said to be in **thermal equilibrium**, and experiment shows that this happens exactly when their temperatures are equal.

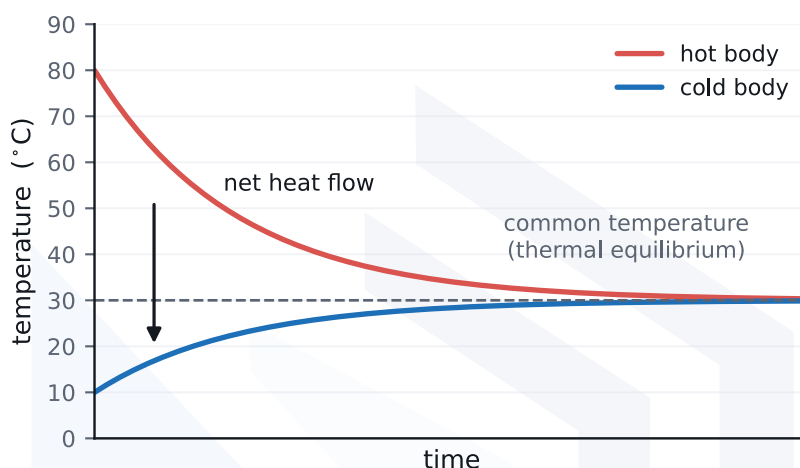


Figure 37. A hot body and a cold body in thermal contact. Energy flows until both reach the same temperature; after that the net flow is zero, though molecular collisions continue.

Now suppose body A is in equilibrium with body C , and body B is separately in equilibrium with the same body C . Is A then in equilibrium with B ? Experiment says yes, always. This result cannot be deduced from the first or second laws, and it is so basic that it was named the **zeroth** law after the others had already been numbered.

ZEROTH LAW OF THERMODYNAMICS

If two bodies A and B are each separately in thermal equilibrium with a third body C , then A and B are in thermal equilibrium with each other.

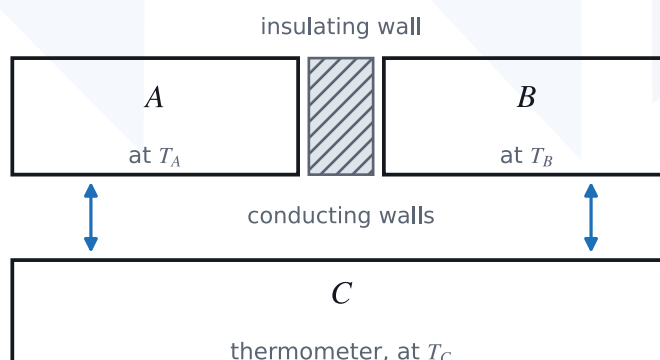


Figure 38. Zeroth law. A and B are separated by an insulating wall but each is in contact with C through a conducting wall. If $T_A = T_C$ and $T_B = T_C$, then $T_A = T_B$.

The zeroth law is the licence to use a thermometer. A thermometer is the third body C . We bring it into contact with A and read it, then with B and read it. If the readings agree, the law guarantees that A and B are in equilibrium with each other — so the reading is a genuine property of the state of each body. That property is what we call **temperature**.

WHAT THE ZEROth LAW REALLY DEFINES

The zeroth law establishes that temperature exists as a measurable physical quantity, and that bodies in thermal equilibrium share a common value of it. Without it, comparing the temperature of Pokhara and Jumla with two different thermometers would have no meaning.

9.3 Thermal equilibrium as working principle of mercury thermometer

Any physical property that changes measurably and reproducibly with temperature can be used to measure temperature. Such a property is called a **thermometric property** — the length of a liquid column, the pressure of a fixed mass of gas, the resistance of a wire, the e.m.f. of a thermocouple.

Thermometer	Thermometric property	Useful range
Mercury-in-glass	length of mercury column	$-39\text{ }^{\circ}\text{C}$ to $357\text{ }^{\circ}\text{C}$
Alcohol-in-glass	length of alcohol column	$-115\text{ }^{\circ}\text{C}$ to $78\text{ }^{\circ}\text{C}$
Constant-volume gas	pressure of fixed mass of gas	$-270\text{ }^{\circ}\text{C}$ to $1500\text{ }^{\circ}\text{C}$
Platinum resistance	electrical resistance	$-200\text{ }^{\circ}\text{C}$ to $1200\text{ }^{\circ}\text{C}$
Thermocouple	thermo-e.m.f.	$-200\text{ }^{\circ}\text{C}$ to $1600\text{ }^{\circ}\text{C}$

A mercury thermometer is a thin-walled glass **bulb** joined to a uniform capillary tube of very fine bore, sealed after the air above the mercury has been removed. The bulb holds the mercury; the fine bore makes a small expansion of the mercury produce a long, easily read movement of the thread.

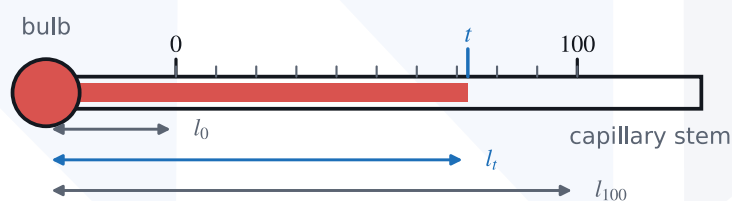


Figure 39. Mercury-in-glass thermometer. The length of the thread measured from the bulb is l_0 at the ice point, l_{100} at the steam point and l_t at the unknown temperature t .

How it works — thermal equilibrium is the whole principle. The bulb is placed in contact with the body whose temperature is wanted. Heat flows between body and bulb until the two reach thermal equilibrium, i.e. until the mercury is at the **same temperature as the body**. The mercury, having a larger expansivity than glass, has by then moved along the stem to a definite position, and by the zeroth law that position is a measure of the body's own temperature. Note the two conditions hidden in this: the thermometer must be left long enough to reach equilibrium, and it must be small enough that the heat it absorbs does not noticeably change the body's temperature.

Calibration. Two fixed points are used:

- **Lower fixed point (ice point):** the temperature of pure melting ice at standard atmospheric pressure, marked $0\text{ }^{\circ}\text{C}$; the thread length is l_0 .
- **Upper fixed point (steam point):** the temperature of steam above water boiling at standard atmospheric pressure, marked $100\text{ }^{\circ}\text{C}$; the thread length is l_{100} .

The interval between them is divided into 100 equal parts, which gives the working formula of every two-fixed-point thermometer.

THE THERMOMETRIC FORMULA FOR ANY THERMOMETER

We start from the assumption that the thermometric property changes uniformly with temperature, and reach the formula that converts a reading into a temperature.

Setting up. Let X be whatever property we measure — the length of a mercury thread, the pressure of a gas, the resistance of a wire. Its values are X_0 at the ice point ($0\text{ }^{\circ}\text{C}$), X_{100} at the steam point ($100\text{ }^{\circ}\text{C}$), and X_t at the unknown temperature t .

Step 1 — assume a straight line. "Uniform change" means X is a linear function of t , so a graph of X against t is a straight line:

$$X = a + bt$$

Step 2 — how much the property changed for the unknown reading. Subtracting the ice point value:

$$X_t - X_0 = (a + bt) - (a + b \times 0) = bt$$

Step 3 — how much it changed over the whole 100 degrees. The same subtraction with $t = 100$:

$$X_{100} - X_0 = b \times 100$$

Step 4 — divide Step 2 by Step 3. We divide so that the unknown constant b cancels, and a has already gone:

$$\frac{X_t - X_0}{X_{100} - X_0} = \frac{bt}{100b} = \frac{t}{100}$$

Step 5 — multiply both sides by 100:

$$t = \frac{X_t - X_0}{X_{100} - X_0} \times 100\text{ }^{\circ}\text{C}$$

Result. For a mercury thermometer, X is the thread length l , so

$$t = \frac{l_t - l_0}{l_{100} - l_0} \times 100\text{ }^{\circ}\text{C}$$

What it means. The unknown constants a and b both dropped out, so we never need to know them — two fixed points are enough to read any temperature in between. The fraction is simply "how far along the scale we are", expressed as a fraction of the whole interval.

Condition used. The property must change linearly with temperature over this range. If it does not, two different thermometers will disagree at the same real temperature even though they agree at 0 and 100 °C.

The three scales in use are related in exactly the same way — each fraction is "how far along its own scale", and equal fractions mean the same temperature:

$$\frac{C}{100} = \frac{F-32}{180} = \frac{K-273.15}{100}$$

WHY MERCURY AND NOT WATER

- It is a **good conductor** of heat, so it reaches equilibrium quickly.
- It expands **uniformly** over a wide range, giving an evenly divided scale.
- It has a **wide range**: freezes at $-39\text{ }^{\circ}\text{C}$, boils at $357\text{ }^{\circ}\text{C}$.
- Its **specific heat capacity is low**, so it takes very little heat from the body being measured.
- It is **opaque and shiny**, so the thread is easy to see.
- It **does not wet glass**, so none is left clinging to the bore.

Water fails on almost every count: it wets glass, is a poor conductor, is transparent, has a small range and — worst — expands anomalously below $4\text{ }^{\circ}\text{C}$, so one length would stand for two temperatures.

A **clinical thermometer** is the same instrument with the range narrowed to about $35\text{ }^{\circ}\text{C}$ – $42\text{ }^{\circ}\text{C}$ so that each division can be $0.1\text{ }^{\circ}\text{C}$, and with a **constriction** in the bore just above the bulb. The constriction lets mercury push past while the thread rises but breaks the thread when it cools, so the reading is held until the thermometer is shaken down.

WORKED EXAMPLE 9.2

Problem. In a mercury thermometer the mercury thread is 5.0 cm long at the ice point and 25.0 cm long at the steam point. (a) What temperature corresponds to a thread length of 15.6 cm? (b) How long is the thread at normal body temperature, $37\text{ }^{\circ}\text{C}$?

Solution. (a) Using the calibration formula,

$$t = \frac{15.6 - 5.0}{25.0 - 5.0} \times 100 = \frac{10.6}{20.0} \times 100 = 53\text{ }^{\circ}\text{C}$$

(b) Rearranging, $l_t = l_0 + \frac{t}{100}(l_{100} - l_0) = 5.0 + 0.37 \times 20.0 = 12.4\text{ cm}$.

WORKED EXAMPLE 9.3

Problem. (a) Convert normal body temperature $37\text{ }^{\circ}\text{C}$ to the Fahrenheit and Kelvin scales. (b) At what temperature do the Celsius and Fahrenheit scales give the same reading?

Solution. (a) From $\frac{C}{100} = \frac{F-32}{180}$ we get $F = \frac{9}{5}C + 32 = \frac{9}{5}(37) + 32 = 66.6 + 32 = 98.6\text{ }^{\circ}\text{F}$, and $K = C + 273.15 = 310.15\text{ K}$.

(b) Put $F = C = x$:

$$x = \frac{9}{5}x + 32 \Rightarrow -\frac{4}{5}x = 32 \Rightarrow x = -40$$

So $-40\text{ }^{\circ}\text{C} = -40\text{ }^{\circ}\text{F}$.

WORKED EXAMPLE 9.4

Problem. A faulty mercury thermometer reads $5\text{ }^{\circ}\text{C}$ in melting ice and $95\text{ }^{\circ}\text{C}$ in steam. What is the true temperature when it reads $59\text{ }^{\circ}\text{C}$?

Solution. The faulty scale has 90 of its own divisions between the two fixed points, so treat its reading like a thread length:

$$t_{\text{true}} = \frac{59-5}{95-5} \times 100 = \frac{54}{90} \times 100 = 60\text{ }^{\circ}\text{C}$$

The divisions are still equal — only the two end marks are misplaced — so the correction is a simple proportion.

✓ Chapter summary

- Internal energy is the total kinetic + potential energy of the molecules; thermal energy is its temperature-dependent (kinetic) part.
- Temperature measures the **average** molecular kinetic energy: $\bar{E}_k = \frac{3}{2}k_B T$ with $k_B = 1.38 \times 10^{-23}\text{ J K}^{-1}$.
- Heat is energy in transit due to a temperature difference. Its cause is the temperature difference and its direction is always hot $\rightarrow$ cold.
- Thermal equilibrium is the state of zero net heat flow, reached when the two temperatures become equal.
- Zeroth law: if A and B are each in equilibrium with C , they are in equilibrium with each other. This is what makes temperature measurable.
- A mercury thermometer works by coming into thermal equilibrium with the body; its thermometric property is the length of the mercury thread, and $t = \frac{l_t - l_0}{l_{100} - l_0} \times 100\text{ }^{\circ}\text{C}$.
- Scale conversion: $\frac{C}{100} = \frac{F-32}{180} = \frac{K-273.15}{100}$.

Practice questions

Group A — Multiple choice (1 mark each)

- Heat flows from body A to body B when placed in contact because A has [1]
 - ☐ a more heat
 - ☐ b greater mass
 - ☐ c higher average molecular kinetic energy
 - ☐ d greater volume
- The law of thermodynamics that makes temperature a measurable quantity is the [1]

- a** zeroth law **b** first law **c** second law **d** third law

3. The temperature at which the Celsius and Fahrenheit scales read the same is [1]

- a** 0° **b** -32° **c** -40° **d** 100°

4. The thermometric property of a mercury-in-glass thermometer is the [1]

- a** pressure of mercury **b** resistance of mercury
c length of the mercury column **d** mass of mercury

5. A temperature of 300 K is approximately [1]

- a** 27°C **b** 300°C **c** 573°C **d** -27°C

ANSWERS TO GROUP A

- (c) — temperature, i.e. average molecular KE, decides the direction; the total heat content does not.
- (a) — the zeroth law guarantees a common value shared by bodies in equilibrium.
- (c) — solving $x = \frac{9}{5}x + 32$ gives $x = -40$.
- (c) — the measured quantity is the length of the thread in the uniform bore.
- (a) — $300 - 273 = 27^\circ\text{C}$.

Group B — Short answer (5 marks each)

- Distinguish between heat and temperature on a molecular basis. Explain why a bucket of warm water can hold more thermal energy than a small flame at a much higher temperature. [5]
- State the zeroth law of thermodynamics and explain how it makes the measurement of temperature possible. [5]
- Explain thermal equilibrium as the working principle of a mercury thermometer, and give four reasons why mercury is preferred to water as a thermometric liquid. [5]
- The mercury thread in a thermometer is 4.0 cm long at the ice point and 24.0 cm long at the steam point. Find
 - the temperature when the thread is 13.0 cm long and
 - the thread length at 65°C . [5]
- Calculate the average translational kinetic energy of a gas molecule at 127°C , and find the temperature in $^\circ\text{C}$ at which it is half this value. [5]

ANSWERS TO GROUP B

- Outline: temperature = average molecular KE (an intensive quantity, per molecule); heat = energy in transit driven by a temperature difference. Thermal energy is the total molecular KE, so it grows with the number of molecules — a bucket of water has of order 10^{27} molecules against a flame's tiny number, so the total wins even though the energy per molecule is much smaller.
- (a) $t = \frac{13.0 - 4.0}{24.0 - 4.0} \times 100 = \frac{9.0}{20.0} \times 100 = 45^\circ\text{C}$. (b) $l = 4.0 + 0.65 \times 20.0 = 17.0\text{ cm}$.
- $T = 127 + 273 = 400\text{ K}$, so $\bar{E}_k = 1.5 \times 1.38 \times 10^{-23} \times 400 = 8.28 \times 10^{-21}\text{ J}$. Half of this needs $T = 200\text{ K} = -73^\circ\text{C}$.

Group C — Long answer (8 marks each)

1. **(a)** Explain, on the molecular model, what is meant by thermal energy, heat and temperature, and state the cause and direction of heat flow. [4]
- (b)** State the zeroth law of thermodynamics and describe the construction and working of a mercury-in-glass thermometer, showing how it is calibrated between the two fixed points. [4]
2. **(a)** Define thermometric property and name the property used in a constant volume gas thermometer and in a platinum resistance thermometer. [2]
- (b)** A faulty thermometer reads 2°C in melting ice and 102°C in steam. What is the true temperature when it reads 77°C ? [3]
- (c)** At what temperature does the Fahrenheit reading become exactly twice the Celsius reading? [3]

ANSWERS TO GROUP C

2. (a) A thermometric property is any measurable physical property that varies continuously, uniformly and reproducibly with temperature. Gas thermometer: the pressure of a fixed mass of gas at constant volume. Platinum thermometer: the electrical resistance of the platinum wire.

(b) The faulty scale spans $102 - 2 = 100$ of its own divisions, so $t_{\text{true}} = \frac{77-2}{102-2} \times 100 = 75^{\circ}\text{C}$.

(c) Put $F = 2C$ in $F = \frac{9}{5}C + 32$: $2C = 1.8C + 32 \Rightarrow 0.2C = 32 \Rightarrow C = 160^{\circ}\text{C}$, and then $F = 320^{\circ}\text{F}$. Check: $\frac{9}{5}(160) + 32 = 288 + 32 = 320$. ✓

UNIT 10

Thermal Expansion

Heat and Thermodynamics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Linear expansion and its measurement
- Cubical expansion, superficial expansion and relation with linear expansion
- Liquid Expansion: Absolute and apparent
- Dulong and Petit method of determining expansivity of liquid

Heat a solid and its molecules vibrate through wider swings about their lattice positions, so their average separation grows and the body gets bigger. That one sentence explains the gaps left between rails on the Janakpur railway, the rollers under a bridge girder, the sag designed into transmission lines, and the fact that a mercury thermometer works at all. In this unit we put numbers on the effect: one coefficient for length, one for area, one for volume, and a special treatment for liquids, which cannot be heated without a container that expands too.

THE THREE COEFFICIENTS ARE NOT INDEPENDENT

α , β and γ describe expansion in one, two and three dimensions, and for an isotropic solid they are in the ratio 1 : 2 : 3. Learn the derivation of $\beta = 2\alpha$ and $\gamma = 3\alpha$ — it is a standard Group B question — and learn why a liquid has two expansivities.

10.1 Linear expansion and its measurement

When a rod of length l_1 at temperature θ_1 is heated to θ_2 , its length becomes l_2 . Experiment shows the increase is proportional to the original length and to the temperature rise.

COEFFICIENT OF LINEAR EXPANSION

The coefficient of linear expansion (linear expansivity) α of a material is the increase in length per unit original length per unit rise in temperature:

$$\alpha = \frac{l_2 - l_1}{l_1(\theta_2 - \theta_1)} \quad \Rightarrow \quad l_2 = l_1[1 + \alpha(\theta_2 - \theta_1)]$$

Its SI unit is K^{-1} (numerically the same as $^{\circ}C^{-1}$) and it has no dimensions of length: $[\alpha] = [K^{-1}]$.

Strictly l_1 should be the length at $0^{\circ}C$, but because α is of order 10^{-5} the difference between "length at $0^{\circ}C$ " and "length at room temperature" changes the answer only in the fifth decimal place, so any convenient starting length is used.

Material	α ($\times 10^{-6} \text{ K}^{-1}$)	Material	α ($\times 10^{-6} \text{ K}^{-1}$)
Invar	1.2	Copper	17
Pyrex glass	3.2	Brass	19
Ordinary glass	9	Aluminium	23
Steel / iron / concrete	12	Lead	29

Invar (an iron–nickel alloy) expands so little that it is used for pendulum rods and survey tapes. Steel and concrete happen to have almost the same α — which is the reason reinforced concrete does not tear itself apart in the Terai summer.

Measurement: Pullinger's apparatus

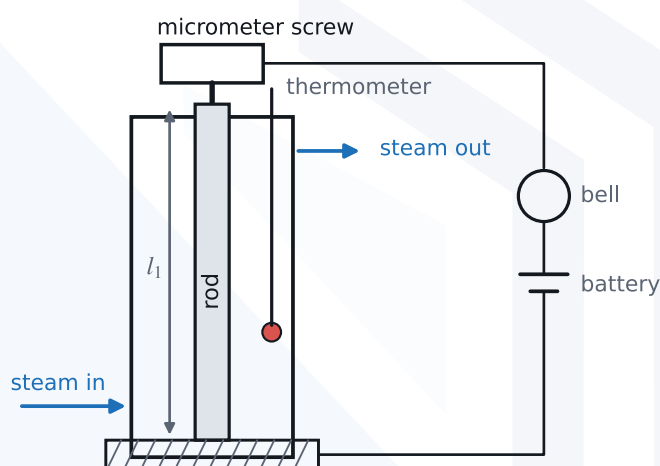


Figure 40. Pullinger's apparatus. The rod expands inside a steam jacket and the expansion is read on a micrometer screw whose contact with the rod completes the bell circuit.

The rod under test stands on a fixed base inside a hollow **steam jacket**. Its upper end is just below the tip of a **micrometer screw** carried on a rigid frame. The screw, the rod and a battery with an electric bell are connected in series, so the bell rings the instant the screw touches the rod.

1. With the rod at room temperature θ_1 (read on the thermometer in the jacket), turn the screw down until the bell just rings; record the reading a_1 . Measure the length l_1 of the rod with a metre rule.
2. Raise the screw well clear of the rod.
3. Pass steam through the jacket until the thermometer is steady at θ_2 . The rod has now expanded.
4. Turn the screw down again until the bell just rings; record a_2 .

The expansion is $\delta l = a_1 - a_2$ (the screw has less distance to travel), so

$$\alpha = \frac{\delta l}{l_1(\theta_2 - \theta_1)}$$

WORKED EXAMPLE 10.1

Problem. A steel railway rail is 30.0 m long at 10 °C. What gap must be left at each joint if the rail temperature can rise to 45 °C? Take $\alpha_{\text{steel}} = 1.2 \times 10^{-5} \text{ K}^{-1}$.

Solution. Temperature rise $\Delta\theta = 45 - 10 = 35 \text{ K}$.

$$\delta l = l_1 \alpha \Delta\theta = 30.0 \times 1.2 \times 10^{-5} \times 35 = 1.26 \times 10^{-2} \text{ m}$$

So a gap of about 12.6 mm is needed. (A gap smaller than this and the rails push against each other and buckle in hot weather.)

WORKED EXAMPLE 10.2

Problem. A pendulum clock with a steel pendulum keeps correct time at 15 °C. How much time does it lose per day when the temperature rises to 35 °C? Take $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$.

Solution. For a simple pendulum $T = 2\pi\sqrt{l/g}$, so $T \propto \sqrt{l}$. With $l' = l(1 + \alpha\Delta\theta)$,

$$\frac{T'}{T} = \sqrt{1 + \alpha\Delta\theta} \approx 1 + \frac{1}{2}\alpha\Delta\theta$$

The fractional gain in the period is $\frac{1}{2}\alpha\Delta\theta$, so in one day of 86400 s the clock loses

$$\Delta t = \frac{1}{2}\alpha\Delta\theta \times 86400 = 0.5 \times 1.2 \times 10^{-5} \times 20 \times 86400 = 10.4 \text{ s}$$

A longer pendulum swings more slowly, so the clock runs **slow** in hot weather.

10.2 Cubical expansion, superficial expansion and relation with linear expansion

SUPERFICIAL AND CUBICAL EXPANSIVITY

Superficial expansivity β is the increase in area per unit original area per unit rise in temperature: $A_2 = A_1(1 + \beta\Delta\theta)$.

Cubical (volume) expansivity γ is the increase in volume per unit original volume per unit rise in temperature: $V_2 = V_1(1 + \gamma\Delta\theta)$.

Both have the unit K^{-1} .

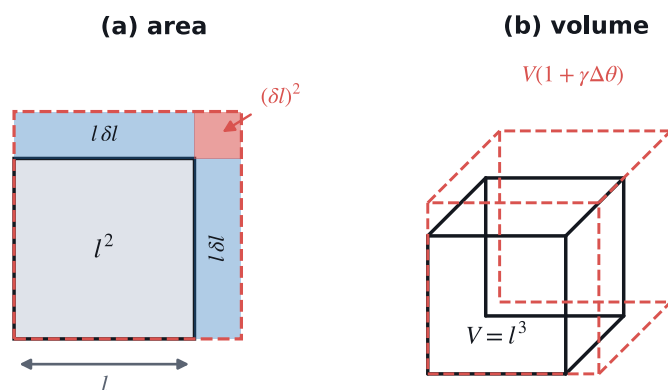


Figure 41. A square of side l heated through $\Delta\theta$. The two shaded strips have total area $2l\delta l$; the corner square $(\delta l)^2$ is far smaller and is neglected, giving $\beta = 2\alpha$.

THE RELATIONS $\beta = 2\alpha$ AND $\gamma = 3\alpha$

We start from the definition of linear expansivity applied to every edge of a body, and reach the two relations connecting the three expansivities.

Setting up. Everything rests on one fact: when the temperature rises by $\Delta\theta$, every length in the body grows by the same fraction, so a length l becomes $l(1 + \alpha\Delta\theta)$.

Part (a): $\beta = 2\alpha$

Step 1 — start with a square plate. Let it have side l , so its area at the start is

$$A_1 = l^2$$

Step 2 — find the new side. Each side is a length, so it grows by the fraction $\alpha\Delta\theta$:

$$l' = l(1 + \alpha\Delta\theta)$$

Step 3 — square it to get the new area:

$$A_2 = l'^2 = l^2(1 + \alpha\Delta\theta)^2$$

Step 4 — expand the bracket. Using $(1 + x)^2 = 1 + 2x + x^2$:

$$A_2 = l^2(1 + 2\alpha\Delta\theta + \alpha^2\Delta\theta^2)$$

Step 5 — throw away the tiny term. For solids α is about 10^{-5} K^{-1} , so α^2 is about 10^{-10} — ten thousand times smaller than α itself. Dropping it changes nothing we could measure:

$$A_2 \approx l^2(1 + 2\alpha\Delta\theta)$$

Step 6 — put A_1 back in, since $l^2 = A_1$:

$$A_2 \approx A_1(1 + 2\alpha\Delta\theta)$$

Step 7 — compare with the definition of β , which says $A_2 = A_1(1 + \beta\Delta\theta)$. The two right-hand sides must agree for every $\Delta\theta$, so the coefficients of $\Delta\theta$ must match:

$$\beta = 2\alpha$$

Part (b): $\gamma = 3\alpha$

Step 8 — start with a cube of side l , so its volume at the start is

$$V_1 = l^3$$

Step 9 — cube the new side $l(1 + \alpha\Delta\theta)$ from Step 2:

$$V_2 = l^3(1 + \alpha\Delta\theta)^3$$

Step 10 — expand, using $(1 + x)^3 = 1 + 3x + 3x^2 + x^3$:

$$V_2 = l^3(1 + 3\alpha\Delta\theta + 3\alpha^2\Delta\theta^2 + \alpha^3\Delta\theta^3)$$

Step 11 — drop the α^2 and α^3 terms for the same reason as in Step 5:

$$V_2 \approx l^3(1 + 3\alpha\Delta\theta) = V_1(1 + 3\alpha\Delta\theta)$$

Step 12 — compare with $V_2 = V_1(1 + \gamma\Delta\theta)$:

$$\gamma = 3\alpha$$

Result.

$$\beta = 2\alpha, \quad \gamma = 3\alpha, \quad \alpha : \beta : \gamma = 1 : 2 : 3, \quad \frac{\alpha}{1} = \frac{\beta}{2} = \frac{\gamma}{3}$$

What it means. The numbers 1, 2, 3 are simply the number of directions involved: a line stretches in one direction, an area in two, a volume in three. So you never need to look up β or γ in a table — one number per material is enough.

Conditions used. $\alpha\Delta\theta \ll 1$, so that the squared and cubed terms can be neglected; and α is the same in all directions (true for metals and glass, not for wood or crystals like calcite). The shape does not matter: any body can be imagined as built from small cubes, each expanding this way, so the result holds equally for a sphere, a rod or an irregular casting.

THE EXAMINER IS LOOKING FOR

1. "Each side becomes $l(1 + \alpha\Delta\theta)$ " — stated, not assumed silently.
2. The binomial expansion written out in full.
3. An explicit sentence saying why the α^2 term is dropped (because $\alpha \approx 10^{-5}$), not just a quiet disappearance.
4. The comparison with the defining equation for β or γ .
5. The assumption that expansion is the same in all directions.

A HOLE EXPANDS, IT DOES NOT CLOSE UP

Students often expect a hole in a heated metal plate to get smaller. It does not. Imagine the disc of metal that was cut out to make the hole: on heating, that disc would grow by $\gamma\Delta\theta$ per unit volume, and the hole must grow with it so that the disc would still fit. **A cavity expands exactly as if it were filled with the material of the body.** This is why a tight metal lid loosens when held under hot water.

WORKED EXAMPLE 10.3

Problem. A brass plate has a circular hole of diameter 5.000 cm at 20 °C. Find the diameter of the hole and the area of the hole at 220 °C. Take $\alpha_{\text{brass}} = 1.9 \times 10^{-5} \text{ K}^{-1}$.

Solution. The hole expands like the metal, with $\Delta\theta = 200 \text{ K}$.

$$d_2 = d_1(1 + \alpha\Delta\theta) = 5.000(1 + 1.9 \times 10^{-5} \times 200) = 5.000(1 + 3.8 \times 10^{-3}) = 5.019 \text{ cm}$$

Original area $A_1 = \pi d_1^2/4 = \pi(5.000)^2/4 = 19.635 \text{ cm}^2$. Using $\beta = 2\alpha = 3.8 \times 10^{-5} \text{ K}^{-1}$,

$$A_2 = A_1(1 + \beta\Delta\theta) = 19.635(1 + 3.8 \times 10^{-5} \times 200) = 19.635 \times 1.0076 = 19.784 \text{ cm}^2$$

10.3 Liquid Expansion: Absolute and apparent

A liquid has no shape of its own, so only its **volume** expansion is defined — there is no α or β for a liquid. But a liquid must be held in a vessel, and when the vessel is heated it expands too. The level in the vessel therefore rises by less than the liquid's true expansion. Two coefficients are needed.

REAL AND APPARENT EXPANSIVITY

Real (absolute) expansivity γ_r is the true increase in volume of the liquid per unit volume per unit rise in temperature.

Apparent expansivity γ_a is the observed increase in volume per unit volume per unit rise in temperature, measured relative to the vessel.

REAL, APPARENT AND VESSEL EXPANSIVITY: $\gamma_r = \gamma_a + \gamma_g$

We start from the fact that the vessel expands too, and reach the correction that must be added to every measured value.

Setting up. A vessel of volume V is filled to the brim with liquid at temperature θ_1 . Both are heated to θ_2 , a rise of $\Delta\theta$. Let γ_r be the real expansivity of the liquid and γ_g the cubical expansivity of the vessel's material.

Step 1 — the true new volume of the liquid. By the definition of γ_r :

$$V_{\text{liquid}} = V(1 + \gamma_r\Delta\theta)$$

Step 2 — the new volume of the vessel. The vessel is made of a solid, and its inside volume expands by its own cubical expansivity:

$$V_{\text{vessel}} = V(1 + \gamma_g \Delta\theta)$$

Step 3 — only the difference can escape. The liquid that will not fit inside the enlarged vessel spills out over the top:

$$\text{overflow} = V_{\text{liquid}} - V_{\text{vessel}} = V(1 + \gamma_r \Delta\theta) - V(1 + \gamma_g \Delta\theta)$$

Step 4 — take out the common factor V and remove the brackets. The two 1's cancel:

$$\text{overflow} = V(\gamma_r \Delta\theta - \gamma_g \Delta\theta)$$

Step 5 — take out $\Delta\theta$ as well:

$$\text{overflow} = V(\gamma_r - \gamma_g) \Delta\theta$$

Step 6 — say what we actually observe. All an experimenter sees is the spilt liquid, and that overflow is exactly what the apparent expansivity is defined by:

$$\text{overflow} = V\gamma_a \Delta\theta$$

Step 7 — set Step 5 equal to Step 6. Both describe the same spilt liquid:

$$V\gamma_a \Delta\theta = V(\gamma_r - \gamma_g) \Delta\theta$$

Step 8 — cancel V and $\Delta\theta$, which appear on both sides:

$$\gamma_a = \gamma_r - \gamma_g$$

Step 9 — rearrange for the real expansivity:

$$\gamma_r = \gamma_a + \gamma_g$$

Step 10 — express the vessel's part through α , since tables list the linear expansivity α_g of glass, and $\gamma = 3\alpha$:

$$\gamma_r = \gamma_a + 3\alpha_g$$

Result.

$$\gamma_r = \gamma_a + \gamma_g = \gamma_a + 3\alpha_g$$

What it means. The measured (apparent) expansion is always smaller than the real one, because the container quietly makes room for part of the liquid's growth. So the correction is always an **addition**. If the vessel could somehow not expand ($\gamma_g = 0$), apparent and real expansion would be the same.

Conditions used. The vessel is full to the brim at the start, nothing evaporates, and the liquid and vessel reach the same temperature.

In the **weight-thermometer** (specific-gravity bottle) method the bulb is filled with liquid of mass M at θ_1 , heated to θ_2 , and the mass m expelled is weighed. Since apparent expansion is measured by the liquid that leaves,

$$\gamma_a = \frac{m}{(M-m) \Delta\theta}$$

where $M-m$ is the mass still filling the bulb at the higher temperature. Adding $3\alpha_g$ for the glass then gives the real expansivity.

Liquid	γ_r (K^{-1})
Mercury	1.8×10^{-4}
Water (near 20 °C)	2.1×10^{-4}
Glycerine	5.3×10^{-4}
Ethanol	1.1×10^{-3}

Because $\gamma_{\text{glass}} = 3 \times (9 \times 10^{-6}) = 2.7 \times 10^{-5} \text{ K}^{-1}$ is small compared with these, the apparent expansivity is usually only a little less than the real one — but the difference is easily measurable and must never be ignored in a calculation.

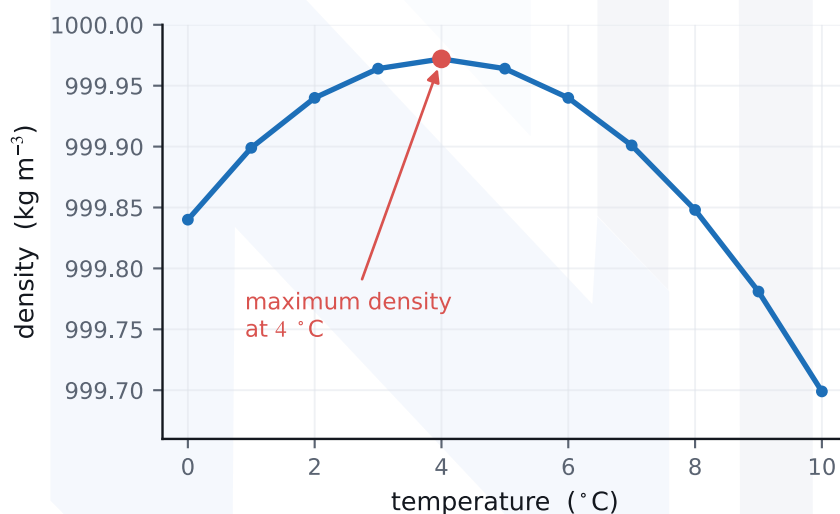


Figure 42. Anomalous expansion of water. Between 0 $^{\circ}\text{C}$ and 4 $^{\circ}\text{C}$ water contracts on heating, so its density is greatest at 4 $^{\circ}\text{C}$.

Water is the famous exception. From 0 $^{\circ}\text{C}$ to 4 $^{\circ}\text{C}$ it contracts on heating, and only above 4 $^{\circ}\text{C}$ does it expand normally. Its density is therefore a maximum at 4 $^{\circ}\text{C}$. This **anomalous expansion** is why ice floats, and why a lake such as Rara freezes from the surface downwards in winter, leaving water at 4 $^{\circ}\text{C}$ at the bottom in which fish survive.

WORKED EXAMPLE 10.4

Problem. A glass flask of volume 1000 cm^3 is filled completely with mercury at 0°C and heated to 100°C . Given $\gamma_{\text{Hg}} = 1.8 \times 10^{-4} \text{ K}^{-1}$ and $\alpha_{\text{glass}} = 9 \times 10^{-6} \text{ K}^{-1}$, find the volume of mercury that overflows.

Solution. Cubical expansivity of glass:

$$\gamma_g = 3\alpha_g = 3 \times 9 \times 10^{-6} = 2.7 \times 10^{-5} \text{ K}^{-1}$$

$$\gamma_a = \gamma_r - \gamma_g = 1.8 \times 10^{-4} - 0.27 \times 10^{-4} = 1.53 \times 10^{-4} \text{ K}^{-1}$$

$$\text{overflow} = V\gamma_a\Delta\theta = 1000 \times 1.53 \times 10^{-4} \times 100 = 15.3 \text{ cm}^3$$

Had we forgotten the flask, we would have said 18.0 cm^3 — an error of almost 18 %.

10.4 Dulong and Petit method of determining expansivity of liquid

Every method that uses a vessel gives the apparent expansivity first, and the real value only after the vessel's own expansion is known. Dulong and Petit (1817) found a way to measure the **real** expansivity directly, using the fact that pressure, not volume, is what balances in a connected liquid.

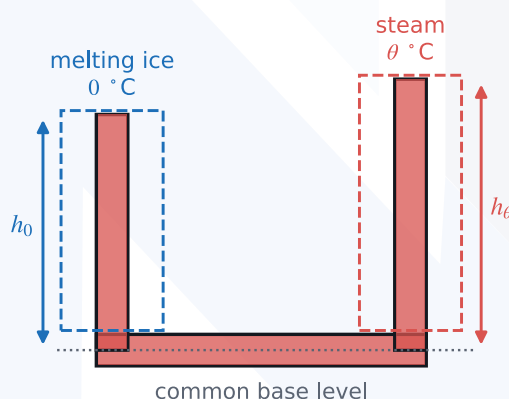


Figure 43. Dulong and Petit's apparatus. The two limbs of liquid balance at the base, so $h_0\rho_0 = h_\theta\rho_\theta$ and the expansion of the glass never enters the result.

Apparatus. The liquid fills a U-shaped glass tube whose two vertical limbs are joined by a narrow horizontal tube at the bottom. One limb is surrounded by a jacket of melting ice at 0°C ; the other is surrounded by a steam jacket at $\theta^\circ\text{C}$. The heights of the two columns above the common horizontal base are measured with a travelling microscope, so nothing touches the liquid.

THE DULONG AND PETIT FORMULA, $\gamma = \frac{h_\theta - h_0}{h_0\theta}$

We start from the balance of pressures in the two columns and reach a formula for the real expansivity in terms of two measured heights.

Setting up. The two vertical columns of the same liquid are joined at the bottom by a horizontal tube. One column is kept at 0°C and stands at height h_0 ; the other is kept at $\theta^\circ\text{C}$ and stands at height h_θ . Let ρ_0 and ρ_θ be the densities at those two temperatures.

Step 1 — the liquid is at rest. Nothing flows through the horizontal tube, so the two columns must push on it equally hard. Pressure of a column of height h and density ρ is $h\rho g$:

$$h_0\rho_0g = h_\theta\rho_\theta g$$

Step 2 — cancel g , which is the same on both sides:

$$h_0\rho_0 = h_\theta\rho_\theta$$

Step 3 — now find ρ_θ . Take a fixed mass M of the liquid. At 0°C it occupies V_0 , so

$$\rho_0 = \frac{M}{V_0}$$

Step 4 — the same mass at θ occupies a bigger volume. By the definition of cubical expansivity, the volume becomes $V_0(1 + \gamma\theta)$:

$$\rho_\theta = \frac{M}{V_0(1 + \gamma\theta)}$$

Step 5 — replace M/V_0 by ρ_0 using Step 3. This is the "density falls because the same mass spreads out" step:

$$\rho_\theta = \frac{\rho_0}{1 + \gamma\theta}$$

Step 6 — substitute this into Step 2:

$$h_0\rho_0 = h_\theta \cdot \frac{\rho_0}{1 + \gamma\theta}$$

Step 7 — cancel ρ_0 from both sides:

$$h_0 = \frac{h_\theta}{1 + \gamma\theta}$$

Step 8 — multiply both sides by $(1 + \gamma\theta)$:

$$h_0(1 + \gamma\theta) = h_\theta$$

Step 9 — expand the left side:

$$h_0 + h_0\gamma\theta = h_\theta$$

Step 10 — move h_0 across:

$$h_0\gamma\theta = h_\theta - h_0$$

Step 11 — divide both sides by $h_0\theta$:

$$\gamma = \frac{h_\theta - h_0}{h_0 \theta}$$

Result.

$$\gamma = \frac{h_\theta - h_0}{h_0 \theta}$$

What it means. Only two heights and one temperature are needed — no masses, no volumes, no weighing. The hot column stands taller because the same mass has become less dense and so needs more height to produce the same pressure.

Conditions used. The liquid is in equilibrium (not flowing), the two columns are at steady, uniform temperatures, and the heights are measured from the level of the horizontal joining tube.

THE EXAMINER IS LOOKING FOR

1. The pressure-balance equation $h_0 \rho_0 g = h_\theta \rho_\theta g$ with a reason.
2. $\rho_\theta = \rho_0 / (1 + \gamma \theta)$, derived from a **fixed mass**.
3. The substitution and the cancellation of ρ_0 .
4. The final rearrangement to $\gamma = (h_\theta - h_0) / h_0 \theta$.
5. The explanation of why **no vessel correction** is needed.

WHY THIS GIVES THE *REAL* EXPANSIVITY

The result contains only heights, temperatures and the liquid's own densities. The glass tube does expand, but its expansion changes neither the density of the liquid nor the measured heights, which are read from outside with a microscope against a fixed scale. So Dulong and Petit's method needs no correction for the vessel and gives γ_r directly — its great advantage over the weight-thermometer method.

WORKED EXAMPLE 10.5

Problem. In a Dulong and Petit experiment the cold column of mercury stands 75.00 cm high at 0 °C while the hot column stands 76.35 cm high at 100 °C. Find the real expansivity of mercury. If the tube is glass with $\alpha = 9 \times 10^{-6} \text{ K}^{-1}$, what apparent expansivity would a glass-vessel experiment have given?

Solution.

$$\gamma_r = \frac{h_\theta - h_0}{h_0 \theta} = \frac{76.35 - 75.00}{75.00 \times 100} = \frac{1.35}{7500} = 1.8 \times 10^{-4} \text{ K}^{-1}$$

For glass, $\gamma_g = 3\alpha = 2.7 \times 10^{-5} \text{ K}^{-1}$, so

$$\gamma_a = \gamma_r - \gamma_g = 1.8 \times 10^{-4} - 0.27 \times 10^{-4} = 1.53 \times 10^{-4} \text{ K}^{-1}$$

✓ Chapter summary

- Linear expansivity $\alpha = \frac{l_2 - l_1}{l_1 \Delta\theta}$, so $l_2 = l_1(1 + \alpha\Delta\theta)$; unit K^{-1} .
- Pullinger's apparatus measures α : rod in a steam jacket, expansion read on a micrometer screw that completes a bell circuit; $\alpha = \delta l / (l_1 \Delta\theta)$.
- $A_2 = A_1(1 + \beta\Delta\theta)$ and $V_2 = V_1(1 + \gamma\Delta\theta)$, with $\beta = 2\alpha$, $\gamma = 3\alpha$ and $\alpha : \beta : \gamma = 1 : 2 : 3$.
- A cavity or hole expands exactly as if it were filled with the surrounding material.
- A liquid has two expansivities: real γ_r and apparent γ_a , with $\gamma_r = \gamma_a + \gamma_g$ where $\gamma_g = 3\alpha_g$ for the vessel. By the weight thermometer, $\gamma_a = m / [(M - m)\Delta\theta]$.
- Water expands anomalously: it contracts from 0 to 4°C and has maximum density at 4°C .
- Dulong and Petit: balancing columns give $h_\theta = h_0(1 + \gamma\theta)$, so $\gamma_r = (h_\theta - h_0)/(h_0\theta)$, with no correction for the vessel.

Practice questions

Group A — Multiple choice (1 mark each)

- For an isotropic solid the ratio $\alpha : \beta : \gamma$ is [1]

☐ a 1 : 2 : 3 ☐ b 3 : 2 : 1 ☐ c 1 : 3 : 2 ☐ d 2 : 3 : 1
- When a metal plate with a hole in it is heated, the hole [1]

☐ a gets smaller ☐ b gets larger
☐ c stays the same ☐ d first shrinks, then grows
- The SI unit of the coefficient of linear expansion is [1]

☐ a m ☐ b K^{-1} ☐ c m K^{-1} ☐ d it has no unit
- The real expansivity of a liquid equals its apparent expansivity plus the [1]

☐ a linear expansivity of the vessel ☐ b cubical expansivity of the vessel
☐ c superficial expansivity of the vessel ☐ d nothing
- Dulong and Petit's method directly determines the [1]

☐ a apparent expansivity ☐ b real expansivity
☐ c linear expansivity ☐ d superficial expansivity

ANSWERS TO GROUP A

- (a) — from $(1 + \alpha\Delta\theta)^n$ for $n = 1, 2, 3$.
- (b) — a cavity expands like the material that would fill it.
- (b) — it is a fractional length change per kelvin, so only K^{-1} is left.
- (b) — $\gamma_r = \gamma_a + \gamma_g$, where $\gamma_g = 3\alpha_g$.
- (b) — it balances pressures, so the vessel's expansion cancels out.

Group B — Short answer (5 marks each)

1. Define the coefficient of linear expansion and show that $\gamma = 3\alpha$ for an isotropic solid. [5]
2. With a labelled diagram, describe how the linear expansivity of a metal rod is determined using Pullinger's apparatus. [5]
3. Distinguish between the real and apparent expansivity of a liquid and show that $\gamma_r = \gamma_a + \gamma_g$. [5]
4. A glass flask of volume 500 cm^3 is completely filled with mercury at 0°C . On heating to 80°C , 6.0 cm^3 of mercury overflows. If $\gamma_{\text{Hg}} = 1.8 \times 10^{-4} \text{ K}^{-1}$, find the coefficient of linear expansion of the glass. [5]
5. A brass scale is correct at 15°C . It is used at 35°C and reads a length of 50.00 cm . What is the true length? Take $\alpha_{\text{brass}} = 1.9 \times 10^{-5} \text{ K}^{-1}$. [5]

ANSWERS TO GROUP B

4. $\gamma_a = \frac{6.0}{500 \times 80} = 1.5 \times 10^{-4} \text{ K}^{-1}$. Then $\gamma_g = \gamma_r - \gamma_a = 1.8 \times 10^{-4} - 1.5 \times 10^{-4} = 3.0 \times 10^{-5} \text{ K}^{-1}$, so $\alpha_g = \gamma_g/3 = 1.0 \times 10^{-5} \text{ K}^{-1}$.

5. At 35°C each scale division has grown, so every division now measures more than it claims. True length = $50.00(1 + 1.9 \times 10^{-5} \times 20) = 50.00(1 + 3.8 \times 10^{-4}) = 50.019 \text{ cm}$.

Group C — Long answer (8 marks each)

1. (a) Derive the relations $\beta = 2\alpha$ and $\gamma = 3\alpha$, stating clearly the approximation used. [4]
 (b) Describe Dulong and Petit's method for the absolute expansivity of a liquid and derive $\gamma = \frac{h_\theta - h_0}{h_0 \theta}$. Explain why no correction for the expansion of the glass is needed. [4]
2. (a) An iron rim of diameter 79.80 cm at 20°C is to be fitted on a wooden cart wheel of diameter 80.00 cm . To what temperature must the rim be heated? Take $\alpha_{\text{iron}} = 1.2 \times 10^{-5} \text{ K}^{-1}$. [4]
 (b) In a Dulong and Petit experiment on a liquid, the column at 0°C is 60.00 cm high and the column at 100°C is 61.20 cm high. Find the real expansivity of the liquid, and its apparent expansivity when it is kept in a glass vessel with $\alpha = 9 \times 10^{-6} \text{ K}^{-1}$. [4]

ANSWERS TO GROUP C

2. (a) The rim must expand from 79.80 to 80.00 cm , so $\Delta d = 0.20 \text{ cm}$ and

$$\Delta\theta = \frac{\Delta d}{d_1 \alpha} = \frac{0.20}{79.80 \times 1.2 \times 10^{-5}} = 208.9 \text{ K}$$

The rim must be heated to $20 + 208.9 \approx 229^\circ\text{C}$. On cooling it grips the wheel tightly — the standard way a wheelwright fits a tyre.

(b) $\gamma_r = \frac{61.20 - 60.00}{60.00 \times 100} = \frac{1.20}{6000} = 2.0 \times 10^{-4} \text{ K}^{-1}$.

$\gamma_g = 3 \times 9 \times 10^{-6} = 2.7 \times 10^{-5} \text{ K}^{-1}$, so $\gamma_a = 2.0 \times 10^{-4} - 0.27 \times 10^{-4} = 1.73 \times 10^{-4} \text{ K}^{-1}$.

UNIT 11

Quantity of Heat

Heat and Thermodynamics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Newton's law of cooling
- Measurement of specific heat capacity of solids and liquids
- Change of phases: Latent heat
- Specific latent heat of fusion and vaporization
- Measurement of specific latent heat of fusion and vaporization
- Triple point

Heat is energy that flows from a hotter body to a colder one because of the temperature difference between them. This unit is about counting that energy: how much heat a body must absorb to warm up, how much more it needs to melt or boil, and how quickly a hot body loses heat to the air around it. The unit of every quantity here is the joule, and almost every numerical problem is solved with one sentence — **heat lost by the hot bodies equals heat gained by the cold bodies**, provided nothing leaks to the surroundings.

THE THREE EQUATIONS THIS UNIT IS BUILT ON

Warming without a phase change: $Q = mc \Delta\theta$. Phase change at constant temperature: $Q = mL$. Heat capacity of a body: $C = mc$, measured in J K^{-1} .

The **water equivalent** W of a calorimeter is the mass of water that would need the same heat for the same temperature rise: $W = mc/c_w$. Adding the calorimeter's water equivalent to the water already inside turns a two-body problem into a one-body problem.

11.1 Newton's law of cooling

Leave a cup of tea on the table in Kathmandu in winter and it cools fast at first, then more and more slowly, finally settling at room temperature. Newton described this behaviour in a law that works well for moderate temperature differences.

NEWTON'S LAW OF COOLING

The rate at which a body loses heat is directly proportional to the excess of its temperature over that of its surroundings, provided the excess is small and the physical conditions (surface area, nature of surface, movement of the surrounding air) remain unchanged.

If the body is at θ and the surroundings at θ_0 ,

$$\frac{dQ}{dt} = -K(\theta - \theta_0)$$

where K depends on the surface area and on the nature of the surface. Since $Q = mc\theta$ for a body of mass m and specific heat capacity c , we may write $dQ/dt = mc (d\theta/dt)$, so

$$\frac{d\theta}{dt} = -k(\theta - \theta_0), \quad k = \frac{K}{mc}$$

The **rate of fall of temperature** is therefore also proportional to the excess temperature. Note carefully that dQ/dt and $d\theta/dt$ are different things: the first is a power in watts, the second a cooling rate in K s^{-1} .

A COOLING BODY FALLS IN TEMPERATURE EXPONENTIALLY

We start from Newton's law of cooling written as a rate equation, and reach a formula that gives the temperature at any later time.

Setting up. Let θ be the body's temperature at time t , θ_0 the temperature of the surroundings, and θ_1 the body's temperature at $t=0$. Newton's law says the rate of cooling is proportional to the excess temperature:

$$\frac{d\theta}{dt} = -k(\theta - \theta_0)$$

The minus sign is there because θ is falling, so $d\theta/dt$ is negative while k and the excess temperature are both positive.

Step 1 — separate the variables. We want all the θ 's on one side and all the t 's on the other, so divide both sides by $(\theta - \theta_0)$ and multiply by dt :

$$\frac{d\theta}{\theta - \theta_0} = -k dt$$

Step 2 — integrate both sides. The left side is of the standard form $\int \frac{dx}{x} = \ln x$, with $x = \theta - \theta_0$ (note θ_0 is a constant, so $d(\theta - \theta_0) = d\theta$):

$$\ln(\theta - \theta_0) = -kt + C$$

Step 3 — find the constant C from the starting condition. At $t=0$ the temperature is θ_1 :

$$\ln(\theta_1 - \theta_0) = -k(0) + C = C$$

Step 4 — put this value of C back:

$$\ln(\theta - \theta_0) = -kt + \ln(\theta_1 - \theta_0)$$

Step 5 — collect the two logarithms on the left:

$$\ln(\theta - \theta_0) - \ln(\theta_1 - \theta_0) = -kt$$

Step 6 — combine them into one, using $\ln A - \ln B = \ln(A/B)$:

$$\ln \frac{\theta - \theta_0}{\theta_1 - \theta_0} = -kt$$

Step 7 — undo the logarithm by taking e to the power of each side:

$$\frac{\theta - \theta_0}{\theta_1 - \theta_0} = e^{-kt}$$

Step 8 — multiply both sides by $(\theta_1 - \theta_0)$:

$$\theta - \theta_0 = (\theta_1 - \theta_0)e^{-kt}$$

Step 9 — add θ_0 to both sides:

$$\theta = \theta_0 + (\theta_1 - \theta_0)e^{-kt}$$

Result.

$$\theta = \theta_0 + (\theta_1 - \theta_0)e^{-kt}$$

What it means. The excess temperature $(\theta - \theta_0)$ dies away **exponentially**, halving in equal intervals of time. The body never quite reaches θ_0 in theory — it only gets closer and closer, which is why cooling looks fast at first and very slow at the end. Step 6 also gives the experimental test: a graph of $\ln(\theta - \theta_0)$ against t is a **straight line of slope $-k$** .

Conditions used. The excess temperature must be small (roughly under 30°C), cooling must be by natural convection in a draught-free room, and the surroundings must stay at a steady θ_0 .

THE EXAMINER IS LOOKING FOR

1. Newton's law written as $d\theta/dt = -k(\theta - \theta_0)$, with the minus sign explained.
2. The separation of variables.
3. The integration giving a logarithm, **with a constant of integration**.
4. Using $t=0$, $\theta = \theta_1$ to evaluate that constant.
5. Taking exponentials to reach $\theta = \theta_0 + (\theta_1 - \theta_0)e^{-kt}$.
6. The remark that $\ln(\theta - \theta_0)$ against t is a straight line of slope $-k$.

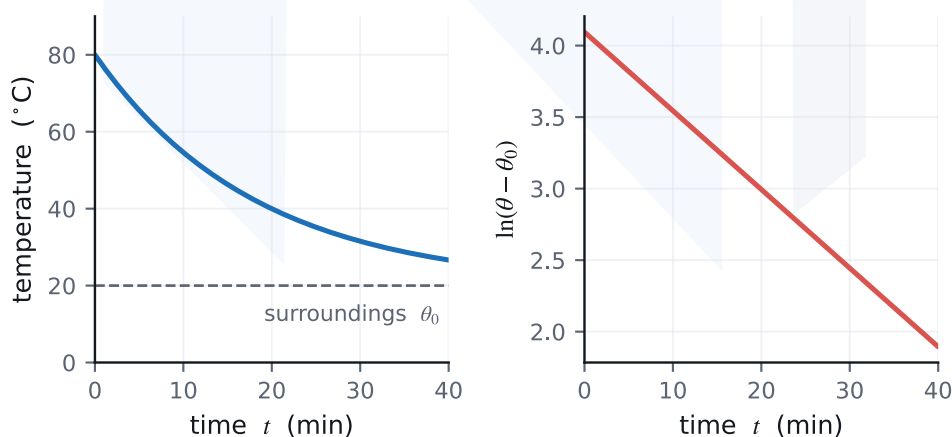


Figure 44. Left: a body at 80°C cooling in surroundings at 20°C . Right: the same data plotted as $\ln(\theta - \theta_0)$ against t gives a straight line of slope $-k$.

For examination work the differential form is usually replaced by an **average form**. If a body cools from θ_1 to θ_2 in time t , and the fall is not too large, we may take the mean temperature during the interval as $(\theta_1 + \theta_2)/2$:

$$\frac{\theta_1 - \theta_2}{t} = k \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

WORKED EXAMPLE 11.1

Problem. A body cools from 60°C to 50°C in 10 minutes. The temperature of the surroundings is 25°C . What will its temperature be after the next 10 minutes?

Solution. Using the average form for the first interval,

$$\frac{60 - 50}{10} = k \left(\frac{60 + 50}{2} - 25 \right) = k(55 - 25) = 30k$$

so $1 = 30k$, giving $k = 1/30$ per minute.

For the second interval, let the final temperature be θ :

$$\frac{50 - \theta}{10} = \frac{1}{30} \left(\frac{50 + \theta}{2} - 25 \right)$$

Multiplying through by 30: $3(50 - \theta) = \frac{50 + \theta}{2} - 25$, i.e. $150 - 3\theta = \theta/2$. Hence $150 = 3.5\theta$ and

$$\theta = \frac{150}{3.5} = 42.9^\circ\text{C}$$

The body fell 10°C in the first ten minutes but only about 7.1°C in the second — cooling slows as the excess temperature drops.

NEWTON'S LAW IS NOT UNIVERSAL

The law holds only for a **small** excess temperature (in practice up to about 30°C) and for a body losing heat mainly by convection in a draught or by forced air flow. For a red-hot body radiation dominates, the loss goes as T^4 (Unit 12), and Newton's law fails badly.

11.2 Measurement of specific heat capacity of solids and liquids

SPECIFIC HEAT CAPACITY

The specific heat capacity c of a substance is the heat required to raise the temperature of **unit mass** of it by **one kelvin**: $c = Q/(m \Delta\theta)$, in $\text{J kg}^{-1} \text{K}^{-1}$.

Water has an unusually large specific heat capacity, $c_w = 4200 \text{ J kg}^{-1} \text{K}^{-1}$, which is why lakes such as Phewa Tal change temperature far more slowly than the land around them.

Substance	$c / \text{J kg}^{-1} \text{K}^{-1}$	Substance	$c / \text{J kg}^{-1} \text{K}^{-1}$
Water	4200	Copper	400
Ice	2100	Aluminium	900
Steam	2010	Iron	470
Kerosene	2100	Brass	380

Method of mixtures (for a solid)

A known mass m_s of the solid is heated in a steam jacket to a steady temperature θ_s and then dropped quickly into water of mass m_w in a calorimeter of mass m_c and specific heat capacity c_c , initially at θ_1 . The mixture is stirred and the highest steady temperature θ_2 is recorded. Then

$$m_s c_s (\theta_s - \theta_2) = (m_w c_w + m_c c_c) (\theta_2 - \theta_1)$$

$$c_s = \frac{(m_w c_w + m_c c_c) (\theta_2 - \theta_1)}{m_s (\theta_s - \theta_2)}$$

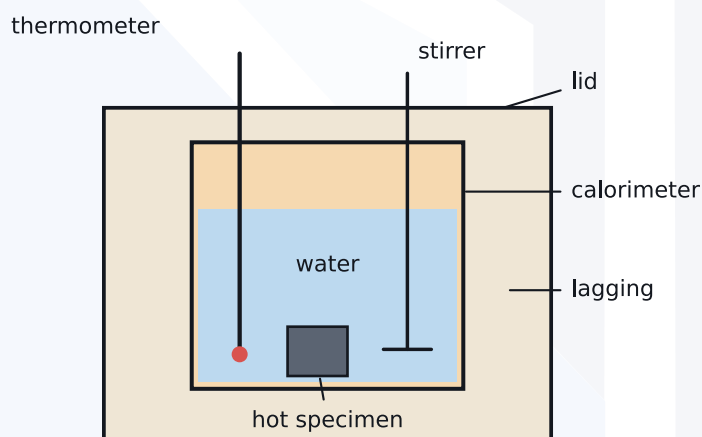


Figure 45. Calorimeter for the method of mixtures. Lagging and the polished outer vessel cut down heat exchange with the room.

Method of mixtures (for a liquid)

The same equation is used the other way round. A solid of known specific heat capacity is heated and dropped into the unknown liquid; c_s is now known and c_{liquid} is the unknown. The liquid must not attack the calorimeter and must not boil.

Electrical method

For a liquid it is often easier to supply a measured amount of electrical energy. A heating coil carrying current I at potential difference V for time t delivers $Q = VIt$ joules, so

$$VIt = (m_l c_l + m_c c_c) (\theta_2 - \theta_1)$$

The advantage is that the heat supplied is known directly from electrical measurements; the disadvantage is that heat losses over a long run can be large, so a **cooling correction** based on Newton's law of cooling is applied.

WORKED EXAMPLE 11.2

Problem. A copper calorimeter of mass 150 g contains 200 g of water at 20 °C. A metal block of mass 100 g at 100 °C is dropped in and the final steady temperature is 25 °C. Find the specific heat capacity of the metal. Take $c_w = 4200$ and $c_{Cu} = 400 \text{ J kg}^{-1} \text{K}^{-1}$.

Solution. Heat gained by water and calorimeter:

$$Q = (0.200 \times 4200 + 0.150 \times 400)(25 - 20) = (840 + 60) \times 5 = 4500 \text{ J}$$

Heat lost by the metal block:

$$Q = 0.100 \times c \times (100 - 25) = 7.5 \ c$$

Equating, $7.5 \ c = 4500$, so

$$c = 600 \text{ J kg}^{-1} \text{K}^{-1}$$

11.3 Change of phases: Latent heat

Supply heat steadily to a block of ice taken from a deep freeze and record the temperature. The graph is not a straight line: it has two flat steps.

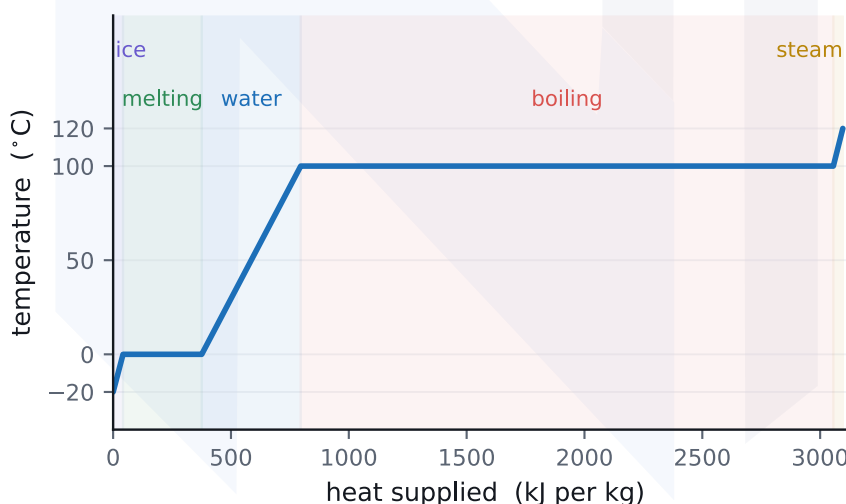


Figure 46. Heating curve for 1 kg of water substance from -20°C ice to 120°C steam. The temperature stands still while a phase change is in progress.

Along the sloping parts the heat raises the kinetic energy of the molecules and the temperature rises. Along the flat parts the heat does something else: it breaks the bonds holding the molecules in place, increasing their potential energy. The temperature does not change at all. Heat absorbed in this way is called **latent** — Latin for hidden — because a thermometer cannot see it.

LATENT HEAT

Latent heat is the heat absorbed or given out by a substance during a change of phase at constant temperature and constant pressure.

Change of phase	Name	Heat
solid → liquid	fusion (melting)	absorbed
liquid → solid	freezing	given out
liquid → gas	vaporisation (boiling)	absorbed
gas → liquid	condensation	given out
solid → gas	sublimation	absorbed

"HEAT ALWAYS RAISES TEMPERATURE" IS FALSE

During melting and boiling the substance absorbs a great deal of heat with **no temperature change at all**. A mistake in the reverse direction is just as common: students forget that condensing steam gives out its latent heat, which is why a steam burn is far worse than a burn from boiling water.

11.4 Specific latent heat of fusion and vaporization**SPECIFIC LATENT HEAT**

The specific latent heat L of a substance is the heat required to change the phase of **unit mass** of it at constant temperature: $Q = mL$, with L in J kg^{-1} .

Two values matter for the NEB paper, both for water at normal atmospheric pressure:

Quantity	Symbol	Value	Temperature
Specific latent heat of fusion of ice	L_f	$3.34 \times 10^5 \text{ J kg}^{-1}$	0°C
Specific latent heat of vaporisation of water	L_v	$2.26 \times 10^6 \text{ J kg}^{-1}$	100°C

L_v is nearly seven times L_f . Melting only loosens the molecules so that they can slide past one another; boiling has to separate them completely against the attractive forces and push back the atmosphere while the vapour expands. Both jobs need much more energy than simply unlocking the lattice.

WORKED EXAMPLE 11.3

Problem. How much heat is needed to convert 50 g of ice at -10°C completely into steam at 100°C ? Take $c_{ice} = 2100$, $c_w = 4200$, $c_{steam} = 2010 \text{ J kg}^{-1} \text{K}^{-1}$, $L_f = 3.34 \times 10^5$ and $L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$.

Solution. Work through the stages one by one with $m = 0.050$ kg.

Warming ice from -10°C to 0°C :

$$Q_1 = mc_{ice}\Delta\theta = 0.050 \times 2100 \times 10 = 1050 \text{ J}$$

Melting the ice at 0°C :

$$Q_2 = mL_f = 0.050 \times 3.34 \times 10^5 = 16\,700 \text{ J}$$

Warming the water from 0°C to 100°C :

$$Q_3 = mc_w\Delta\theta = 0.050 \times 4200 \times 100 = 21\,000 \text{ J}$$

Boiling the water at 100°C :

$$Q_4 = mL_v = 0.050 \times 2.26 \times 10^6 = 113\,000 \text{ J}$$

Total heat required:

$$Q = 1050 + 16\,700 + 21\,000 + 113\,000 = 1.52 \times 10^5 \text{ J}$$

Almost three quarters of it goes into the last step alone.

11.5 Measurement of specific latent heat of fusion and vaporization

Both measurements use the method of mixtures, with one extra precaution in each case.

Latent heat of fusion of ice

Small pieces of ice at 0°C are dried on blotting paper — **wet ice carries water at 0°C and makes L_f come out too small** — and added to warm water in a calorimeter. The water is started a few degrees above room temperature and finishes a few degrees below it, so that the heat gained from the room in the second half roughly cancels the heat lost in the first half.

If m_i is the mass of ice, W the water equivalent of the calorimeter, θ_1 the initial and θ_2 the final temperature:

$$(m_w + W)c_w(\theta_1 - \theta_2) = m_iL_f + m_ic_w(\theta_2 - 0)$$

$$L_f = \frac{(m_w + W)c_w(\theta_1 - \theta_2)}{m_i} - c_w\theta_2$$

Latent heat of vaporisation of water

Dry steam at 100°C is passed into cold water in a calorimeter. A **steam trap** between the boiler and the calorimeter catches the water droplets carried over with the steam; without it the measured L_v is too small. The mass m_s of steam condensed is found by reweighing the calorimeter at the end. With θ_1 and θ_2 the initial and final temperatures,

$$m_sL_v + m_sc_w(100 - \theta_2) = (m_w + W)c_w(\theta_2 - \theta_1)$$

$$L_v = \frac{(m_w + W)c_w(\theta_2 - \theta_1)}{m_s} - c_w(100 - \theta_2)$$

WORKED EXAMPLE 11.4

Problem. A calorimeter of water equivalent 20 g contains 180 g of water at 35.0°C . When 23.4 g of dry ice at 0°C is stirred in, the steady temperature becomes 23.0°C . Find the specific latent heat of fusion of ice. ($c_w = 4200 \text{ J kg}^{-1}\text{K}^{-1}$.)

Solution. The calorimeter behaves like an extra 20 g of water, so the cooling mass is $180 + 20 = 200 \text{ g} = 0.200 \text{ kg}$.

Heat given out by the water and calorimeter:

$$Q = 0.200 \times 4200 \times (35.0 - 23.0) = 840 \times 12 = 10\,080 \text{ J}$$

Heat taken by the ice, in two stages — melting, then warming the melt water from 0°C to 23.0°C :

$$Q = m_i L_f + m_i c_w \theta_2 = 0.0234 L_f + 0.0234 \times 4200 \times 23.0$$

The second term is $0.0234 \times 96\,600 = 2260 \text{ J}$. Therefore

$$0.0234 L_f = 10\,080 - 2260 = 7820 \text{ J}$$

$$L_f = \frac{7820}{0.0234} = 3.34 \times 10^5 \text{ J kg}^{-1}$$

WHERE THE MARKS ARE IN A CALORIMETRY QUESTION

Write the heat-lost side and the heat-gained side as two separate labelled expressions before you equate them. Examiners give method marks for the correct energy balance even when the arithmetic goes wrong, and they cannot give them if they cannot see which term is which.

11.6 Triple point

The temperature at which a substance melts and the temperature at which it boils both depend on pressure. Plotting pressure against temperature gives the **phase diagram**, whose three curves meet at a single point.

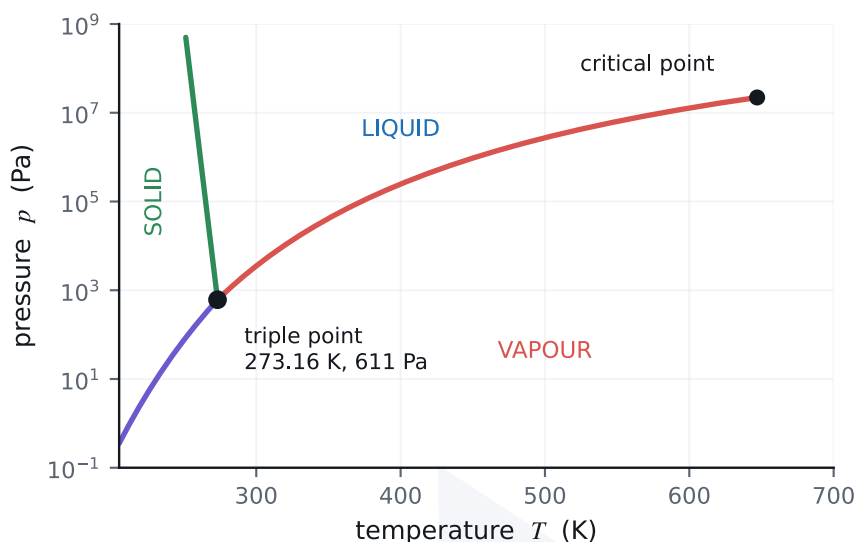


Figure 47. Phase diagram of water (pressure on a logarithmic scale). The fusion curve leans backwards, so extra pressure lowers the melting point of ice.

The three curves divide the diagram into solid, liquid and vapour regions. The **fusion curve** separates solid from liquid, the **vapourisation curve** separates liquid from vapour and ends at the critical point, and the **sublimation curve** separates solid from vapour. They meet at one point only.

TRIPLE POINT

The triple point of a substance is the single pressure and temperature at which its solid, liquid and vapour phases can coexist in thermal equilibrium.

For water the triple point is at

$$T = 273.16 \text{ K} = 0.01 \text{ }^{\circ}\text{C}, \quad p = 611.66 \text{ Pa}$$

which is about 0.006 atmosphere. Below that pressure liquid water cannot exist at all: ice warmed at low pressure sublimates straight to vapour. That is exactly how freeze-drying works.

Because the triple point is fixed by nature and can be reproduced in a sealed glass triple-point cell to better than a thousandth of a kelvin, it was chosen as the single fixed point of the Kelvin scale. From 1954 to 2019 the kelvin was **defined** as $1/273.16$ of the thermodynamic temperature of the triple point of water. Since the SI revision of 2019 the kelvin is instead fixed by giving the Boltzmann constant the exact value $1.380649 \times 10^{-23} \text{ J K}^{-1}$, and 273.16 K has become a very accurately measured quantity rather than a defined one.

THE TRIPLE POINT IS NOT THE MELTING POINT

Ice melts at $0 \text{ }^{\circ}\text{C}$ under **1 atmosphere** of pressure. The triple point is at $0.01 \text{ }^{\circ}\text{C}$ under only 611 Pa. The two differ because dissolved air and the much higher pressure of the atmosphere each lower the melting point slightly.

✓ Chapter summary

- Heat to change temperature: $Q = mc\Delta\theta$; heat capacity $C = mc$; water equivalent $W = mc/c_w$.
- Newton's law of cooling: $dQ/dt = -K(\theta - \theta_0)$, hence $\theta = \theta_0 + (\theta_1 - \theta_0)e^{-kt}$. Valid only for small excess temperature and unchanged surroundings.
- Working form for problems: $(\theta_1 - \theta_2)/t = k[(\theta_1 + \theta_2)/2 - \theta_0]$.
- Specific heat capacity is measured by the method of mixtures, $m_s c_s (\theta_s - \theta_2) = (m_w c_w + m_c c_c)(\theta_2 - \theta_1)$, or electrically, $VIt = (m_l c_l + m_c c_c)\Delta\theta$.
- Latent heat is absorbed at constant temperature during a phase change: $Q = mL$. For water $L_f = 3.34 \times 10^5$ and $L_v = 2.26 \times 10^6$ J kg⁻¹.
- L_f is measured with dried ice and L_v with dry steam from a steam trap; both use the same energy balance.
- The triple point is the unique (p, T) at which solid, liquid and vapour coexist. For water: 273.16 K and 611.66 Pa.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of specific latent heat is [1]

☐ a J kg⁻¹ K⁻¹ ☐ b J kg⁻¹ ☐ c J K⁻¹ ☐ d J
- While a solid is melting, the heat supplied to it [1]

☐ a raises its temperature
☐ b raises its kinetic energy only
☐ c increases the potential energy of its molecules
☐ d is destroyed
- Newton's law of cooling is best obeyed when [1]

☐ a the excess temperature is small ☐ b the body is red hot
☐ c the body is in vacuum ☐ d the surroundings are heated
- The triple point of water occurs at [1]

☐ a 273.15 K and 1 atm ☐ b 273.16 K and 611 Pa
☐ c 373.15 K and 1 atm ☐ d 273.16 K and 1 atm
- The water equivalent of a calorimeter of mass m and specific heat capacity c is [1]

☐ a mc ☐ b mc_w ☐ c mc/c_w ☐ d c_w/mc

ANSWERS TO GROUP A

- (b) — $L = Q/m$, so joule per kilogram; there is no temperature change, hence no K in the unit.
- (c) — the bonds are broken, so molecular potential energy rises while the temperature stays fixed.
- (a) — the law is a linear approximation valid only for a small excess temperature.

4. (b) — $0.01\text{ }^{\circ}\text{C}$ and 611.66 Pa .
5. (c) — the mass of water having the same heat capacity: $mc = Wc_w$.

Group B — Short answer (5 marks each)

1. State Newton's law of cooling and show that the temperature of a cooling body falls exponentially towards that of its surroundings. [5]
2. A block of copper of mass 0.4 kg at $100\text{ }^{\circ}\text{C}$ is dropped into 0.2 kg of water at $20\text{ }^{\circ}\text{C}$ in a container of negligible heat capacity. Find the final temperature. ($c_{Cu} = 400$, $c_w = 4200\text{ J kg}^{-1}\text{K}^{-1}$.) [5]
3. Define specific latent heat of fusion and of vaporisation, and explain why L_v is much larger than L_f for the same substance. [5]
4. Steam at $100\text{ }^{\circ}\text{C}$ is passed into 0.5 kg of water at $20\text{ }^{\circ}\text{C}$ in a calorimeter of water equivalent 0.02 kg until the temperature reaches $60\text{ }^{\circ}\text{C}$. Find the mass of steam condensed. ($L_v = 2.26 \times 10^6$, $c_w = 4200\text{ J kg}^{-1}\text{K}^{-1}$.) [5]

ANSWERS TO GROUP B

1. Outline: state the law; write $dQ/dt = -K(\theta - \theta_0)$; put $dQ = mc\,d\theta$ to get $d\theta/dt = -k(\theta - \theta_0)$; separate variables and integrate to $\ln(\theta - \theta_0) = -kt + C$; apply $\theta = \theta_1$ at $t = 0$ to obtain $\theta = \theta_0 + (\theta_1 - \theta_0)e^{-kt}$.
2. Heat lost $= 0.4 \times 400 \times (100 - \theta) = 160(100 - \theta)$; heat gained $= 0.2 \times 4200 \times (\theta - 20) = 840(\theta - 20)$. So $16\,000 - 160\theta = 840\theta - 16\,800$, giving $1000\theta = 32\,800$ and $\theta = 32.8\text{ }^{\circ}\text{C}$.
3. Definitions as in §11.4. $L_v > L_f$ because melting only has to loosen the lattice enough for molecules to slide, whereas vaporisation must separate the molecules completely against their mutual attraction and, in addition, do work pushing back the atmosphere as the vapour expands.
4. Heat gained $= (0.5 + 0.02) \times 4200 \times (60 - 20) = 2184 \times 40 = 87\,360\text{ J}$. Heat given out by mass m of steam $= m[2.26 \times 10^6 + 4200 \times 40] = m(2.428 \times 10^6)$. Hence $m = 87\,360 / 2.428 \times 10^6 = 0.036\text{ kg} = 36\text{ g}$.

Group C — Long answer (8 marks each)

1. (a) Define specific latent heat of vaporisation. [2]
 (b) Describe, with the necessary theory and two precautions, how the specific latent heat of vaporisation of water is measured by the method of mixtures. [6]
2. 100 g of ice at $0\text{ }^{\circ}\text{C}$ is dropped into 300 g of water at $50\text{ }^{\circ}\text{C}$ contained in a calorimeter of water equivalent 25 g .
 (a) Show that all the ice melts. [3]
 (b) Find the final temperature of the mixture. [5] ($L_f = 3.34 \times 10^5$, $c_w = 4200\text{ J kg}^{-1}\text{K}^{-1}$.)

ANSWERS TO GROUP C

1. (a) The heat needed to change unit mass of a liquid into vapour at its boiling point without any change of temperature. (b) Steam from a boiler passes through a steam trap into a weighed calorimeter holding a known mass of cold water; record θ_1 , pass steam until the rise is about $20\text{ }^{\circ}\text{C}$, record θ_2 , reweigh to get m_s . Theory: $m_s L_v + m_s c_w (100 - \theta_2) = (m_w + W)c_w(\theta_2 - \theta_1)$. Precautions: use a

steam trap so that only **dry** steam enters; start below and finish above room temperature so the heat exchange with the room cancels; stir continuously; lag the calorimeter.

2. (a) Heat available from the water and calorimeter as they cool to 0°C is $(0.300 + 0.025) \times 4200 \times 50 = 0.325 \times 4200 \times 50 = 68\,250\text{ J}$. The heat needed to melt all the ice is $0.100 \times 3.34 \times 10^5 = 33\,400\text{ J}$. Since $68\,250 > 33\,400$, all the ice melts and some heat is left over.

(b) Surplus heat $= 68\,250 - 33\,400 = 34\,850\text{ J}$. This warms the whole mixture — 300 g of original water, 100 g of melt water and the 25 g water equivalent, i.e. 0.425 kg — from 0°C :

$$\theta = 34\,850 / (0.425 \times 4200) = 34\,850 / 1785 = 19.5^\circ\text{C}.$$



UNIT 12

Rate of heat flow

Heat and Thermodynamics • 5 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Conduction: Thermal conductivity and measurement
- Convection
- Radiation: Ideal radiator
- Black-body radiation
- Stefan-Boltzmann law

Unit 11 asked how much heat. This unit asks how fast. Heat travels from a hot place to a cold one by three mechanisms — conduction, convection and radiation — and each has its own law. Conduction passes energy along without moving the material, convection carries it bodily in a moving fluid, and radiation sends it across empty space as electromagnetic waves. A thermos flask is the standard exam object because it is designed to defeat all three at once.

THREE MECHANISMS, THREE SIGNATURES

Mechanism	Needs matter?	Matter moves?	Governing law
Conduction	yes	no	$H = kA \Delta\theta/L$
Convection	yes (fluid only)	yes	$H \propto \Delta\theta$ (Newton)
Radiation	no	no	$H = e\sigma AT^4$ (Stefan)

12.1 Conduction: Thermal conductivity and measurement

Hold one end of an iron rod in a fire and the other end soon becomes too hot to hold. The atoms at the hot end vibrate more violently, collide with their neighbours and pass the energy along the lattice; in a metal the free electrons carry most of it, which is why metals conduct far better than non-metals. No part of the rod moves as a whole.

THE CONDUCTION EQUATION $H = \frac{kA(\theta_1 - \theta_2)}{L}$

We start from three facts that experiment gives us, and reach the single equation that governs all steady conduction.

Setting up. A slab has cross-sectional area A and thickness L . Its two faces are held at steady temperatures θ_1 and θ_2 , with $\theta_1 > \theta_2$. Let $H = dQ/dt$ be the heat current — the joules crossing the slab each second. We wait until the **steady state**: every cross-section then carries the same H , because no part of the slab is still warming up.

Step 1 — H depends on the area. Two identical slabs side by side conduct twice as much as one, and they are the same thing as one slab of double the area. So

$$H \propto A$$

Step 2 — H depends on the temperature difference. Nothing flows when the two faces are equally hot, and experiment shows the flow doubles when the difference doubles:

$$H \propto (\theta_1 - \theta_2)$$

Step 3 — H depends inversely on the thickness. A thicker slab is a longer journey for the heat. Two slabs one behind the other (thickness $2L$) carry half the current of one:

$$H \propto \frac{1}{L}$$

Step 4 — combine the three proportionalities. When a quantity is separately proportional to several others, it is proportional to their product:

$$H \propto \frac{A(\theta_1 - \theta_2)}{L}$$

Step 5 — replace " $\propto$ " by an equals sign and a constant. Call that constant k ; it depends only on the material:

$$H = \frac{kA(\theta_1 - \theta_2)}{L}$$

Step 6 — the differential form. In the steady state the temperature falls evenly across the slab, so $(\theta_1 - \theta_2)/L$ is just the size of the **temperature gradient** $d\theta/dx$. Writing the gradient with its own sign:

$$H = -kA \frac{d\theta}{dx}$$

Result.

$$H = \frac{kA(\theta_1 - \theta_2)}{L} = -kA \frac{d\theta}{dx}$$

What it means. The minus sign in the last form says heat flows down the temperature gradient — from hot to cold, never the other way. The constant k is the **thermal conductivity**: a large k (copper) means a big heat current for a small temperature difference, a small k (wood, still air) means a good insulator.

Conditions used. The steady state has been reached (temperatures no longer changing); the flow is one-dimensional, i.e. no heat escapes from the sides, which is why rods are **lagged**; and the slab is made of one uniform material.

THERMAL CONDUCTIVITY

The thermal conductivity of a material is the rate of flow of heat per unit area per unit temperature gradient, in the steady state. Its SI unit is $\text{W m}^{-1} \text{K}^{-1}$, and its dimensional formula is $[MLT^{-3}\Theta^{-1}]$.

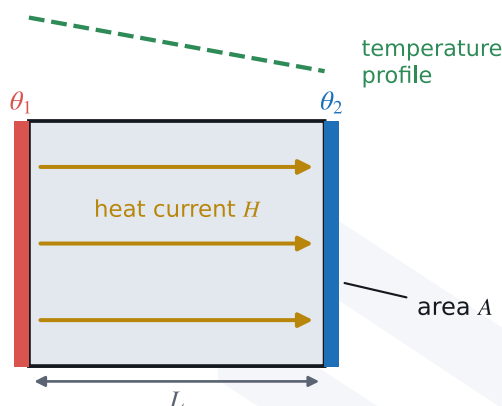


Figure 48. Steady conduction through a slab. In the steady state the temperature falls uniformly from θ_1 to θ_2 across the thickness L .

It is often convenient to write the law like Ohm's law. Defining the **thermal resistance** $R = L/(kA)$,

$$H = \frac{\theta_1 - \theta_2}{R}$$

so that temperature difference plays the part of potential difference and heat current the part of electric current. Slabs one behind the other are in **series**, and their thermal resistances simply add: $R = R_1 + R_2 + \dots$. This single idea handles every composite-wall problem.

Material	$k / \text{W m}^{-1} \text{K}^{-1}$	Material	$k / \text{W m}^{-1} \text{K}^{-1}$
Silver	420	Glass	0.8
Copper	390	Brick	0.6
Aluminium	240	Water	0.6
Brass	110	Wood	0.15
Steel	50	Air (still)	0.024

Measuring k of a good conductor: Searle's method

A thick lagged bar of the metal is heated at one end by a steam chest and cooled at the other by water flowing steadily through a spiral. Two thermometers θ_1 and θ_2 sit in holes a measured distance x apart along the bar; two more read the inlet and outlet temperatures θ_3 and θ_4 of the water.

THE WORKING FORMULA OF SEARLE'S METHOD

We start from the fact that in the steady state the water carries away exactly the heat the bar delivers, and reach a formula for k in terms of things we can measure.

Setting up. A is the cross-sectional area of the bar, x the distance between the two thermometers in it, c_w the specific heat capacity of water, and m the mass of water collected in time t .

Step 1 — the heat conducted along the bar between the two thermometers. Apply the conduction equation to the length x of bar between them:

$$H_{\text{conducted}} = \frac{kA(\theta_1 - \theta_2)}{x}$$

Step 2 — the heat gained by the water. A mass m of water rises in temperature from θ_3 to θ_4 , so it absorbs heat $Q = mc_w(\theta_4 - \theta_3)$. Dividing by the time t gives the rate:

$$H_{\text{absorbed}} = \frac{mc_w(\theta_4 - \theta_3)}{t}$$

Step 3 — set them equal. In the steady state nothing in the bar is getting hotter, so none of the heat stays behind, and the lagging stops any escaping sideways. Everything conducted in is carried out by the water:

$$\frac{kA(\theta_1 - \theta_2)}{x} = \frac{mc_w(\theta_4 - \theta_3)}{t}$$

Step 4 — multiply both sides by x :

$$kA(\theta_1 - \theta_2) = \frac{mc_w(\theta_4 - \theta_3) x}{t}$$

Step 5 — divide both sides by $A(\theta_1 - \theta_2)$:

$$k = \frac{mc_w(\theta_4 - \theta_3) x}{A(\theta_1 - \theta_2) t}$$

Result.

$$k = \frac{mc_w(\theta_4 - \theta_3) x}{A(\theta_1 - \theta_2) t}$$

What it means. Every symbol on the right can be measured in the laboratory: m by weighing the collected water, t by a stopwatch, the four temperatures by thermometers, and x and A by a ruler and a micrometer. Note which temperature difference goes where — $(\theta_1 - \theta_2)$ belongs to the **bar**, $(\theta_4 - \theta_3)$ to the **water**.

Conditions used. A true steady state (all four thermometer readings constant), perfect lagging so no heat leaks from the sides, and a steady flow of water.

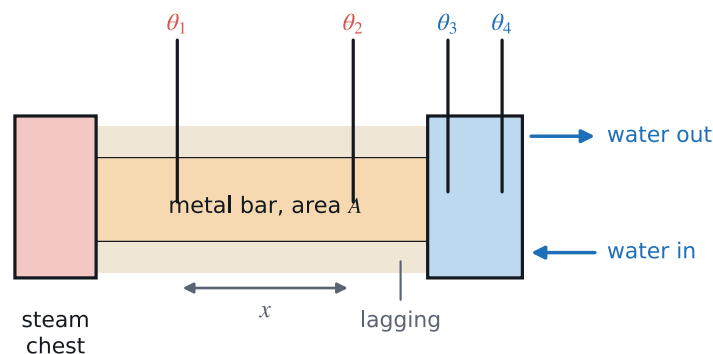


Figure 49. Searle's apparatus. The heat conducted between the two thermometers in the bar is collected by the flowing water.

For a **bad** conductor such as glass, cardboard or rubber, Searle's bar is useless: the flow would be far too small to measure. **Lee's disc method** is used instead. The specimen is made a thin disc of large area sandwiched between a steam chamber and a thick brass disc, so that the heat flows through a short distance and a wide area.

LAGGING IS NOT DECORATION

The derivation of $H = kA\Delta\theta/L$ assumes that **all** the heat entering one face leaves the other. If the bar is not lagged, heat escapes sideways, the temperature gradient stops being uniform, and k comes out too small. In every diagram you draw, show the lagging and say what it is for.

WORKED EXAMPLE 12.1

Problem. A glass window measures $1.5 \text{ m} \times 1.0 \text{ m}$ and is 4.0 mm thick. The inner surface is at 25°C and the outer at 5°C . Taking $k_{\text{glass}} = 0.80 \text{ W m}^{-1} \text{K}^{-1}$, find the rate of heat loss.

Solution. $A = 1.5 \times 1.0 = 1.5 \text{ m}^2$, $L = 4.0 \times 10^{-3} \text{ m}$ and $\Delta\theta = 20 \text{ K}$.

$$H = \frac{kA \Delta\theta}{L} = \frac{0.80 \times 1.5 \times 20}{4.0 \times 10^{-3}} = \frac{24}{4.0 \times 10^{-3}} = 6.0 \times 10^3 \text{ W}$$

Six kilowatts through one window is absurdly large, and it tells us the assumption is wrong: the glass surfaces are not at room and outdoor temperature. Thin films of still air on each side, with k only 0.024, provide nearly all the real thermal resistance.

WORKED EXAMPLE 12.2

Problem. A copper rod ($k = 390$) and a brass rod ($k = 110 \text{ W m}^{-1} \text{K}^{-1}$) of equal length and equal cross-section are joined end to end. The free end of the copper is kept at 100°C and the free end of the brass at 0°C . Find the steady temperature of the junction. The sides are perfectly lagged.

Solution. In the steady state the same heat current passes through both rods. With equal A and equal L , and junction temperature θ :

$$\frac{k_1 A (100 - \theta)}{L} = \frac{k_2 A (\theta - 0)}{L} \implies 390(100 - \theta) = 110\theta$$

$$39\,000 = 500\theta \implies \theta = 78\text{ }^{\circ}\text{C}$$

In general $\theta = (k_1\theta_1 + k_2\theta_2)/(k_1 + k_2)$ for equal rods: the junction sits closer to the temperature of the **better** conductor.

12.2 Convection

CONVECTION

Convection is the transfer of heat through a fluid by the bulk movement of the heated fluid itself.

Warm a fluid from below and it expands, becomes less dense, and is pushed up by the denser cold fluid around it, which then takes its place and is warmed in turn. The resulting circulation is a **convection current**. Because the material itself has to move, convection is impossible in solids and impossible in a vacuum.

	Natural convection	Forced convection
Cause of motion	density difference and gravity	pump, fan or blower
Speed	slow	fast
Example	air above a heater, boiling water	radiator fan, hair dryer, blood circulation

The rate of convective loss from a surface is roughly proportional to the excess temperature, $H = hA(\theta - \theta_0)$, where h is the convection coefficient. This is precisely Newton's law of cooling from Unit 11 — Newton's law is a law of convection, which is why it fails when radiation takes over at high temperature.

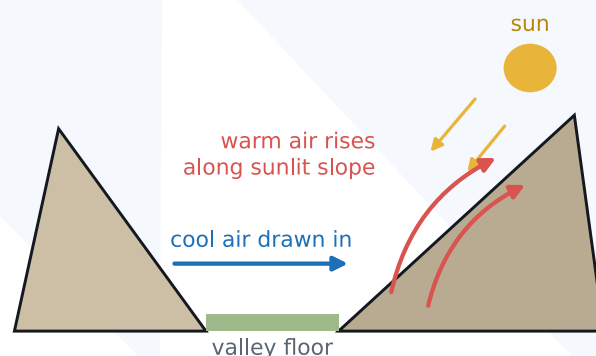


Figure 50. Daytime valley wind. The sunlit slope heats the air against it, which rises, and cooler air flows up the valley to replace it.

Convection explains a long list of everyday things: the up-valley wind that blows through Pokhara and Kathmandu on a sunny afternoon and reverses at night; land and sea breezes; the draught up a chimney; the circulation of water in a solar water heater on a rooftop, which needs no pump because hot water rises on its own; why a room heater is placed low and an air conditioner high; and why the trapped air in a woollen sweater keeps you warm — the wool stops the air from convecting away.

12.3 Radiation: Ideal radiator

The Earth receives energy from the Sun across 150 million kilometres of vacuum, so there must be a third mechanism that needs no matter at all.

THERMAL RADIATION

Thermal radiation is energy emitted by every body on account of its temperature, carried by electromagnetic waves — mainly infrared for bodies near room temperature.

Radiation travels at $3 \times 10^8 \text{ m s}^{-1}$, obeys the inverse-square law, and can be reflected and refracted like light. **Prevost's theory of heat exchange** states that every body above absolute zero radiates continuously, whatever its surroundings; a body appears to cool only because it emits more than it absorbs, and at thermal equilibrium the two rates are equal.

When radiation falls on a body, part is absorbed, part reflected and part transmitted. Writing each as a fraction of the incident energy,

$$a + r + t = 1$$

PERFECT BLACK BODY (IDEAL RADIATOR)

A perfect black body is one that absorbs **all** the radiation of every wavelength falling on it, so that $a = 1$ and $r = t = 0$. At a given temperature it is also the best possible emitter at every wavelength.

That the best absorber is also the best emitter is **Kirchhoff's law**: at a given temperature and wavelength, the ratio of emissive power to absorptive power is the same for all bodies and equals the emissive power of a black body. The everyday proof is a glazed white cup with a coloured pattern: heat it in a furnace and the pattern, which absorbs best, glows brightest.

No real surface is perfectly black — lamp black absorbs about 96 %. A practical black body is made instead as a cavity: **Ferry's black body** is a double-walled copper sphere, blackened with lamp black inside, with a small hole and a conical projection opposite it. Radiation entering the hole is reflected many times inside and absorbed almost completely before it can find its way out, so the hole behaves as a black surface. Heat the sphere and the same hole becomes an ideal radiator.

The **emissivity** e of a real surface is the ratio of its emissive power to that of a black body at the same temperature, so $0 \leq e \leq 1$, with $e = 1$ for a black body. By Kirchhoff's law $e = a$.

12.4 Black-body radiation

A black body does not radiate at one wavelength only. Measuring how the emitted power is distributed over wavelength gives the **black-body spectrum**, first measured accurately by Lummer and Pringsheim.

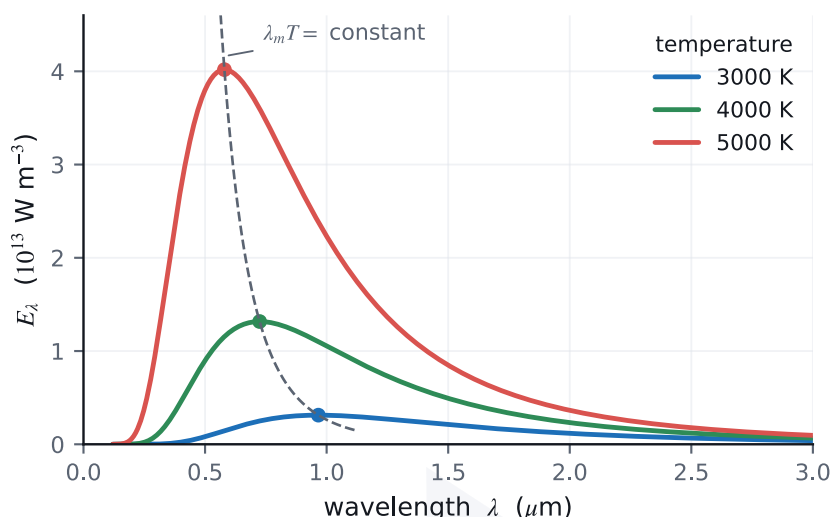


Figure 51. Black-body spectra at three temperatures. As T rises the peak grows, the total area grows much faster, and λ_m moves to shorter wavelengths.

Four features of these curves must be quoted:

1. The spectrum is **continuous**: energy is emitted at every wavelength.
2. Each curve rises to a single maximum at a wavelength λ_m and falls away on both sides; the emission is very small at very short and very long wavelengths.
3. As the temperature rises, λ_m shifts towards **shorter** wavelengths. This is **Wien's displacement law**:

$$\lambda_m T = b = 2.898 \times 10^{-3} \text{ m K}$$

1. The **area** under a curve is the total power radiated per unit area, and it increases very rapidly with temperature — as T^4 , which is the next section.

Wien's law is why a heated iron bar glows dull red, then orange, then white as it gets hotter, and why the blue star Rigel is hotter than the red star Betelgeuse. Classical physics could not reproduce these curves at all; it predicted that the emission should grow without limit at short wavelengths (the ultraviolet catastrophe). Planck fixed it in 1900 by assuming that energy is emitted in discrete quanta $E = h\nu$ — the beginning of quantum theory.

WORKED EXAMPLE 12.3

Problem. The Sun radiates most strongly at $\lambda_m = 500 \text{ nm}$. Treating it as a black body, find (a) its surface temperature and (b) the power radiated per square metre of its surface. Take $b = 2.898 \times 10^{-3} \text{ m K}$ and $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

Solution. (a) From Wien's displacement law,

$$T = \frac{b}{\lambda_m} = \frac{2.898 \times 10^{-3}}{500 \times 10^{-9}} = 5.80 \times 10^3 \text{ K}$$

(b) From Stefan's law, with $T = 5796 \text{ K}$,

$$E = \sigma T^4 = 5.67 \times 10^{-8} \times (5796)^4 = 5.67 \times 10^{-8} \times 1.129 \times 10^{15}$$

$$E = 6.4 \times 10^7 \text{ W m}^{-2}$$

12.5 Stefan-Boltzmann law

STEFAN-BOLTZMANN LAW

The total energy radiated per second per unit area by a perfect black body is directly proportional to the fourth power of its absolute temperature:

$$E = \sigma T^4$$

where $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$ is the Stefan-Boltzmann constant.

For a body of surface area A and emissivity e the total power radiated is $P = e\sigma AT^4$. But the body is also absorbing from its surroundings at T_0 , at a rate $e\sigma AT_0^4$ by Prevost's theory. The **net** rate of loss is therefore

$$P_{\text{net}} = e\sigma A(T^4 - T_0^4)$$

KELVIN, ALWAYS KELVIN

T in Stefan's law is the **absolute** temperature. Putting degrees Celsius into T^4 is the single commonest error in this unit — and because of the fourth power, the answer is not slightly wrong, it is wrong by orders of magnitude. Convert first, every time.

NEWTON'S LAW OF COOLING FOLLOWS FROM STEFAN'S LAW

We start from the net rate of radiation loss and reach Newton's law of cooling, valid when the body is only a little hotter than its surroundings.

Step 1 — write the body's temperature as surroundings plus a small excess. Let $\Delta T = T - T_0$ be that excess:

$$T = T_0 + \Delta T$$

Step 2 — put this into the net-loss formula:

$$P_{\text{net}} = e\sigma A[(T_0 + \Delta T)^4 - T_0^4]$$

Step 3 — take T_0 outside the first bracket so we can use the binomial expansion:

$$P_{\text{net}} = e\sigma AT_0^4 \left[\left(1 + \frac{\Delta T}{T_0}\right)^4 - 1 \right]$$

Step 4 — expand by the binomial theorem. Since $\Delta T/T_0$ is small, we keep only the first power of it:

$$\left(1 + \frac{\Delta T}{T_0}\right)^4 \approx 1 + 4 \frac{\Delta T}{T_0}$$

Step 5 — substitute that back. The two T_0 's cancel:

$$P_{net} \approx e\sigma AT_0^4 \cdot 4 \frac{\Delta T}{T_0}$$

Step 6 — cancel one factor of T_0 :

$$P_{net} \approx 4e\sigma AT_0^3 \Delta T$$

Step 7 — recognise that everything except ΔT is constant. e , σ , A and T_0 all stay fixed while the body cools, so

$$P_{net} \propto \Delta T = (T - T_0)$$

Result. The rate of heat loss, and therefore the rate of fall of temperature, is proportional to the excess temperature — which is exactly **Newton's law of cooling**, with k identified as $4e\sigma AT_0^3/(mc)$.

What it means. Newton's law is not a separate law of nature; it is Stefan's T^4 law seen through a narrow window of small temperature differences.

Condition used. $\Delta T \ll T_0$ — this is the only reason we were allowed to stop the binomial expansion after the first term. For a large excess temperature the T^4 law must be used in full, and Newton's law fails.

WORKED EXAMPLE 12.4

Problem. A blackened metal sphere of radius 5.0 cm is maintained at 327°C in a room at 27°C . Taking it as a perfect black body, find the net rate at which it loses energy by radiation.

Solution. Convert to kelvin: $T = 327 + 273 = 600\text{ K}$ and $T_0 = 27 + 273 = 300\text{ K}$.

Surface area of the sphere:

$$A = 4\pi r^2 = 4\pi(0.050)^2 = 3.14 \times 10^{-2} \text{ m}^2$$

Fourth powers: $T^4 = (600)^4 = 1.296 \times 10^{11}$ and $T_0^4 = (300)^4 = 8.10 \times 10^9$, so $T^4 - T_0^4 = 1.215 \times 10^{11} \text{ K}^4$.

$$P_{net} = \sigma A(T^4 - T_0^4) = 5.67 \times 10^{-8} \times 3.14 \times 10^{-2} \times 1.215 \times 10^{11}$$

$$P_{net} = 216 \text{ W}$$

Note that doubling the absolute temperature multiplied the emission by $2^4 = 16$; the surroundings contribute only about 6 % of what the sphere emits.

A vacuum flask now makes complete sense. The double wall with the space evacuated stops conduction and convection; the silvered surfaces have a very low emissivity and so radiate and absorb very little; and the narrow insulating stopper blocks the only remaining conduction path.

✓ Chapter summary

- Steady conduction: $H = kA(\theta_1 - \theta_2)/L = -kA \, d\theta/dx$; thermal conductivity k is in $\text{W m}^{-1} \text{K}^{-1}$.
- Thermal resistance $R = L/(kA)$; resistances of slabs in series add, and $H = \Delta\theta/R$.
- Searle's method measures k for good conductors: $k = mc_w(\theta_4 - \theta_3)x/[A(\theta_1 - \theta_2)t]$; Lee's disc is used for bad conductors.
- Convection needs a moving fluid; natural convection is driven by density differences, forced convection by a fan or pump. Newton's law of cooling is a convection law.
- Radiation needs no medium. $a + r + t = 1$; a perfect black body has $a = e = 1$; Kirchhoff's law says good absorbers are good emitters; Prevost's theory says emission never stops.
- Black-body curves are continuous with one maximum; Wien's displacement law gives $\lambda_m T = 2.898 \times 10^{-3} \text{ m K}$.
- Stefan-Boltzmann law: $E = \sigma T^4$ with $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{K}^{-4}$; net loss $P = e\sigma A(T^4 - T_0^4)$, with T in kelvin.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of thermal conductivity is [1]
 - $\text{W m}^{-1} \text{K}^{-1}$
 - $\text{W m}^{-2} \text{K}^{-4}$
 - $\text{J kg}^{-1} \text{K}^{-1}$
 - W m K
- If the absolute temperature of a black body is doubled, the energy it radiates per second becomes [1]
 - 2 times
 - 4 times
 - 8 times
 - 16 times
- A good absorber of radiation is also a good emitter. This is [1]
 - Stefan's law
 - Wien's law
 - Kirchhoff's law
 - Prevost's theory
- Heat transfer by convection cannot take place in [1]
 - water
 - air
 - a solid
 - any fluid
- The wavelength of maximum emission from a black body at 2000 K is about 1450 nm. At 4000 K it will be about [1]
 - 5800 nm
 - 2900 nm
 - 725 nm
 - 1450 nm

ANSWERS TO GROUP A

- (a) — from $H = kA\Delta\theta/L$, k has units $\text{W m}^{-1} \text{K}^{-1}$.
- (d) — $E \propto T^4$, and $2^4 = 16$.
- (c) — Kirchhoff's law, $e = a$.
- (c) — convection needs the material itself to move, which a solid cannot do.
- (c) — $\lambda_m \propto 1/T$, so doubling T halves λ_m .

Group B — Short answer (5 marks each)

1. Define thermal conductivity and derive $H = kA(\theta_1 - \theta_2)/L$ for steady conduction through a slab. Why must the slab be lagged? [5]
2. A wall of area 10 m^2 is made of 12 cm of brick ($k = 0.60 \text{ W m}^{-1}\text{K}^{-1}$) covered on the outside with 4.0 cm of insulating board ($k = 0.050 \text{ W m}^{-1}\text{K}^{-1}$). The inside surface is at 25°C and the outside surface at 5°C . Find the rate of heat flow and the temperature of the junction. [5]
3. Distinguish between natural and forced convection, and explain why a thermos flask has a vacuum between its walls and silvered surfaces. [5]
4. A black body of surface area 20 cm^2 is kept at 527°C in surroundings at 27°C . Find the net rate of loss of heat by radiation. [5]

ANSWERS TO GROUP B

1. Outline: define k as in §12.1; state $H \propto A$ and $H \propto \Delta\theta/L$ in the steady state; combine to $H = kA\Delta\theta/L$. Lagging ensures no heat escapes through the sides, so the heat current is the same at every cross-section and the gradient is uniform.

2. $R_1 = L_1/(k_1A) = 0.12/(0.60 \times 10) = 0.020 \text{ K W}^{-1}$ and $R_2 = 0.040/(0.050 \times 10) = 0.080 \text{ K W}^{-1}$, so $R = 0.100 \text{ K W}^{-1}$. Then $H = \Delta\theta/R = 20/0.100 = 200 \text{ W}$. Junction temperature $\theta = 25 - HR_1 = 25 - 200 \times 0.020 = 21^\circ \text{C}$. (Check: $(21 - 5)/0.080 = 200 \text{ W}$.)

3. Natural convection is driven by the density change of the fluid itself in a gravitational field; forced convection is driven by an external pump or fan and is much faster. In a flask the vacuum removes the matter that conduction and convection need, and the silvering gives a very low emissivity so that little is radiated or absorbed.

4. $A = 20 \times 10^{-4} = 2.0 \times 10^{-3} \text{ m}^2$, $T = 800 \text{ K}$, $T_0 = 300 \text{ K}$. $T^4 - T_0^4 = 4.096 \times 10^{11} - 8.1 \times 10^9 = 4.015 \times 10^{11}$. So $P = 5.67 \times 10^{-8} \times 2.0 \times 10^{-3} \times 4.015 \times 10^{11} = 45.5 \text{ W}$.

Group C — Long answer (8 marks each)

1. (a) Describe Searle's method for determining the thermal conductivity of a good conductor, with a labelled diagram, and derive the working formula. [6]
(b) State two reasons why the method is unsuitable for glass. [2]
2. (a) Draw the energy distribution curves of black-body radiation for three temperatures and state the conclusions drawn from them. [4]
(b) A star radiates most strongly at 350 nm and has a radius of $7.0 \times 10^8 \text{ m}$. Find its surface temperature and its total power output, treating it as a black body. [4]

ANSWERS TO GROUP C

1. (a) Diagram and theory as in §12.1; the working formula is $k = mc_w(\theta_4 - \theta_3)x/[A(\theta_1 - \theta_2)t]$, obtained by equating the heat conducted between the two bar thermometers to the heat carried away by the water. (b) Glass has a very small k , so the heat collected by the water in a reasonable time would be too small to measure, and the temperature difference $\theta_4 - \theta_3$ would be lost in the experimental error; also, a long thin glass bar would lose most of the heat sideways despite lagging.

2. (a) Curves as in Figure 4, with the four conclusions of §12.4. (b)
 $T = b/\lambda_m = 2.898 \times 10^{-3} / (350 \times 10^{-9}) = 8280 \text{ K}$. Surface area $A = 4\pi R^2 = 4\pi(7.0 \times 10^8)^2 = 6.16 \times 10^{18} \text{ m}^2$.
 With $T^4 = (8280)^4 = 4.70 \times 10^{15}$, $P = \sigma AT^4 = 5.67 \times 10^{-8} \times 6.16 \times 10^{18} \times 4.70 \times 10^{15} = 1.6 \times 10^{27} \text{ W}$.



UNIT 13

Ideal gas

Heat and Thermodynamics • 8 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Ideal gas equation
- Molecular properties of matter
- Kinetic-molecular model of an ideal gas
- Derivation of pressure exerted by gas
- Average translational kinetic energy of gas molecule
- Boltzmann constant; root mean square speed
- Heat capacities: gases and solids

A gas is the simplest state of matter to model, because its molecules are so far apart that for most of the time they do not feel one another at all. That single simplification lets us start from Newton's laws applied to a swarm of tiny balls and end up with the gas laws, with an exact meaning for temperature, and with a prediction of the specific heat capacities of gases and solids. This unit is the bridge between mechanics and heat, and it carries the longest derivation in the Grade 11 course.

WHAT THE EXAMINER ASKS FROM THIS UNIT

Two things appear almost every year: the derivation of $p = \frac{1}{3}\rho\overline{c^2}$ (an 8-mark Group C standard), and a numerical on root mean square speed or on $C_p - C_v = R$. Learn the derivation as a sequence of labelled steps — marks are given step by step.

13.1 Ideal gas equation

Three experimental laws describe how a fixed mass of gas behaves.

Law	Held constant	Statement	Form
Boyle's law	T	volume varies inversely as pressure	$pV = \text{constant}$
Charles's law	p	volume varies directly as absolute temperature	$V/T = \text{constant}$
Pressure (Gay-Lussac) law	V	pressure varies directly as absolute temperature	$p/T = \text{constant}$

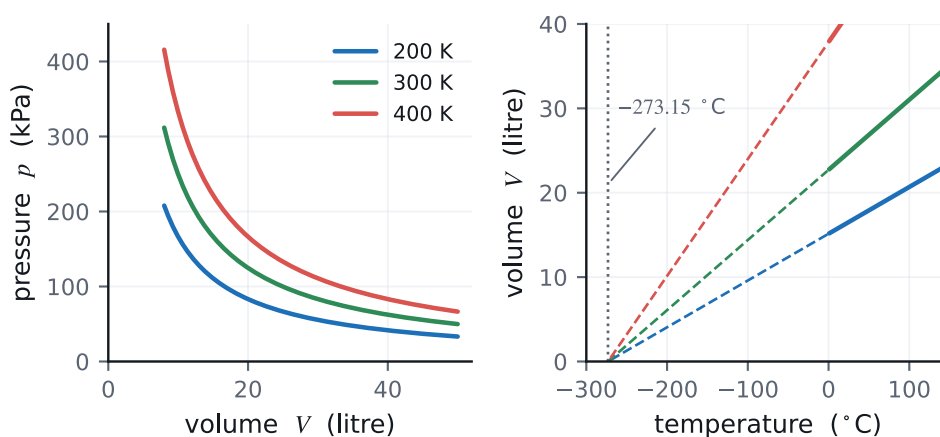


Figure 52. Left: Boyle's law isotherms for one mole. Right: Charles's law. Every line, extrapolated back, meets the temperature axis at -273.15°C .

Combining the three gives, for a fixed mass of gas,

$$\frac{pV}{T} = \text{constant}, \quad \text{so} \quad \frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Avogadro's law fixes the constant: equal volumes of all gases at the same temperature and pressure contain equal numbers of molecules. Hence for n moles

$$pV = nRT$$

where $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ is the **universal gas constant**, the same for every gas. Writing $n = N/N_A$ for N molecules gives the molecular form $pV = NkT$, and writing $n = m/M$ gives $p = \rho RT/M$.

IDEAL GAS

An ideal gas is one that obeys $pV = nRT$ exactly at all pressures and temperatures. A real gas approaches this behaviour at low pressure and at temperatures well above its boiling point, where the molecules are far apart and their own volume and mutual attraction can be ignored.

KELVIN AND CONSISTENT UNITS

T in $pV = nRT$ is always absolute temperature in kelvin, never $^{\circ}\text{C}$. And if you use $R = 8.314$, then p must be in pascal and V in cubic metres — mixing in litres or kPa without converting is the standard way to lose a numerical mark.

WORKED EXAMPLE 13.1

Problem. A weather balloon of volume 8.0 m^3 is released from Kathmandu, where the pressure is 86 kPa and the temperature 27°C . Find its volume at a height where the pressure is 43 kPa and the temperature -33°C . Assume the gas inside is ideal and none escapes.

Solution. Convert temperatures: $T_1 = 27 + 273 = 300 \text{ K}$ and $T_2 = -33 + 273 = 240 \text{ K}$. Since the mass is fixed,

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \implies V_2 = V_1 \times \frac{p_1}{p_2} \times \frac{T_2}{T_1}$$

$$V_2 = 8.0 \times \frac{86}{43} \times \frac{240}{300} = 8.0 \times 2.0 \times 0.80 = 12.8 \text{ m}^3$$

The pressure drop alone would have doubled the volume; the fall in temperature pulls it back to 1.6 times the original.

13.2 Molecular properties of matter

All matter is made of molecules that are in ceaseless motion and that attract one another when a little way apart but repel strongly when pushed very close together. The separation r_0 at which the force is zero — about $3 \times 10^{-10} \text{ m}$ — is the normal spacing in a solid.

Property	Solid	Liquid	Gas
Molecular spacing	$\approx r_0$	slightly more than r_0	$\approx 10r_0$
Intermolecular force	strong	moderate	negligible
Motion	vibration about fixed sites	vibration plus slow drift	free, rapid, random
Shape / volume	both fixed	volume fixed, shape not	neither fixed

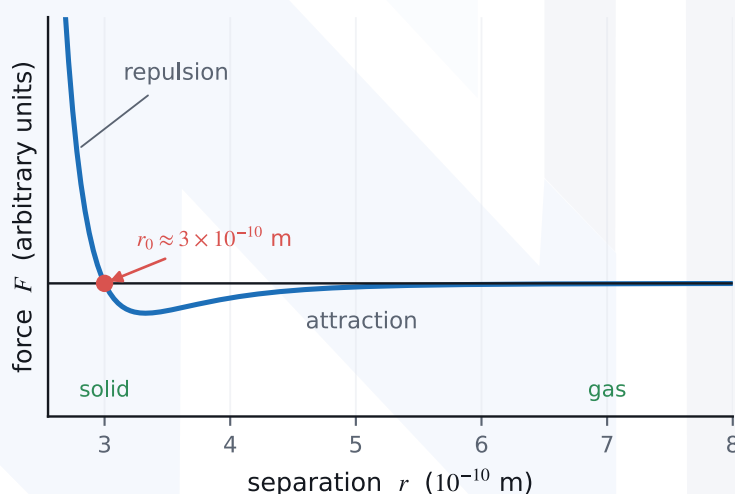


Figure 53. Force between two molecules. It is strongly repulsive when they are pushed closer than r_0 and weakly attractive beyond it.

Two numbers are worth memorising. The **Avogadro constant** $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ is the number of molecules in one mole, and one mole of any gas occupies 22.4 litres at STP (273.15 K, 101.325 kPa). The mass of a single molecule is $m = M/N_A$, where M is the molar mass in kilograms per mole.

Molecules are not visible, but their motion is. In 1827 Robert Brown saw pollen grains suspended in water jiggling in a random zig-zag that never stopped. This **Brownian motion** is caused by the unbalanced bombardment of the grain by the surrounding molecules, and the smaller the grain and the higher the temperature, the more violent the dance — direct evidence that molecules are real and are always moving.

The **mean free path** is the average distance a molecule travels between collisions,

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

where d is the molecular diameter and n the number of molecules per unit volume. For air at room conditions $\lambda \approx 10^{-7}$ m — about 300 molecular diameters, which is why a gas molecule suffers some 10^9 collisions every second yet still spends nearly all its time moving freely.

13.3 Kinetic-molecular model of an ideal gas

THE SIX ASSUMPTIONS

1. A gas consists of a very large number of **identical** molecules in continuous **random** motion.
2. The total volume of the molecules is **negligible** compared with the volume of the container.
3. Intermolecular forces are **negligible** except during a collision.
4. Collisions with each other and with the walls are **perfectly elastic**, so kinetic energy is conserved.
5. The duration of a collision is negligible compared with the time between collisions.
6. Between collisions a molecule moves in a **straight line** with constant velocity, obeying Newton's laws.

Two consequences of the model give physical meaning to the two quantities in the gas equation. **Pressure** is the average force per unit area produced by the continual bombardment of the walls by molecules. **Temperature** turns out to be a direct measure of the average kinetic energy of the molecules — a result we shall prove in §13.5.

Because the motion is random, no direction is preferred. If u , v , w are the velocity components of a molecule and $c^2 = u^2 + v^2 + w^2$, then averaged over all the molecules

$$\overline{u^2} = \overline{v^2} = \overline{w^2} = \frac{1}{3}\overline{c^2}$$

13.4 Derivation of pressure exerted by gas

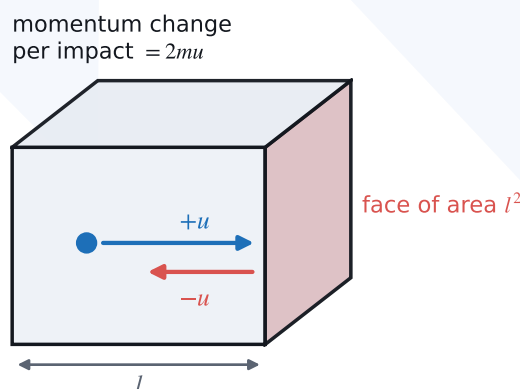


Figure 54. A molecule of mass m rebounding elastically from the shaded face of a cube of side l . Only the x -component of its velocity is reversed.

PRESSURE OF AN IDEAL GAS, $p = \frac{1}{3}\rho \overline{c^2}$

We start from one molecule bouncing off one wall, and reach the pressure of the whole gas. The plan is: momentum given in one hit $\rightarrow$ how often the hits come $\rightarrow$ force from one molecule $\rightarrow$ force from all of them $\rightarrow$ pressure.

Setting up. A cube of side l contains N molecules, each of mass m . Take one molecule with velocity components u, v, w along the three axes, and watch the face perpendicular to the x -axis.

Step 1 — velocity after the bounce. The collision is perfectly elastic and the wall is smooth, so only the x -component is reversed:

$$u \longrightarrow -u$$

Step 2 — the molecule's change in momentum. Momentum after minus momentum before:

$$\Delta p_{\text{molecule}} = (-mu) - (mu) = -2mu$$

Step 3 — the momentum given to the wall. By Newton's third law, whatever the molecule loses the wall gains:

$$\Delta p_{\text{wall}} = 2mu$$

Step 4 — how far the molecule travels before it hits this same face again. It must cross to the opposite face and come back:

$$\text{distance} = 2l$$

Step 5 — the time between two hits on this face. Only the x -motion matters for reaching this face, and u does not change in size:

$$\Delta t = \frac{2l}{u}$$

Step 6 — the average force from this one molecule. Newton's second law in momentum form says force is rate of change of momentum:

$$f = \frac{\Delta p_{\text{wall}}}{\Delta t} = \frac{2mu}{2l/u}$$

Step 7 — simplify that fraction. Dividing by $2l/u$ is the same as multiplying by $u/2l$:

$$f = 2mu \times \frac{u}{2l} = \frac{mu^2}{l}$$

Step 8 — add up over all N molecules. Let their x -components be $u_1, u_2, \dots, u_N$. Forces on the same wall in the same direction simply add:

$$F = \frac{m}{l} (u_1^2 + u_2^2 + \dots + u_N^2)$$

Step 9 — write the sum using a mean. By definition the mean of the squares is $\overline{u^2} = \frac{u_1^2 + \dots + u_N^2}{N}$, so the sum equals $N\overline{u^2}$:

$$F = \frac{mN\overline{u^2}}{l}$$

Step 10 — turn force into pressure. The face has area l^2 :

$$p = \frac{F}{l^2} = \frac{mN\overline{u^2}}{l \times l^2} = \frac{mN\overline{u^2}}{l^3}$$

Step 11 — replace l^3 by the volume V of the cube:

$$p = \frac{mN\overline{u^2}}{V}$$

Step 12 — bring in the other two directions. For any one molecule, Pythagoras in three dimensions gives $c^2 = u^2 + v^2 + w^2$. Taking means over all molecules:

$$\overline{c^2} = \overline{u^2} + \overline{v^2} + \overline{w^2}$$

Step 13 — use the randomness of the motion. No direction is special, so on average the molecules move as much along x as along y or z :

$$\overline{u^2} = \overline{v^2} = \overline{w^2}$$

Step 14 — combine Steps 12 and 13. The right-hand side of Step 12 becomes three equal terms:

$$\overline{c^2} = 3 \overline{u^2} \quad \implies \quad \overline{u^2} = \frac{1}{3} \overline{c^2}$$

Step 15 — substitute into Step 11:

$$p = \frac{1}{3} \cdot \frac{mN}{V} \cdot \overline{c^2}$$

Step 16 — recognise the density. The total mass of the gas is mN , so mN/V is the density ρ :

$$p = \frac{1}{3} \rho \overline{c^2}$$

Result.

$$p = \frac{1}{3} \rho \overline{c^2} \quad \text{or, multiplying by } V, \quad pV = \frac{1}{3} mN \overline{c^2}$$

What it means. Pressure is nothing more than the drumming of molecular impacts. It rises if the molecules are heavier, if there are more of them in the same space, or — most strongly — if they move faster, since the speed enters **squared**.

Conditions used. Collisions are perfectly elastic; molecules have negligible volume and no forces between them except during collisions; motion is completely random (Step 13); and the container walls are smooth and rigid. The cube is only for convenience — the result holds for a vessel of any shape.

THE EXAMINER IS LOOKING FOR

1. Momentum change per impact $= 2mu$, justified by Newton's third law.
2. Time between impacts $= 2l/u$, with the distance $2l$ explained.
3. Force from one molecule $= mu^2/l$ using $F = \Delta p/\Delta t$.
4. Summation over N molecules and the introduction of $\overline{u^2}$.
5. Division by area l^2 and the substitution $V = l^3$.
6. The randomness argument giving $\overline{u^2} = \frac{1}{3}\overline{c^2}$ — **this step alone often carries a mark.**
7. The final statement, plus the list of assumptions.

$\overline{c^2}$ **IS A MEAN OF SQUARES, NOT THE SQUARE OF A MEAN**

For molecules of speeds 1, 2 and 3 units the mean speed is 2 but $\overline{c^2} = (1 + 4 + 9)/3 = 4.67$, whose square root is 2.16. The rms speed is always a little larger than the mean speed; never square the average.

13.5 Average translational kinetic energy of gas molecule

AVERAGE TRANSLATIONAL KINETIC ENERGY $\overline{E} = \frac{3}{2}kT$

We start from the kinetic-theory result for pV and the ideal gas equation, and reach the statement that temperature is molecular kinetic energy.

Step 1 — the kinetic theory result from section 13.4:

$$pV = \frac{1}{3}mN\overline{c^2}$$

Step 2 — reshape the right side so that a kinetic energy appears. We want $\frac{1}{2}m\overline{c^2}$ inside, so write $\frac{1}{3} = \frac{2}{3} \times \frac{1}{2}$:

$$pV = \frac{2}{3}N\left(\frac{1}{2}m\overline{c^2}\right)$$

Step 3 — name the bracket. $\frac{1}{2}m\overline{c^2}$ is the average translational kinetic energy of one molecule; call it $\overline{E}$:

$$pV = \frac{2}{3}N\overline{E}$$

Step 4 — bring in the experimental gas law. For N molecules, $pV = NkT$ where k is the Boltzmann constant:

$$NkT = \frac{2}{3}N\overline{E}$$

Step 5 — cancel N from both sides:

$$kT = \frac{2}{3}\bar{E}$$

Step 6 — multiply both sides by $\frac{3}{2}$:

$$\bar{E} = \frac{3}{2}kT$$

Result.

$$\bar{E} = \frac{1}{2}m\bar{c}^2 = \frac{3}{2}kT$$

What it means. The right-hand side contains no m , no p and no V — only the temperature. So at the same temperature a hydrogen molecule and a carbon-dioxide molecule have the **same** average kinetic energy; the heavy one simply moves more slowly. This is what absolute temperature really measures.

Conditions used. The gas is ideal, and only translational energy is counted (rotation and vibration are extra, and are why C_v differs between gases).

ROOT MEAN SQUARE SPEED $c_{rms} = \sqrt{3RT/M}$

We start from the same two equations, but written for **one mole**, and reach a formula for molecular speed.

Step 1 — take one mole. Then $N = N_A$, and the total mass mN_A is the molar mass M :

$$pV = \frac{1}{3}M\bar{c}^2$$

Step 2 — write the gas equation for one mole:

$$pV = RT$$

Step 3 — set the two expressions for pV equal:

$$\frac{1}{3}M\bar{c}^2 = RT$$

Step 4 — multiply both sides by 3:

$$M\bar{c}^2 = 3RT$$

Step 5 — divide both sides by M :

$$\bar{c}^2 = \frac{3RT}{M}$$

Step 6 — take the square root. The root of the mean of the squares is, by definition, the **rms speed**:

$$c_{rms} = \sqrt{\frac{3RT}{M}}$$

Result. Two other forms follow at once, by using $p = \frac{1}{3}\rho\overline{c^2}$ and by dividing R/M by Avogadro's number:

$$c_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3p}{\rho}}$$

What it means. $c_{rms} \propto \sqrt{T}$, so to double the molecular speed you must **quadruple** the absolute temperature. And $c_{rms} \propto 1/\sqrt{M}$, which is why light gases such as hydrogen and helium leak out of a balloon — and out of a planet's atmosphere — faster than heavy ones.

Conditions used. Ideal-gas behaviour; T in kelvin and M in kg mol^{-1} .

TEMPERATURE IS MOLECULAR KINETIC ENERGY

The average translational kinetic energy of a gas molecule depends **only** on the absolute temperature — not on the pressure, the volume or the kind of gas. Heavy molecules simply move more slowly. For one mole the total translational kinetic energy is $\frac{3}{2}RT$, and at $T=0$ it would vanish, which is the kinetic-theory meaning of absolute zero.

WORKED EXAMPLE 13.2

Problem. Find the average translational kinetic energy of one molecule of any gas at 27°C , and the total translational kinetic energy of one mole at the same temperature. Take $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$ and $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

Solution. $T = 300 \text{ K}$. For one molecule,

$$\overline{E} = \frac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21} \text{ J}$$

For one mole,

$$E = \frac{3}{2}RT = 1.5 \times 8.314 \times 300 = 3.74 \times 10^3 \text{ J}$$

That is about 3.7 kJ per mole — the same for helium, nitrogen or carbon dioxide at this temperature.

13.6 Boltzmann constant; root mean square speed

BOLTZMANN CONSTANT

The Boltzmann constant is the gas constant per molecule:

$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

It converts absolute temperature into energy per molecule.

The **root mean square speed** is the square root of the mean of the squares of the molecular speeds. It was derived step by step in section 13.5:

$$c_{rms} = \sqrt{\overline{c^2}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3p}{\rho}}$$

Use the first form when you know the temperature and the molar mass, and the last when you know the pressure and density. Two conclusions follow at once: $c_{rms} \propto \sqrt{T}$, so to double the rms speed you must **quadruple** the absolute temperature; and $c_{rms} \propto 1/\sqrt{M}$ at a given temperature, so hydrogen molecules move four times as fast as oxygen molecules.

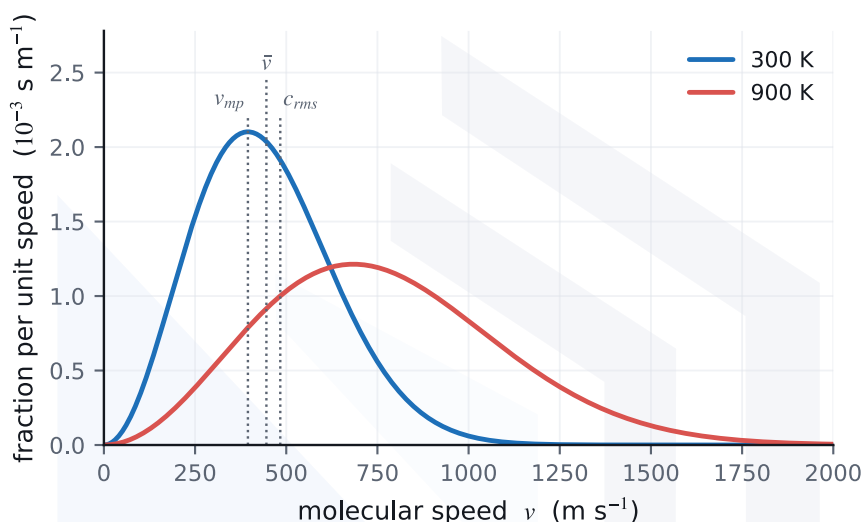


Figure 55. Maxwell distribution of molecular speeds for oxygen. Raising the temperature flattens and broadens the curve and shifts it to higher speeds.

The speeds are spread over a wide range described by the **Maxwell distribution**. Three speeds are quoted from it: the most probable speed $v_{mp} = \sqrt{2kT/m}$, the mean speed $\bar{v} = \sqrt{8kT/\pi m}$ and the rms speed $c_{rms} = \sqrt{3kT/m}$, always in that increasing order, in the ratio 1 : 1.128 : 1.225.

WORKED EXAMPLE 13.3

Problem. Calculate the rms speed of oxygen molecules at 27 °C ($M = 32 \text{ g mol}^{-1}$). At what temperature will hydrogen molecules ($M = 2 \text{ g mol}^{-1}$) have this same rms speed?

Solution. With $T = 300 \text{ K}$ and $M = 0.032 \text{ kg mol}^{-1}$,

$$c_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 8.314 \times 300}{0.032}} = \sqrt{2.338 \times 10^5} = 484 \text{ m s}^{-1}$$

For equal rms speeds, $3RT_H/M_H = 3RT_O/M_O$, so T is proportional to M :

$$T_H = T_O \times \frac{M_H}{M_O} = 300 \times \frac{2}{32} = 18.8 \text{ K}$$

Hydrogen must be cooled almost to liquid-helium temperatures to slow it down to the speed oxygen has at room temperature.

13.7 Heat capacities: gases and solids

A gas has two molar heat capacities, because it can be heated in two different ways.

MOLAR HEAT CAPACITIES

C_v is the heat needed to raise the temperature of one mole by 1 K at constant **volume**; C_p is the heat needed for the same rise at constant **pressure**. Both are in $\text{J mol}^{-1} \text{K}^{-1}$.

C_p is always the larger. At constant volume the gas cannot expand, so all the heat goes into internal energy. At constant pressure the gas must also push back the surroundings, and the extra heat supplies that work.

MAYER'S RELATION, $C_p - C_v = R$

We start from the same small temperature rise dT given to one mole of gas in two different ways, and reach the relation between the two molar heat capacities.

Setting up. Take exactly one mole. Heat it once at constant volume and once at constant pressure, in both cases raising its temperature by the same small amount dT . The internal energy of an ideal gas depends only on its temperature, so the same dT always produces the same change dU , whichever path we take. That single fact is the key to the whole derivation.

Step 1 — heat it at constant volume. By the definition of C_v , the heat needed is

$$dQ_v = C_v dT$$

Step 2 — no work is done at constant volume. Work is $p dV$, and $dV = 0$ because the gas cannot expand. So by the first law, $dQ = dU + p dV$ reduces to

$$dU = C_v dT$$

Step 3 — now heat it at constant pressure. By the definition of C_p :

$$dQ_p = C_p dT$$

Step 4 — apply the first law to this second path. Now the gas does expand, pushing back the surroundings:

$$C_p dT = dU + p dV$$

Step 5 — replace dU using Step 2. This is the step licensed by the "setting up" remark: the same dT means the same dU :

$$C_p dT = C_v dT + p dV$$

Step 6 — find $p dV$ from the gas law. For one mole, $pV = RT$. Differentiating both sides with p held constant:

$$p dV = R dT$$

Step 7 — substitute this into Step 5:

$$C_p dT = C_v dT + R dT$$

Step 8 — divide every term by dT :

$$C_p = C_v + R$$

Step 9 — rearrange:

$$C_p - C_v = R$$

Result.

$$C_p - C_v = R$$

What it means. C_p is always the larger of the two, because heating at constant pressure has to pay for the expansion work as well as the temperature rise. The extra amount is exactly R — the same for every ideal gas, monatomic or diatomic, because the work done in expanding depends only on the gas law and not on the molecule's structure.

Conditions used. One mole of an **ideal** gas, and internal energy a function of temperature alone. For a mass of n moles the relation becomes $C_p - C_v = nR$ if C means total heat capacity; per unit mass it becomes $c_p - c_v = R/M$.

THE EXAMINER IS LOOKING FOR

1. Both processes applied to the **same** dT and the same one mole.
2. $dU = C_v dT$, with the reason "no work is done at constant volume".
3. The first law written as $dQ = dU + p dV$ for the constant-pressure path.
4. $p dV = R dT$ obtained by differentiating $pV = RT$ at constant p .
5. The cancellation of dT and the final statement $C_p - C_v = R$.

The ratio $\gamma = C_p/C_v$ identifies the kind of molecule, because the internal energy depends on how many independent ways a molecule can store energy.

Gas	C_v	C_p	$\gamma = C_p/C_v$	Example
Monatomic	$\frac{3}{2}R = 12.5$	$\frac{5}{2}R = 20.8$	1.67	He, Ne, Ar
Diatomic	$\frac{5}{2}R = 20.8$	$\frac{7}{2}R = 29.1$	1.40	H ₂ , O ₂ , N ₂

(Values in J mol⁻¹ K⁻¹.) A monatomic molecule is a single point and can only move in three directions, so its energy is entirely translational, $U = \frac{3}{2}RT$ per mole, giving $C_v = \frac{3}{2}R$. A diatomic molecule is a dumbbell that can also rotate about two axes, storing $\frac{1}{2}RT$ more per mole in each, so $U = \frac{5}{2}RT$ and $C_v = \frac{5}{2}R$.

Solids. In a solid the atoms cannot travel; they vibrate about fixed sites. Each atom vibrating in three dimensions stores $3kT$ of energy on average — half kinetic and half potential in each of the three directions — so one mole stores $U = 3RT$ and the molar heat capacity is

$$C = 3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}$$

This is the **law of Dulong and Petit**: all simple solids have about the same molar heat capacity. It works well: for copper ($M = 63.5 \text{ g mol}^{-1}$) it predicts $c = 3R/M = 24.9/0.0635 = 392 \text{ J kg}^{-1} \text{ K}^{-1}$, against a measured 385. It fails badly at low temperatures, where every solid's heat capacity falls towards zero — a purely quantum effect that classical physics cannot explain.

WORKED EXAMPLE 13.4

Problem. For a certain gas $C_p = 29.1$ and $C_v = 20.8 \text{ J mol}^{-1} \text{ K}^{-1}$. (a) Find R and γ and say whether the gas is monatomic or diatomic. (b) How much heat is needed to raise the temperature of 2.0 mol of this gas by 50 K at constant pressure, and how much at constant volume? (c) Account for the difference.

Solution. (a) $R = C_p - C_v = 29.1 - 20.8 = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, and $\gamma = 29.1/20.8 = 1.40$, so the gas is **diatomic**.

(b) At constant pressure:

$$Q_p = nC_p \Delta T = 2.0 \times 29.1 \times 50 = 2910 \text{ J}$$

At constant volume:

$$Q_v = nC_v \Delta T = 2.0 \times 20.8 \times 50 = 2080 \text{ J}$$

(c) The difference is $2910 - 2080 = 830 \text{ J}$. This is exactly the work done by the gas in expanding at constant pressure:

$$W = p\Delta V = nR\Delta T = 2.0 \times 8.3 \times 50 = 830 \text{ J}$$

✓ Chapter summary

- Gas laws combine into the ideal gas equation $pV = nRT = NkT$, with $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ and T in kelvin.
- Kinetic theory assumes many identical molecules of negligible volume in random motion, with no forces except during elastic collisions.
- Pressure of an ideal gas: $p = \frac{1}{3}\rho\overline{c^2}$, equivalently $pV = \frac{1}{3}mN\overline{c^2}$.
- Average translational kinetic energy per molecule is $\overline{E} = \frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$, and per mole $\frac{3}{2}RT$ — independent of the nature of the gas.
- Boltzmann constant $k = R/N_A = 1.38 \times 10^{-23} \text{ J K}^{-1}$.
- Root mean square speed $c_{rms} = \sqrt{3RT/M} = \sqrt{3kT/m} = \sqrt{3p/\rho}$, so $c_{rms} \propto \sqrt{T}$ and $c_{rms} \propto 1/\sqrt{M}$.
- Mayer's relation $C_p - C_v = R$; $\gamma = 1.67$ for monatomic and 1.40 for diatomic gases.
- Dulong and Petit: the molar heat capacity of a simple solid is about $3R = 25 \text{ J mol}^{-1} \text{ K}^{-1}$.

Practice questions

Group A — Multiple choice (1 mark each)

- At constant temperature the pressure of a fixed mass of ideal gas is proportional to [1]
 a V b $1/V$ c V^2 d $\sqrt{V}$
- The rms speed of the molecules of an ideal gas is proportional to [1]
 a T b T^2 c $\sqrt{T}$ d $1/T$
- The average translational kinetic energy of a molecule at absolute temperature T is [1]
 a kT b $\frac{3}{2}kT$ c $\frac{3}{2}RT$ d $\frac{1}{2}kT$
- For a diatomic gas the ratio C_p/C_v is about [1]
 a 1.33 b 1.40 c 1.67 d 1.00
- According to the law of Dulong and Petit, the molar heat capacity of a simple solid is about [1]
 a R b $\frac{3}{2}R$ c $3R$ d $\frac{5}{2}R$

ANSWERS TO GROUP A

- (b) — Boyle's law, pV constant.
- (c) — $c_{rms} = \sqrt{3RT/M}$.
- (b) — per **molecule**; $\frac{3}{2}RT$ would be per mole.
- (b) — $C_p/C_v = 3.5R/2.5R = 1.40$.
- (c) — $3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}$.

Group B — Short answer (5 marks each)

- State the assumptions of the kinetic molecular model of an ideal gas, and explain on this model what pressure and temperature mean. [5]
- A closed cylinder contains 2.0 g of hydrogen ($M = 2.0 \text{ g mol}^{-1}$) at 27°C . Find (i) the number of molecules present and (ii) the total translational kinetic energy of the gas. [5]
- Define the Boltzmann constant and show that the average translational kinetic energy of a gas molecule is $\frac{3}{2}kT$. [5]
- Derive Mayer's relation $C_p - C_v = R$ and hence find γ for a monatomic gas. [5]

ANSWERS TO GROUP B

- The six assumptions of §13.3. Pressure is the average force per unit area from molecular bombardment of the walls; temperature is a direct measure of the average translational kinetic energy of the molecules, $\bar{E} = \frac{3}{2}kT$.
- (i) $n = 2.0/2.0 = 1.0 \text{ mol}$, so $N = nN_A = 6.02 \times 10^{23} \text{ molecules}$. (ii)
 $E = \frac{3}{2}nRT = 1.5 \times 1.0 \times 8.314 \times 300 = 3.74 \times 10^3 \text{ J}$.
- Define $k = R/N_A$. Then from kinetic theory $pV = \frac{1}{3}mN\overline{c^2} = \frac{2}{3}N(\frac{1}{2}m\overline{c^2})$; equating to $pV = NkT$ gives $\frac{1}{2}m\overline{c^2} = \frac{3}{2}kT$.

4. Derivation as in §13.7. For a monatomic gas $C_v = \frac{3}{2}R$, so $C_p = C_v + R = \frac{5}{2}R$ and $\gamma = \frac{5}{2}R \div \frac{3}{2}R = 5/3 = 1.67$.

Group C — Long answer (8 marks each)

- (a)** State the assumptions of the kinetic theory of gases and derive the expression $p = \frac{1}{3}\rho\overline{c^2}$ for the pressure of an ideal gas. [6]

(b) Hence show that the average translational kinetic energy of a molecule is $\frac{3}{2}kT$. [2]
- A vessel of volume 0.0249 m^3 contains oxygen at 300 K and a pressure of $1.0 \times 10^5 \text{ Pa}$. Taking $M = 32 \text{ g mol}^{-1}$, find

 - the number of moles,
 - the number of molecules,
 - the density of the gas and
 - the rms speed of its molecules. [8]

ANSWERS TO GROUP C

- Full derivation as set out in §13.4, Steps 1 to 6, followed by the argument of §13.5.
- (a) $n = pV/RT = (1.0 \times 10^5 \times 0.0249)/(8.314 \times 300) = 2490/2494 = 1.00 \text{ mol}$.

(b) $N = nN_A = 6.02 \times 10^{23} \text{ molecules}$.

(c) Mass $= nM = 1.00 \times 0.032 = 0.032 \text{ kg}$, so $\rho = 0.032/0.0249 = 1.29 \text{ kg m}^{-3}$.

(d) $c_{rms} = \sqrt{3p/\rho} = \sqrt{(3 \times 1.0 \times 10^5)/1.29} = \sqrt{2.33 \times 10^5} = 483 \text{ m s}^{-1}$. (The same answer follows from $\sqrt{3RT/M}$, as it must.)

UNIT 14

Reflection at curved mirror

Waves and Optics • 2 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Real and Virtual images
- Mirror formula

A curved mirror is a polished part of a sphere. Because the surface curves, the normals at different points of it point in different directions, so a bundle of parallel rays is brought together at a point or spread out from a point. That one fact is behind the shaving mirror, the dentist's mirror, the headlight reflector and the wide-angle mirror on a Sajha bus. This unit fixes the language (pole, centre of curvature, focus), separates real images from virtual ones, and derives the single equation that ties object distance, image distance and focal length together.

WHAT THE EXAM ASKS

Two things, almost every year: state the nature of the image for a given object position, and use the mirror formula with correct signs. Marks are lost on signs far more often than on arithmetic.

14.1 Real and virtual images

The words you must use correctly

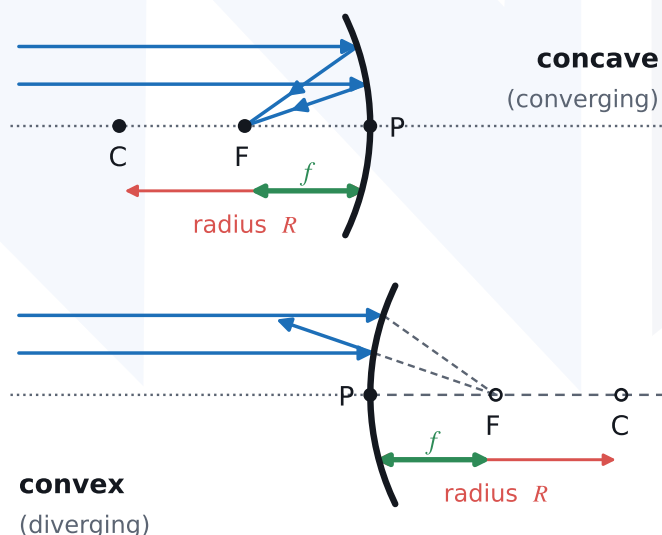


Figure 56. Terms used for spherical mirrors. For a concave mirror F and C lie in front of the mirror; for a convex mirror they lie behind it and are only virtual points.

Term	Meaning
Pole P	the geometrical centre of the reflecting surface
Centre of curvature C	the centre of the sphere of which the mirror is a part
Radius of curvature R	the distance PC
Principal axis	the straight line through P and C
Principal focus F	the point on the axis where paraxial rays parallel to the axis converge (concave) or from which they appear to diverge (convex)
Focal length f	the distance PF , and $f = R/2$
Aperture	the effective diameter of the reflecting surface

Paraxial rays are rays close to the principal axis and nearly parallel to it. Everything in this unit assumes paraxial rays and a small aperture; otherwise the mirror suffers from spherical aberration and the rays do not meet at one point.

THE FOCAL LENGTH OF A SPHERICAL MIRROR IS HALF ITS RADIUS, $f = R/2$

We start from the law of reflection at one point on the mirror, and reach a relation between the focal length $f = PF$ and the radius of curvature $R = PC$.

Setting up. P is the pole, C the centre of curvature and F the principal focus. A ray parallel to the axis strikes the mirror at M , a small height h above the axis, and after reflection passes through F .

Step 1 — identify the normal. The normal to a spherical surface at any point passes through the centre of curvature, so the normal at M is the line MC .

Step 2 — the angle of incidence. The incident ray is parallel to the axis PC , and MC cuts both. Alternate angles between parallel lines are equal, so if the angle of incidence is θ , then

$$\angle MCP = \theta$$

Step 3 — the angle of reflection. By the law of reflection the reflected ray makes the same angle θ with the normal MC :

$$\angle CMF = \theta$$

Step 4 — use the exterior-angle rule on triangle MCF . The exterior angle at F equals the sum of the two opposite interior angles, which are both θ :

$$\angle MFP = \theta + \theta = 2\theta$$

Step 5 — drop a perpendicular from M to the axis. Call the foot N , so $MN = h$. From the right-angled triangle MNC :

$$\tan \theta = \frac{h}{NC}$$

Step 6 — do the same for triangle MNF :

$$\tan 2\theta = \frac{h}{NF}$$

Step 7 — use the paraxial approximation. The ray is close to the axis, so θ is small and, in radians, $\tan \theta \approx \theta$ and $\tan 2\theta \approx 2\theta$. Also N lies practically at the pole P , so $NC \approx PC$ and $NF \approx PF$:

$$\theta \approx \frac{h}{PC}, \quad 2\theta \approx \frac{h}{PF}$$

Step 8 — divide the second by the first so that h and θ both cancel:

$$\frac{2\theta}{\theta} = \frac{h/PF}{h/PC} = \frac{PC}{PF}$$

Step 9 — simplify the left side:

$$2 = \frac{PC}{PF}$$

Step 10 — rearrange, and rename $PC = R$, $PF = f$:

$$PF = \frac{1}{2}PC \quad \implies \quad f = \frac{R}{2}$$

Result.

$$f = \frac{R}{2} \quad \text{equivalently} \quad R = 2f$$

What it means. The focus sits exactly halfway between the pole and the centre of curvature. Notice that h dropped out in Step 8 — so every paraxial ray, whatever its height, crosses the axis at the same point F . That is precisely why a mirror forms a sharp image.

Conditions used. Only paraxial rays (small aperture, $h \ll R$). Rays far from the axis cross the axis nearer the mirror — the defect called **spherical aberration** — and for them $f = R/2$ is no longer true.

Real and virtual images

REAL AND VIRTUAL IMAGE

A **real image** is formed when reflected rays actually meet at a point. Light energy reaches that point, so a real image can be caught on a screen. A **virtual image** is formed when reflected rays only appear to come from a point, because their backward extensions meet there. No light reaches that point, so a virtual image cannot be caught on a screen — it can only be seen by an eye looking into the mirror.

	Real image	Virtual image
Formed by	actual intersection of rays	intersection of backward extensions
Screen test	can be caught on a screen	cannot be caught on a screen
Orientation	inverted (for a single mirror)	erect
Position	in front of the mirror	behind the mirror
Sign of v	negative (new Cartesian)	positive
Sign of m	negative	positive

To locate an image by drawing, use any two of these four standard rays:

1. A ray **parallel to the principal axis** reflects through F (concave), or appears to come from F (convex).
2. A ray **through F** reflects parallel to the principal axis.
3. A ray **through C** strikes the mirror normally and retraces its path.
4. A ray **striking the pole P** reflects making an equal angle with the axis.

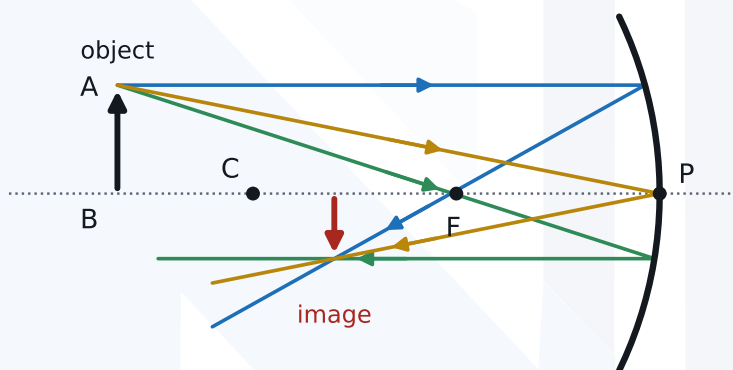


Figure 57. Concave mirror with the object beyond C . The three reflected rays actually cross in front of the mirror, so the image is real, inverted and diminished.

The same construction repeated for every object position gives the table every NEB candidate should know by heart.

Object position (concave mirror)	Image position	Nature	Size
At infinity	At F	Real, inverted	Point-sized
Beyond C	Between F and C	Real, inverted	Diminished
At C	At C	Real, inverted	Same size
Between C and F	Beyond C	Real, inverted	Magnified
At F	At infinity	Real, inverted	Highly magnified
Between F and P	Behind the mirror	Virtual, erect	Magnified

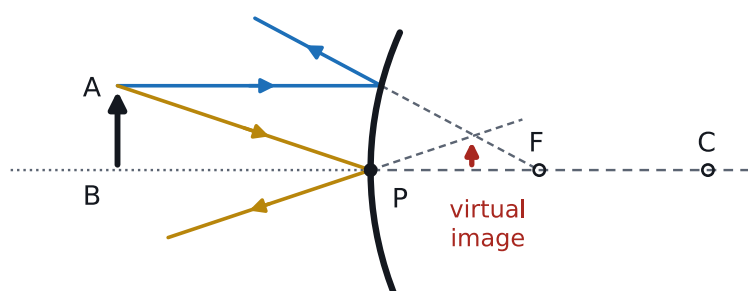


Figure 58. Convex mirror. The reflected rays diverge; only their backward extensions (dashed) meet, behind the mirror, so the image is virtual, erect and diminished.

14.2 Mirror formula

The new Cartesian sign convention

Before any formula can be used, the signs must be fixed.

THE FOUR RULES OF THE NEW CARTESIAN CONVENTION

1. All distances are measured **from the pole** P .
2. Distances measured **in the direction of the incident light** are **positive**; those measured against it are **negative**.
3. Heights **above** the principal axis are positive, heights **below** it are negative.
4. Light is always drawn as travelling from left to right, so a real object in front of the mirror always has u **negative**.

It follows that for a **concave** mirror f and R are negative, and for a **convex** mirror they are positive. A real image has v negative; a virtual image has v positive.

Deriving the formula

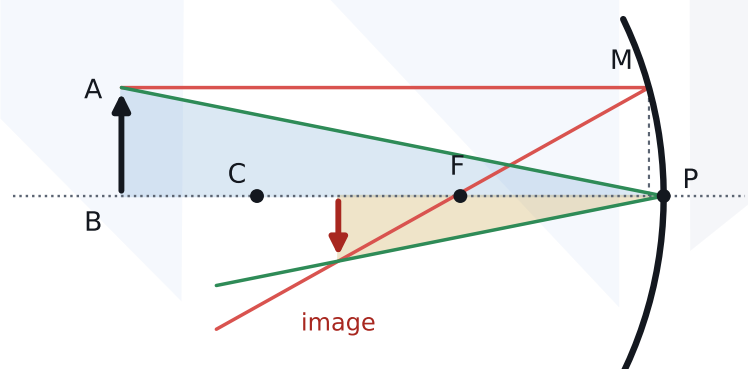


Figure 59. Geometry for the mirror formula. The two shaded triangles about the pole are similar; so are the triangles that meet at F , because $MP \approx AB$ for paraxial rays.

THE MIRROR FORMULA $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

We start from two pairs of similar triangles in the ray diagram, and reach the relation between object distance, image distance and focal length.

Setting up. An object AB stands upright on the axis in front of a concave mirror, and its real inverted image is $A'B'$. The strategy is important: first do all the geometry with **plain lengths**, which are all positive numbers, and only at the end put in the signs of the Cartesian convention. Mixing the two is where most marks are lost.

Step 1 — the first pair of similar triangles. The ray AP hits the pole and reflects along PA' , making equal angles with the axis (the axis is the normal at P). So the right-angled triangles ABP and $A'B'P$ have equal angles at P , and are similar.

Step 2 — write the ratio of corresponding sides:

$$\frac{A'B'}{AB} = \frac{B'P}{BP} \quad (1)$$

Step 3 — the second pair of similar triangles. The ray AM travels parallel to the axis and so reflects through the focus F , going on to A' . Triangles MPF and $A'B'F$ are similar, because they have equal vertically opposite angles at F and a right angle each:

$$\frac{A'B'}{MP} = \frac{B'F}{PF}$$

Step 4 — replace MP by AB . For paraxial rays the point M lies practically at the pole, so the vertical height MP is the same as the object height AB :

$$\frac{A'B'}{AB} = \frac{B'F}{PF} \quad (2)$$

Step 5 — equate (1) and (2). Both are equal to the same ratio $A'B'/AB$:

$$\frac{B'P}{BP} = \frac{B'F}{PF}$$

Step 6 — split $B'F$ into lengths we have names for. From the diagram, F lies between P and B' , so

$$B'F = B'P - PF$$

Step 7 — substitute that in:

$$\frac{B'P}{BP} = \frac{B'P - PF}{PF} \quad (3)$$

Step 8 — now put in the signs. All three points B , B' and F lie in front of the mirror, i.e. on the side from which light comes, so all three distances are negative in the new Cartesian convention:

$$BP = -u, \quad B'P = -v, \quad PF = -f$$

Step 9 — substitute these into (3):

$$\frac{-v}{-u} = \frac{-v - (-f)}{-f}$$

Step 10 — tidy both sides. On the left the two minus signs cancel; on the right the numerator becomes $-v + f$:

$$\frac{v}{u} = \frac{f - v}{-f}$$

Step 11 — take the minus sign out of the denominator on the right:

$$\frac{v}{u} = \frac{v - f}{f}$$

Step 12 — cross-multiply:

$$vf = u(v - f)$$

Step 13 — remove the bracket:

$$vf = uv - uf$$

Step 14 — divide every single term by uvf :

$$\frac{vf}{uvf} = \frac{uv}{uvf} - \frac{uf}{uvf}$$

Step 15 — cancel in each term:

$$\frac{1}{u} = \frac{1}{f} - \frac{1}{v}$$

Step 16 — move $1/v$ to the left:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Result.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

What it means. This is the **mirror formula**. Although it was derived for one particular case — a concave mirror forming a real image — it holds for concave and convex mirrors and for images of either kind, provided the sign convention is obeyed throughout. That is the whole point of having a sign convention: one equation instead of four.

Conditions used. Paraxial rays only (Step 4 needs $MP \approx AB$), a mirror of small aperture, and all distances measured from the pole with the new Cartesian convention.

THE EXAMINER IS LOOKING FOR

1. A labelled ray diagram with P , F , C , the object and the image marked.
2. The first similar-triangle pair and equation (1).
3. The second similar-triangle pair, **with the statement** $MP \approx AB$ **for paraxial rays** — a mark is usually reserved for this.
4. The substitution $B'F = B'P - PF$.
5. The sign-convention line $BP = -u$, $B'P = -v$, $PF = -f$, applied after the geometry.
6. Division by uvf to reach the final form.

Since $f = R/2$, the formula may also be written

$$\frac{1}{v} + \frac{1}{u} = \frac{2}{R}$$

Linear magnification

From equation (1) above, $A'B'/AB = B'P/BP$. Putting in signs, the image height h_i is negative (image inverted) while h_o is positive, $B'P = -v$ and $BP = -u$, so

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Using the mirror formula to eliminate u or v gives two useful alternatives:

$$m = \frac{f-v}{f} = \frac{f}{f-u}$$

SIGNS, NOT SIZES

$m = -v/u$, **not** v/u . A negative m means the image is inverted, and for a single mirror that also means it is real. A positive m means erect and virtual. Students who drop the minus sign report a real image as erect and lose the "nature of image" mark even when the numbers are right.

WORKED EXAMPLE 14.1

Problem. An object 4 cm tall is placed 25 cm in front of a concave mirror of radius of curvature 30 cm. Find the position, nature and size of the image.

Solution. Concave mirror: $R = -30$ cm, so $f = R/2 = -15$ cm. The object is in front, so $u = -25$ cm.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-15} - \frac{1}{-25} = \frac{-5+3}{75} = -\frac{2}{75}$$

so $v = -37.5$ cm. The negative sign means the image is 37.5 cm in front of the mirror, i.e. **real**.

$$m = -\frac{v}{u} = -\frac{-37.5}{-25} = -1.5$$

The image is **inverted** and 1.5 times as large: $h_i = m h_o = -1.5 \times 4 = -6$ cm, i.e. 6 cm tall and below the axis.

WORKED EXAMPLE 14.2

Problem. The rear-view mirror of a bus is convex with radius of curvature 4.0 m. A motorcycle is 6.0 m behind the bus. Where is its image and how large is it compared with the motorcycle?

Solution. Convex mirror: $R = +4.0$ m, $f = +2.0$ m; $u = -6.0$ m.

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{2.0} + \frac{1}{6.0} = \frac{3+1}{6.0} = \frac{2}{3} \text{ m}^{-1}$$

so $v = +1.5$ m; the image is 1.5 m **behind** the mirror and therefore **virtual** and **erect**.

$$m = -\frac{v}{u} = -\frac{1.5}{-6.0} = +0.25$$

The image is one quarter of the size of the motorcycle. This is exactly why convex mirrors are used on vehicles: a wide field of view, at the price of objects looking smaller (and so further away) than they really are.

WORKED EXAMPLE 14.3

Problem. A concave mirror forms, on a screen, a real image three times the size of the object. The screen and the object are 80 cm apart. Find the focal length and the radius of curvature of the mirror.

Solution. The image is real and magnified three times, so $m = -3$:

$$-\frac{v}{u} = -3 \quad \Rightarrow \quad v = 3u$$

Both object and image are in front of the mirror. If $u = -d$ then $v = -3d$, and the object-to-screen separation is $|v| - |u| = 3d - d = 2d = 80$ cm, giving $d = 40$ cm. Hence $u = -40$ cm and $v = -120$ cm.

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = -\frac{1}{120} - \frac{1}{40} = -\frac{1+3}{120} = -\frac{1}{30}$$

So $f = -30$ cm and $R = 2f = -60$ cm: a concave mirror of focal length 30 cm and radius of curvature 60 cm.

CHECKING AN ANSWER IN ONE LINE

After solving, test the answer against the table in §14.1. Here the object at 40 cm lies between C (60 cm) and F (30 cm), and the table says the image must be beyond C , real, inverted and magnified — which is exactly what $v = -120$ cm, $m = -3$ says. If the two disagree, a sign has gone wrong.

✓ Chapter summary

- For a spherical mirror the pole P , centre of curvature C and focus F lie on the principal axis, and $f = R/2$ (derived from the paraxial approximation).
- A **real** image is formed where reflected rays actually meet; it can be caught on a screen and is inverted. A **virtual** image is formed where the backward extensions meet; it cannot be caught on a screen and is erect.
- New Cartesian convention: distances from P , positive along the incident light. Concave $f < 0$, convex $f > 0$; real image $v < 0$, virtual image $v > 0$.
- Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$.
- Magnification: $m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f-v}{f} = \frac{f}{f-u}$. Negative m = inverted and real; positive m = erect and virtual.
- A concave mirror gives a real image for every object beyond F , and a magnified virtual image only when the object is between F and P .
- A convex mirror gives a virtual, erect, diminished image for **every** position of a real object.

Practice questions

Group A — Multiple choice (1 mark each)

- The focal length of a concave mirror of radius of curvature 40 cm is [1]
 - 10 cm
 - 20 cm
 - 40 cm
 - 80 cm
- An object is placed between the focus and the pole of a concave mirror. The image is [1]
 - real, inverted, diminished
 - real, erect, magnified
 - virtual, erect, magnified
 - virtual, inverted, diminished
- For a real image formed by a concave mirror, the linear magnification is [1]
 - positive
 - negative
 - always +1
 - always zero
- A convex mirror of focal length f forms, of a real object, an image that is always [1]
 - real and inverted
 - real and erect
 - virtual and inverted
 - virtual and erect
- An object is placed at the centre of curvature of a concave mirror. The image is formed [1]
 - at F
 - at C
 - at infinity
 - behind the mirror

ANSWERS TO GROUP A

- (b) — $f = R/2 = 40/2 = 20$ cm.
- (c) — inside the focus a concave mirror acts as a magnifying mirror.
- (b) — a real image has v and u of the same sign, so $m = -v/u < 0$.
- (d) — the reflected rays always diverge, so the image is virtual, erect and diminished.

5. (b) — the ray through C retraces itself; $u = -R$ gives $v = -R$.

Group B — Short answer (5 marks each)

1. Distinguish between a real image and a virtual image, giving one example of each from everyday life. [5]
2. Show that the focal length of a spherical mirror is half its radius of curvature for paraxial rays. [5]
3. An object 4 cm tall is placed 20 cm from a concave mirror of focal length 15 cm. Find the position, nature and size of the image. [5]
4. A convex rear-view mirror has a radius of curvature of 2.0 m. A car is 4.0 m behind it. Find the position of the image and the magnification. [5]
5. State the new Cartesian sign convention and use it to explain why the focal length of a convex mirror is taken as positive. [5]

ANSWERS TO GROUP B

1. Real: rays actually meet, screen-catchable, inverted, formed in front of the mirror (the image on a cinema screen, or on the film of a reflecting telescope). Virtual: only the backward extensions meet, not screen-catchable, erect, formed behind the mirror (your face in a plane or convex mirror).
2. See the derivation in §14.1: $PC \approx h/\theta$, $PF \approx h/2\theta$, hence $f = R/2$.
3. $u = -20$ cm, $f = -15$ cm. $\frac{1}{v} = \frac{1}{-15} + \frac{1}{20} = \frac{-4+3}{60} = -\frac{1}{60}$, so $v = -60$ cm: real, 60 cm in front of the mirror. $m = -v/u = -(-60)/(-20) = -3$, so the image is inverted and $3 \times 4 = 12$ cm tall.
4. $f = +1.0$ m, $u = -4.0$ m. $\frac{1}{v} = \frac{1}{1.0} + \frac{1}{4.0} = 1.25 \text{ m}^{-1}$, so $v = +0.8$ m — virtual, 0.8 m behind the mirror. $m = -0.8/(-4.0) = +0.2$: erect and one-fifth the size.
5. Rules as in §14.2. Light is drawn travelling left to right and strikes the convex surface from the left; its focus lies **behind** the mirror, i.e. further along the direction of the incident light, so PF is a positive distance and $f > 0$.

Group C — Long answer (8 marks each)

1. (a) Define real and virtual images and state the new Cartesian sign convention. [3]
 (b) Derive the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for a concave mirror forming a real image, and hence obtain an expression for the linear magnification. [5]
2. A concave mirror throws on a wall an image of a lamp filament magnified four times. The wall is 3.0 m from the mirror. Find
 - (a) the distance of the filament from the mirror,
 - (b) the focal length and radius of curvature of the mirror, and
 - (c) the nature, position and magnification of the image if the filament is moved to 0.40 m from the mirror. [8]

ANSWER TO GROUP C QUESTION 2

(a) The image on a wall is real and magnified $4\times$, so $m = -4 = -v/u$, giving $v = 4u$. With $v = -3.0$ m we get $u = -0.75$ m; the filament is 75 cm from the mirror.

(b) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = -\frac{1}{3.0} - \frac{1}{0.75} = -\frac{1}{3} - \frac{4}{3} = -\frac{5}{3} \text{ m}^{-1}$, so $f = -0.60$ m and $R = 2f = -1.2$ m. The mirror has focal length 60 cm and radius of curvature 1.2 m.

(c) Now $u = -0.40$ m, which is inside the focus. $\frac{1}{v} = \frac{1}{-0.60} + \frac{1}{0.40} = -1.6667 + 2.5 = 0.8333 \text{ m}^{-1}$, so $v = +1.2$ m. The image is virtual, 1.2 m behind the mirror, with $m = -1.2/(-0.40) = +3$ — erect and three times as large. It can no longer be caught on the wall.



UNIT 15

Refraction at plane surfaces

Waves and Optics • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Laws of refraction: Refractive index
- Relation between refractive indices
- Lateral shift
- Total internal reflection

Light changes speed when it crosses from one transparent medium into another, and wherever it meets the boundary obliquely that change of speed makes it change direction. This bending is **refraction**. It is why a straw in a glass of water looks broken, why a fish in the Bagmati is never quite where it seems to be, why a glass slab shifts an image sideways, and — through total internal reflection — why a thread of glass thinner than a hair can carry telephone calls across Nepal. This unit develops the laws of refraction, the relations between refractive indices, lateral shift and total internal reflection.

THE ONE EQUATION BEHIND THE WHOLE UNIT

Snell's law $n_1 \sin i = n_2 \sin r$. Every formula in this chapter — relative refractive index, apparent depth, lateral shift, critical angle — is Snell's law with different things known.

15.1 Laws of refraction: Refractive index

When a ray of light travelling in medium 1 meets a plane boundary with medium 2, part of it is reflected back and part enters medium 2 with a changed direction. The angle between the incident ray and the normal is the **angle of incidence** i ; the angle between the refracted ray and the normal is the **angle of refraction** r .

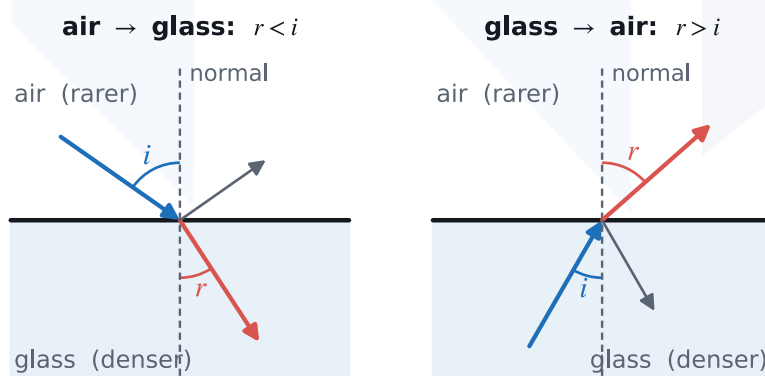


Figure 60. Refraction at a plane boundary. Going into the denser medium the ray bends towards the normal; coming out of it the ray bends away from the normal. The faint grey ray is the partially reflected light.

THE LAWS OF REFRACTION

First law. The incident ray, the refracted ray and the normal to the surface at the point of incidence all lie in the same plane.

Second law (Snell's law). For two given media and light of a given colour, the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant:

$$\frac{\sin i}{\sin r} = {}_1n_2 = \text{constant}$$

This constant is the **refractive index of medium 2 with respect to medium 1**.

Why light bends: refractive index and speed

Refraction happens because light travels at different speeds in different media. If v_1 and v_2 are the speeds in the two media, it can be shown that

$${}_1n_2 = \frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

The **frequency does not change** on refraction — it is fixed by the source — so if the speed falls, the wavelength must fall in the same ratio.

ABSOLUTE REFRACTIVE INDEX

The **absolute refractive index** n of a medium is the refractive index of that medium with respect to vacuum:

$$n = \frac{c}{v}$$

where $c = 3 \times 10^8 \text{ m s}^{-1}$ is the speed of light in vacuum and v is its speed in the medium. Since $v < c$ always, $n > 1$ for every material medium. n has no unit.

In terms of absolute refractive indices, Snell's law takes its most useful form:

$$n_1 \sin i = n_2 \sin r$$

Medium	Absolute refractive index n	Speed of light (10^8 m s^{-1})
Vacuum	1 (exactly)	3.00
Air (at STP)	1.0003	3.00
Ice	1.31	2.29
Water	1.33 ($= 4/3$)	2.26
Crown glass	1.52	1.97
Dense flint glass	1.65	1.82
Diamond	2.42	1.24

A medium of larger n is called **optically denser**. Optical density is not the same as mass density: turpentine ($n = 1.47$) is optically denser than water but less dense in kg m^{-3} .

"BENDS TOWARDS THE NORMAL" — WHICH WAY?

Going from rarer to denser (air $\rightarrow$ glass) the ray slows down and bends **towards** the normal, so $r < i$. Going from denser to rarer (glass $\rightarrow$ air) it speeds up and bends **away** from the normal, so $r > i$. A ray striking the surface normally ($i = 0$) is not bent at all, though its speed still changes.

WORKED EXAMPLE 15.1

Problem. A ray of light of wavelength 600 nm in air strikes a glass surface at 60° to the normal. The refractive index of the glass is 1.5. Find (a) the angle of refraction, (b) the speed of light in the glass, (c) the wavelength and frequency of the light inside the glass.

Solution.

(a) By Snell's law, $1 \times \sin 60^\circ = 1.5 \sin r$, so

$$\sin r = \frac{0.8660}{1.5} = 0.5774 \quad \Rightarrow \quad r = 35.3^\circ$$

(b) $n = c/v$, so

$$v = \frac{c}{n} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m s}^{-1}$$

(c) The wavelength shrinks in the same ratio as the speed:

$$\lambda_{\text{glass}} = \frac{\lambda_{\text{air}}}{n} = \frac{600}{1.5} = 400 \text{ nm}$$

The frequency is unchanged: $f = c/\lambda_{\text{air}} = (3 \times 10^8)/(600 \times 10^{-9}) = 5 \times 10^{14} \text{ Hz}$. Check: $v/\lambda_{\text{glass}} = (2 \times 10^8)/(400 \times 10^{-9}) = 5 \times 10^{14} \text{ Hz}$ — the same.

15.2 Relation between refractive indices

Three relations follow directly from the definitions and are asked as standard five-mark questions.

THE THREE RELATIONS BETWEEN REFRACTIVE INDICES

We start from Snell's law and the principle of reversibility, and reach three results that let you convert freely between any pair of media.

Part (a): ${}_1n_2 = n_2/n_1 = v_1/v_2$

Step 1 — Snell's law in its two-medium form. For a ray passing from medium 1 (angle i) into medium 2 (angle r):

$$n_1 \sin i = n_2 \sin r$$

Step 2 — make $\sin i/\sin r$ the subject. Divide both sides by $n_1 \sin r$:

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$

Step 3 — name the left-hand side. By definition, $\sin i/\sin r$ for this pair of media is the refractive index of 2 with respect to 1:

$${}_1n_2 = \frac{n_2}{n_1}$$

Step 4 — bring in the speeds. Absolute index is defined as $n = c/v$, so $n_1 = c/v_1$ and $n_2 = c/v_2$. Substituting:

$${}_1n_2 = \frac{c/v_2}{c/v_1}$$

Step 5 — cancel c and flip the divided fraction:

$${}_1n_2 = \frac{v_1}{v_2}$$

Part (b): ${}_1n_2 \times {}_2n_1 = 1$

Step 6 — state the principle of reversibility. If the direction of a ray is exactly reversed, it retraces its own path. So send a ray back from medium 2 along the old refracted ray. It now strikes the boundary at angle r and must emerge along the old incident ray, at angle i .

Step 7 — apply the definition to this reversed journey. The ray now goes from 2 to 1, so the angle in medium 2 is the incidence angle and the angle in medium 1 is the refraction angle:

$${}_2n_1 = \frac{\sin r}{\sin i}$$

Step 8 — compare with ${}_1n_2 = \sin i/\sin r$ from Step 3. The right-hand side of Step 7 is the reciprocal of the right-hand side of Step 3:

$${}_2n_1 = \frac{1}{{}_1n_2}$$

Step 9 — multiply both sides by ${}_1n_2$:

$${}_1n_2 \times {}_2n_1 = 1$$

Part (c): the chain relation

Step 10 — write each relative index as a ratio of absolute indices, using Part (a):

$${}_1n_2 \times {}_2n_3 = \frac{n_2}{n_1} \times \frac{n_3}{n_2}$$

Step 11 — cancel n_2 , which appears once on top and once below:

$${}_1n_2 \times {}_2n_3 = \frac{n_3}{n_1}$$

Step 12 — recognise the right-hand side as ${}_1n_3$, again by Part (a):

$${}_1n_2 \times {}_2n_3 = {}_1n_3$$

Step 13 — multiply both sides by ${}_3n_1$ and use Part (b) on the right, since ${}_1n_3 \times {}_3n_1 = 1$:

$${}_1n_2 \times {}_2n_3 \times {}_3n_1 = 1$$

Result.

$${}_1n_2 = \frac{n_2}{n_1} = \frac{v_1}{v_2}, \quad {}_1n_2 \times {}_2n_1 = 1, \quad {}_1n_2 \times {}_2n_3 \times {}_3n_1 = 1$$

What it means. You only ever need a table of absolute indices; any relative index is a ratio of two of them. Part (b) says glass-with-respect-to-water is the exact reciprocal of water-with-respect-to-glass, and Part (c) says a chain of media closing back on itself multiplies to one — like walking round a loop and returning to the same height.

Conditions used. A single wavelength (since n depends on colour), and media that are transparent and uniform.

READING THE SUBSCRIPTS

${}_1n_2$ always means "the refractive index of 2 **with respect to** 1", and equals n_2/n_1 — the second subscript goes on top. Getting this upside down turns 1.5 into 0.67 and is the commonest slip in this section.

Refractive index from real and apparent depth

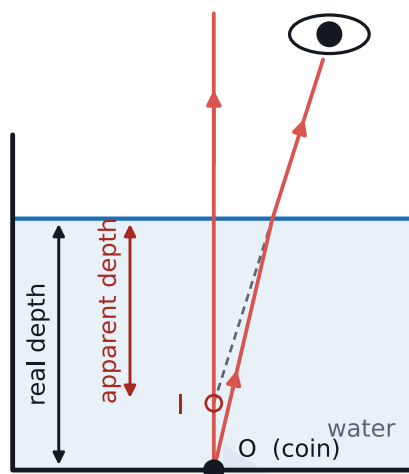


Figure 61. Real and apparent depth. Rays from the coin at O bend away from the normal as they leave the water, so to the eye above they seem to come from the shallower point I .

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

We start from Snell's law applied to a ray leaving the water, and reach a way of measuring n with nothing but a ruler.

Setting up. A coin lies at O , a depth OA vertically below A on the flat surface of water of refractive index n . Take two rays from the coin: OA straight up along the normal, which passes out undeviated, and OQ making a small angle θ with the normal.

Step 1 — apply Snell's law at Q . The ray goes from water into air, so the water term carries n and the air term carries 1. Let ϕ be the angle in air:

$$n \sin \theta = 1 \times \sin \phi$$

Step 2 — make n the subject:

$$n = \frac{\sin \phi}{\sin \theta}$$

Step 3 — locate the image. The emergent ray bends away from the normal (air is rarer, so $\phi > \theta$). Produced backwards, it meets the normal OA at a point I above O . The eye receives both rays as if they came from I , so IA is the **apparent depth**.

Step 4 — use the right-angled triangle OAQ , right-angled at A , with θ at O :

$$\tan \theta = \frac{AQ}{OA}$$

Step 5 — use the right-angled triangle IAQ , right-angled at A , with ϕ at I (the angle at I equals ϕ because IQ is the emergent ray and IA is parallel to the normal at Q):

$$\tan \phi = \frac{AQ}{IA}$$

Step 6 — restrict to a nearly normal view. If we look almost straight down, θ and ϕ are both small, and for small angles $\sin x \approx \tan x$:

$$\sin \theta \approx \tan \theta, \quad \sin \phi \approx \tan \phi$$

Step 7 — replace the sines in Step 2 by tangents:

$$n \approx \frac{\tan \phi}{\tan \theta}$$

Step 8 — substitute Steps 4 and 5:

$$n \approx \frac{AQ/IA}{AQ/OA}$$

Step 9 — divide the fractions. AQ cancels, and dividing by AQ/OA means multiplying by OA/AQ :

$$n = \frac{OA}{IA}$$

Step 10 — name the two lengths:

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$

Step 11 — how far the coin seems to rise. The shift is the difference of the two depths, and from Step 10, $IA = OA/n$:

$$\text{shift} = OA - IA = OA - \frac{OA}{n}$$

Step 12 — take out the common factor OA :

$$\text{shift} = OA\left(1 - \frac{1}{n}\right)$$

Result.

$$n = \frac{\text{real depth}}{\text{apparent depth}}, \quad \text{apparent shift} = \text{real depth}\left(1 - \frac{1}{n}\right)$$

What it means. Because $n > 1$, the apparent depth is always less than the real depth — a pond always looks shallower than it is, which is a genuine danger. Both lengths are measured with a ruler, so this gives a simple laboratory method for n .

Conditions used. The viewing must be **nearly normal** (Step 6). If you look at the water at a slant, the small-angle step fails and the simple ratio no longer holds. The surface must also be flat and the liquid uniform.

For several liquid layers stacked one above the other, each layer contributes its own apparent depth, so the total apparent depth is $\sum t_i/n_i$.

WORKED EXAMPLE 15.2

Problem. A tank contains water ($n = 4/3$) to a depth of 12 cm, and on top of the water floats a layer of oil ($n = 1.6$) of thickness 8 cm. A coin lies at the bottom. By how much does the coin appear to be raised when viewed from vertically above?

Solution. Each layer is treated separately.

$$\text{apparent depth of water layer} = \frac{12}{\frac{4}{3}} = 9 \text{ cm}$$

$$\text{apparent depth of oil layer} = \frac{8}{1.6} = 5 \text{ cm}$$

Total apparent depth = $9 + 5 = 14 \text{ cm}$, while the real depth is $12 + 8 = 20 \text{ cm}$.

$$\text{apparent shift} = 20 - 14 = 6 \text{ cm}$$

The coin appears raised by 6 cm.

15.3 Lateral shift

A **parallel-sided glass slab** (a glass block, a window pane) has two plane faces that are parallel. A ray entering it is refracted twice — once in, once out — and the two refractions cancel in direction but not in position.

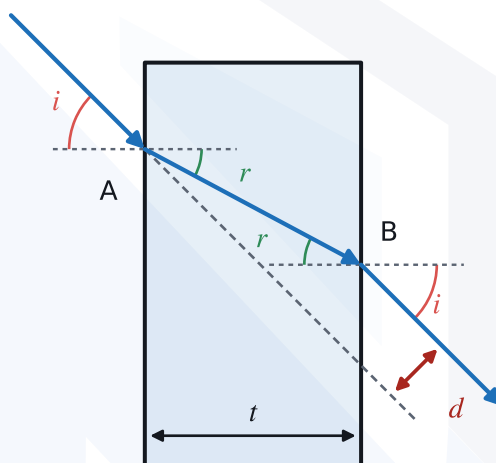


Figure 62. A ray through a parallel-sided slab of thickness t . The emergent ray is parallel to the incident ray but displaced sideways by the lateral shift d .

Because the faces are parallel, the normal at B is parallel to the normal at A . The angle of incidence inside the glass at B equals r , so by Snell's law at the second face $\sin i' = n \sin r = \sin i$, giving $i' = i$: **the emergent ray is parallel to the incident ray**. The slab does not change the direction of the light, only its line of travel. That sideways displacement is the **lateral shift**.

THE LATERAL SHIFT $d = t \frac{\sin(i-r)}{\cos r}$

We start by showing the emergent ray is parallel to the incident ray, then find how far sideways it has been pushed.

Setting up. The slab has thickness t and refractive index n . The ray enters at A with angle of incidence i and angle of refraction r , travels inside along AB , and leaves at B with angle of incidence r (inside) and angle of emergence e .

Step 1 — Snell's law at the first face, air into glass:

$$\frac{\sin i}{\sin r} = n$$

Step 2 — why the angle inside at B is also r . The two faces are parallel, so their normals are parallel. AB cuts both normals, and alternate angles between parallel lines are equal. So the angle of incidence inside the glass at B equals r .

Step 3 — Snell's law at the second face, glass into air:

$$\frac{\sin e}{\sin r} = n$$

Step 4 — compare Steps 1 and 3. Both left-hand sides equal n , so

$$\frac{\sin e}{\sin r} = \frac{\sin i}{\sin r}$$

Step 5 — cancel $\sin r$ and take the inverse sine:

$$e = i$$

So the emergent ray is **parallel** to the incident ray: the slab changes the ray's position but not its direction. That is the first part of the Group C question.

Now the shift itself.

Step 6 — define the shift. Produce the incident ray straight on, as if the glass were not there. Drop the perpendicular BN from B onto this undeviated line. Then

$$d = BN$$

Step 7 — find the angle at B in triangle ABN . AN lies along the original direction, which makes angle i with the normal at A ; AB makes angle r with the same normal. The angle between them is therefore their difference:

$$\angle BAN = i - r$$

Step 8 — use the right-angled triangle ABN , right-angled at N :

$$\sin(i - r) = \frac{BN}{AB} = \frac{d}{AB}$$

Step 9 — make d the subject:

$$d = AB \sin(i - r)$$

Step 10 — now find AB . Drop the perpendicular AM from A onto the second face; its length is the thickness of the slab, $AM = t$. In the right-angled triangle ABM the angle at A is r (both AM and the normal at A are perpendicular to the faces):

$$\cos r = \frac{AM}{AB} = \frac{t}{AB}$$

Step 11 — make AB the subject:

$$AB = \frac{t}{\cos r}$$

Step 12 — substitute this into Step 9:

$$d = \frac{t \sin(i - r)}{\cos r}$$

Result.

$$e = i \quad \text{and} \quad d = \frac{t \sin(i - r)}{\cos r}$$

What it means. A window pane does not distort the direction of what you see, only shifts it a little sideways — which is why you can read through a glass block at an angle without the scene tilting. The shift grows with the thickness t and with the angle of incidence.

Conditions used. The two faces must be **parallel** (Step 2 fails otherwise — that is why a prism does deviate light), and the same medium (air) lies on both sides.

THE EXAMINER IS LOOKING FOR

1. Snell's law written at **both** faces.
2. The statement that the angle inside at the second face is r , because the faces are parallel.
3. The conclusion $e = i$, i.e. emergent ray parallel to incident ray.
4. The construction of the perpendicular BN and the identification $\angle BAN = i - r$.
5. $AB = t/\cos r$ from the second triangle.
6. The final substitution giving $d = t \sin(i - r)/\cos r$.

Two special cases are worth noting:

- At **normal incidence** $i = r = 0$, so $d = 0$: a slab looked through straight-on shifts nothing sideways.
- At **grazing incidence** $i \rightarrow 90^\circ$, $r \rightarrow C$ (the critical angle) and $d \rightarrow t$: the shift is greatest and approaches the thickness of the slab.

For a slab viewed almost normally there is still a shift **along** the line of sight — the **normal shift**. It is the same calculation as apparent depth:

$$\text{normal shift} = t \left(1 - \frac{1}{n} \right)$$

so an object seen through a 6 cm glass block ($n = 1.5$) appears $6(1 - 1/1.5) = 2$ cm nearer than it really is.

WORKED EXAMPLE 15.3

Problem. A ray of light strikes a glass slab of thickness 6 cm and refractive index 1.5 at an angle of incidence of 60° . Calculate the angle of refraction and the lateral shift.

Solution. From Snell's law,

$$\sin r = \frac{\sin 60^\circ}{1.5} = \frac{0.8660}{1.5} = 0.5774 \quad \Rightarrow \quad r = 35.26^\circ$$

Then $i - r = 60^\circ - 35.26^\circ = 24.74^\circ$, so $\sin(i - r) = 0.4183$ and $\cos r = 0.8165$.

$$d = \frac{t \sin(i - r)}{\cos r} = \frac{6 \times 0.4183}{0.8165} = 3.07 \text{ cm}$$

The emergent ray is parallel to the incident ray but displaced sideways by about 3.1 cm.

15.4 Total internal reflection

When light goes from a **denser** medium to a **rarer** one it bends away from the normal, so $r > i$. Increase i and r grows faster, until at one particular angle of incidence r reaches 90° and the refracted ray grazes along the boundary. Beyond that angle there is no refracted ray at all: **all** the light is reflected back into the denser medium.

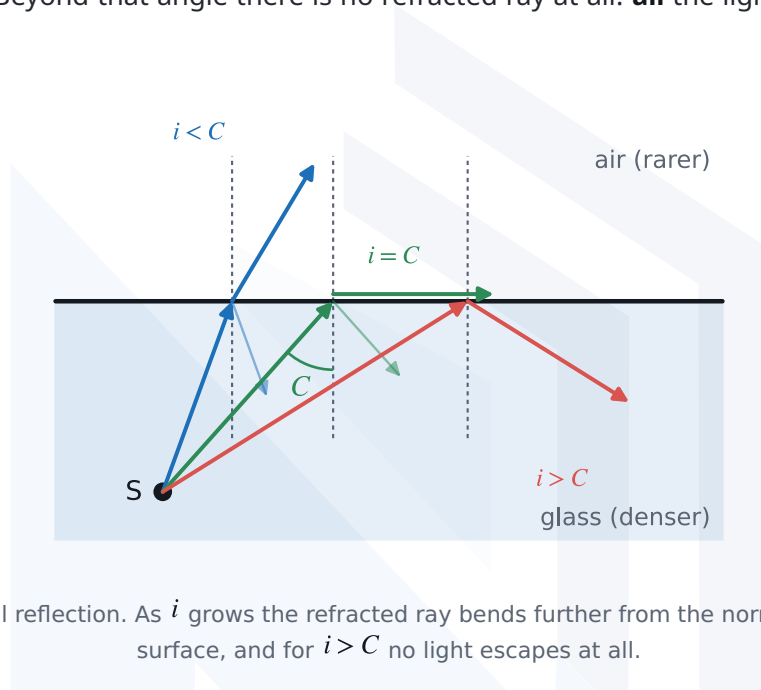


Figure 63. Total internal reflection. As i grows the refracted ray bends further from the normal; at $i = C$ it grazes the surface, and for $i > C$ no light escapes at all.

CRITICAL ANGLE AND TOTAL INTERNAL REFLECTION

The **critical angle** C for a pair of media is the angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° .

Total internal reflection is the complete reflection of light back into the denser medium, which occurs when light travelling in the denser medium strikes the boundary at an angle of incidence greater than the critical angle.

THE TWO CONDITIONS — BOTH MUST HOLD

1. Light must travel from the **denser** medium towards the **rarer** medium.
2. The angle of incidence must be **greater than the critical angle**.

THE CRITICAL ANGLE: $\sin C = 1/n$

We start from Snell's law at the one special angle where the refracted ray just grazes the boundary, and reach a relation between the critical angle and the refractive index.

Setting up. Light travels in a denser medium of absolute index n_1 and meets a rarer medium of index n_2 . By the definition of the critical angle, when the angle of incidence is C the angle of refraction is exactly 90° .

Step 1 — write Snell's law for this boundary:

$$n_1 \sin i = n_2 \sin r$$

Step 2 — substitute the critical condition $i = C$ **and** $r = 90^\circ$:

$$n_1 \sin C = n_2 \sin 90^\circ$$

Step 3 — evaluate $\sin 90^\circ = 1$:

$$n_1 \sin C = n_2$$

Step 4 — divide both sides by n_1 :

$$\sin C = \frac{n_2}{n_1}$$

Step 5 — take the usual case of a denser medium in air. Then $n_2 = 1$ and we write $n_1 = n$:

$$\sin C = \frac{1}{n}$$

Step 6 — invert to get n **from a measured** C :

$$n = \frac{1}{\sin C}$$

Result.

$$\sin C = \frac{n_2}{n_1}, \quad \text{and against air} \quad \sin C = \frac{1}{n} \quad \text{or} \quad n = \frac{1}{\sin C}$$

What it means. A bigger n makes $1/n$ smaller, so a denser material has a smaller critical angle and traps light more easily — this is exactly why diamond sparkles. Step 4 also explains why total internal reflection can never happen from rarer to denser: then $n_2/n_1 > 1$, and no angle has a sine greater than 1, so no critical angle exists.

Conditions used. Light must be going from denser to rarer, and one wavelength at a time (since n , and hence C , depends slightly on colour).

Denser medium (against air)	n	Critical angle C
Ice	1.31	49.8°
Water	1.33	48.6°
Crown glass	1.52	41.1°
Dense flint glass	1.65	37.3°
Diamond	2.42	24.4°

The smaller the critical angle, the easier it is to trap light. Diamond has the smallest C of common materials, so light entering a cut diamond is reflected many times inside before it escapes — that is its "fire".

Applications

Application	How it uses total internal reflection
Optical fibre	A glass core of higher n is coated with cladding of lower n ; light striking the core-cladding boundary above C is reflected over and over and travels kilometres with little loss. Used for internet backbones and endoscopes.
Totally reflecting prism	A 45° – 45° – 90° glass prism has $C = 41.8^\circ < 45^\circ$, so it turns light through 90° or 180° with no silvering and no loss. Used in periscopes and binoculars.
Mirage	Hot air just above a road is less dense, so its n is smaller. Light from the sky curves, is totally internally reflected in the air layers, and the road looks wet.
Shine of a diamond	$C = 24.4^\circ$, so most rays entering are trapped and emerge through the top facets.
Optical levelling and light pipes	Light follows a bent glass or plastic rod.

TOTAL INTERNAL REFLECTION IS NOT "PARTIAL REFLECTION"

At every boundary a little light is reflected, whatever the angle. That is ordinary partial reflection. Total internal reflection is different: past the critical angle **100 %** of the light is reflected because a refracted ray is geometrically impossible. It also cannot happen when light goes from rarer to denser — there is no i for which r can reach 90° .

WORKED EXAMPLE 15.4

Problem. A fish is at a depth of 2.0 m in Phewa Lake ($n_{\text{water}} = 4/3$). Looking up, the fish sees the whole sky compressed into a bright circular window on the surface. Find (a) the critical angle for water and (b) the radius of that window.

Solution.

(a) $\sin C = 1/n = 3/4 = 0.75$, so

$$C = \sin^{-1}(0.75) = 48.6^\circ$$

(b) Light can enter the water from the sky only within the cone of half-angle C about the vertical through the fish. On the surface this cone cuts a circle of radius R where

$$\tan C = \frac{R}{h} \quad \Rightarrow \quad R = h \tan C$$

With $\sin C = 0.75$, $\cos C = \sqrt{1 - 0.5625} = 0.6614$, so $\tan C = 0.75/0.6614 = 1.134$. Hence

$$R = 2.0 \times 1.134 = 2.27 \text{ m}$$

The window is about 4.5 m across; outside it the fish sees only the reflected lake bed.

✓ Chapter summary

- Laws of refraction: incident ray, refracted ray and normal are coplanar; and $\frac{\sin i}{\sin r} = {}_1n_2$ (Snell's law), usually written $n_1 \sin i = n_2 \sin r$.
- Absolute refractive index $n = c/v$; also ${}_1n_2 = n_2/n_1 = v_1/v_2 = \lambda_1/\lambda_2$. Frequency never changes on refraction.
- Reversibility gives ${}_1n_2 = 1/{}_2n_1$, and the chain rule gives ${}_1n_2 \times {}_2n_3 \times {}_3n_1 = 1$.
- $n = \frac{\text{real depth}}{\text{apparent depth}}$ for near-normal viewing; apparent shift $= t(1 - 1/n)$, and layers add: apparent depth $= \sum t_i/n_i$.
- A parallel-sided slab leaves the direction unchanged but displaces the ray laterally by $d = \frac{t \sin(i - r)}{\cos r}$, which is zero at normal incidence.
- Total internal reflection needs light going denser $\rightarrow$ rarer with $i > C$, where $\sin C = n_2/n_1$, or $\sin C = 1/n$ against air.
- Optical fibres, totally reflecting prisms, mirages and the sparkle of diamond are all total internal reflection.

Practice questions

Group A — Multiple choice (1 mark each)

- The refractive index of a medium in which light travels at $2.0 \times 10^8 \text{ m s}^{-1}$ is [1]
 a 1.33 b 1.5 c 1.6 d 2.0
- When a ray of light passes from glass into air it [1]
 a bends towards the normal b bends away from the normal
 c is not deviated d is always totally reflected
- The critical angle for a glass of refractive index 1.5 is about [1]
 a 30° b 42° c 49° d 60°
- The lateral shift produced by a parallel-sided glass slab is zero when the angle of incidence is [1]
 a 0° b 30° c 45° d 90°
- A coin lies at the bottom of a beaker containing water ($n = 4/3$) 8 cm deep. Seen from directly above it appears to be at a depth of [1]
 a 4 cm b 6 cm c 8 cm d 10.7 cm
- On passing from air into glass, the quantity that does **not** change is the [1]
 a speed b wavelength c frequency d direction

ANSWERS TO GROUP A

- (b) — $n = c/v = (3 \times 10^8)/(2 \times 10^8) = 1.5$.

2. (b) — denser to rarer, so $r > i$.
3. (b) — $\sin C = 1/1.5 = 0.667$, $C = 41.8^\circ$.
4. (a) — $d = t \sin(i - r) / \cos r$ and $i = r = 0$ gives $d = 0$.
5. (b) — apparent depth $= 8/(4/3) = 6$ cm.
6. (c) — frequency is set by the source; speed and wavelength both fall by the factor n .

Group B — Short answer (5 marks each)

1. State the laws of refraction of light. Define absolute refractive index and relative refractive index, and write the relation between them. [5]
2. Using the principle of reversibility of light, show that ${}_1n_2 = 1/{}_2n_1$, and hence prove ${}_1n_2 \times {}_2n_3 \times {}_3n_1 = 1$. [5]
3. Derive the relation $n = \frac{\text{real depth}}{\text{apparent depth}}$ for a liquid viewed normally from above. [5]
4. A ray of light falls at 45° on a glass slab 8 cm thick of refractive index 1.5. Calculate the angle of refraction and the lateral shift. [5]
5. Define critical angle and derive $\sin C = 1/n$. Explain with a diagram how an optical fibre carries light around bends. [5]

ANSWERS TO GROUP B

1. Laws and definitions as in §15.1; ${}_1n_2 = n_2/n_1$.
2. See §15.2 (b) and (c). Reversing the ray interchanges i and r , so ${}_2n_1 = \sin r / \sin i = 1/{}_1n_2$. Writing each relative index as a ratio of absolute indices, ${}_1n_2 \cdot {}_2n_3 \cdot {}_3n_1 = \frac{n_2}{n_1} \cdot \frac{n_3}{n_2} \cdot \frac{n_1}{n_3} = 1$.
3. See the derivation in §15.2.
4. $\sin r = \sin 45^\circ / 1.5 = 0.7071 / 1.5 = 0.4714$, so $r = 28.13^\circ$. Then $i - r = 16.87^\circ$, $\sin(i - r) = 0.2903$, $\cos r = 0.8819$, and

$$d = \frac{8 \times 0.2903}{0.8819} = 2.63 \text{ cm}$$

5. Definition and derivation as in §15.4. In a fibre the core has a larger refractive index than the cladding, so light that enters nearly along the axis always meets the core-cladding wall at an angle greater than C and is totally reflected, again and again, following the fibre even round bends.

Group C — Long answer (8 marks each)

1. (a) Show that a ray emerging from a parallel-sided glass slab is parallel to the incident ray, and derive an expression for the lateral shift. [5]
 (b) A slab of glass of refractive index 1.5 is 10 cm thick. Find the lateral shift when light strikes it at 30° . [3]
2. (a) Define total internal reflection and state the conditions under which it occurs. Derive the relation between the critical angle and the refractive indices of the two media. [4]
 (b) Calculate the critical angle for a glass–water interface, given $n_{\text{glass}} = 1.5$ and $n_{\text{water}} = 1.33$. [2]

- (c) An optical fibre has a core of refractive index 1.52 and cladding of refractive index 1.48. Find the critical angle at the core-cladding boundary. [2]

ANSWERS TO GROUP C

1. (a) See §15.3. (b) $\sin r = \sin 30^\circ / 1.5 = 0.5 / 1.5 = 0.3333$, so $r = 19.47^\circ$. Then $i - r = 10.53^\circ$, $\sin(i - r) = 0.1828$ and $\cos r = 0.9428$, giving

$$d = \frac{10 \times 0.1828}{0.9428} = 1.94 \text{ cm}$$

2. (a) See §15.4; $\sin C = n_2/n_1$ from $n_1 \sin C = n_2 \sin 90^\circ$.

(b) $\sin C = n_{\text{water}}/n_{\text{glass}} = 1.33/1.5 = 0.8867$, so $C = 62.5^\circ$.

(c) $\sin C = 1.48/1.52 = 0.9737$, so $C = 76.8^\circ$. Any ray meeting the wall at more than 76.8° to the normal stays inside the core.

UNIT 16

Refraction through prisms

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Minimum deviation condition
- Relation between angle of prism, minimum deviation and refractive index
- Deviation in small angle prism

A prism is a block of glass with two plane faces inclined to each other. Because the two faces are not parallel, the two refractions no longer cancel as they did in a slab: the ray leaves in a new direction, bent towards the thick part of the glass. That bending is the **angle of deviation**, and it turns out to be smallest when the light passes through the prism symmetrically. Measuring that smallest deviation is the standard laboratory method of finding the refractive index of a transparent solid, and it is the basis of the spectrometer.

THE FOUR-FORMULA UNIT

$A = r_1 + r_2$, $\delta = i_1 + i_2 - A$, $n = \frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$ and $\delta = (n-1)A$ for a thin prism. Learn the derivations, not just the results — NEB awards marks for each labelled step.

16.1 Minimum deviation condition

The two faces through which light passes are the **refracting surfaces**; the angle between them is the **refracting angle** or **angle of the prism**, A . Their line of intersection is the **refracting edge** and the third face is the **base**. A section perpendicular to the refracting edge is the **principal section** — every diagram in this chapter is a principal section.

Symbol	Name	Where it is measured
A	refracting angle (angle of the prism)	between the two refracting faces
i_1	angle of incidence	first face, in air
r_1	angle of refraction	first face, in glass
r_2	angle of incidence inside	second face, in glass
i_2	angle of emergence	second face, in air
δ	angle of deviation	between incident and emergent directions

ANGLE OF DEVIATION

The **angle of deviation** of a prism is the angle between the direction of the incident ray produced forward and the direction of the emergent ray produced backward. It measures how far the prism has turned the light from its original path.

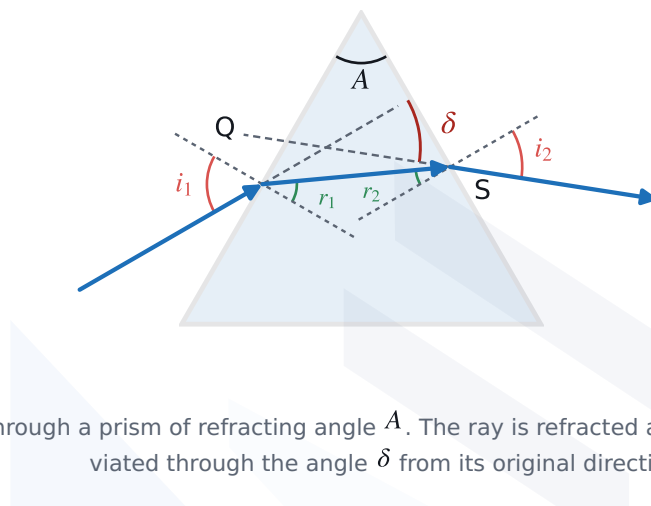


Figure 64. Refraction through a prism of refracting angle A . The ray is refracted at Q and again at S , and leaves deviated through the angle δ from its original direction.

THE TWO PRISM RELATIONS, $A = r_1 + r_2$ AND $\delta = i_1 + i_2 - A$

We start from pure geometry in the ray diagram — no optics at all in part (a) — and reach the two equations that every prism question is built on.

Setting up. The ray strikes the first face at Q with angle of incidence i_1 and is refracted at r_1 . Inside the glass it travels to S on the second face, meets the normal there at r_2 , and emerges at i_2 . Let N be the point where the two normals (at Q and at S) cross. A is the refracting angle of the prism, at the vertex where the two faces meet.

Part (a): $A = r_1 + r_2$

Step 1 — look at the quadrilateral $AQNS$. NQ is the normal at Q , so it is perpendicular to the first face; NS is the normal at S , so it is perpendicular to the second face. Therefore

$$\angle AQN = 90^\circ, \quad \angle ASN = 90^\circ$$

Step 2 — add up the four angles of the quadrilateral. They total 360° :

$$A + 90^\circ + \angle QNS + 90^\circ = 360^\circ$$

Step 3 — subtract 180° from both sides:

$$A + \angle QNS = 180^\circ$$

Step 4 — now look at triangle QNS . Its three angles are r_1 at Q , r_2 at S , and $\angle QNS$ at N , and they add to 180° :

$$r_1 + r_2 + \angle QNS = 180^\circ$$

Step 5 — compare Steps 3 and 4. Both left-hand sides equal 180° :

$$A + \angle QNS = r_1 + r_2 + \angle QNS$$

Step 6 — cancel $\angle QNS$ from both sides:

$$A = r_1 + r_2$$

Part (b): $\delta = i_1 + i_2 - A$

Step 7 — how much the ray turns at the first face. It arrives along a direction making i_1 with the normal and leaves making r_1 with the same normal, so it has been turned through

$$\delta_1 = i_1 - r_1$$

Step 8 — how much it turns at the second face. Same argument, in the same rotational sense:

$$\delta_2 = i_2 - r_2$$

Step 9 — add the two turns. Both bend the ray the same way, so the total deviation δ (the exterior angle of the triangle formed by the incident ray produced and the emergent ray produced backwards) is their sum:

$$\delta = (i_1 - r_1) + (i_2 - r_2)$$

Step 10 — group the i 's and the r 's:

$$\delta = (i_1 + i_2) - (r_1 + r_2)$$

Step 11 — replace $(r_1 + r_2)$ by A , using Part (a):

$$\delta = i_1 + i_2 - A$$

Result.

$$A = r_1 + r_2, \quad \delta = i_1 + i_2 - A$$

What it means. The first relation is purely geometrical — it holds whatever the glass is made of. The second says the prism's total bending is shared between the two faces. Together they reduce four unknown angles to two, which is why every prism numerical is solvable.

Conditions used. The ray must lie in a plane perpendicular to the refracting edge (the "principal section"), and it must actually emerge at the second face — if r_2 exceeds the critical angle, there is no i_2 and these relations cannot be used.

How the deviation varies with the angle of incidence

Keep the prism and the colour of light fixed and slowly increase i_1 . Each value of i_1 fixes r_1 , hence $r_2 = A - r_1$, hence i_2 , hence δ . Plotting δ against i_1 gives a curve with a single minimum.

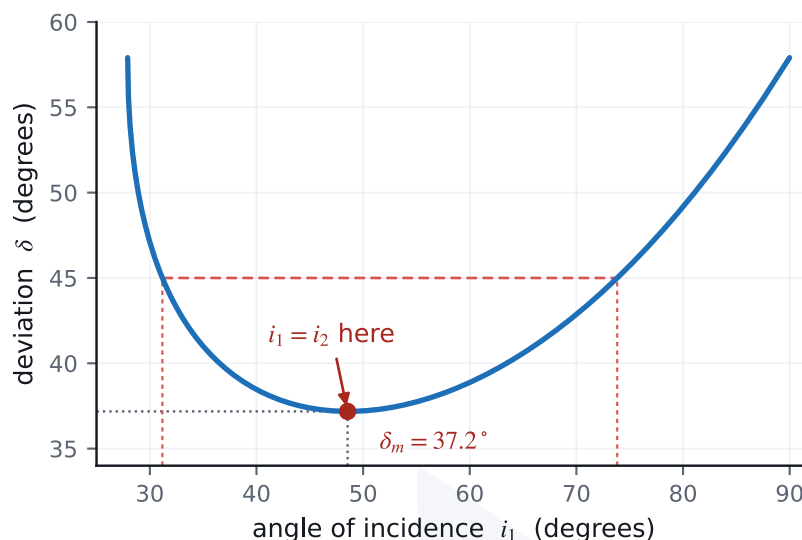


Figure 65. Deviation against angle of incidence for an equilateral prism with $n = 1.5$. Any deviation above the minimum is produced by two different angles of incidence; at the minimum the two coincide.

AT MINIMUM DEVIATION THE RAY PASSES SYMMETRICALLY

We start from the shape of the δ - i_1 graph together with the principle of reversibility, and reach the symmetry condition $i_1 = i_2$, $r_1 = r_2$.

Step 1 — read the graph. Draw any horizontal line above the lowest point of the curve. It cuts the curve **twice**. So for every deviation δ greater than the minimum there are **two** different angles of incidence that produce the very same deviation. Call them i and i' .

Step 2 — apply the principle of reversibility. If a ray enters at i_1 and emerges at i_2 , then reversing its direction sends a ray in at i_2 which emerges at i_1 — along the same path, so with the same deviation δ .

Step 3 — identify the two roots. Step 2 gives us two angles of incidence with the same δ , namely i_1 and i_2 . Step 1 says there are only two. So the pair in Step 1 must be

$$i = i_1, \quad i' = i_2$$

Step 4 — now slide the horizontal line down. As δ decreases towards δ_m , the two intersection points move towards each other. At the minimum itself the line touches the curve at exactly **one** point, so the two roots have merged:

$$i_1 = i_2$$

Step 5 — carry this inside the prism. Snell's law at each face gives $\sin i_1 = n \sin r_1$ and $\sin i_2 = n \sin r_2$. Equal i 's therefore force equal r 's:

$$r_1 = r_2$$

Result. At minimum deviation the ray passes symmetrically, $i_1 = i_2$ and $r_1 = r_2$, and therefore the refracted ray inside the prism runs **parallel to the base**.

What it means. Symmetry is not an extra assumption we make for convenience — it is forced on us by the shape of the curve. This is why the prism formula of the next section is exact, not approximate.

Conditions used. One prism, one colour, and a ray that actually emerges from the second face.

THE MINIMUM DEVIATION CONDITION

Deviation is a minimum when the ray passes through the prism **symmetrically**:

$$i_1 = i_2 = i, \quad r_1 = r_2 = r = \frac{A}{2}$$

and the refracted ray inside the prism is then **parallel to the base** of the prism.

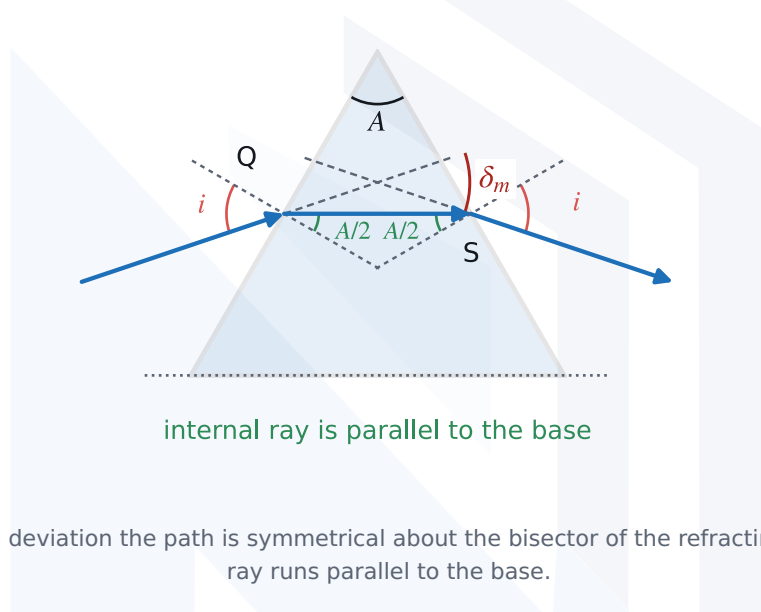


Figure 66. At minimum deviation the path is symmetrical about the bisector of the refracting angle, and the internal ray runs parallel to the base.

MINIMUM DEVIATION IS NOT ZERO DEVIATION

A prism always deviates light — δ can never be made zero (unlike a parallel-sided slab, where it always is). "Minimum" means the smallest value the deviation can be pushed down to, and it is reached at one particular angle of incidence only.

When no light emerges

At the second face the internal ray meets the glass–air boundary at r_2 . If r_2 exceeds the critical angle C the ray is totally internally reflected and never leaves through that face. Since r_1 can never exceed C (its largest value, at grazing incidence $i_1 = 90^\circ$, is exactly C), the smallest possible r_2 is $A - C$. So light can emerge only if $A - C \leq C$, that is

$$A \leq 2C$$

For crown glass, $C = 41.8^\circ$, so no light emerges through the second face of a prism whose refracting angle exceeds about 83.6° .

16.2 Relation between angle of prism, minimum deviation and refractive index

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

We start from the two prism relations, add the fact that at minimum deviation the ray passes symmetrically, and reach a formula for n from two measurable angles.

Setting up. At minimum deviation the path through the prism is **symmetrical**: the ray inside runs parallel to the base, the angle of incidence equals the angle of emergence, and the two internal angles are equal. So

$$i_1 = i_2 = i, \quad r_1 = r_2 = r, \quad \delta = \delta_m$$

Step 1 — put the equal internal angles into $A = r_1 + r_2$:

$$A = r + r = 2r$$

Step 2 — divide both sides by 2:

$$r = \frac{A}{2}$$

Step 3 — put the equal external angles into $\delta = i_1 + i_2 - A$:

$$\delta_m = i + i - A = 2i - A$$

Step 4 — add A to both sides:

$$A + \delta_m = 2i$$

Step 5 — divide both sides by 2:

$$i = \frac{A + \delta_m}{2}$$

Step 6 — apply Snell's law at the first face, where light goes from air into glass:

$$n = \frac{\sin i}{\sin r}$$

Step 7 — substitute the expressions for i and r from Steps 5 and 2:

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

Result.

$$n = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

What it means. This is the **prism formula**, and it is the standard accurate method for finding n . Everything on the right is an angle, and a spectrometer measures angles to within a few minutes of arc — far more precisely than we could measure a length or a depth. Note also that a larger δ_m

for the same prism means a larger n , which is why violet light (deviated most) has the largest refractive index.

Conditions used. The prism must be set at **minimum deviation**, so that the path is symmetrical; monochromatic light must be used, since n and δ_m both depend on colour; and the surrounding medium must be air.

THE EXAMINER IS LOOKING FOR

1. A labelled diagram with i_1 , r_1 , r_2 , i_2 , A and δ marked.
2. The two prism relations quoted or derived.
3. The statement of the symmetry condition at minimum deviation, $i_1 = i_2$ and $r_1 = r_2$ — usually worth a mark on its own.
4. $r = A/2$ and $i = (A + \delta_m)/2$, each shown as a separate line.
5. Snell's law at the first face, and the final substitution.

THE LABORATORY ROUTINE

On a spectrometer, rotate the prism table slowly while watching the refracted image through the telescope. The image moves one way, stops, and turns back. The turning point is the position of minimum deviation — that "stop and go back" moment is what the examiner wants described.

ORDER OF ATTACK FOR ANY PRISM NUMERICAL

1. From i_1 find r_1 using $\sin i_1 = n \sin r_1$.
2. From $A = r_1 + r_2$ find r_2 .
3. From $n \sin r_2 = \sin i_2$ find i_2 .
4. From $\delta = i_1 + i_2 - A$ find the deviation.

Reverse the chain if you are given δ_m : $r = A/2$, $i = (A + \delta_m)/2$, then $n = \sin i / \sin r$. Never mix the two chains in one question.

WORKED EXAMPLE 16.1

Problem. The angle of minimum deviation of an equilateral glass prism is 30° . Find the refractive index of the glass and the angle of incidence at minimum deviation.

Solution. An equilateral prism has $A = 60^\circ$, and $\delta_m = 30^\circ$.

$$n = \frac{\sin\left(\frac{60^\circ + 30^\circ}{2}\right)}{\sin\left(\frac{60^\circ}{2}\right)} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{0.7071}{0.5} = 1.414$$

So $n = \sqrt{2} \approx 1.41$. The angle of incidence is

$$i = \frac{A + \delta_m}{2} = \frac{60^\circ + 30^\circ}{2} = 45^\circ$$

and the angle of refraction inside is $r = A/2 = 30^\circ$.

WORKED EXAMPLE 16.2

Problem. An equilateral prism is made of glass of refractive index 1.5. Calculate its angle of minimum deviation.

Solution. Here $A = 60^\circ$ and $n = 1.5$. Rearranging the prism formula,

$$\sin\left(\frac{A + \delta_m}{2}\right) = n \sin \frac{A}{2} = 1.5 \times \sin 30^\circ = 1.5 \times 0.5 = 0.75$$

$$\frac{A + \delta_m}{2} = \sin^{-1}(0.75) = 48.59^\circ$$

$$\delta_m = 2 \times 48.59^\circ - 60^\circ = 37.18^\circ \approx 37.2^\circ$$

The light must strike the first face at 48.6° for this to happen.

WORKED EXAMPLE 16.3

Problem. A ray strikes one refracting face of an equilateral prism ($n = 1.5$) at an angle of incidence of 45° . Find the angle of refraction at the first face, the angle of incidence at the second face, the angle of emergence, and the deviation. Compare the deviation with the minimum value found in Example 16.2.

Solution. At the first face,

$$\sin r_1 = \frac{\sin 45^\circ}{1.5} = \frac{0.7071}{1.5} = 0.4714 \quad \Rightarrow \quad r_1 = 28.13^\circ$$

Using $A = r_1 + r_2$,

$$r_2 = 60^\circ - 28.13^\circ = 31.87^\circ$$

At the second face, glass to air,

$$\sin i_2 = n \sin r_2 = 1.5 \times \sin 31.87^\circ = 1.5 \times 0.5280 = 0.7921$$

$$i_2 = 52.38^\circ$$

The deviation is

$$\delta = i_1 + i_2 - A = 45^\circ + 52.38^\circ - 60^\circ = 37.38^\circ$$

This is slightly **greater** than $\delta_m = 37.18^\circ$, as it must be — and it is only 0.2° greater even though i_1 is 3.6° away from the symmetric value. Near the minimum the curve is very flat, which is exactly why the minimum position is easy to find on a spectrometer.

16.3 Deviation in small angle prism

A prism whose refracting angle is only a few degrees is called a **thin** or **small-angle** prism. For such a prism a beautifully simple result holds.

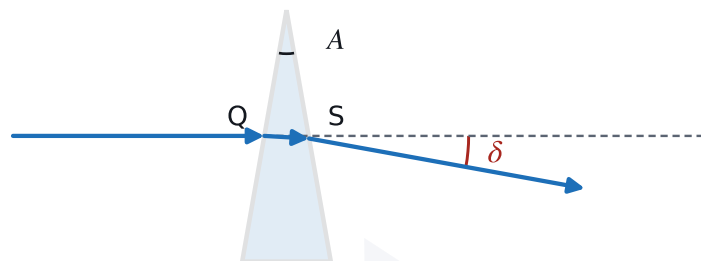


Figure 67. A thin prism of refracting angle $A = 20^\circ$. The emergent ray is turned through $\delta \approx (n - 1)A$ from the incident direction, almost independently of how the ray enters.

$\delta = (n - 1)A$ **FOR A THIN PRISM**

We start from Snell's law at the two faces with all angles small, and reach a deviation formula that no longer contains the angle of incidence at all.

Setting up. The refracting angle A is small (a few degrees) and the ray enters nearly normally. Then $A = r_1 + r_2$ forces r_1 and r_2 to be small, and Snell's law then makes i_1 and i_2 small too. All four are small angles measured in radians.

Step 1 — Snell's law at the first face:

$$\frac{\sin i_1}{\sin r_1} = n$$

Step 2 — use the small-angle approximation $\sin \theta \approx \theta$:

$$\frac{i_1}{r_1} = n$$

Step 3 — make i_1 the subject:

$$i_1 = nr_1$$

Step 4 — repeat the same three steps at the second face:

$$i_2 = nr_2$$

Step 5 — substitute both into $\delta = i_1 + i_2 - A$:

$$\delta = nr_1 + nr_2 - A$$

Step 6 — take out the common factor n :

$$\delta = n(r_1 + r_2) - A$$

Step 7 — replace $(r_1 + r_2)$ **by** A , using the first prism relation:

$$\delta = nA - A$$

Step 8 — take out the common factor A :

$$\delta = (n - 1)A$$

Result.

$$\delta = (n - 1)A$$

What it means. The angle of incidence has vanished from the answer. For a thin prism the deviation is fixed by the prism itself — its angle and its glass — so you do not need to hunt for the minimum-deviation position. And because n depends on colour, so does δ : this one line is the seed of the whole theory of dispersion.

Conditions used. A must be small (in practice under about 10°) and the ray must strike nearly normally, so that $\sin \theta \approx \theta$ is safe in Step 2. For a thick prism or a slanting ray the formula fails and the full prism formula must be used.

Three consequences matter:

1. **δ does not depend on i_1 .** For a thin prism the deviation is essentially the same however the ray is aimed (within small angles), so there is no need to set it at minimum deviation.
2. **δ depends on n , and n depends on colour.** Violet light, with the larger n , is deviated more than red. This is the origin of dispersion (Unit 18), where the angular dispersion is $\delta_V - \delta_R = (n_V - n_R)A$.
3. **Thin prisms add.** Two thin prisms placed together with their refracting edges the same way deviate light by $\delta_1 + \delta_2$; reversing one subtracts. This is how achromatic prism combinations are designed, and how optometrists in Kathmandu correct squint with "prism dioptres".

WORKED EXAMPLE 16.4

Problem. A thin prism of refracting angle 5° is made of glass of refractive index 1.52. Find the deviation it produces. What angle of prism would be needed to produce the same deviation with glass of $n = 1.65$?

Solution. For a thin prism,

$$\delta = (n - 1)A = (1.52 - 1) \times 5^\circ = 0.52 \times 5^\circ = 2.6^\circ$$

For the second glass, requiring the same $\delta = 2.6^\circ$,

$$A' = \frac{\delta}{n' - 1} = \frac{2.6^\circ}{1.65 - 1} = \frac{2.6^\circ}{0.65} = 4^\circ$$

A denser glass needs a smaller refracting angle for the same bending — which is why high-index glass is used where a prism must be kept thin and light.

✓ Chapter summary

- For a prism of refracting angle A : $A = r_1 + r_2$ and $\delta = i_1 + i_2 - A$.
- The graph of δ against i_1 has one minimum. Above the minimum, two angles of incidence give the same deviation; at the minimum they coincide.
- At minimum deviation the passage is symmetrical: $i_1 = i_2$, $r_1 = r_2 = A/2$, and the internal ray is parallel to the base.
- Prism formula: $n = \frac{\sin(\frac{A+\delta_m}{2})}{\sin(\frac{A}{2})}$, with $i = \frac{A+\delta_m}{2}$ at minimum deviation.
- Light can emerge through the second face only if $A \leq 2C$, where C is the critical angle; otherwise total internal reflection occurs inside the prism.
- For a thin prism, $\delta = (n - 1)A$, independent of the angle of incidence, and thin-prism deviations simply add.

Practice questions

Group A — Multiple choice (1 mark each)

- For a prism of refracting angle A at minimum deviation δ_m , the angle of incidence is [1]
 - $A/2$
 - $\frac{A + \delta_m}{2}$
 - $\delta_m/2$
 - $A + \delta_m$
- At minimum deviation the ray inside the prism is [1]
 - parallel to the base
 - normal to the base
 - parallel to a refracting face
 - along the base
- The deviation produced by a thin prism of refracting angle 6° made of glass of $n = 1.5$ is [1]
 - 2°
 - 3°
 - 4°
 - 9°
- The refractive index of the material of an equilateral prism whose minimum deviation is 30° is [1]
 - 1.33
 - 1.41
 - 1.50
 - 1.73
- In a prism, the sum of the two angles of refraction $r_1 + r_2$ is equal to [1]
 - the angle of incidence
 - the angle of the prism
 - the angle of deviation
 - 90°
- A ray cannot emerge from the second face of a prism if the refracting angle A is [1]
 - less than C
 - equal to C
 - greater than $2C$
 - less than $2C$

ANSWERS TO GROUP A

- (b) — from $\delta_m = 2i - A$.
- (a) — symmetrical passage makes the internal ray parallel to the base.
- (b) — $\delta = (n - 1)A = 0.5 \times 6^\circ = 3^\circ$.
- (b) — $n = \sin 45^\circ / \sin 30^\circ = 1.414$.

5. (b) — $A = r_1 + r_2$.

6. (c) — then $r_2 = A - r_1 > C$ for every r_1 , so the ray is totally internally reflected.

Group B — Short answer (5 marks each)

1. Define the refracting angle of a prism, the angle of deviation and the angle of minimum deviation. With a labelled diagram, show that $\delta = i_1 + i_2 - A$. [5]
2. Derive the relation $n = \frac{\sin(\frac{A + \delta_m}{2})}{\sin(\frac{A}{2})}$ between the angle of a prism, its angle of minimum deviation and the refractive index of its material. [5]
3. Show that for a prism of small refracting angle the deviation is $\delta = (n - 1)A$ and is independent of the angle of incidence. [5]
4. Calculate the angle of minimum deviation of an equilateral prism made of glass of refractive index 1.6. [5]
5. The angle of minimum deviation of a prism of refracting angle 60° is 40° . Find the refractive index of its material and the angle of incidence at minimum deviation. [5]

ANSWERS TO GROUP B

1. Definitions and derivation as in §16.1.
2. See the derivation in §16.2.
3. See the derivation in §16.3.
4. $\sin\left(\frac{60^\circ + \delta_m}{2}\right) = 1.6 \sin 30^\circ = 0.8$, so $\frac{60^\circ + \delta_m}{2} = 53.13^\circ$ and $\delta_m = 106.26^\circ - 60^\circ = 46.26^\circ \approx 46.3^\circ$.
5. $n = \frac{\sin\left(\frac{60^\circ + 40^\circ}{2}\right)}{\sin 30^\circ} = \frac{\sin 50^\circ}{0.5} = \frac{0.7660}{0.5} = 1.532$. The angle of incidence is $i = \frac{A + \delta_m}{2} = 50^\circ$.

Group C — Long answer (8 marks each)

1. (a) Draw the path of a monochromatic ray through a glass prism and prove that $A = r_1 + r_2$ and $\delta = i_1 + i_2 - A$. [4]
 (b) Sketch the graph of deviation against angle of incidence, and use it to show that at minimum deviation the ray passes symmetrically. Hence derive the prism formula. [4]
2. A ray of light is incident at 40° on one refracting face of an equilateral prism of refractive index 1.5. Calculate
 (a) the angle of refraction at the first face,
 (b) the angle of incidence at the second face,
 (c) the angle of emergence, and
 (d) the angle of deviation. Show that this deviation is greater than the minimum deviation of the same prism. [8]

ANSWER TO GROUP C QUESTION 2

- (a) $\sin r_1 = \frac{\sin 40^\circ}{1.5} = \frac{0.6428}{1.5} = 0.4285$, so $r_1 = 25.37^\circ$.
- (b) $r_2 = A - r_1 = 60^\circ - 25.37^\circ = 34.63^\circ$.

(c) $\sin i_2 = 1.5 \sin 34.63^\circ = 1.5 \times 0.5682 = 0.8522$, so $i_2 = 58.47^\circ$.

(d) $\delta = i_1 + i_2 - A = 40^\circ + 58.47^\circ - 60^\circ = 38.47^\circ$.

For the same prism, $\sin\left(\frac{60^\circ + \delta_m}{2}\right) = 1.5 \sin 30^\circ = 0.75$, giving $\delta_m = 37.18^\circ$. Since $38.47^\circ > 37.18^\circ$, the deviation here is indeed above the minimum — as it must be for any angle of incidence other than 48.59° .



UNIT 17

Lenses

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Spherical lenses, angular magnification
- Lens maker's formula
- Power of a lens

A lens is a piece of transparent material bounded by two surfaces, at least one of which is curved. Because the two surfaces refract light one after the other, a lens can gather a diverging bundle of rays back to a point, or spread a parallel bundle out. Spectacles, the camera in a phone, a hand lens, the objective of a telescope on a Pokhara rooftop — all of them are the same three formulas applied over and over.

WHAT THE EXAMINER ASKS FOR

Three things carry almost all the marks in this unit: a correctly drawn ray diagram with the sign convention respected, the derivation of the lens maker's formula from refraction at two spherical surfaces, and numerical work with $P = 1/f$ in dioptres. Learn those and the unit is done.

17.1 Spherical lenses, angular magnification

Types and basic terms

A **spherical lens** is bounded by two surfaces that are parts of spheres (a flat face is a sphere of infinite radius). Lenses fall into two families.

CONVERGING (convex)



Bi-convex

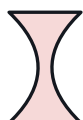


Plano-convex



Convex meniscus

DIVERGING (concave)



Bi-concave



Plano-concave



Concave meniscus

Figure 68. The six common thin lenses. A lens thicker at the centre than at the rim converges light; one thinner at the centre diverges it.

Term	Meaning
Centres of curvature C_1, C_2	centres of the spheres of which the two faces are parts
Radii of curvature R_1, R_2	radii of those spheres
Principal axis	the line joining C_1 and C_2
Optical centre O	point inside the lens through which a ray passes undeviated
Principal focus F	point where a beam parallel to the axis converges (convex) or appears to diverge from (concave)
Focal length f	distance OF
Aperture	effective diameter of the lens

A lens has **two** foci, one on each side, at equal distances from O when the same medium lies on both sides.

SIGN CONVENTION (NEW CARTESIAN)

All distances are measured from the optical centre. Distances measured **in the direction of the incident light are positive**, those measured against it are negative. Heights **above** the principal axis are positive, below it negative. Hence a real object on the left always has u negative; f is positive for a convex lens and negative for a concave lens.

Image formation

Only two of these three standard rays are needed for any construction:

1. A ray parallel to the principal axis passes, after refraction, through the second focus F_2 (convex) or appears to come from F_1 (concave).
2. A ray through the optical centre goes straight on, undeviated.
3. A ray through the first focus F_1 emerges parallel to the principal axis.

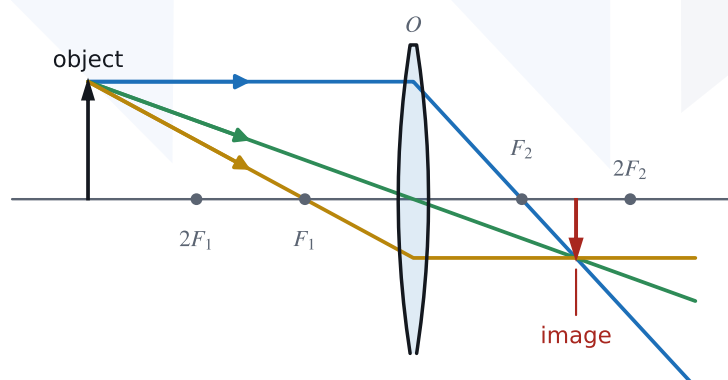


Figure 69. Convex lens with the object beyond $2F_1$. The three standard rays meet at the image, which is real, inverted and diminished.

The results are summarised by the **thin lens formula**, valid with the sign convention above:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

and the **linear (transverse) magnification**

$$m = \frac{h_i}{h_o} = \frac{v}{u}$$

A negative m means the image is inverted (and therefore real); a positive m means erect and virtual.

Position of object (convex lens)	Image position	Nature	Size
At infinity	at F_2	real, inverted	point
Beyond $2F_1$	between F_2 and $2F_2$	real, inverted	diminished
At $2F_1$	at $2F_2$	real, inverted	same size
Between F_1 and $2F_1$	beyond $2F_2$	real, inverted	magnified
At F_1	at infinity	real, inverted	highly magnified
Between F_1 and O	same side as object	virtual, erect	magnified

A **concave lens** is simpler: for every real object it gives a virtual, erect, diminished image between the focus and the lens.

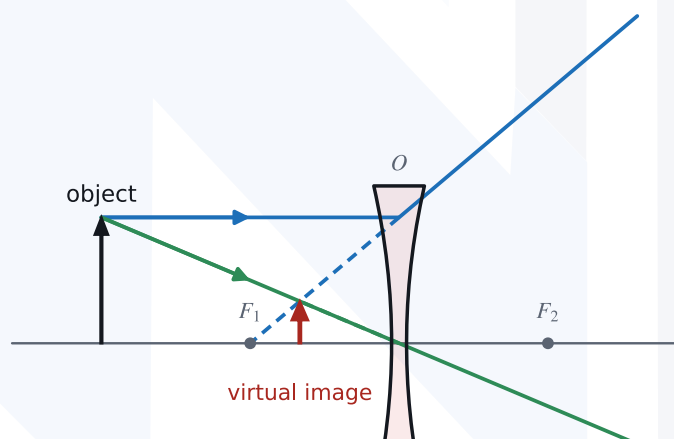


Figure 70. Concave lens. The emergent rays diverge; their backward extensions (dashed) meet at a virtual, erect, diminished image between F_1 and the lens.

WORKED EXAMPLE 17.1

Problem. An object 3 cm tall stands 30 cm in front of a converging lens of focal length 20 cm. Find the position, nature and size of the image.

Solution. With the sign convention, $u = -30$ cm, $f = +20$ cm.

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{20} - \frac{1}{30} = \frac{3-2}{60} = \frac{1}{60}$$

So $v = +60$ cm; the image is 60 cm behind the lens, and because v is positive it is **real**.

$$m = \frac{v}{u} = \frac{+60}{-30} = -2$$

The image is inverted and twice as tall: $h_i = -2 \times 3 = -6$ cm, i.e. 6 cm tall and upside down.

Angular magnification

How big something looks depends not on its real size but on the angle it subtends at the eye. A lens used as a magnifier does not change the object; it lets you bring the object closer than the near point while still seeing it sharply, so the angle grows.

ANGULAR MAGNIFICATION (MAGNIFYING POWER)

The angular magnification M of an optical instrument is the ratio of the angle β subtended at the eye by the final image to the angle α subtended at the unaided eye by the object placed at the least distance of distinct vision D ($D = 25$ cm for a normal eye):

$$M = \frac{\beta}{\alpha}$$

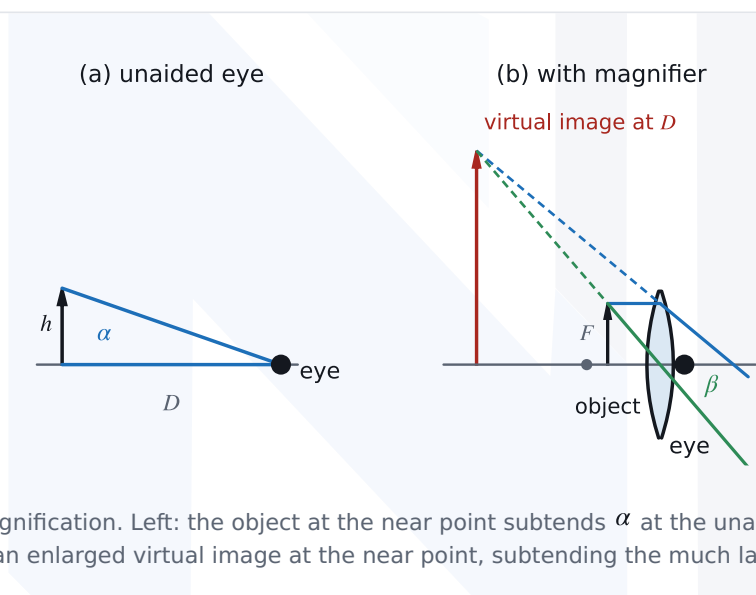


Figure 71. Angular magnification. Left: the object at the near point subtends α at the unaided eye. Right: a convex lens forms an enlarged virtual image at the near point, subtending the much larger angle β .

MAGNIFYING POWER OF A SIMPLE MICROSCOPE

We start from the definition of angular magnification and reach the two standard formulas, one for each way of using a magnifying glass.

Setting up. A simple microscope is a single convex lens of short focal length f . The object has height h , and the eye is held close behind the lens. D is the least distance of distinct vision (the near-point distance, 25 cm for a normal eye). The object is placed between the lens and its focus, so the image is virtual, erect and enlarged.

Step 1 — the angle without the lens. To see the object as big as possible with the bare eye, you bring it to the near point, a distance D away. The angle it then subtends is small, so

$$\alpha \approx \frac{h}{D}$$

Case 1 — final image at the near point (maximum magnification).

Step 2 — where the object must be. The image must be at the near point, so with the sign convention $v = -D$. Let the object distance be $u = -u_0$.

Step 3 — the angle with the lens. The eye is at the lens, so it receives the rays at the angle the object subtends at the lens:

$$\beta \approx \frac{h}{u_0}$$

Step 4 — form the ratio, by the definition of magnifying power:

$$M = \frac{\beta}{\alpha} = \frac{h/u_0}{h/D}$$

Step 5 — cancel h and simplify the divided fraction:

$$M = \frac{D}{u_0}$$

Step 6 — now find u_0 from the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, putting $v = -D$ and $u = -u_0$:

$$\frac{1}{-D} - \frac{1}{-u_0} = \frac{1}{f}$$

Step 7 — clean up the signs:

$$-\frac{1}{D} + \frac{1}{u_0} = \frac{1}{f}$$

Step 8 — move $1/D$ across:

$$\frac{1}{u_0} = \frac{1}{f} + \frac{1}{D}$$

Step 9 — multiply every term by D , because Step 5 needs D/u_0 :

$$\frac{D}{u_0} = \frac{D}{f} + \frac{D}{D}$$

Step 10 — simplify the last term and use Step 5:

$$M = 1 + \frac{D}{f}$$

Case 2 — final image at infinity (relaxed eye).

Step 11 — where the object must be. For the image to go to infinity, the object must sit exactly at the focus, so $u_0 = f$.

Step 12 — the angle with the lens is then

$$\beta \approx \frac{h}{f}$$

Step 13 — form the ratio again:

$$M = \frac{\beta}{\alpha} = \frac{h/f}{h/D}$$

Step 14 — cancel h :

$$M = \frac{D}{f}$$

Result.

$$M_{\text{near point}} = 1 + \frac{D}{f}, \quad M_{\text{infinity}} = \frac{D}{f}$$

What it means. A shorter focal length gives a bigger magnification, which is why a magnifying glass is a strongly curved little lens. The near-point setting gives exactly **one** extra unit of magnification, but the eye must accommodate, which tires it; the infinity setting is more comfortable for long use.

Conditions used. The eye is close to the lens, all angles are small (so that $\tan \theta \approx \theta$ in Steps 1, 3 and 12), and $D = 25$ cm for a normal eye.

The near-point setting gives one extra unit of magnification but the eye must accommodate, which tires it; the infinity setting is more comfortable.

WORKED EXAMPLE 17.2

Problem. A magnifying glass has focal length 5 cm. Taking $D = 25$ cm, find its magnifying power when the final image is (a) at the near point and (b) at infinity. Where must the object be placed in case (a)?

Solution.

(a) $M = 1 + D/f = 1 + 25/5 = 6.$

(b) $M = D/f = 25/5 = 5.$

For (a), $\frac{1}{u_0} = \frac{1}{f} + \frac{1}{D} = \frac{1}{5} + \frac{1}{25} = \frac{6}{25}$, so $u_0 = 25/6 = 4.17$ cm — just inside the focus, as expected.

17.2 Lens maker's formula

A lens refracts light twice, once at each face. So we first need the result for refraction at **one** spherical surface.

REFRACTION AT A SINGLE SPHERICAL SURFACE

We start from Snell's law at one curved boundary and reach the equation that a single spherical refracting surface obeys. This one result, used twice, gives the whole lens theory.

Setting up. Light travels from a medium of index n_1 into one of index n_2 across a spherical surface of radius R , with pole P and centre of curvature C . A paraxial ray from the axial object O meets the surface at A , a small height h above the axis, and after refraction crosses the axis at I .

Let α , β , γ be the (small) angles that AO , AC and AI make with the axis. Note AC is the **normal** at A , since a radius is always normal to a sphere.

Step 1 — find the angle of incidence. In triangle AOC , the angle at C is the exterior angle rule applied the other way round: the exterior angle at A (which is i) equals the sum of the two opposite interior angles:

$$i = \alpha + \beta$$

Step 2 — find the angle of refraction. In triangle AIC , β is the exterior angle, so it equals the sum of the two opposite interior angles r and γ :

$$\beta = r + \gamma$$

Step 3 — make r the subject:

$$r = \beta - \gamma$$

Step 4 — write Snell's law at A :

$$n_1 \sin i = n_2 \sin r$$

Step 5 — use the paraxial approximation. A is close to the axis, so i and r are small and $\sin \theta \approx \theta$:

$$n_1 i = n_2 r$$

Step 6 — substitute Steps 1 and 3:

$$n_1(\alpha + \beta) = n_2(\beta - \gamma)$$

Step 7 — express the three angles as ratios. For a small angle, angle $\approx$ opposite side over adjacent side, and A is practically at P . With the Cartesian convention (u negative for a real object):

$$\alpha = \frac{h}{-u}, \quad \beta = \frac{h}{R}, \quad \gamma = \frac{h}{v}$$

Step 8 — substitute these into Step 6:

$$n_1 \left(\frac{h}{-u} + \frac{h}{R} \right) = n_2 \left(\frac{h}{R} - \frac{h}{v} \right)$$

Step 9 — cancel h , which multiplies every term. (This is the important moment: the answer does not depend on the ray's height, which is why a sharp image forms at all.)

$$-\frac{n_1}{u} + \frac{n_1}{R} = \frac{n_2}{R} - \frac{n_2}{v}$$

Step 10 — collect the v and u terms on the left and the R terms on the right:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2}{R} - \frac{n_1}{R}$$

Step 11 — combine the right-hand side over the common denominator R :

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

Result.

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

What it means. One equation covers every single curved refracting boundary — the front of a lens, the back of a lens, the cornea of an eye — provided you put in the right n_1 , n_2 and the right sign for R . Applied twice in succession it gives the lens maker's formula.

Conditions used. Paraxial rays only (Steps 5 and 7), a surface of small aperture, and all distances measured from the pole P using the new Cartesian convention.

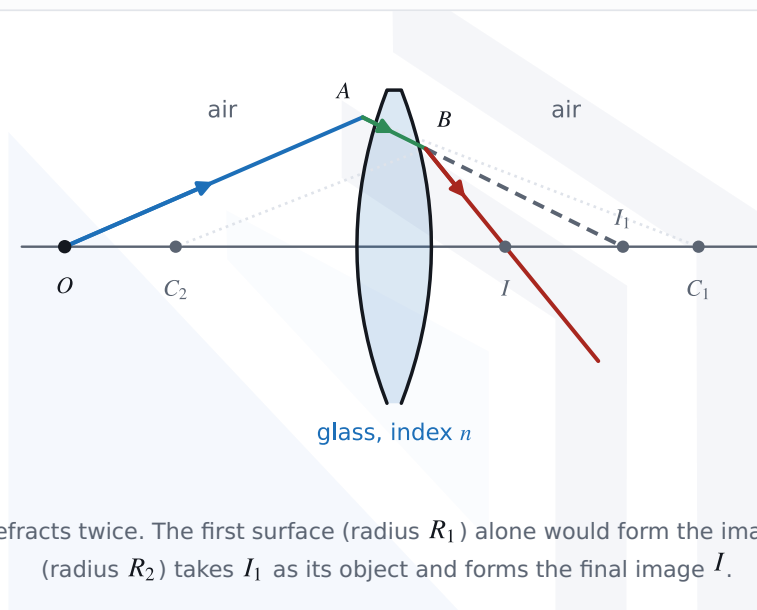


Figure 72. A thin lens refracts twice. The first surface (radius R_1) alone would form the image I_1 ; the second surface (radius R_2) takes I_1 as its object and forms the final image I .

LENS MAKER'S FORMULA

We start from the single-surface equation just derived, apply it twice, and reach the formula that connects a lens's focal length to its shape and its glass.

Setting up. A thin lens of material index n sits in air. Its faces have radii R_1 (the first surface the light meets) and R_2 . A real object is at u ; the first surface alone would form an image at v_1 , which the second surface then uses as its object to form the final image at v .

Step 1 — apply the single-surface equation to the first face. Light goes from air into glass, so $n_1 = 1$, $n_2 = n$, radius R_1 , image distance v_1 :

$$\frac{n}{v_1} - \frac{1}{u} = \frac{n-1}{R_1} \quad (1)$$

Step 2 — apply it again to the second face. Now light goes from glass into air, so $n_1 = n$, $n_2 = 1$, radius R_2 , and the object is I_1 . Because the lens is **thin**, the distance of I_1 from the second surface is still v_1 :

$$\frac{1}{v} - \frac{n}{v_1} = \frac{1-n}{R_2} \quad (2)$$

Step 3 — add equations (1) and (2). We add rather than subtract because the n/v_1 terms have opposite signs and so will cancel:

$$\frac{n}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{n}{v_1} = \frac{n-1}{R_1} + \frac{1-n}{R_2}$$

Step 4 — cancel the two n/v_1 terms on the left. The intermediate image has now disappeared from the problem entirely:

$$\frac{1}{v} - \frac{1}{u} = \frac{n-1}{R_1} + \frac{1-n}{R_2}$$

Step 5 — rewrite $(1-n)$ as $-(n-1)$ so that the same factor appears in both terms:

$$\frac{1}{v} - \frac{1}{u} = \frac{n-1}{R_1} - \frac{n-1}{R_2}$$

Step 6 — take out the common factor $(n-1)$:

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Step 7 — now specialise to an object at infinity. Then $1/u=0$, and by the definition of focal length the image forms at $v=f$:

$$\frac{1}{f} - 0 = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Step 8 — write it out:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Step 9 — recover the thin lens formula. The right-hand side of Step 6 is now known to equal $1/f$, so

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Result.

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

What it means. A lens's power comes from two things only: how different its glass is from the surroundings ($n-1$) and how sharply its faces are curved. Make either bigger and f gets shorter. Notice also that a lens immersed in a liquid of the same index ($n-1 \rightarrow 0$) has $f=\infty$ — it becomes invisible and does nothing.

Conditions used. The lens is **thin** (Step 2 needs the two surfaces to be at effectively the same place), rays are paraxial, light is monochromatic, and the surrounding medium is air.

THE EXAMINER IS LOOKING FOR

1. The single-surface equation quoted correctly.
2. Its application to the **first** surface, with $n_1 = 1$, $n_2 = n$.
3. Its application to the **second** surface, with $n_1 = n$, $n_2 = 1$, **and the sentence explaining that the lens is thin** so the intermediate image distance is unchanged.
4. Adding the two equations and cancelling n/v_1 .
5. Putting $u = \infty$, $v = f$ to reach the boxed result.

If the lens of index n_l sits in a medium of index n_m the same derivation gives

$$\frac{1}{f} = \left(\frac{n_l}{n_m} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

DO NOT FORGET THE SIGN OF R_2

For a bi-convex lens, R_1 is positive and R_2 is **negative**, so $1/R_1 - 1/R_2$ is a sum of two positive terms. Students who write $1/R_1 - 1/R_2 = 1/20 - 1/20 = 0$ for an equiconvex lens have forgotten the convention and get $f = \infty$.

WORKED EXAMPLE 17.3

Problem. An equiconvex lens of glass ($n = 1.5$) has faces of radius 20 cm. Find its focal length in air, and its focal length when the whole lens is immersed in water ($n_w = 4/3$).

Solution. For an equiconvex lens $R_1 = +20$ cm and $R_2 = -20$ cm, so

$$\frac{1}{R_1} - \frac{1}{R_2} = \frac{1}{20} + \frac{1}{20} = \frac{1}{10} \text{ cm}^{-1}$$

In air:

$$\frac{1}{f} = (1.5 - 1) \times \frac{1}{10} = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

In water:

$$\frac{1}{f_w} = \left(\frac{1.5}{4/3} - 1 \right) \times \frac{1}{10} = (1.125 - 1) \times \frac{1}{10} = 0.0125 \text{ cm}^{-1}$$

so $f_w = 80$ cm. The lens is four times weaker in water, because the glass-water index contrast is much smaller than the glass-air one. This is exactly why your eyes cannot focus under water.

17.3 Power of a lens

A short focal length bends light strongly. It is therefore convenient to grade lenses by the reciprocal of the focal length.

POWER OF A LENS

The power of a lens is the reciprocal of its focal length expressed in metres:

$$P = \frac{1}{f \text{ (in m)}}$$

Its SI unit is the **diopetre** (D); $1 \text{ D} = 1 \text{ m}^{-1}$. A convex lens has positive power, a concave lens negative power.

So $f = 50 \text{ cm} = 0.5 \text{ m}$ gives $P = +2 \text{ D}$, and $f = -25 \text{ cm}$ gives $P = -4 \text{ D}$. Combining with the lens maker's formula,

$$P = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

with the radii in metres.

LENSES IN CONTACT

We start by tracing the light through the two lenses one at a time, and reach the focal length of the pair treated as one lens.

Setting up. Two thin lenses of focal lengths f_1 and f_2 are placed in contact, so the distance between them is zero. An object is at u . The first lens forms an intermediate image at v_1 ; the second lens then uses that image as its own object and forms the final image at v . Let F be the focal length of the equivalent single lens.

Step 1 — apply the lens formula to the first lens:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad (1)$$

Step 2 — apply it to the second lens. Its object is I_1 , and because the lenses are in contact the object distance for the second lens is the same v_1 :

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad (2)$$

Step 3 — add (1) and (2). Again we add because the $1/v_1$ terms carry opposite signs:

$$\frac{1}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_1} + \frac{1}{f_2}$$

Step 4 — cancel the two $1/v_1$ terms:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2}$$

Step 5 — compare with the lens formula for the equivalent lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$. The same object gives the same final image, so the left-hand sides are identical:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Step 6 — restate in terms of power, since $P = 1/f$:

$$P = P_1 + P_2$$

Result.

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}, \quad P = P_1 + P_2, \quad m = m_1 \times m_2$$

What it means. Powers add; focal lengths do not. Two +2 D lenses in contact make a +4 D lens. Because a diverging lens has negative power, a +5 D lens combined with a -2 D lens gives +3 D — this is how an optician builds any prescription out of a small stock of lenses. The magnifications multiply, not add, because each lens magnifies what the previous one produced.

Conditions used. Both lenses are **thin** and in **contact** (zero separation), rays are paraxial, and the two lenses share the same principal axis. If they are separated by a distance d the extra term below is needed.

If the two lenses are separated by a distance d , the result becomes

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}, \quad P = P_1 + P_2 - d P_1 P_2$$

POWERS ADD, FOCAL LENGTHS DO NOT

In any numerical on lens combinations in contact, convert every focal length to a power in dioptres first, add them algebraically (watch the minus signs), and convert back at the end. It is faster and far less error-prone than adding reciprocals.

WORKED EXAMPLE 17.4

Problem. A student in Biratnagar can read comfortably only when the page is at least 50 cm from her eye. What spectacle lens will let her read at the normal near point of 25 cm? If a protective lens of power -0.5 D is cemented to it, what is the power and focal length of the pair?

Solution. The spectacle lens must take an object at 25 cm and form a virtual image at her own near point, 50 cm, on the same side. So $u = -25$ cm and $v = -50$ cm.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = -\frac{1}{50} + \frac{1}{25} = \frac{-1+2}{50} = \frac{1}{50}$$

so $f = +50$ cm = 0.50 m and

$$P = \frac{1}{0.50} = +2.0 \text{ D}$$

a converging lens, the standard correction for **hypermetropia** (long sight).

With the protective lens in contact,

$$P = P_1 + P_2 = 2.0 + (-0.5) = +1.5 \text{ D}, \quad F = \frac{1}{1.5} = 0.667 \text{ m} = 66.7 \text{ cm}$$

✓ Chapter summary

- Thin lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with linear magnification $m = h_i/h_o = v/u$, in the New Cartesian convention.
- A convex lens gives a real inverted image for any object beyond F_1 , and a virtual magnified one inside F_1 ; a concave lens always gives a virtual, erect, diminished image.
- Angular magnification $M = \beta/\alpha$. For a simple microscope, $M = 1 + D/f$ (image at the near point) and $M = D/f$ (image at infinity), with $D = 25$ cm.
- Refraction at one spherical surface: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$.
- Lens maker's formula: $\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$; in a medium, replace n by n_l/n_m .
- Power $P = 1/f$ (f in metres), unit dioptre; convex positive, concave negative.
- Lenses in contact: $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$, i.e. $P = P_1 + P_2$ and $m = m_1 m_2$; separated by d , $P = P_1 + P_2 - dP_1 P_2$.

Practice questions

Group A — Multiple choice (1 mark each)

- The power of a lens of focal length -25 cm is [1]

a +4 D b -4 D c $+0.25$ D d -0.25 D
- A convex lens of focal length 10 cm is used as a magnifying glass with the image at the near point. Its magnifying power is [1]

a 1.5 b 2.5 c 3.5 d 10
- An equiconvex glass lens ($n = 1.5$) has focal length 20 cm in air. Immersed in water ($n = 4/3$), its focal length becomes [1]

a 20 cm b 40 cm c 60 cm d 80 cm
- For a real object, a concave lens always forms an image that is [1]

a real and inverted b virtual, erect and diminished
c virtual, erect and magnified d real and erect
- A ray directed at the optical centre of a thin lens [1]

a passes through the focus b emerges parallel to the axis
c emerges undeviated d is totally reflected

ANSWERS TO GROUP A

- (b) — $P = 1/(-0.25 \text{ m}) = -4 \text{ D}$.
- (c) — $M = 1 + D/f = 1 + 25/10 = 3.5$.
- (d) — $(1.5/1.333 - 1)/(1.5 - 1) = 0.25$, so f is multiplied by 4.
- (b) — the emergent rays always diverge, so the image is virtual, erect and smaller.
- (c) — the two faces are locally parallel there, so the ray is only laterally shifted, and for a thin lens that shift is negligible.

Group B — Short answer (5 marks each)

1. Define optical centre, principal focus and focal length of a thin lens. State the three rules used to construct ray diagrams and draw the diagram for an object placed between F_1 and $2F_1$ of a convex lens. [5]
2. An object 4 cm high is placed 25 cm from a converging lens of focal length 15 cm. Find the position, nature and size of the image. [5]
3. Define the power of a lens and its unit. Two thin lenses of powers +5 D and -3 D are placed in contact. Find the focal length of the combination and state whether it converges or diverges. [5]
4. Define angular magnification and show that, for a simple microscope with the final image at the least distance of distinct vision, $M = 1 + D/f$. [5]
5. A double convex lens has faces of radii 20 cm and 30 cm and is made of glass of refractive index 1.5. Calculate its focal length and power. [5]

ANSWERS TO GROUP B

2. $u = -25$ cm, $f = +15$ cm. $\frac{1}{v} = \frac{1}{15} - \frac{1}{25} = \frac{5-3}{75} = \frac{2}{75}$, so $v = +37.5$ cm (real, behind the lens). $m = v/u = 37.5/(-25) = -1.5$, so $h_i = -1.5 \times 4 = -6$ cm: real, inverted, 6 cm tall.

3. $P = P_1 + P_2 = 5 - 3 = +2$ D, so $F = 1/2 = 0.5$ m = 50 cm. The power is positive, so the combination **converges**.

4. See the derivation in §17.3 — the marks are for $M = \beta/\alpha = D/u_0$, for substituting $v = -D$ in the lens formula, and for the final step $D/u_0 = 1 + D/f$.

5. Taking the light to meet the 20 cm face first, $R_1 = +20$ cm, $R_2 = -30$ cm:

$$\frac{1}{f} = (1.5 - 1)\left(\frac{1}{20} + \frac{1}{30}\right) = 0.5 \times \frac{5}{60} = \frac{1}{24} \text{ cm}^{-1}$$

So $f = 24$ cm = 0.24 m and $P = 1/0.24 = +4.17$ D.

Group C — Long answer (8 marks each)

1. (a) Derive the lens maker's formula $\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ for a thin lens, starting from the formula for refraction at a single spherical surface. [5]
 (b) A plano-convex lens of glass of refractive index 1.5 has a focal length of 30 cm. Find the radius of curvature of its curved face. [3]
2. (a) Define angular magnification and derive expressions for the magnifying power of a simple microscope when the final image is formed (i) at the near point and (ii) at infinity. [5]
 (b) A hand lens of focal length 5 cm is used by a normal eye. Calculate its magnifying power in both cases and comment on which setting is more comfortable. [3]

ANSWERS TO GROUP C

1.(b) Let light strike the flat face first: $R_1 = \infty$ and $R_2 = -R$. Then

$$\frac{1}{30} = (1.5 - 1)\left(0 + \frac{1}{R}\right) = \frac{0.5}{R}$$

so $R = 0.5 \times 30 = 15$ cm. (The same answer follows if the curved face is met first, with $R_1 = +R$, $R_2 = \infty$.)

2.(b) Near point: $M = 1 + D/f = 1 + 25/5 = 6$. Infinity: $M = D/f = 25/5 = 5$. The near-point setting gives the larger magnification, but the eye must accommodate fully and tires quickly; the infinity setting ($M = 5$) is used for prolonged viewing because the eye stays relaxed.



UNIT 18

Dispersion

Waves and Optics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Pure spectrum and dispersive power
- Chromatic and spherical aberration
- Achromatism and its applications

White light is a mixture. A glass prism bends violet light a little more than red light, so the mixture comes apart into a band of colours — this is **dispersion**. The same effect that gives a rainbow over the Kathmandu valley also spoils the image in a cheap lens, and most of this unit is about understanding the spoiling and then curing it.

THE ONE FACT BEHIND THE WHOLE UNIT

Refractive index depends on wavelength: $n_{\text{violet}} > n_{\text{red}}$. Every result here — angular dispersion, dispersive power, chromatic aberration, the achromatic doublet — is that single fact worked through.

18.1 Pure spectrum and dispersive power

Dispersion by a prism

For a prism of small refracting angle A , the deviation of a ray is

$$\delta = (n - 1)A$$

Since n is different for each colour, each colour is deviated by a different amount and the emergent beam is spread into a **spectrum** with red deviated least and violet most.

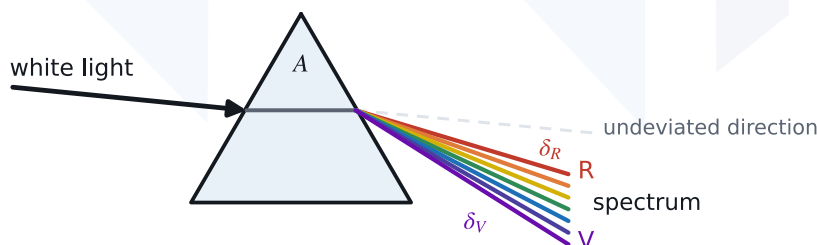


Figure 73. Dispersion of white light by a prism. Violet is deviated most, red least; the angular dispersion is $\theta = \delta_V - \delta_R$.

The **angular dispersion** between the extreme colours is

$$\theta = \delta_V - \delta_R = (n_V - 1)A - (n_R - 1)A = (n_V - n_R)A$$

and the **mean deviation** is taken as that of yellow, the middle of the spectrum:

$$\delta_Y = (n_Y - 1)A$$

DISPERSIVE POWER

The dispersive power ω of the material of a prism is the ratio of the angular dispersion between two chosen colours to the mean deviation:

$$\omega = \frac{\theta}{\delta_Y} = \frac{\delta_V - \delta_R}{\delta_Y} = \frac{n_V - n_R}{n_Y - 1}$$

It is a **pure number** (no unit, no dimensions) and depends only on the material, not on the refracting angle of the prism.

DISPERSIVE POWER IS DIMENSIONLESS AND INDEPENDENT OF THE PRISM ANGLE

We start from the thin-prism deviation formula and reach an expression for ω in which the prism's own angle A has disappeared.

Step 1 — write the deviation of each colour using $\delta = (n - 1)A$ for the same thin prism. The angle A is the same for all three, only n changes:

$$\delta_V = (n_V - 1)A, \quad \delta_R = (n_R - 1)A, \quad \delta_Y = (n_Y - 1)A$$

Step 2 — form the angular dispersion, the spread between the extreme colours:

$$\theta = \delta_V - \delta_R = (n_V - 1)A - (n_R - 1)A$$

Step 3 — remove the brackets. The two $-A$ terms cancel:

$$\theta = n_V A - n_R A$$

Step 4 — take out the common factor A :

$$\theta = (n_V - n_R)A$$

Step 5 — form the ratio that defines ω :

$$\omega = \frac{\theta}{\delta_Y} = \frac{(n_V - n_R)A}{(n_Y - 1)A}$$

Step 6 — cancel A , which appears on the top and on the bottom:

$$\omega = \frac{n_V - n_R}{n_Y - 1}$$

Result.

$$\theta = (n_V - n_R)A, \quad \omega = \frac{n_V - n_R}{n_Y - 1}$$

What it means. Two things follow from Step 6. First, ω is a **ratio of refractive indices**, and refractive index is itself a ratio of two speeds, so ω has no unit and no dimensions. Second, A cancelled, so ω is a property of the **glass alone**: a 5° prism and a 60° prism of the same glass have the same dispersive power, even though their spectra are very different in width.

Conditions used. A thin prism, so that $\delta = (n - 1)A$ applies; the same three chosen colours throughout; and yellow taken as the mean colour.

Typical values: crown glass $\omega \approx 0.017 - 0.020$, flint glass $\omega \approx 0.03 - 0.05$. Flint glass disperses roughly twice as strongly as crown glass for the same mean deviation — the fact on which every achromatic lens is built.

ω HAS NOTHING TO DO WITH THE PRISM ANGLE

A cancels when you form the ratio. Two prisms of the same glass but angles 5° and 60° have the same dispersive power even though their spectra are very different in width.

Impure and pure spectrum

If white light from a wide slit simply falls on a prism and then on a screen, the coloured bands overlap: each point of the slit produces its own complete spectrum, displaced from the others. That is an **impure spectrum**.

PURE SPECTRUM

A pure spectrum is one in which the light at each point of the spectrum is of a single wavelength, so that no two colours overlap.

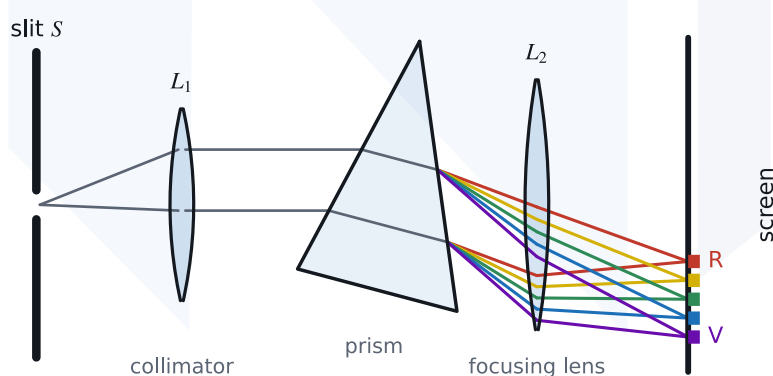


Figure 74. Apparatus for a pure spectrum: narrow slit at the focus of the collimating lens L_1 , prism at minimum deviation, and lens L_2 focusing each parallel colour beam to its own sharp line on the screen.

FOUR CONDITIONS FOR A PURE SPECTRUM

1. The source slit must be **narrow**.
2. Light must fall on the prism as a **parallel beam** (slit at the focus of the collimating lens).
3. The prism must be set in the **minimum deviation** position.
4. A converging lens must bring each parallel emergent beam to its **own focus** on the screen, with the screen in the focal plane.

WORKED EXAMPLE 18.1

Problem. A thin prism of angle 5° is made of glass for which $n_V = 1.532$, $n_Y = 1.523$ and $n_R = 1.514$. Find (a) the mean deviation, (b) the angular dispersion between violet and red, and (c) the dispersive power.

Solution.

$$(a) \delta_Y = (n_Y - 1)A = (1.523 - 1) \times 5^\circ = 0.523 \times 5^\circ = 2.615^\circ$$

$$(b) \theta = (n_V - n_R)A = (1.532 - 1.514) \times 5^\circ = 0.018 \times 5^\circ = 0.090^\circ$$

$$(c) \omega = \frac{\theta}{\delta_Y} = \frac{0.090}{2.615} = 0.0344$$

Equivalently $\omega = \frac{n_V - n_R}{n_Y - 1} = \frac{0.018}{0.523} = 0.0344$ — the same number, and notice A has vanished.

18.2 Chromatic and spherical aberration

An **aberration** is any departure of a real image from the perfect, sharp, undistorted image predicted by the paraxial theory.

Chromatic aberration

From the lens maker's formula, $\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$. Since $n_V > n_R$, we get $f_V < f_R$: a lens focuses violet nearer than red. There is no single image plane, and a white object gives an image fringed with colour.

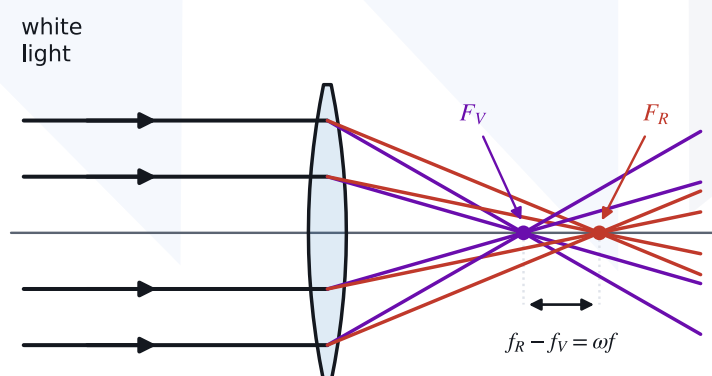


Figure 75. Chromatic aberration. A converging lens brings violet to a focus at F_V and red at F_R ; the separation $f_R - f_V$ is the longitudinal chromatic aberration.

LONGITUDINAL CHROMATIC ABERRATION EQUALS ωf

We start from the lens maker's formula applied separately to violet and red light, and reach the remarkably simple result that the gap between the two foci is just ωf .

Setting up. The lens has a fixed shape, so write

$$K = \frac{1}{R_1} - \frac{1}{R_2}$$

which is the same number for every colour. Only n changes with colour. Let f_V , f_R , f_Y be the focal lengths for violet, red and (mean) yellow light.

Step 1 — lens maker's formula for violet and for red:

$$\frac{1}{f_V} = (n_V - 1)K, \quad \frac{1}{f_R} = (n_R - 1)K$$

Step 2 — subtract the second from the first:

$$\frac{1}{f_V} - \frac{1}{f_R} = (n_V - 1)K - (n_R - 1)K$$

Step 3 — simplify the right side. The two $-K$ terms cancel, then take out K :

$$\frac{1}{f_V} - \frac{1}{f_R} = (n_V - n_R)K$$

Step 4 — combine the left side over a common denominator:

$$\frac{f_R - f_V}{f_R f_V} = (n_V - n_R)K$$

Step 5 — approximate the denominator. For any usable lens f_V and f_R differ by less than about one per cent, so their product is very close to the square of the mean focal length:

$$f_R f_V \approx f_Y^2$$

Step 6 — substitute that in:

$$\frac{f_R - f_V}{f_Y^2} = (n_V - n_R)K$$

Step 7 — multiply both sides by f_Y^2 :

$$f_R - f_V = (n_V - n_R)K f_Y^2$$

Step 8 — get rid of K . The lens maker's formula for yellow gives $\frac{1}{f_Y} = (n_Y - 1)K$, so

$$K f_Y = \frac{1}{n_Y - 1}$$

Step 9 — split one f_Y off in Step 7 and use Step 8:

$$f_R - f_V = (n_V - n_R) \cdot (K f_Y) \cdot f_Y = \frac{n_V - n_R}{n_Y - 1} f_Y$$

Step 10 — recognise the fraction as the dispersive power ω :

$$f_R - f_V = \omega f_Y$$

Result.

$$f_R - f_V = \omega f_Y$$

What it means. The separation of the violet and red foci depends on nothing but the dispersive power of the glass and the mean focal length. So the defect cannot be cured by grinding the surfaces differently — only by changing the glass, or by combining two glasses. A crown-glass lens of $f = 20$ cm with $\omega = 0.018$ has its two foci 0.36 cm apart.

Conditions used. A thin lens, paraxial rays, and $f_V \approx f_R \approx f_Y$ so that Step 5 is safe.

Two kinds are distinguished: **longitudinal** (axial) chromatic aberration, the distance $f_R - f_V$ along the axis, and **lateral** (transverse) chromatic aberration, the difference in the sizes of the violet and red images on a given screen, which shows up as coloured fringes at the edges of the image.

Spherical aberration

A spherical surface is easy to grind but is not the ideal shape for focusing. Rays striking the **marginal** (outer) zone of a lens are bent more than the **paraxial** rays near the axis, so they cross the axis closer to the lens. There is no point at which all rays meet; the smallest patch of light, the **circle of least confusion**, is where the image is sharpest.

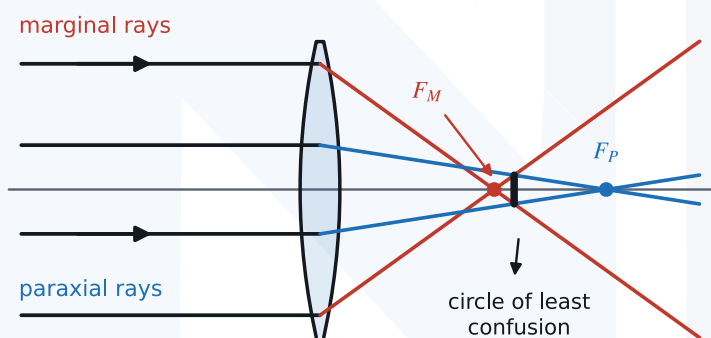


Figure 76. Spherical aberration. Marginal rays cross the axis at F_M , paraxial rays at F_P . The narrowest waist between them is the circle of least confusion.

	Chromatic aberration	Spherical aberration
Cause	n depends on wavelength	spherical surface, wide aperture
Occurs with mono-chromatic light?	No	Yes
Occurs in mirrors?	No	Yes
Depends on	dispersive power ω	aperture and shape of the lens
Remedy	achromatic doublet (crown + flint)	stops, crossed lens, plano-convex pair sharing the deviation, parabolic mirror, aplanatic lenses

WORKED EXAMPLE 18.2

Problem. A convex lens of mean focal length 20 cm is made of glass of dispersive power 0.03. Find its longitudinal chromatic aberration, and the focal lengths for violet and red light.

Solution. The longitudinal chromatic aberration is

$$f_R - f_V = \omega f_Y = 0.03 \times 20 = 0.60 \text{ cm}$$

Taking $f_Y = 20$ cm as the mean of the two,

$$f_R = 20 + 0.30 = 20.30 \text{ cm}, \quad f_V = 20 - 0.30 = 19.70 \text{ cm}$$

The violet focus lies 6 mm closer to the lens than the red one.

18.3 Achromatism and its applications

ACHROMATISM

A combination of lenses (or prisms) is said to be achromatic when it brings two chosen colours — usually red and violet — to the same focus, so that the chromatic aberration between them is eliminated.

CONDITION FOR ACHROMATISM OF TWO THIN LENSES IN CONTACT

We start from the rule for two lenses in contact and demand that the combination have the same focal length for violet as for red. That single demand gives the condition.

Setting up. The two lenses have mean focal lengths f_1, f_2 and dispersive powers ω_1, ω_2 . Write K_1 and K_2 for their shape factors, $K = \frac{1}{R_1} - \frac{1}{R_2}$.

Step 1 — lenses in contact add their powers:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

Step 2 — write this for violet and for red separately:

$$\frac{1}{F_V} = \frac{1}{f_{1V}} + \frac{1}{f_{2V}}, \quad \frac{1}{F_R} = \frac{1}{f_{1R}} + \frac{1}{f_{2R}}$$

Step 3 — impose the achromatic condition. "Achromatic" means the two colours focus at the same place, so $F_V = F_R$, and therefore $1/F_V = 1/F_R$:

$$\frac{1}{f_{1V}} + \frac{1}{f_{2V}} = \frac{1}{f_{1R}} + \frac{1}{f_{2R}}$$

Step 4 — move everything to the left and pair up each lens with itself:

$$\left(\frac{1}{f_{1V}} - \frac{1}{f_{1R}}\right) + \left(\frac{1}{f_{2V}} - \frac{1}{f_{2R}}\right) = 0$$

Step 5 — work out the first bracket. Apply the lens maker's formula to lens 1 for each colour and subtract, exactly as in the previous derivation:

$$\frac{1}{f_{1V}} - \frac{1}{f_{1R}} = (n_{1V} - n_{1R})K_1$$

Step 6 — replace the index difference using the definition of dispersive power. Since $\omega_1 = \frac{n_{1V} - n_{1R}}{n_{1Y} - 1}$, multiplying across gives $n_{1V} - n_{1R} = \omega_1(n_{1Y} - 1)$:

$$\frac{1}{f_{1V}} - \frac{1}{f_{1R}} = \omega_1(n_{1Y} - 1)K_1$$

Step 7 — recognise $(n_{1Y} - 1)K_1$ as $1/f_1$, by the lens maker's formula for the mean colour:

$$\frac{1}{f_{1V}} - \frac{1}{f_{1R}} = \frac{\omega_1}{f_1}$$

Step 8 — repeat Steps 5–7 for the second lens:

$$\frac{1}{f_{2V}} - \frac{1}{f_{2R}} = \frac{\omega_2}{f_2}$$

Step 9 — substitute both results into Step 4:

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

Result.

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

What it means. Dispersive power is always a positive number, so the only way the two terms can cancel is for f_1 and f_2 to have **opposite signs**: one lens must converge and the other diverge. The colour spread introduced by the first is then undone by the second, while a useful net power survives because the two glasses have different ω .

Conditions used. Both lenses are thin and in contact, and only two colours (violet and red) are brought together — other colours are still slightly out of focus, a residual defect called the secondary spectrum.

Two consequences follow immediately.

- Dispersive powers are always positive, so f_1 and f_2 must have **opposite signs**: one lens must converge and the other diverge.
- $\frac{f_2}{f_1} = -\frac{\omega_2}{\omega_1}$. To leave a useful net power the two dispersive powers must differ, which is why the pair is made of **crown glass** (low ω , convex) cemented to **flint glass** (high ω , concave).

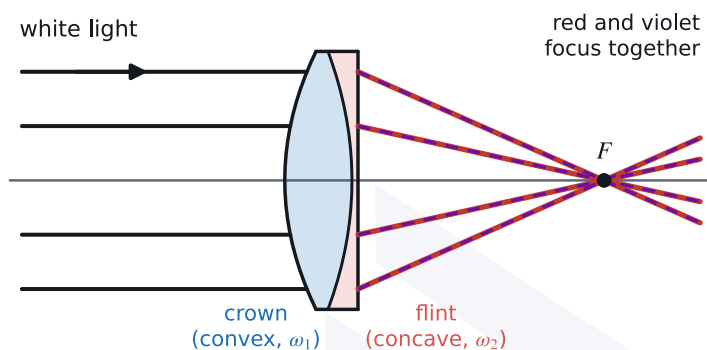


Figure 77. Achromatic doublet: a strong crown-glass convex lens cemented to a weaker flint-glass concave lens. The dispersion of the second cancels that of the first, so red and violet meet at the same focus F .

Achromatic prism combinations

The same idea applied to prisms gives two useful devices.

Device	What it must do	Condition
Achromatic prism combination	deviation without dispersion	$(n_V - n_R)A + (n'_V - n'_R)A' = 0$
Direct-vision prism	dispersion without deviation	$(n_Y - 1)A + (n'_Y - 1)A' = 0$

In both, the second prism is inverted with respect to the first, so its contribution has the opposite sign.

WHICH CONDITION IS WHICH

"Achromatic" always means the colours must not separate, so the condition is written in terms of the **dispersions**. "Direct vision" means the beam must come out straight, so the condition is written in terms of the **deviations**.

Applications

- **Achromatic doublet objectives** in refracting telescopes, microscopes, binoculars and camera lenses — the standard cure for coloured fringes.
- **Direct-vision spectroscopes**, in which a train of crown and flint prisms spreads the spectrum while letting the observer look straight at the source.
- **Reflecting telescopes** avoid the problem completely: a mirror has no dispersion at all, which is why all large research telescopes use mirrors.
- Spectacle and projector lenses, where coloured edges would be objectionable.

WORKED EXAMPLE 18.3

Problem. An achromatic doublet of focal length 20 cm is to be made by cementing a crown-glass lens ($\omega_1 = 0.020$) to a flint-glass lens ($\omega_2 = 0.040$). Find the focal length and the power of each lens.

Solution. The achromatism condition gives

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \Rightarrow \frac{0.020}{f_1} = -\frac{0.040}{f_2} \Rightarrow f_2 = -2f_1$$

Substituting into $\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$ with $F = 20$ cm:

$$\frac{1}{20} = \frac{1}{f_1} - \frac{1}{2f_1} = \frac{1}{2f_1} \Rightarrow f_1 = 10 \text{ cm}$$

and therefore $f_2 = -20$ cm.

Check: $\frac{1}{10} - \frac{1}{20} = \frac{1}{20}$, so $F = 20$ cm.

Powers: $P_1 = 1/0.10 = +10$ D (crown, convex) and $P_2 = 1/(-0.20) = -5$ D (flint, concave), giving $P = +5$ D.

WORKED EXAMPLE 18.4

Problem. A crown-glass prism of refracting angle 10° ($n_Y = 1.52$) is to be combined with a flint-glass prism ($n'_Y = 1.65$) so that white light passes through the pair without net deviation. Find the refracting angle of the flint prism.

Solution. For no net deviation the two deviations must be equal and opposite, so in magnitude

$$(n_Y - 1)A = (n'_Y - 1)A'$$

$$(1.52 - 1) \times 10^\circ = (1.65 - 1) \times A'$$

$$5.2^\circ = 0.65 A' \Rightarrow A' = 8^\circ$$

The flint prism is placed inverted with respect to the crown prism. Because flint has the larger dispersive power, the emergent light is still dispersed — the combination gives **dispersion without deviation**, which is exactly what a direct-vision spectroscope needs.

✓ Chapter summary

- Dispersion happens because n depends on wavelength, with $n_V > n_R$; for a thin prism $\delta = (n - 1)A$.
- Angular dispersion $\theta = (n_V - n_R)A$; mean deviation $\delta_Y = (n_Y - 1)A$.
- Dispersive power $\omega = \frac{n_V - n_R}{n_Y - 1}$ — dimensionless, a property of the material only. Flint glass has roughly twice the ω of crown glass.
- A pure spectrum needs a narrow slit, a parallel incident beam, the prism at minimum deviation, and a lens focusing each colour separately on the screen.
- Chromatic aberration: $f_V < f_R$; longitudinal chromatic aberration $= \omega f$. Spherical aberration: marginal rays focus nearer than paraxial rays, leaving a circle of least confusion.
- Condition for achromatism of two thin lenses in contact: $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$, so one lens must be convex (crown) and the other concave (flint).
- Prism pairs: $(n_V - n_R)A + (n'_V - n'_R)A' = 0$ gives deviation without dispersion; $(n_Y - 1)A + (n'_Y - 1)A' = 0$ gives dispersion without deviation.

Practice questions

Group A — Multiple choice (1 mark each)

- The dispersive power of the material of a prism depends on [1]
 - a the refracting angle
 - b the angle of incidence
 - c the material only
 - d the height of the prism
- For a prism of small refracting angle A , the deviation is [1]
 - a $(n - 1)A$
 - b nA
 - c $A/(n - 1)$
 - d $(n + 1)A$
- The condition for achromatism of two thin lenses in contact is [1]
 - a $\omega_1 f_1 + \omega_2 f_2 = 0$
 - b $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$
 - c $\frac{f_1}{\omega_1} + \frac{f_2}{\omega_2} = 0$
 - d $\omega_1 f_2 = \omega_2 f_1$
- Chromatic aberration in a lens is caused by [1]
 - a the spherical shape of the surfaces
 - b a large aperture
 - c the dependence of refractive index on wavelength
 - d the thickness of the lens
- Spherical aberration of a lens can be reduced by [1]
 - a using monochromatic light
 - b placing a stop to cut off marginal rays
 - c increasing the aperture
 - d cementing a flint lens to it

ANSWERS TO GROUP A

- (c) — A cancels in $\omega = (n_V - n_R)/(n_Y - 1)$.

2. (a) — the small-angle result derived from the prism formula.
3. (b) — obtained by setting $F_V = F_R$ for lenses in contact.
4. (c) — $n_V > n_R$ makes $f_V < f_R$; it persists even with perfect surfaces.
5. (b) — a stop blocks the marginal rays that focus too near the lens. (Option (d) cures chromatic, not spherical, aberration.)

Group B — Short answer (5 marks each)

1. What is dispersion of light? Define dispersive power and show that it is a dimensionless quantity independent of the angle of the prism. [5]
2. The refractive indices of flint glass for violet, yellow and red light are 1.663, 1.649 and 1.635. Calculate the dispersive power of the glass, and the angular dispersion produced by a flint prism of angle 6° . [5]
3. Distinguish between a pure and an impure spectrum, and state the conditions necessary for obtaining a pure spectrum. [5]
4. Distinguish between chromatic and spherical aberration. State one method of reducing each. [5]
5. A convex lens of focal length 25 cm is made of glass of dispersive power 0.024. Calculate its longitudinal chromatic aberration and the focal lengths for red and violet light. [5]

ANSWERS TO GROUP B

$$2. \omega = \frac{n_V - n_R}{n_Y - 1} = \frac{1.663 - 1.635}{1.649 - 1} = \frac{0.028}{0.649} = 0.0431.$$

$$\text{Angular dispersion } \theta = (n_V - n_R)A = 0.028 \times 6^\circ = 0.168^\circ.$$

$$5. f_R - f_V = \omega f = 0.024 \times 25 = 0.60 \text{ cm. Hence } f_R = 25.30 \text{ cm and } f_V = 24.70 \text{ cm.}$$

Group C — Long answer (8 marks each)

1. (a) What is achromatism? Derive the condition $\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$ for two thin lenses in contact to form an achromatic combination, and explain why one lens must be converging and the other diverging. [5]
- (b) An achromatic doublet of focal length 30 cm is made from crown glass of dispersive power 0.018 and flint glass of dispersive power 0.036. Find the focal length of each lens. [3]
2. (a) What is chromatic aberration? Show that the longitudinal chromatic aberration of a thin lens is ωf . [5]
- (b) For a certain lens the focal lengths for violet and red light are 19.6 cm and 20.4 cm. Calculate the dispersive power of the glass. [3]

ANSWERS TO GROUP C

$$1.(b) \text{ From } \frac{0.018}{f_1} = -\frac{0.036}{f_2} \text{ we get } f_2 = -2f_1. \text{ Then}$$

$$\frac{1}{30} = \frac{1}{f_1} - \frac{1}{2f_1} = \frac{1}{2f_1}$$

$$\text{so } f_1 = +15 \text{ cm (crown, convex) and } f_2 = -30 \text{ cm (flint, concave).}$$

2.(b) The mean focal length is $f_Y \approx \sqrt{f_V f_R} = \sqrt{19.6 \times 20.4} = \sqrt{399.84} = 20.0$ cm. Then

$$\omega = \frac{f_R - f_V}{f_Y} = \frac{20.4 - 19.6}{20.0} = \frac{0.8}{20.0} = 0.04$$



UNIT 19

Electric Charges

Electricity and Magnetism • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Electric charges
- Charging by induction
- Coulomb's law - Force between two point charges
- Force between multiple electric charges

Rub a plastic comb on dry hair in winter and it picks up small pieces of paper. That small experiment contains the whole of this unit: charge can be separated, it comes in two kinds, and charges exert forces on one another across empty space. Everything in the rest of electricity — fields, potential, capacitors, circuits — is built on the inverse-square force law we meet here.

WHAT THIS UNIT IS REALLY ABOUT

One law, $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$, plus one principle, superposition. Most Group B numericals are just these two applied to two or three charges with careful attention to **direction**.

19.1 Electric charges

ELECTRIC CHARGE

Electric charge is the intrinsic property of matter because of which it experiences and exerts electrostatic force. There are two kinds, called **positive** and **negative** (the names are due to Benjamin Franklin). Like charges repel; unlike charges attract.

A neutral atom has equal numbers of protons and electrons. Charging a body means **transferring electrons** — never protons, which are locked in the nucleus. A body that has lost electrons is positively charged; one that has gained electrons is negatively charged.

Properties of charge

Property	Statement
Quantisation	Charge exists only in integral multiples of the elementary charge: $q = \pm ne$, with $e = 1.6 \times 10^{-19}$ C and $n = 1, 2, 3, \dots$
Conservation	The total charge of an isolated system never changes; charge can only be transferred, not created or destroyed
Additivity	Charge is a scalar; total charge is the algebraic sum of the individual charges
Invariance	The charge on a body does not change with its speed, unlike its mass

The SI unit of charge is the **coulomb** (C): the charge carried past a point in one second by a steady current of one ampere, so $1 \text{ C} = 1 \text{ A s}$. One coulomb is an enormous charge — it corresponds to

$$n = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons}$$

which is why laboratory charges are usually quoted in microcoulombs ($1 \mu\text{C} = 10^{-6} \text{ C}$) or nanocoulombs.

Conductors and insulators

	Conductor	Insulator (dielectric)
Free electrons	many	practically none
Charge given to it	spreads over the whole outer surface	stays where it is put
Examples	metals, graphite, human body, earth, salt solutions	glass, dry wood, ebonite, plastic, dry air

Three ways of charging a body: by **friction** (rubbing glass with silk, ebonite with fur), by **conduction** (touching it with an already charged body — it gains charge of the same sign), and by **induction** (§19.2 — it gains charge of the opposite sign).

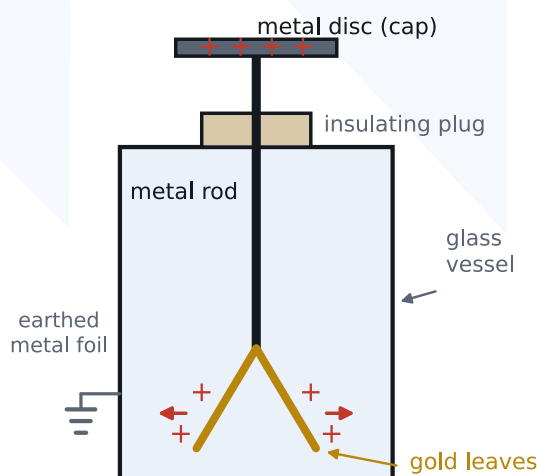


Figure 78. Gold-leaf electroscope. Charge given to the disc spreads down the rod to the leaves, which repel each other and diverge; the divergence measures the charge.

WORKED EXAMPLE 19.1

Problem. A polythene rod acquires a charge of $-3.2 \mu\text{C}$ when rubbed with wool. (a) How many electrons has it gained? (b) What is the charge on the wool? (c) By how much has the mass of the rod changed? Take $e = 1.6 \times 10^{-19} \text{ C}$ and $m_e = 9.1 \times 10^{-31} \text{ kg}$.

Solution.

(a) By quantisation, $q = ne$, so

$$n = \frac{q}{e} = \frac{3.2 \times 10^{-6}}{1.6 \times 10^{-19}} = 2 \times 10^{13} \text{ electrons}$$

(b) By conservation of charge the wool must carry $+3.2 \mu\text{C}$: the electrons were transferred from the wool to the rod, nothing was created.

(c) The rod gains the mass of those electrons:

$$\Delta m = n m_e = 2 \times 10^{13} \times 9.1 \times 10^{-31} = 1.82 \times 10^{-17} \text{ kg}$$

— utterly unmeasurable, which is why we treat charging as a mass-free process.

19.2 Charging by induction

A charged body can produce a charge in a neutral conductor **without touching it**. The process is called electrostatic induction and it depends on the free electrons of the conductor being able to move.

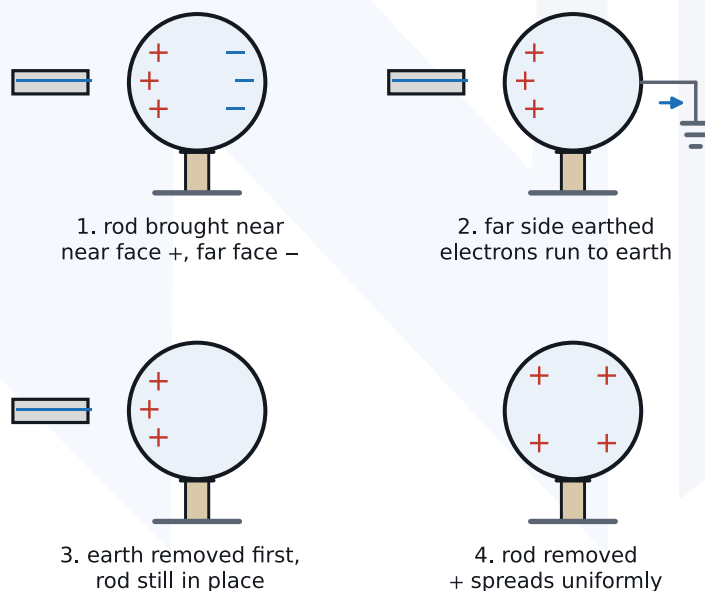


Figure 79. Charging a metal sphere positively by induction with a negatively charged rod. The sphere ends up with a charge opposite to that of the rod, and the rod itself loses nothing.

The four steps, in the order the examiner wants them:

- 1. Bring the charged rod near.** The free electrons of the sphere are repelled by the negative rod and collect on the far face, leaving the near face positive. The sphere as a whole is still neutral.

2. **Earth the sphere** (touch the far side, or connect it to earth). Electrons on the far face are pushed down to earth. The near-face positive charge stays, held by the attraction of the rod.
3. **Remove the earth connection first**, while the rod is still in place.
4. **Remove the rod**. The positive charge, no longer bound, spreads uniformly over the sphere.

ORDER MATTERS

If you remove the rod before breaking the earth connection, electrons flow back up from the earth and the sphere ends up neutral. In the examination you lose marks for writing the steps in the wrong order.

Three points to notice:

- The induced charge is **opposite in sign** to the inducing charge.
- The inducing body **loses no charge at all** — the same rod can charge any number of spheres.
- The magnitude of the induced charge is at most equal to the inducing charge.

Induction also explains why a charged comb attracts neutral bits of paper: the comb induces an opposite charge on the near face of each piece and a like charge on the far face. The near charge is closer, so by the inverse-square law the attraction beats the repulsion and the paper jumps.

19.3 Coulomb's law — force between two point charges

Coulomb measured this force in 1785 with a torsion balance.

COULOMB'S LAW

The force of attraction or repulsion between two stationary point charges is directly proportional to the product of the magnitudes of the charges, inversely proportional to the square of the distance between them, and acts along the line joining them.

$$F \propto \frac{q_1 q_2}{r^2} \quad \Rightarrow \quad F = k \frac{q_1 q_2}{r^2}$$

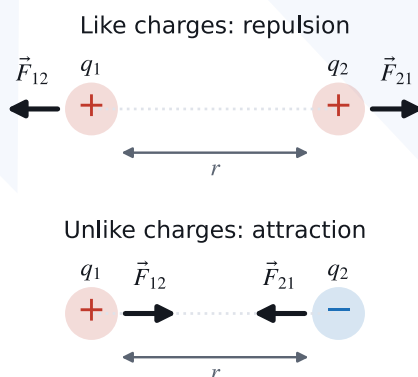


Figure 80. Coulomb forces are equal, opposite and directed along the line joining the charges — outwards for like charges, inwards for unlike charges.

In SI units the constant is written $k = \frac{1}{4\pi\epsilon_0}$, so

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$$

where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$ is the **permittivity of free space** and

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$$

THE COULOMB, DEFINED FROM COULOMB'S LAW

One coulomb is that charge which, placed 1 m from an equal charge in vacuum, repels it with a force of $9 \times 10^9 \text{ N}$.

In a material medium of relative permittivity (dielectric constant) ϵ_r ,

$$F_{med} = \frac{1}{4\pi\epsilon_0\epsilon_r} \cdot \frac{q_1 q_2}{r^2} = \frac{F_{vac}}{\epsilon_r}, \quad \epsilon_r = \frac{F_{vac}}{F_{med}}$$

so a medium always **weakens** the force ($\epsilon_r \geq 1$; for water $\epsilon_r \approx 80$).

Vector form. If $\hat{r}_{21}$ is the unit vector pointing from charge 1 to charge 2, the force on q_2 due to q_1 is

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2} \hat{r}_{21}$$

Substituting signed values of the charges automatically gives the right direction: $q_1 q_2 > 0$ (like charges) makes $\vec{F}_{21}$ point along $\hat{r}_{21}$, i.e. away from q_1 . Newton's third law holds: $\vec{F}_{12} = -\vec{F}_{21}$.

	Coulomb force	Gravitational force
Depends on	product of charges	product of masses
Distance dependence	$1/r^2$	$1/r^2$
Sign	attractive or repulsive	always attractive
Affected by medium	yes, reduced by ϵ_r	no
Relative strength	about 10^{36} times stronger	very weak

WORKED EXAMPLE 19.2

Problem. Two point charges of $+2 \mu\text{C}$ and $-3 \mu\text{C}$ are placed 30 cm apart in air. (a) Find the force between them. (b) What would the force become if the charges were immersed in a liquid of dielectric constant 4 at the same separation?

Solution.

(a) $q_1 = 2 \times 10^{-6} \text{ C}$, $q_2 = 3 \times 10^{-6} \text{ C}$, $r = 0.30 \text{ m}$.

$$F = \frac{9 \times 10^9 \times (2 \times 10^{-6}) \times (3 \times 10^{-6})}{(0.30)^2} = \frac{9 \times 10^9 \times 6 \times 10^{-12}}{0.09}$$

$$F = \frac{5.4 \times 10^{-2}}{0.09} = 0.6 \text{ N}$$

Since the charges are unlike, the force is one of **attraction**.

(b) $F_{med} = F_{vac}/\epsilon_r = 0.6/4 = 0.15 \text{ N}$, still attractive.

WORKED EXAMPLE 19.3

Problem. Compare the electrostatic and gravitational forces between two protons. Take $e = 1.6 \times 10^{-19} \text{ C}$, $m_p = 1.67 \times 10^{-27} \text{ kg}$ and $G = 6.67 \times 10^{-11} \text{ N m}^2\text{kg}^{-2}$.

Solution. Both forces vary as $1/r^2$, so the separation cancels in the ratio:

$$\frac{F_e}{F_g} = \frac{ke^2/r^2}{Gm_p^2/r^2} = \frac{ke^2}{Gm_p^2}$$

Numerator: $9 \times 10^9 \times (1.6 \times 10^{-19})^2 = 9 \times 10^9 \times 2.56 \times 10^{-38} = 2.304 \times 10^{-28}$.

Denominator: $6.67 \times 10^{-11} \times (1.67 \times 10^{-27})^2 = 6.67 \times 10^{-11} \times 2.789 \times 10^{-54} = 1.860 \times 10^{-64}$.

$$\frac{F_e}{F_g} = \frac{2.304 \times 10^{-28}}{1.860 \times 10^{-64}} \approx 1.24 \times 10^{36}$$

Gravity is utterly negligible inside an atom — which is why we ignore it in every electrostatics problem.

19.4 Force between multiple electric charges

PRINCIPLE OF SUPERPOSITION

The force exerted on a charge by another charge is **unaffected** by the presence of any other charges. The resultant force on any one charge is therefore the **vector sum** of the forces exerted on it by each of the other charges taken one at a time:

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \cdots + \vec{F}_{1n}$$

Written out for n charges,

$$\vec{F}_1 = \frac{q_1}{4\pi\epsilon_0} \sum_{j=2}^n \frac{q_j}{r_{1j}^2} \hat{r}_{1j}$$

where $\hat{r}_{1j}$ is the unit vector pointing from q_j towards q_1 , in keeping with the notation of §19.3. For two forces F_a and F_b with angle θ between them, the parallelogram law gives

$$F = \sqrt{F_a^2 + F_b^2 + 2F_aF_b\cos\theta}$$

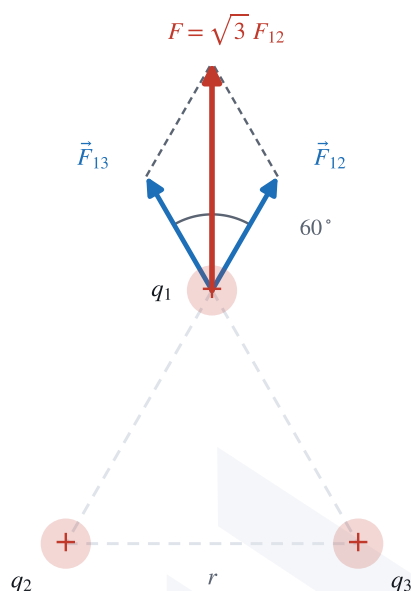


Figure 81. Superposition. The two Coulomb forces on q_1 are combined by the parallelogram law; for three equal charges at the corners of an equilateral triangle the resultant is $\sqrt{3}F$ directed away from the centroid.

A RELIABLE METHOD FOR SUPERPOSITION NUMERICALS

1. Draw the charges and mark each force **on** the chosen charge with an arrow, using only the signs to decide the direction.
2. Compute each magnitude from $F = kq_1q_2/r^2$, ignoring signs.
3. Resolve into x and y components, add, then recombine with $F = \sqrt{F_x^2 + F_y^2}$ and $\tan \alpha = F_y/F_x$.

WORKED EXAMPLE 19.4

Problem. Three charges of $+2 \mu\text{C}$ each are fixed at the corners of an equilateral triangle of side 20 cm. Find the magnitude and direction of the resultant force on any one of them.

Solution. Each pair gives the same magnitude of force, with $q = 2 \times 10^{-6} \text{ C}$ and $r = 0.20 \text{ m}$:

$$\begin{aligned} F_{12} = F_{13} &= \frac{9 \times 10^9 \times (2 \times 10^{-6})^2}{(0.20)^2} = \frac{9 \times 10^9 \times 4 \times 10^{-12}}{0.04} \\ &= \frac{3.6 \times 10^{-2}}{0.04} = 0.9 \text{ N} \end{aligned}$$

Both are repulsive, and the angle between them is 60° . By the parallelogram law,

$$\begin{aligned} F &= \sqrt{F_{12}^2 + F_{13}^2 + 2F_{12}F_{13}\cos 60^\circ} = F\sqrt{1 + 1 + 2(0.5)} = 0.9\sqrt{3} \\ F &= 1.56 \text{ N} \end{aligned}$$

Because the two forces are equal, the resultant bisects the angle between them, i.e. it points **directly away from the centroid** of the triangle.

NEVER ADD COULOMB FORCES AS NUMBERS

Forces are vectors. Adding $0.9 + 0.9 = 1.8 \text{ N}$ in the example above is the single commonest error in this chapter; the correct answer, $0.9\sqrt{3} = 1.56 \text{ N}$, is smaller because the forces are 60° apart.

WHERE THE RESULTANT FORCE ON A THIRD CHARGE IS ZERO

We start from the condition that two Coulomb forces balance, and reach a general formula for the position of the **null point** between two like charges.

Setting up. Charges q_1 and q_2 are fixed a distance d apart, both of the same sign. A test charge q_0 is placed on the line joining them, a distance x from q_1 and so $(d - x)$ from q_2 .

Step 1 — why the point must lie between them. Only between the two charges do the two forces on q_0 point in opposite directions. Outside, both push the same way and can never cancel. So $0 < x < d$.

Step 2 — write the two force magnitudes from Coulomb's law:

$$F_1 = \frac{kq_0q_1}{x^2}, \quad F_2 = \frac{kq_0q_2}{(d-x)^2}$$

Step 3 — set them equal. They already point opposite ways, so equal magnitudes means zero resultant:

$$\frac{kq_0q_1}{x^2} = \frac{kq_0q_2}{(d-x)^2}$$

Step 4 — cancel k and q_0 , which appear on both sides. This is the key moment: the test charge has vanished, so the answer cannot depend on its size or its sign.

$$\frac{q_1}{x^2} = \frac{q_2}{(d-x)^2}$$

Step 5 — cross-multiply:

$$q_1(d-x)^2 = q_2x^2$$

Step 6 — divide both sides by q_2x^2 :

$$\frac{q_1}{q_2} = \frac{x^2}{(d-x)^2}$$

Step 7 — take the positive square root. We keep only the positive root because Step 1 tells us x and $(d - x)$ are both positive:

$$\sqrt{\frac{q_1}{q_2}} = \frac{x}{d-x}$$

Step 8 — write $\sqrt{q_1/q_2} = s$ for short, and cross-multiply:

$$x = s(d - x) = sd - sx$$

Step 9 — collect the x terms:

$$x + sx = sd$$

Step 10 — take out x and divide:

$$x = \frac{sd}{1+s} = \frac{d\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}}$$

Result.

$$\frac{x}{d-x} = \sqrt{\frac{q_1}{q_2}}, \quad x = \frac{d\sqrt{q_1}}{\sqrt{q_1} + \sqrt{q_2}}$$

What it means. The distances are in the ratio of the **square roots** of the charges, not the charges themselves. So the null point always lies **nearer the smaller charge** — it has to be closer to make up for being weaker. And because q_0 cancelled in Step 4, the same point works for any third charge, positive or negative, large or small.

Conditions used. Point charges, both of the **same sign**, and the test point on the straight line joining them. If the two charges have opposite signs the null point lies outside the pair, on the far side of the smaller charge, and Step 1 must be redone.

WORKED EXAMPLE 19.5

Problem. Charges of $+9 \mu\text{C}$ and $+4 \mu\text{C}$ are fixed 50 cm apart. At what point on the line joining them is the resultant force on a third charge zero?

Solution. Between the two charges the forces on a third charge are in opposite directions, so the null point lies there. Let it be at a distance x from the $9 \mu\text{C}$ charge, so that it is $(0.50 - x)$ from the $4 \mu\text{C}$ charge. For the magnitudes to be equal,

$$\frac{kq_0(9 \mu\text{C})}{x^2} = \frac{kq_0(4 \mu\text{C})}{(0.50 - x)^2}$$

$$\frac{9}{x^2} = \frac{4}{(0.50 - x)^2} \implies \frac{3}{x} = \frac{2}{0.50 - x}$$

(taking positive square roots, since $0 < x < 0.50$)

$$3(0.50 - x) = 2x \implies 1.5 = 5x \implies x = 0.30 \text{ m}$$

The null point is 30 cm from the $9 \mu\text{C}$ charge and 20 cm from the $4 \mu\text{C}$ charge — nearer the smaller charge, as it must be. Note the answer does not depend on the third charge at all.

✓ Chapter summary

- Charge is quantised ($q = \pm ne$, $e = 1.6 \times 10^{-19}$ C), conserved, additive and independent of speed. Charging transfers electrons only.
- $1 \text{ C} = 1 \text{ A s} = 6.25 \times 10^{18}$ electronic charges.
- Charging by induction gives a charge **opposite** to the inducing charge; the earth connection must be broken **before** the rod is taken away.
- Coulomb's law: $F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$ with $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$.
- In a medium the force is reduced: $F_{\text{med}} = F_{\text{vac}}/\epsilon_r$.
- Vector form $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{21}$; the forces obey Newton's third law.
- Superposition: the resultant force is the **vector** sum of the individual pairwise forces, $\vec{F}_1 = \sum_j \vec{F}_{1j}$; combine two forces with $F = \sqrt{F_a^2 + F_b^2 + 2F_a F_b \cos \theta}$.

Practice questions

Group A — Multiple choice (1 mark each)

- Which of the following is **not** a property of electric charge? [1]

a quantisation	b conservation
c additivity	d variation with speed
- The number of electronic charges in one coulomb is about [1]

a 1.6×10^{19}	b 6.25×10^{18}	c 6.02×10^{23}	d 9×10^9
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- A body charged by induction acquires a charge [1]

a of the same sign as the inducing charge
b of the opposite sign
c that is always zero
d equal to twice the inducing charge
- Two charges repel each other with a force F in air. Placed at the same separation in a medium of dielectric constant ϵ_r , the force becomes [1]

a $4F$	b $F/2$	c $F/4$	d $F/16$
---------------	----------------	----------------	-----------------
- If the distance between two point charges is halved, the force between them [1]

a is halved	b is doubled
c becomes four times	d becomes one fourth

ANSWERS TO GROUP A

- (d) — charge is invariant; it is the mass that changes with speed.
- (b) — $1/(1.6 \times 10^{-19}) = 6.25 \times 10^{18}$.
- (b) — electrons are driven away or attracted, leaving the opposite charge behind.

4. (c) — $F_{med} = F_{vac}/\epsilon_r$.
5. (c) — $F \propto 1/r^2$, so halving r multiplies F by 4.

Group B — Short answer (5 marks each)

1. State Coulomb's law and express it in vector form. Hence define the coulomb. [5]
2. Explain, with labelled diagrams, how a metal sphere standing on an insulating base can be given a positive charge by induction. Why must the earth connection be removed before the charging rod? [5]
3. Two point charges of $+5 \mu\text{C}$ and $+10 \mu\text{C}$ are placed 20 cm apart in vacuum. Calculate the force between them, and state its nature. [5]
4. State the principle of quantisation and conservation of charge. How many electrons must be removed from a body to give it a charge of $+2 \mu\text{C}$? [5]
5. Two identical metal spheres carrying charges of $+6 \mu\text{C}$ and $-2 \mu\text{C}$ are brought into contact and then separated to a distance of 10 cm. Find the force between them. [5]

ANSWERS TO GROUP B

3. $F = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 10 \times 10^{-6}}{(0.20)^2} = \frac{9 \times 10^9 \times 5 \times 10^{-11}}{0.04} = \frac{0.45}{0.04} = 11.25 \text{ N}$, repulsive (both charges positive).

4. $n = \frac{q}{e} = \frac{2 \times 10^{-6}}{1.6 \times 10^{-19}} = 1.25 \times 10^{13}$ electrons must be **removed**.

5. Identical spheres in contact share the total charge equally. Total = $+6 - 2 = +4 \mu\text{C}$, so each carries $+2 \mu\text{C}$. Then

$$F = \frac{9 \times 10^9 \times (2 \times 10^{-6})^2}{(0.10)^2} = \frac{3.6 \times 10^{-2}}{0.01} = 3.6 \text{ N}$$

repulsive, since both spheres are now positive.

Group C — Long answer (8 marks each)

1. (a) State Coulomb's law, write it in vector form, and state the principle of superposition of electrostatic forces. [4]
- (b) Three point charges of $+3 \mu\text{C}$ each are placed at the corners of an equilateral triangle of side 30 cm. Calculate the magnitude and direction of the resultant force on one of them. [4]
2. (a) What is meant by charging by induction? Describe, with diagrams, how two metal spheres in contact can be given equal and opposite charges in a single operation. [4]
- (b) Two point charges of $+9 \mu\text{C}$ and $+4 \mu\text{C}$ are fixed 50 cm apart in air. Find the position on the line joining them at which a third charge would experience no resultant force, and explain why the answer does not depend on the size or sign of that third charge. [4]

ANSWERS TO GROUP C

1.(b) With $q = 3 \times 10^{-6} \text{ C}$ and $r = 0.30 \text{ m}$, each pairwise force is

$$F_1 = \frac{9 \times 10^9 \times (3 \times 10^{-6})^2}{(0.30)^2} = \frac{9 \times 10^9 \times 9 \times 10^{-12}}{0.09} = \frac{8.1 \times 10^{-2}}{0.09} = 0.9 \text{ N}$$

The two forces on the chosen charge are each 0.9 N, repulsive, with 60° between them:

$$F = \sqrt{0.9^2 + 0.9^2 + 2(0.9)(0.9)\cos 60^\circ} = 0.9\sqrt{3} = 1.56 \text{ N}$$

directed along the bisector, i.e. radially outward from the centroid of the triangle.

2.(a) Outline: place the two spheres in contact on insulating stands; bring a negatively charged rod near sphere A; electrons are repelled to sphere B; **separate the spheres while the rod is still in position**; then remove the rod. A is left positive and B negative, with equal magnitudes (charge is conserved, and the pair was neutral to begin with).

2.(b) Let the null point be x from the $9 \mu\text{C}$ charge, on the line between the two charges. Equating magnitudes,

$$\frac{9}{x^2} = \frac{4}{(0.50 - x)^2} \Rightarrow \frac{3}{x} = \frac{2}{0.50 - x}$$

$$1.5 - 3x = 2x \Rightarrow x = 0.30 \text{ m}$$

The point lies 30 cm from the $9 \mu\text{C}$ charge (and 20 cm from the $4 \mu\text{C}$ charge). The test charge q_0 appears as a common factor on both sides of the equation and cancels, so neither its magnitude nor its sign affects the position of the null point.

UNIT 20

Electric field

Electricity and Magnetism • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Electric field due to point charges; Field lines
- Gauss Law: Electric Flux
- Application of Gauss law: field of a charged sphere, line charge, charged plane conductor

Coulomb's law tells you the force one charge exerts on another, but it says nothing about how the push gets across empty space. The modern answer is the **field**: a charge alters the space around it, and any second charge placed in that space responds to the local condition it finds there. This unit builds the field of a point charge, draws it, and then introduces Gauss's law — a shortcut that reduces a page of integration to three lines whenever the charge distribution is symmetric.

WHAT THE EXAM ASKS

Group B nearly always wants one Gauss-law derivation (sphere, line, or plane) and one numerical on the combined field of two point charges. Learn the three derivations as one family: they differ only in the Gaussian surface you choose.

20.1 Electric field due to point charges; Field lines

Place a small positive **test charge** q_0 at a point and measure the force $\vec{F}$ on it. The **electric field intensity** at that point is

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

ELECTRIC FIELD INTENSITY

The electric field intensity at a point is the force experienced per unit positive charge placed at that point. It is a vector, directed along the force on a positive charge, with SI unit newton per coulomb (N C^{-1}), equivalently volt per metre (V m^{-1}).

The limit $q_0 \rightarrow 0$ matters: a large test charge would itself push the source charges around and change the very field we are trying to measure.

For a point charge Q , Coulomb's law gives $F = \frac{1}{4\pi\epsilon_0} \frac{Qq_0}{r^2}$, so dividing by q_0 ,

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} = \frac{kQ}{r^2}$$

with $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N m}^2\text{C}^{-2}$ and $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$. The field points **away** from a positive charge and **towards** a negative one.

Because forces add as vectors, so do fields. This is the **principle of superposition**: the field of a group of charges is the vector sum of the fields each would produce alone,

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots = \sum_i \frac{kq_i}{r_i^2} \hat{r}_i$$

Electric field lines

A **field line** is a curve drawn so that the tangent at every point gives the direction of $\vec{E}$ there. Faraday invented them to make an invisible field visible, and their rules are examined directly.

- They start on positive charge and end on negative charge (or run to infinity).
- The tangent at any point gives the direction of the field at that point.
- They never cross. If two lines crossed, the field would have two directions at one point, which is impossible.
- Where the lines crowd together the field is strong; where they spread out it is weak. The number of lines per unit cross-sectional area measures E .
- They are always perpendicular to the surface of a conductor in equilibrium, and there are **no lines inside** a conductor — the field there is zero.
- They do not form closed loops in electrostatics, and they are continuous curves with no breaks in charge-free space.

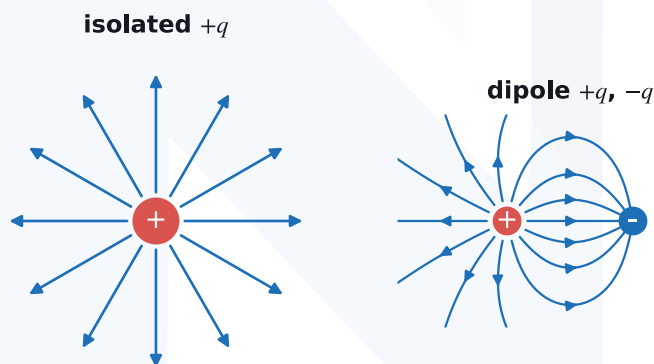


Figure 82. Field lines of an isolated positive charge (radial, outward) and of an electric dipole. Lines leave $+q$ and terminate on $-q$.

FIELD LINES ARE NOT TRAJECTORIES

A charge released from rest does follow the line at first, but a charge with sideways velocity does **not** travel along a field line. The line gives the direction of the force, not of the motion.

WORKED EXAMPLE 20.1

Problem. Point charges $q_1 = +9 \mu\text{C}$ and $q_2 = -4 \mu\text{C}$ are held 1 m apart. Find the point on the line joining them where the resultant field is zero.

Solution. Between the charges both fields point from q_1 towards q_2 , so they add and cannot cancel. Outside, on the side of the larger charge, the field of q_1 always dominates because it is both bigger and nearer. So the null point lies beyond the smaller charge q_2 .

Let it be a distance d beyond q_2 , i.e. $(1 + d)$ from q_1 . Then

$$\frac{k(9 \times 10^{-6})}{(1 + d)^2} = \frac{k(4 \times 10^{-6})}{d^2} \implies \frac{3}{1 + d} = \frac{2}{d}$$

Cross-multiplying, $3d = 2 + 2d$, so $d = 2$ m. The field vanishes 2 m beyond the $-4 \mu\text{C}$ charge, i.e. 3 m from the $+9 \mu\text{C}$ charge on the far side.

WORKED EXAMPLE 20.2

Problem. Two equal charges $q = +2 \mu\text{C}$ sit at $A(-3 \text{ cm}, 0)$ and $B(+3 \text{ cm}, 0)$. Find the field at $P(0, 4 \text{ cm})$.

Solution. $AP = BP = \sqrt{3^2 + 4^2} = 5 \text{ cm} = 0.05 \text{ m}$. Each charge gives

$$E_1 = E_2 = \frac{kq}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{(0.05)^2} = 7.2 \times 10^6 \text{ N C}^{-1}$$

By symmetry the horizontal components are equal and opposite and cancel. The vertical components each carry a factor $\cos \theta = 4/5 = 0.8$, so

$$E = 2E_1 \cos \theta = 2(7.2 \times 10^6)(0.8) = 1.15 \times 10^7 \text{ N C}^{-1}$$

directed along $+y$, i.e. straight away from AB .

20.2 Gauss Law: Electric Flux

Electric flux measures how much field "flows" through a surface. For a flat area A in a uniform field $\vec{E}$, with θ the angle between $\vec{E}$ and the outward **normal** to the area,

$$\Phi_E = \vec{E} \cdot \vec{A} = EA \cos \theta$$

Flux is a **scalar**, with SI unit $\text{N m}^2 \text{C}^{-1}$ (equivalently V m). It is maximum (EA) when the surface faces the field squarely, and zero when the surface lies edge-on, $\theta = 90^\circ$. For a curved surface in a non-uniform field we chop it into elements $d\vec{A}$ and add:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

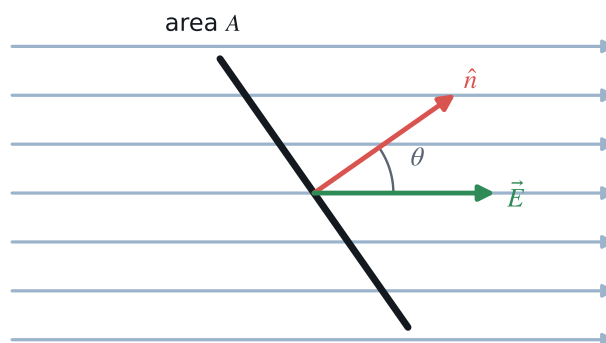


Figure 83. Flux through a plane area seen edge-on. Only the component $E \cos \theta$ along the normal $\hat{n}$ contributes, so $\Phi_E = EA \cos \theta$.

GAUSS'S LAW

The net electric flux through any **closed** surface is $1/\epsilon_0$ times the total charge enclosed by that surface:

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

The imaginary closed surface used is called a **Gaussian surface**.

PROOF OF GAUSS'S LAW FOR A POINT CHARGE

We start from Coulomb's law and reach Gauss's law, by computing the total flux out of a sphere drawn around a single charge.

Setting up. Put a charge q at the **centre** of an imaginary sphere of radius r . The sphere is chosen with the charge at the centre because that is the shape the field respects.

Step 1 — the field on the sphere. By Coulomb's law, at every point of the sphere

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

and this has the **same value everywhere** on the sphere, because every point is the same distance r away.

Step 2 — the direction of the field. The field of a point charge is radial, and the outward normal $d\vec{A}$ of a sphere is also radial. So the angle between them is zero everywhere:

$$\theta = 0, \quad \cos \theta = 1$$

Step 3 — write the flux integral:

$$\Phi = \oint \vec{E} \cdot d\vec{A} = \oint E \, dA \cos \theta$$

Step 4 — put $\cos \theta = 1$ and take the constant E outside the integral. We may do this because Step 1 showed E has the same value at every point:

$$\Phi = E \oint dA$$

Step 5 — do the remaining integral. $\oint dA$ is just the total surface area of the sphere:

$$\oint dA = 4\pi r^2$$

Step 6 — substitute Steps 1 and 5:

$$\Phi = \frac{q}{4\pi\epsilon_0 r^2} \times 4\pi r^2$$

Step 7 — cancel $4\pi r^2$, which appears on top and bottom:

$$\Phi = \frac{q}{\epsilon_0}$$

Result.

$$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

What it means. r disappeared in Step 7, so the flux is the same through a big sphere as through a small one. Physically, the field weakens as $1/r^2$ but the area grows as r^2 , and the two effects cancel exactly. Since the same field lines cross every closed surface drawn round the charge, the result holds for a surface of **any** shape, not just a sphere — which is what makes Gauss's law useful.

Conditions used. The medium is vacuum or air (otherwise ϵ_0 becomes ϵ); the charge is a point charge at the centre; and the surface is closed.

Two consequences are worth stating separately. First, only the charge inside the surface counts: a charge outside contributes equal inward and outward flux, netting zero. Second, if $q_{enc} = 0$ the net flux is zero — but that does not mean $E = 0$ everywhere on the surface.

WORKED EXAMPLE 20.3

Problem. A point charge of $8.85 \mu\text{C}$ is placed at the centre of a cube of side 20 cm. Find (a) the total flux through the cube and (b) the flux through one face.

Solution.

(a) $\Phi = \frac{q}{\epsilon_0} = \frac{8.85 \times 10^{-6}}{8.85 \times 10^{-12}} = 1.0 \times 10^6 \text{ N m}^2\text{C}^{-1}.$

(b) By symmetry the six faces share the flux equally:

$$\Phi_{\text{face}} = \frac{1.0 \times 10^6}{6} = 1.67 \times 10^5 \text{ N m}^2\text{C}^{-1}$$

The side of the cube never entered the calculation — flux depends only on the charge enclosed.

20.3 Application of Gauss law: field of a charged sphere, line charge, charged plane conductor

The method is always the same three steps: **(i)** identify the symmetry, **(ii)** choose a Gaussian surface on which E is constant and either parallel or perpendicular to $d\vec{A}$, **(iii)** equate EA to q_{enc}/ϵ_0 .

(a) Uniformly charged conducting sphere

Let a conducting sphere of radius R carry charge Q , which spreads uniformly over its outer surface.

FIELD OF A UNIFORMLY CHARGED CONDUCTING SPHERE

We start from Gauss's law and reach the field at three places: outside the sphere, at its surface, and inside it.

Setting up. A conducting sphere of radius R carries charge Q . Because it is a conductor, all the charge sits on the **outer surface**, spread evenly. The surface charge density is $\sigma = Q/(4\pi R^2)$.

Case 1 — outside, $r > R$.

Step 1 — choose the Gaussian surface. Take a sphere of radius r concentric with the charged sphere. The charge distribution is spherically symmetric, so the field must be radial and must have the same magnitude everywhere on this surface.

Step 2 — write the flux. Field parallel to the outward normal, constant magnitude, total area $4\pi r^2$:

$$\Phi = E \times 4\pi r^2$$

Step 3 — find the enclosed charge. The Gaussian sphere is bigger than the conductor, so it contains all of it:

$$q_{enc} = Q$$

Step 4 — apply Gauss's law, $\Phi = q_{enc}/\epsilon_0$:

$$E \times 4\pi r^2 = \frac{Q}{\epsilon_0}$$

Step 5 — divide both sides by $4\pi r^2$:

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2}$$

So outside, the sphere behaves **exactly like a point charge Q at its centre.**

Case 2 — at the surface, $r = R$.

Step 6 — put $r = R$ in the result of Step 5:

$$E_s = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{R^2}$$

Step 7 — rewrite using the surface charge density. From $\sigma = Q/(4\pi R^2)$ we get $Q = 4\pi R^2 \sigma$. Substituting:

$$E_s = \frac{4\pi R^2 \sigma}{4\pi \epsilon_0 R^2} = \frac{\sigma}{\epsilon_0}$$

This is the largest value the field reaches anywhere.

Case 3 — inside, $r < R$.

Step 8 — choose a Gaussian sphere of radius r inside the metal. Its flux expression is the same as before:

$$\Phi = E \times 4\pi r^2$$

Step 9 — find the enclosed charge. All the charge lies on the outer surface, at radius R , which is outside this Gaussian sphere. So it encloses nothing:

$$q_{enc} = 0$$

Step 10 — apply Gauss's law:

$$E \times 4\pi r^2 = \frac{0}{\epsilon_0} = 0$$

Step 11 — divide by $4\pi r^2$, which is not zero:

$$E = 0$$

Result.

$$E = \frac{Q}{4\pi \epsilon_0 r^2} \quad (r > R), \quad E_s = \frac{\sigma}{\epsilon_0} \quad (r = R), \quad E = 0 \quad (r < R)$$

What it means. The field is zero everywhere inside a charged conductor, jumps suddenly to σ/ϵ_0 at the surface, then dies away as $1/r^2$. The zero inside is the principle of **electrostatic shielding** — why you are safe inside a car in a thunderstorm, and why sensitive instruments are put in metal boxes.

Conditions used. The sphere is a **conductor** in electrostatic equilibrium (so all charge is on the outer surface and $E=0$ within the metal), the charge is spread uniformly, and the surrounding medium is air.

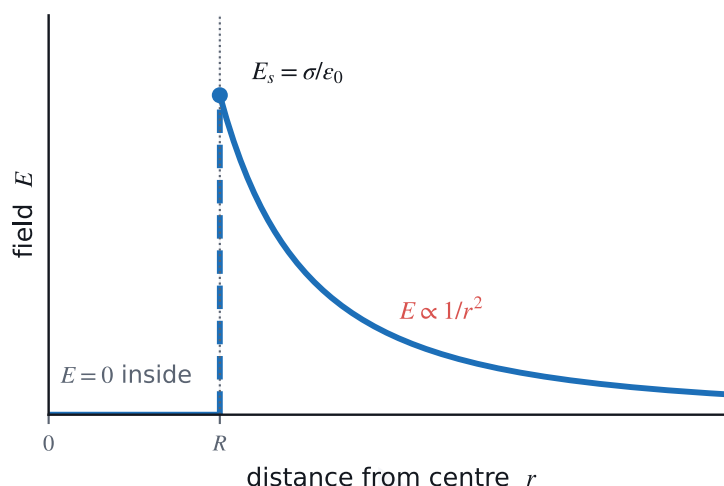


Figure 84. Field of a uniformly charged conducting sphere of radius R . $E = 0$ inside, jumps to σ/ϵ_0 at the surface, then falls as $1/r^2$.

(b) Infinitely long straight line charge

Let charge be spread along a long straight wire with **linear charge density** λ (charge per metre).

FIELD OF AN INFINITELY LONG STRAIGHT LINE CHARGE

We start from Gauss's law with a cylindrical surface and reach the field at distance r from a long charged wire.

Setting up. Charge is spread along a long straight wire with linear charge density λ (coulombs per metre). We want E at perpendicular distance r from the wire.

Step 1 — use the symmetry to guess the field's direction. The wire looks the same from every direction around it and the same from every point along it, so the field can only point straight out from the wire (radially), and its magnitude can depend only on r .

Step 2 — choose the Gaussian surface to match that symmetry. Take a cylinder of radius r and length l , with the wire along its axis.

Step 3 — flux through the two flat ends. There the outward normal points along the wire, while $\vec{E}$ points perpendicular to the wire. Perpendicular vectors give no flux:

$$\Phi_{ends} = 0$$

Step 4 — flux through the curved surface. Here $\vec{E}$ is parallel to the outward normal everywhere, and has the same magnitude everywhere (Step 1). The curved area is $2\pi rl$:

$$\Phi_{curved} = E \times 2\pi rl$$

Step 5 — total flux is the sum of Steps 3 and 4:

$$\Phi = E \times 2\pi rl$$

Step 6 — find the enclosed charge. A length l of wire lies inside the cylinder, and each metre carries λ :

$$q_{enc} = \lambda l$$

Step 7 — apply Gauss's law:

$$E \times 2\pi r l = \frac{\lambda l}{\epsilon_0}$$

Step 8 — cancel l from both sides. The length of cylinder we chose was arbitrary, so it had better not appear in the answer — and it does not:

$$E \times 2\pi r = \frac{\lambda}{\epsilon_0}$$

Step 9 — divide both sides by $2\pi r$:

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

Step 10 — write it with $k = 1/(4\pi\epsilon_0)$. Since $\frac{1}{2\pi\epsilon_0} = \frac{2}{4\pi\epsilon_0} = 2k$:

$$E = \frac{2k\lambda}{r}$$

Result.

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$$

What it means. The field falls off as $1/r$, **not** as $1/r^2$. A line of charge is a "fatter" source than a point charge: as you move away, the wire still stretches out beside you, so the field dies more slowly. It is directed away from the wire if λ is positive, towards it if negative.

Conditions used. The wire must be effectively **infinite** — in practice, r much smaller than the length of the wire, and the point not near either end. Otherwise Step 1's symmetry fails.

Note the field falls off as $1/r$, not $1/r^2$ — a line of charge is a "fatter" source than a point.

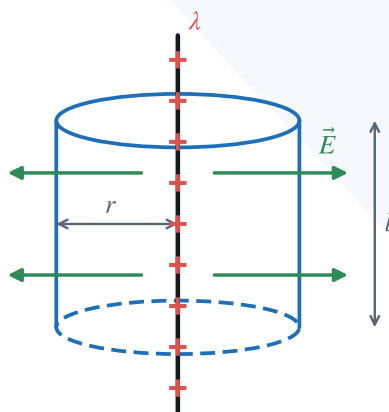


Figure 85. Coaxial cylindrical Gaussian surface of radius r and length l around a line charge λ . Flux crosses only the curved surface.

(c) Charged plane conductor

FIELD JUST OUTSIDE A CHARGED PLANE CONDUCTOR

We start from Gauss's law with a flat "pill box" and reach the field close to a large charged plate — and then see why a non-conducting sheet gives half as much.

Setting up. A large flat conductor carries surface charge density σ (coulombs per square metre). We want E at a point just outside its surface. Because the plate is large and flat, the field there must be **perpendicular** to the surface and the same at all nearby points.

Step 1 — choose the Gaussian surface. Take a short cylinder (a "pill box") of cross-sectional area A , with one flat face just outside the surface and the other buried **inside** the metal.

Step 2 — flux through the inner face is zero. Inside a conductor in equilibrium $E=0$, so nothing crosses that face:

$$\Phi_{inner} = 0$$

Step 3 — flux through the curved side is zero. There $\vec{E}$ lies along the surface, parallel to the curved wall, so it crosses nothing:

$$\Phi_{curved} = 0$$

Step 4 — flux through the outer face. Here $\vec{E}$ is perpendicular to the face, i.e. parallel to its outward normal, and constant across it:

$$\Phi_{outer} = EA$$

Step 5 — total flux is the sum of Steps 2, 3 and 4:

$$\Phi = EA$$

Step 6 — find the enclosed charge. The pill box cuts out a patch of surface of area A , and each square metre carries σ :

$$q_{enc} = \sigma A$$

Step 7 — apply Gauss's law:

$$EA = \frac{\sigma A}{\epsilon_0}$$

Step 8 — cancel A from both sides:

$$E = \frac{\sigma}{\epsilon_0}$$

Now the thin non-conducting sheet.

Step 9 — what changes. A thin insulating sheet has charge on it but no interior in which E must vanish. The field now comes out of **both** faces of the pill box, so the flux is doubled:

$$\Phi = EA + EA = 2EA$$

Step 10 — the enclosed charge is the same σA , so Gauss's law gives

$$2EA = \frac{\sigma A}{\epsilon_0}$$

Step 11 — cancel A and divide by 2:

$$E = \frac{\sigma}{2\epsilon_0}$$

Result.

$$E = \frac{\sigma}{\epsilon_0} \text{ (charged conductor),} \quad E = \frac{\sigma}{2\epsilon_0} \text{ (thin non-conducting sheet)}$$

What it means. r never appeared, so the field near a large plate does **not** weaken as you move away — it is uniform. That is exactly why a parallel-plate capacitor has a uniform field between its plates. The factor of two is not a formula to memorise but a consequence of how many faces the flux comes out of.

Conditions used. The plate is large compared with the distance of the point from it (so edge effects can be ignored), and the charge is spread uniformly.

σ/ϵ_0 **OR** $\sigma/2\epsilon_0$?

Use σ/ϵ_0 just outside a **charged conductor** (all the charge is on one face and there is no field inside the metal). Use $\sigma/2\epsilon_0$ for a **thin non-conducting sheet**, which radiates field from both sides. Mixing them up is the single most common Group B error in this unit.

Charge distribution	Gaussian surface	Field	Dependence
Point charge q	sphere	$\frac{kq}{r^2}$	$1/r^2$
Conducting sphere, $r > R$	sphere	$\frac{kQ}{r^2}$	$1/r^2$
Conducting sphere, $r < R$	sphere	0	—
Long line charge	cylinder	$\frac{\lambda}{2\pi\epsilon_0 r}$	$1/r$
Charged conductor surface	pill box	$\frac{\sigma}{\epsilon_0}$	independent of r
Thin non-conducting sheet	pill box	$\frac{\sigma}{2\epsilon_0}$	independent of r

WORKED EXAMPLE 20.4

Problem. A hollow metal sphere of radius 10 cm carries a charge of $5 \mu\text{C}$. Find the field at $r = 5 \text{ cm}$, at the surface, and at $r = 20 \text{ cm}$ from the centre. Also find the surface charge density.

Solution.

At $r = 5 \text{ cm}$ (inside the conductor): $E = 0$.

At the surface, $r = R = 0.10 \text{ m}$:

$$E = \frac{9 \times 10^9 \times 5 \times 10^{-6}}{(0.10)^2} = 4.5 \times 10^6 \text{ N C}^{-1}$$

At $r = 0.20 \text{ m}$:

$$E = \frac{9 \times 10^9 \times 5 \times 10^{-6}}{(0.20)^2} = 1.125 \times 10^6 \text{ N C}^{-1}$$

Surface charge density: $\sigma = \frac{Q}{4\pi R^2} = \frac{5 \times 10^{-6}}{4\pi(0.10)^2} = 3.98 \times 10^{-5} \text{ C m}^{-2}$.

As a check, $\sigma/\epsilon_0 = 3.98 \times 10^{-5}/8.85 \times 10^{-12} = 4.5 \times 10^6 \text{ N C}^{-1}$, matching the surface field.

✓ Chapter summary

- $\vec{E} = \vec{F}/q_0$, measured in $\text{N C}^{-1} = \text{V m}^{-1}$; for a point charge $E = kQ/r^2$ with $k = 1/4\pi\epsilon_0 = 9 \times 10^9 \text{ SI units}$.
- Fields superpose vectorially: $\vec{E} = \sum \vec{E}_i$.
- Field lines run from $+$ to $-$, never cross, are densest where E is largest, meet conductors at right angles and do not exist inside a conductor.
- Electric flux $\Phi_E = EA \cos \theta$, a scalar with unit $\text{N m}^2 \text{ C}^{-1}$.
- Gauss's law: $\oint \vec{E} \cdot d\vec{A} = q_{\text{enc}}/\epsilon_0$; only enclosed charge contributes.
- Charged conducting sphere: $E = kQ/r^2$ outside, σ/ϵ_0 at the surface, 0 inside.
- Long line charge: $E = \lambda/2\pi\epsilon_0 r$ (falls as $1/r$).
- Charged conductor surface: $E = \sigma/\epsilon_0$; thin non-conducting sheet: $E = \sigma/2\epsilon_0$. Both are uniform.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of electric field intensity is [1]
 - N C
 - N C^{-1}
 - C N^{-1}
 - $\text{N m}^2 \text{ C}^{-1}$
- The electric field at a point inside a charged hollow metal sphere is [1]
 - kQ/r^2
 - σ/ϵ_0
 - zero
 - infinite
- The net electric flux through a closed surface enclosing an electric dipole is [1]
 - q/ϵ_0
 - $2q/\epsilon_0$
 - zero
 - $q/2\epsilon_0$
- For an infinitely long charged straight wire, E varies with distance r as [1]
 - $1/r^2$
 - $1/r$
 - r
 - independent of r
- Two electric field lines can never intersect because [1]

- a they are parallel
- b the field would have two directions at one point
- c charge is quantised
- d flux would be infinite

ANSWERS TO GROUP A

1. (b) — $E = F/q$, so newton per coulomb.
2. (c) — the Gaussian surface inside encloses no charge.
3. (c) — $q_{enc} = +q - q = 0$.
4. (b) — the cylindrical Gaussian surface gives $E = \lambda/2\pi\epsilon_0 r$.
5. (b) — a unique tangent means a unique field direction.

Group B — Short answer (5 marks each)

1. Define electric field intensity and electric flux. State Gauss's law and name its SI units. [5]
2. Using Gauss's law, derive an expression for the electric field due to an infinitely long straight uniformly charged wire. [5]
3. Two point charges $+9 \mu\text{C}$ and $-4 \mu\text{C}$ are placed 1 m apart in air. Locate the point where the resultant electric field is zero. [5]
4. A charge of $17.7 \mu\text{C}$ is placed at the centre of a cube of side 10 cm . Calculate the electric flux through one face of the cube. [5]
5. A metal sphere of radius 5 cm carries a charge of $2 \mu\text{C}$. Find the field at 2 cm , at 5 cm and at 10 cm from its centre. [5]

ANSWERS TO GROUP B

3. The point lies beyond the $-4 \mu\text{C}$ charge. With d measured from it, $9/(1+d)^2 = 4/d^2$ gives $3d = 2 + 2d$, so $d = 2 \text{ m}$ — that is, 3 m from the $+9 \mu\text{C}$ charge on the side away from it.
4. Total flux $= q/\epsilon_0 = 17.7 \times 10^{-6}/8.85 \times 10^{-12} = 2.0 \times 10^6 \text{ N m}^2\text{C}^{-1}$. One face gets one sixth: $3.33 \times 10^5 \text{ N m}^2\text{C}^{-1}$. The side length is irrelevant.
5. At 2 cm (inside): $E = 0$. At the surface: $E = 9 \times 10^9 \times 2 \times 10^{-6}/(0.05)^2 = 7.2 \times 10^6 \text{ N C}^{-1}$. At 10 cm : $E = 9 \times 10^9 \times 2 \times 10^{-6}/(0.10)^2 = 1.8 \times 10^6 \text{ N C}^{-1}$.

Group C — Long answer (8 marks each)

1. (a) State Gauss's law and prove it for a point charge enclosed by a spherical surface. [3]
 (b) Using it, obtain the electric field due to a uniformly charged conducting sphere at a point outside, on and inside the sphere, and sketch E against r . [5]
2. (a) Apply Gauss's law to find the electric field near the surface of a charged plane conductor, and explain why a thin non-conducting sheet of the same charge density gives half that value. [4]
 (b) Two large parallel metal plates carry surface charge densities $+3.54 \times 10^{-6} \text{ C m}^{-2}$ and $-3.54 \times 10^{-6} \text{ C m}^{-2}$. Find the field between the plates and the force on an electron placed there. [4]

ANSWERS TO GROUP C

1. (a) Put q at the centre of a sphere of radius r . $E = kq/r^2$ is constant and parallel to $d\vec{A}$, so $\oint \vec{E} \cdot d\vec{A} = E(4\pi r^2) = q/\epsilon_0$. (b) Outside, $E = kQ/r^2$; at the surface, $E = kQ/R^2 = \sigma/\epsilon_0$; inside, the enclosed charge is zero so $E = 0$. The graph is flat at zero up to R , jumps to σ/ϵ_0 , then decays as $1/r^2$.

2. (b) Between the plates, $E = \sigma/\epsilon_0 = 3.54 \times 10^{-6}/8.85 \times 10^{-12} = 4.0 \times 10^5 \text{ V m}^{-1}$, directed from the positive to the negative plate. Force on an electron: $F = eE = 1.6 \times 10^{-19} \times 4.0 \times 10^5 = 6.4 \times 10^{-14} \text{ N}$, directed towards the positive plate.



UNIT 21

Potential, potential difference and potential energy

Electricity and Magnetism • 4 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Potential difference, Potential due to a point charge, potential energy, electron volt
- Equipotential lines and surfaces
- Potential gradient

The electric field tells you the force on a charge. This unit tells you the **energy**. Because the electrostatic force is conservative — the work it does between two points does not depend on the path taken — we can attach a single number, the potential, to every point in space. Potential is a scalar, so it adds by ordinary arithmetic, which makes it far easier to handle than the field vector. Almost every practical device, from a torch cell to a cathode-ray tube, is described in volts rather than in newtons per coulomb.

SCALAR, NOT VECTOR

The one advantage you must exploit: potentials from several charges add as **signed numbers**, with no angles and no components. If a question can be done with V instead of $\vec{E}$, do it with V .

21.1 Potential difference, Potential due to a point charge, potential energy, electron volt

Move a charge q slowly from point A to point B in an electric field. The external work needed, W_{AB} , is the same for every path (this is what "conservative" means). Divide it by the charge and the result depends only on the two points:

POTENTIAL DIFFERENCE

The potential difference between two points is the work done per unit positive charge in carrying it from one point to the other without acceleration:

$$V_B - V_A = \frac{W_{AB}}{q}$$

Its SI unit is the **volt**: $1 \text{ V} = 1 \text{ J C}^{-1}$.

Taking A at infinity, where we agree the potential is zero, gives the **absolute potential** at a point: the work done per unit positive charge in bringing it from infinity to that point.

Potential due to a point charge

POTENTIAL AT A POINT DUE TO A POINT CHARGE, $V = \frac{kQ}{r}$

We start from Coulomb's law for the force on a test charge, and we reach a formula for the potential at a point a distance r from an isolated charge Q .

Setting up. A charge Q is fixed at the point O . The point P lies a distance r from O . We bring a small positive test charge q_0 from infinity (where we agree the potential is zero) up to P , moving it so slowly that it never picks up any kinetic energy. By definition, the potential at P is the work the outside agent must do, per unit charge.

Step 1 — Coulomb's law gives the force at every point on the way. When the test charge is at a distance x from O , the field of Q pushes it outward with a force

$$F = \frac{kQq_0}{x^2}$$

This force is not constant. It grows as x gets smaller, so we cannot use "work = force $\times$ distance". We must add up the work in tiny steps, which means an integral.

Step 2 — write the work done by the field over one tiny step. Measure x outward from O . The force acts along the direction of increasing x , so for a small displacement dx the work done by the field is

$$dW_{field} = F dx = \frac{kQq_0}{x^2} dx$$

Over a step this small the force does not change, so here "work = force $\times$ distance" is safe.

Step 3 — add up all the tiny steps from infinity down to P . The charge travels from $x = \infty$ to $x = r$, and those become the limits of the integral:

$$W_{field} = \int_{\infty}^r \frac{kQq_0}{x^2} dx$$

Step 4 — take the constants outside. The quantities k , Q and q_0 do not change while the charge moves, so only $1/x^2$ stays inside:

$$W_{field} = kQq_0 \int_{\infty}^r \frac{dx}{x^2}$$

Step 5 — carry out the integration. The integral of x^{-2} is $-x^{-1}$:

$$W_{field} = kQq_0 \left[-\frac{1}{x} \right]_{\infty}^r$$

Step 6 — substitute the limits: value at the top limit minus value at the bottom limit.

$$W_{field} = kQq_0 \left[\left(-\frac{1}{r} \right) - \left(-\frac{1}{\infty} \right) \right]$$

Step 7 — the infinity term is zero. One divided by a very large number is as good as zero, and this is exactly why infinity is a convenient place to call "zero potential":

$$W_{\text{field}} = -\frac{kQq_0}{r}$$

The minus sign is correct and physical. Two positive charges repel, so the field pushes backwards during the inward journey and does negative work.

Step 8 — find the work done by the outside agent. The agent pushes with a force equal and opposite to the field's force at every instant, which is what keeps the charge from speeding up. Equal and opposite force along the same path means equal and opposite work:

$$W_{\text{ext}} = -W_{\text{field}} = +\frac{kQq_0}{r}$$

Step 9 — use the definition of potential: work per unit charge.

$$V = \frac{W_{\text{ext}}}{q_0} = \frac{kQq_0}{q_0 r}$$

Step 10 — cancel q_0 from top and bottom.

$$V = \frac{kQ}{r} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

Result. The absolute potential a distance r from a point charge Q is $V = kQ/r$, where $k = 1/4\pi\epsilon_0 = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$.

What it means. The test charge q_0 has cancelled out, so V belongs to the source charge Q and the point P alone — it describes the space around Q whether or not anything is placed there. The potential is a scalar, so it carries the sign of Q and nothing else: positive around a positive charge, negative around a negative one.

Conditions used. The charge Q is a point charge (or a uniformly charged sphere, seen from outside) at rest in vacuum or air; the potential is taken as zero at infinity; and the test charge is moved slowly, so no kinetic energy is gained or lost.

THE EXAMINER IS LOOKING FOR

1. A statement of the definition being used: $V = W_{\text{ext}}/q_0$, with zero potential at infinity.
2. Coulomb's law written for a general distance x , with a clear reason for integrating (the force varies).
3. The integral with the correct limits ∞ to r .
4. The integration and substitution of limits, including $1/\infty = 0$.
5. The step from W_{field} to W_{ext} (opposite sign), and the cancellation of q_0 giving $V = kQ/r$.

Note carefully: $V \propto 1/r$ while $E \propto 1/r^2$. The potential is **positive** near a positive charge and **negative** near a negative charge, and it carries that sign into every sum. For several charges,

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots \right)$$

which is an ordinary algebraic sum — no resolution into components.

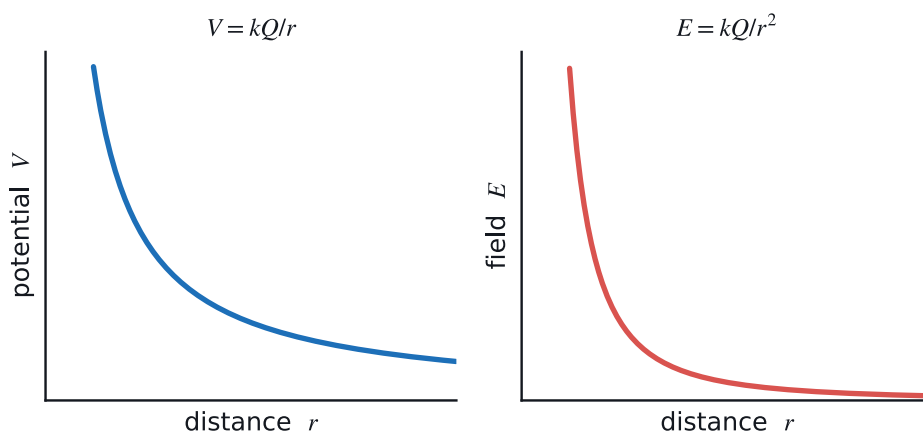


Figure 86. For a point charge, $V \propto 1/r$ falls off more slowly than $E \propto 1/r^2$. Both are taken as zero at infinity.

Electric potential energy

The **potential energy** of a charge q at a point where the potential is V is simply $U = qV$ — the work stored in putting it there. For two point charges a distance r apart,

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

This is positive for like charges (you had to push them together, energy is stored) and negative for unlike charges (the field did the work for you, so you must supply energy to separate them). For a system of three charges, add the energy of each **pair**:

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right)$$

	Potential V	Potential energy U
Belongs to	a point in space	a charge, or a system of charges
Defining relation	$V = W/q$	$U = qV$
SI unit	volt (J C^{-1})	joule
Point charge	kQ/r	$kq_1 q_2 / r$ (a pair)
Type	scalar	scalar

Potential of a charged conducting sphere

A conducting sphere of radius R carrying charge Q behaves outside itself exactly like a point charge, so for $r \geq R$ the potential is $V = kQ/r$. Inside the metal the field is zero, so by $E = -dV/dr$ the potential cannot change: the whole sphere, surface and interior alike, sits at the single value

$$V_{\text{inside}} = V_{\text{surface}} = \frac{kQ}{R}$$

Two useful consequences follow. First, for a given charge a **small** sphere reaches a much higher potential than a large one, which is why charge sprayed onto a small conductor leaks away readily. Second, if two charged spheres are joined by a wire, charge flows until their potentials — not their charges — are equal. The smaller sphere ends with less charge but a larger surface density, and therefore a stronger

surface field. This is the physics behind the pointed lightning conductors on the tall temples and towers of the Kathmandu valley: the sharp tip has a very small radius of curvature, so the field there becomes large enough to ionise the air and quietly discharge the cloud.

The electron volt

Charges on the atomic scale are tiny, so the joule is a clumsy unit.

ELECTRON VOLT

One electron volt is the kinetic energy gained by an electron accelerated through a potential difference of one volt:

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ C} \times 1 \text{ V} = 1.6 \times 10^{-19} \text{ J}$$

It is a unit of **energy**, not of potential. Useful multiples: $1 \text{ keV} = 10^3 \text{ eV}$, $1 \text{ MeV} = 10^6 \text{ eV}$. You will meet it again in nuclear physics.

POTENTIAL IS NOT POTENTIAL ENERGY

"The potential at P is 100 J " is meaningless. Potential is measured in volts, potential energy in joules, and they are linked by $U = qV$. Likewise the electron volt is an energy, so never write "a p.d. of 5 eV ".

WORKED EXAMPLE 21.1

Problem. Three charges of $+2 \mu\text{C}$ each are placed at the corners of an equilateral triangle of side 10 cm . Find (a) the potential at the centroid and (b) the potential energy of the system.

Solution.

(a) The centroid is $r = a/\sqrt{3} = 0.10/1.732 = 0.0577 \text{ m}$ from each charge. Potentials add algebraically:

$$V = 3 \times \frac{kq}{r} = 3 \times \frac{9 \times 10^9 \times 2 \times 10^{-6}}{0.0577} = 9.35 \times 10^5 \text{ V}$$

(b) There are three identical pairs, each a distance $a = 0.10 \text{ m}$ apart:

$$U = 3 \times \frac{kq^2}{a} = 3 \times \frac{9 \times 10^9 \times (2 \times 10^{-6})^2}{0.10} = 3 \times 0.36 = 1.08 \text{ J}$$

WORKED EXAMPLE 21.2

Problem. Charges $+4 \mu\text{C}$ and $-4 \mu\text{C}$ are fixed at $A(0, 0)$ and $B(0.4 \text{ m}, 0)$. Find (a) the potential at $P(0, 0.3 \text{ m})$ and (b) the work needed to bring a charge of $+2 \mu\text{C}$ from infinity to P .

Solution.

(a) $AP = 0.3 \text{ m}$ and $BP = \sqrt{0.4^2 + 0.3^2} = 0.5 \text{ m}$, so

$$V = kq\left(\frac{1}{0.3} - \frac{1}{0.5}\right) = 9 \times 10^9 \times 4 \times 10^{-6} \times (3.333 - 2.000) = 4.8 \times 10^4 \text{ V}$$

(b) $W = qV = 2 \times 10^{-6} \times 4.8 \times 10^4 = 9.6 \times 10^{-2} \text{ J} = 0.096 \text{ J}.$

WORKED EXAMPLE 21.3

Problem. An electron starting from rest is accelerated through a potential difference of 400 V. Find its kinetic energy in eV and in joules, and its final speed. Take $m_e = 9.1 \times 10^{-31} \text{ kg}$.

Solution. The work done on the electron is $W = eV$, so its kinetic energy is 400 eV, that is

$$KE = 400 \times 1.6 \times 10^{-19} = 6.4 \times 10^{-17} \text{ J}$$

Then from $KE = \frac{1}{2}m_ev^2$,

$$v = \sqrt{\frac{2 \times 6.4 \times 10^{-17}}{9.1 \times 10^{-31}}} = \sqrt{1.41 \times 10^{14}} = 1.19 \times 10^7 \text{ m s}^{-1}$$

21.2 Equipotential lines and surfaces

EQUIPOTENTIAL SURFACE

An equipotential surface is a surface on which the electric potential has the same value at every point. Its intersection with the plane of a diagram is an equipotential line.

Their properties follow directly from $W = q(V_A - V_B)$:

- **No work is done** in moving a charge between two points of the same equipotential surface, because $V_A - V_B = 0$. This is true for any path on the surface.
- Equipotential surfaces are **always perpendicular to field lines**. If $\vec{E}$ had a component along the surface, moving a charge along the surface would do work, contradicting the previous point.
- Two equipotential surfaces **never intersect**, since a point cannot have two potentials at once.
- Where the surfaces are **closely spaced the field is strong**, because the same drop in V occurs over a shorter distance.
- The **surface of any charged conductor in equilibrium is an equipotential**, and so is its whole interior volume, since $E = 0$ inside.

The first two properties are the ones examiners ask you to prove, so here they are in full.

NO WORK IS DONE IN MOVING A CHARGE OVER AN EQUIPOTENTIAL SURFACE

We start from the definition of potential difference and reach two results: the work is zero for any path on the surface, and the field must meet the surface at a right angle.

Setting up. Let A and B be any two points lying on the same equipotential surface, with potentials V_A and V_B . A charge q is carried from A to B along any path that stays on the surface.

Step 1 — write the work in terms of potential difference. Potential difference is defined as work done per unit charge, so the work done against the field in taking q from A to B is

$$W = q (V_B - V_A)$$

Step 2 — use the definition of an equipotential surface. Every point of the surface has the same potential, and A and B are both on it, so

$$V_A = V_B$$

Step 3 — substitute this into the expression for the work.

$$W = q (V_A - V_A)$$

Step 4 — the bracket is zero, so the work is zero.

$$W = q \times 0 = 0$$

Nothing in Steps 1 to 4 mentioned the shape of the path, so the answer is zero for every path on the surface, long or short, straight or curved.

Step 5 — now write the same work the other way, using force and displacement. The field exerts a force qE on the charge. If the charge is moved a distance s along the surface and the angle between $\vec{E}$ and that displacement is θ , then

$$W = qE s \cos \theta$$

Step 6 — set this equal to zero, the answer from Step 4.

$$qE s \cos \theta = 0$$

Step 7 — decide which factor is the zero one. We have a real charge ($q \neq 0$), a real field ($E \neq 0$) and a real displacement ($s \neq 0$), so the only factor left that can vanish is the cosine:

$$\cos \theta = 0$$

Step 8 — solve for the angle.

$$\theta = 90^\circ$$

Result. No work is done in moving a charge between any two points of an equipotential surface, $W = 0$; and the electric field is everywhere perpendicular to the equipotential surface.

What it means. Moving along an equipotential is like walking around a hill at a fixed height — you neither climb nor descend, so gravity takes and gives nothing. The field, like gravity on the hillside, points straight "downhill", at right angles to the contour.

Conditions used. The charges producing the field are at rest (an electrostatic field), and the charge q is moved slowly, so no kinetic energy is gained. In Step 7 the field must be non-zero; where $E = 0$, as inside a conductor, the angle is simply undefined and the work is zero anyway.

The shape follows the source: concentric spheres around a point charge, parallel planes in the uniform field between charged plates, and coaxial cylinders around a long charged wire.

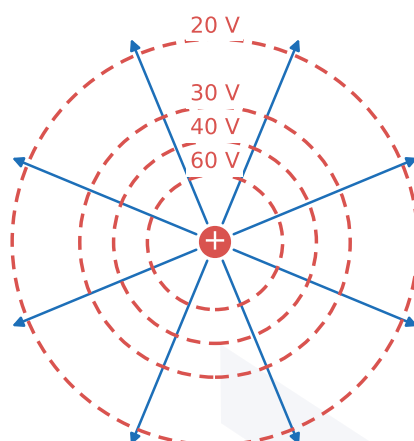


Figure 87. Equipotentials (dashed) and field lines (solid) around a positive point charge. Equal steps of 20 V lie further and further apart because $V \propto 1/r$.

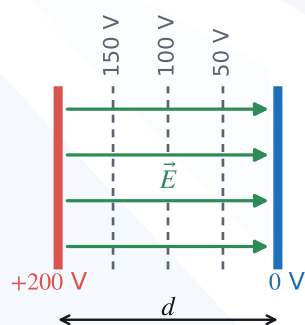


Figure 88. Uniform field between parallel plates. The equipotentials are flat planes parallel to the plates, and $E = V/d$ is the same everywhere between them.

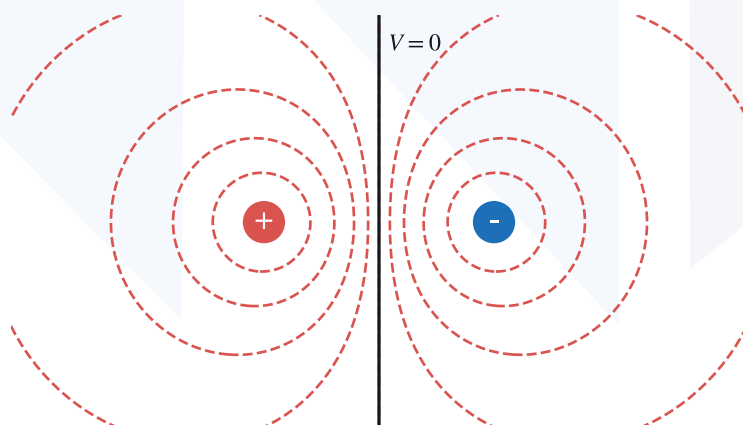


Figure 89. Equipotential contours of a dipole. The line midway between the charges is the $V=0$ equipotential, on which no work is done.

21.3 Potential gradient

The rate at which potential changes with distance is the **potential gradient**. It is directly related to the field.

FIELD IS MINUS THE POTENTIAL GRADIENT, AND HENCE $E = V/d$

We start from the definition of potential difference as work per unit charge and reach a relation that connects the field at a point to how fast the potential is changing there. We then apply it to the uniform field between two plates.

Setting up. Take two nearby points A and B on a field line, a small distance dx apart, with B further along the direction of $\vec{E}$. Let the potential be V at A and $V + dV$ at B . A positive charge q is moved from A to B . Because dx is small, the field is effectively constant over the journey.

Step 1 — write the force the field exerts on the charge. From the definition of electric field strength, force = charge $\times$ field:

$$F = qE$$

Step 2 — work out the work done by the field over this small step. The force and the displacement point the same way, so work = force $\times$ distance:

$$dW_{\text{field}} = qE \, dx$$

Step 3 — write the same work again, this time from the definition of potential difference. Work done by the field = charge $\times$ (potential at the start – potential at the end):

$$dW_{\text{field}} = q[V - (V + dV)]$$

Step 4 — simplify the bracket. The V and $+V$ cancel and only $-dV$ survives:

$$dW_{\text{field}} = -q \, dV$$

Step 5 — the two expressions describe the same work, so set them equal.

$$qE \, dx = -q \, dV$$

Step 6 — cancel q from both sides. It is a real, non-zero charge, so this is allowed — and it shows the result does not depend on what we moved:

$$E \, dx = -dV$$

Step 7 — divide both sides by dx .

$$E = -\frac{dV}{dx}$$

This is the general result. The rest applies it to parallel plates.

Step 8 — say what "uniform field" means for this equation. Between two large parallel plates the field E has the same value everywhere, so dV/dx is a constant. A constant rate of change

means the potential falls **linearly** with distance, and the small changes dV and dx may be replaced by the total changes measured across the whole gap:

$$E = -\frac{\Delta V}{\Delta x}$$

Step 9 — put in the values for the gap. Let the plates be a distance d apart, the positive plate at potential V and the earthed plate at 0. Going across the gap in the direction of the field, the potential changes from V to 0, so $\Delta V = 0 - V = -V$ and $\Delta x = d$:

$$E = -\frac{(-V)}{d}$$

Step 10 — cancel the two minus signs.

$$E = \frac{V}{d}$$

Result. The field at a point equals minus the potential gradient there, $E = -dV/dx$; and for the uniform field between parallel plates a distance d apart with potential difference V , the magnitude is $E = V/d$.

What it means. The minus sign says $\vec{E}$ points the way V decreases fastest, so a positive charge released from rest always slides "downhill" in potential. It also explains the unit: V m^{-1} and N C^{-1} are the same thing. Closely packed equipotentials mean a steep gradient, and therefore a strong field.

Conditions used. Static charges; dx small enough that E is constant over it; and for $E = V/d$ the field must be uniform, which means large plates and a small separation so that the edge effects can be ignored.

THE EXAMINER IS LOOKING FOR

1. The two independent expressions for the same work: $dW = qE dx$ from force $\times$ distance, and $dW = -q dV$ from the definition of potential difference.
2. Equating them and cancelling q .
3. The final form $E = -dV/dx$, **with the minus sign** and a sentence saying what it means.
4. For the V/d part: a clear statement that the field is uniform, so the gradient is constant and dV/dx becomes $-V/d$.

For the **uniform** field between two parallel plates a distance d apart held at a potential difference V ,

$$E = \frac{V}{d}$$

is the working formula for every parallel-plate and cathode-ray-tube problem. For a point charge you can check the general relation directly: $V = kQ/r$ gives $-dV/dr = kQ/r^2 = E$, as it must.

The same relation run backwards recovers the potential from the field: since $dV = -E dx$, the **change in potential between two points is minus the area under the E - x graph** between them. For a charged conducting sphere, for instance, $E = 0$ inside contributes no area, so V stays flat at kQ/R right up to the surface and only then begins to fall.

Arrangement	Shape of equipotential surfaces	Field
Isolated point charge	concentric spheres	radial, kQ/r^2
Long charged wire	coaxial cylinders	radial, $\lambda/2\pi\epsilon_0 r$
Parallel charged plates	parallel planes	uniform, V/d
Any charged conductor	its own surface, and every interior point	zero inside

READING A GRAPH OF V AGAINST r

E is minus the **slope** of the V - r graph. Where the graph is flat (inside a charged hollow conductor, where V is constant) the field is zero; where the graph is steep, the field is large.

WORKED EXAMPLE 21.4

Problem. Two parallel plates 2 cm apart are connected to a 200 V supply. Find (a) the field between them, (b) the force on a charge of 5 nC placed between them, and (c) the work done in moving that charge from one plate to the other.

Solution.

(a) $E = \frac{V}{d} = \frac{200}{0.02} = 1.0 \times 10^4 \text{ V m}^{-1}$.

(b) $F = qE = 5 \times 10^{-9} \times 1.0 \times 10^4 = 5 \times 10^{-5} \text{ N}$.

(c) $W = qV = 5 \times 10^{-9} \times 200 = 1.0 \times 10^{-6} \text{ J}$.

Check: $W = Fd = 5 \times 10^{-5} \times 0.02 = 1.0 \times 10^{-6} \text{ J}$. The two routes agree, as they must in a uniform field.

✓ Chapter summary

- Potential difference $V_B - V_A = W_{AB}/q$, in volts (J C^{-1}); absolute potential takes $V=0$ at infinity.
- Point charge: $V = kQ/r$ — a **scalar** that carries the sign of the charge, so potentials add algebraically.
- Potential energy: $U = qV$; for a pair of charges $U = kq_1q_2/r$; for a system add over all pairs.
- $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ — a unit of energy, not of potential.
- An equipotential surface has constant V : no work is done moving a charge on it, and it is everywhere perpendicular to the field lines.
- Conductors in equilibrium are equipotential volumes; field lines meet their surfaces at right angles.
- $E = -dV/dx$: the field is the negative potential gradient, measured in $\text{V m}^{-1} = \text{N C}^{-1}$. For parallel plates, $E = V/d$.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of electric potential is [1]
 - J C
 - J C^{-1}
 - N C^{-1}
 - C V^{-1}
- The work done in moving a charge between two points of an equipotential surface is [1]
 - qV
 - infinite
 - zero
 - q/V
- One electron volt is equal to [1]
 - $1.6 \times 10^{-19} \text{ J}$
 - $1.6 \times 10^{-19} \text{ V}$
 - $9 \times 10^9 \text{ J}$
 - 1 J
- The electric field and the potential gradient are related by [1]
 - $E = dV/dx$
 - $E = -dV/dx$
 - $V = -dE/dx$
 - $E = -Vx$
- At the midpoint of the line joining two equal and opposite charges, the potential and the field are respectively [1]
 - zero and zero
 - zero and non-zero
 - non-zero and zero
 - both non-zero

ANSWERS TO GROUP A

- (b) — $V = W/q$, so joule per coulomb.
- (c) — $V_A - V_B = 0$, hence $W = q(V_A - V_B) = 0$.
- (a) — $1 \text{ eV} = e \times 1 \text{ V}$.
- (b) — the field points down the potential gradient.
- (b) — the two potentials cancel, but the two fields point the same way and add.

Group B — Short answer (5 marks each)

- Define electric potential and potential difference. Derive an expression for the potential at a distance r from a point charge Q . [5]
- What is an equipotential surface? State four of its properties and show that it must be perpendicular to the electric field. [5]
- Establish the relation $E = -dV/dx$ between electric field and potential gradient, and hence show that $E = V/d$ for a parallel-plate arrangement. [5]
- Calculate the work done in moving a charge of $2 \mu\text{C}$ from a point 10 cm away from a charge of $+3 \mu\text{C}$ to a point 5 cm away from it. [5]
- An electron initially at rest is accelerated through a potential difference of 1000 V . Find its final kinetic energy in joules and its speed. [5]

ANSWERS TO GROUP B

- $V_1 = \frac{9 \times 10^9 \times 3 \times 10^{-6}}{0.10} = 2.7 \times 10^5 \text{ V}$ and $V_2 = \frac{9 \times 10^9 \times 3 \times 10^{-6}}{0.05} = 5.4 \times 10^5 \text{ V}$. The work done by the external agent is $W = q(V_2 - V_1) = 2 \times 10^{-6} \times 2.7 \times 10^5 = 0.54 \text{ J}$.

5. $KE = eV = 1.6 \times 10^{-19} \times 1000 = 1.6 \times 10^{-16} \text{ J}$ (that is 1 keV). Then
 $v = \sqrt{2 \times 1.6 \times 10^{-16} / 9.1 \times 10^{-31}} = \sqrt{3.52 \times 10^{14}} = 1.88 \times 10^7 \text{ m s}^{-1}.$

Group C — Long answer (8 marks each)

- (a) Define absolute electric potential and derive $V = kQ/r$ for a point charge. [4]

(b) Define an equipotential surface, prove that no work is done in moving a charge over it, and deduce the relation $E = -dV/dr$. [4]
- Two point charges $+5 \mu\text{C}$ and $-3 \mu\text{C}$ are placed 0.4 m apart in air. Calculate
 - the potential at the midpoint of the line joining them,
 - the potential energy of the system, and
 - the work required to bring a charge of $+2 \mu\text{C}$ from infinity to that midpoint. [8]

ANSWER TO GROUP C QUESTION 2

Each charge is 0.2 m from the midpoint.

(a) $V = 9 \times 10^9 \left(\frac{5 \times 10^{-6}}{0.2} - \frac{3 \times 10^{-6}}{0.2} \right) = 9 \times 10^9 \times 1 \times 10^{-5} = 9 \times 10^4 \text{ V}.$

(b) $U = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times (-3 \times 10^{-6})}{0.4} = -0.3375 \text{ J}.$ It is negative because the charges attract, so external work would be needed to pull them apart.

(c) $W = qV = 2 \times 10^{-6} \times 9 \times 10^4 = 0.18 \text{ J}.$

UNIT 22

Capacitor

Electricity and Magnetism • 7 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Capacitance and capacitor
- Parallel plate capacitor
- Combination of capacitors
- Energy of charged capacitor
- Effect of a dielectric: Polarization and displacement

A capacitor is the simplest device that stores electrical energy in a field rather than in a chemical reaction. Two conductors, a thin insulating gap, and you have something that can hold charge, smooth a power supply, time a circuit, or deliver the huge pulse that fires a camera flash. This is the longest unit in the electricity section, and it rewards careful work: the derivations are short, the network problems are routine once you know the two combination rules, and almost every NEB paper carries a capacitor numerical.

THREE FORMULAS DO MOST OF THE WORK

$C = Q/V$ defines capacitance, $C = \epsilon_0 A/d$ describes the standard parallel-plate capacitor, and $U = \frac{1}{2} CV^2$ gives the stored energy. Everything else in this unit is an application of one of these three.

22.1 Capacitance and capacitor

Give an isolated conductor a charge Q and its potential rises to V . Experiment shows the two are proportional, $Q \propto V$, so the ratio is a constant of the conductor:

CAPACITANCE

The capacitance of a conductor is the charge required to raise its potential by one unit:

$$C = \frac{Q}{V}$$

Its SI unit is the **farad** (F), where $1 \text{ F} = 1 \text{ C V}^{-1}$.

The farad is enormous, so practical values are given in microfarad ($1 \mu\text{F} = 10^{-6} \text{ F}$), nanofarad (10^{-9} F) or picofarad (10^{-12} F).

For an isolated sphere of radius R we already know $V = kQ/R$, so

$$C = \frac{Q}{V} = \frac{Q}{kQ/R} = 4\pi\epsilon_0 R$$

Capacitance therefore depends only on **geometry and the surrounding medium**, never on how much charge you happen to have put on. (As a scale check, the whole Earth, $R = 6.4 \times 10^6$ m, has $C = 6.4 \times 10^6 / 9 \times 10^9 = 711 \mu\text{F}$ — smaller than a capacitor you can buy in Kathmandu for a few rupees.)

CAPACITOR

A capacitor is an arrangement of two conductors separated by an insulator, designed to store charge and electrical energy. The conductors carry equal and opposite charges $+Q$ and $-Q$, and Q means the magnitude on **one** plate.

Bringing an earthed conductor close to a charged plate lowers its potential without changing its charge, and so **raises** its capacitance. That is the whole idea of a practical capacitor: two plates close together hold far more charge per volt than either plate alone.

22.2 Parallel plate capacitor

Two flat metal plates of area A a distance d apart, with d small compared with the size of the plates, form the standard capacitor.

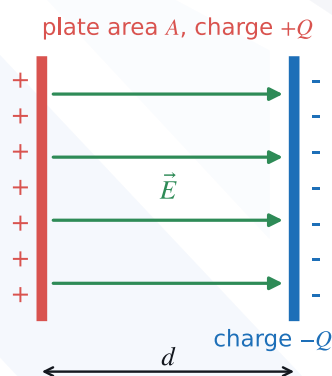


Figure 90. Parallel plate capacitor. The field between the plates is uniform, $E = \sigma/\epsilon_0$, and the potential difference is $V = Ed$.

CAPACITANCE OF A PARALLEL PLATE CAPACITOR, $C = \frac{\epsilon_0 A}{d}$

We start from the field of a charged sheet and reach a formula for the capacitance in terms of the plate area and the separation only.

Setting up. Two flat metal plates, each of area A , face each other a distance d apart with air (or vacuum) between them. One carries charge $+Q$, the other $-Q$. The plates are large and d is small, so we may treat the field between them as uniform and ignore the bulging of the field at the edges.

Step 1 — express the charge as a surface charge density. Spreading Q evenly over the area A gives

$$\sigma = \frac{Q}{A}$$

Step 2 — write the field of one plate alone. From Gauss's law, a thin sheet with charge density σ produces a field on each side of

$$E_{one} = \frac{\sigma}{2\epsilon_0}$$

Step 3 — add the two fields in the region between the plates. The positive plate pushes a test charge away from itself, and the negative plate pulls it towards itself. Between the plates both effects point the same way, from + to -, so the two fields add:

$$E = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0}$$

(Outside the plates the same two fields point in opposite directions and cancel, which is why a capacitor keeps its field neatly inside itself.)

Step 4 — add the fractions.

$$E = \frac{\sigma}{\epsilon_0}$$

Step 5 — put back $\sigma = Q/A$ from Step 1.

$$E = \frac{Q}{\epsilon_0 A}$$

Step 6 — convert the field into a potential difference. The field is uniform, so the potential gradient $E = V/d$ from Unit 21 applies, and

$$V = Ed$$

Step 7 — substitute the field found in Step 5.

$$V = \frac{Qd}{\epsilon_0 A}$$

Step 8 — use the definition of capacitance, $C = Q/V$.

$$C = \frac{Q}{\frac{Qd}{\epsilon_0 A}}$$

Step 9 — dividing by a fraction is multiplying by its reciprocal.

$$C = Q \times \frac{\epsilon_0 A}{Qd}$$

Step 10 — cancel Q from top and bottom.

$$C = \frac{\epsilon_0 A}{d}$$

Result. The capacitance of a parallel plate air capacitor is $C = \epsilon_0 A/d$.

What it means. The charge Q cancelled, so capacitance is a property of the shape and size of the capacitor, not of how much charge you happen to put on it. Bigger plates or a smaller gap store more charge per volt.

Conditions used. Uniform field (large plates, small separation, edge effects ignored) and vacuum or air between the plates.

So capacitance rises if you **increase the plate area** or **decrease the separation**, and it does not depend on Q or V . Filling the gap with an insulating material of relative permittivity ϵ_r multiplies it:

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

CAPACITANCE WITH A DIELECTRIC SLAB OF THICKNESS t IN THE GAP

We start from the same two plates and reach the capacitance when a slab of dielectric of thickness t (with $t < d$) fills only part of the gap.

Setting up. The gap of width d now contains a slab of thickness t and dielectric constant ϵ_r ; the remaining thickness $d - t$ is still air. Let $E_0 = \sigma/\epsilon_0$ be the field in the air part, which is the same as before because the charge on the plates is unchanged.

Step 1 — write the field inside the slab. The polarised molecules of the dielectric set up their own opposing field, which weakens the original field by the factor ϵ_r :

$$E_{slab} = \frac{E_0}{\epsilon_r}$$

Step 2 — the total potential difference is the sum of the drops across the two regions. Potential drop equals field $\times$ distance in each uniform region, so

$$V = E_0(d - t) + E_{slab} t$$

Step 3 — substitute the slab field from Step 1.

$$V = E_0(d - t) + \frac{E_0}{\epsilon_r} t$$

Step 4 — take E_0 out as a common factor.

$$V = E_0 \left(d - t + \frac{t}{\epsilon_r} \right)$$

Step 5 — replace E_0 by $Q/\epsilon_0 A$, using Step 5 of the previous derivation.

$$V = \frac{Q}{\epsilon_0 A} \left(d - t + \frac{t}{\epsilon_r} \right)$$

Step 6 — apply $C = Q/V$ and cancel Q .

$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{\epsilon_r}}$$

Result. With a partly filling slab,

$$C = \frac{\epsilon_0 A}{d - t + \frac{t}{\epsilon_r}}$$

What it means. The bracket in the denominator is smaller than d , because $t/\epsilon_r < t$. A smaller denominator means a larger C , so inserting a dielectric always raises the capacitance. Two checks: putting $t=0$ gives back $\epsilon_0 A/d$, and putting $t=d$ gives $\epsilon_0 \epsilon_r A/d$, exactly as expected.

Conditions used. The slab lies parallel to the plates and fills the whole area A ; the charge Q on the plates is unchanged; edge effects ignored.

THE EXAMINER IS LOOKING FOR

1. $\sigma = Q/A$ and the field of a single sheet, $\sigma/2\epsilon_0$.
2. The reason the two fields **add** between the plates (and cancel outside).
3. $V = Ed$, with a statement that the field is uniform.
4. The definition $C = Q/V$ and the cancellation of Q — this is the step that shows capacitance is independent of charge.
5. For the slab: the potential difference written as a **sum of two drops**, $V = E_0(d-t) + (E_0/\epsilon_r)t$.

WORKED EXAMPLE 22.1

Problem. A parallel plate capacitor has plates of area 100 cm^2 separated by 1 mm of air. Find (a) its capacitance, (b) the charge stored at 100 V , (c) the field between the plates, and (d) the new capacitance if the gap is filled with mica of $\epsilon_r = 6$.

Solution. $A = 100 \text{ cm}^2 = 1.0 \times 10^{-2} \text{ m}^2$ and $d = 1.0 \times 10^{-3} \text{ m}$.

(a) $C = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 1.0 \times 10^{-2}}{1.0 \times 10^{-3}} = 8.85 \times 10^{-11} \text{ F} = 88.5 \text{ pF}$.

(b) $Q = CV = 8.85 \times 10^{-11} \times 100 = 8.85 \times 10^{-9} \text{ C} = 8.85 \text{ nC}$.

(c) $E = \frac{V}{d} = \frac{100}{1.0 \times 10^{-3}} = 1.0 \times 10^5 \text{ V m}^{-1}$.

(d) $C' = \epsilon_r C = 6 \times 88.5 = 531 \text{ pF}$.

22.3 Combination of capacitors

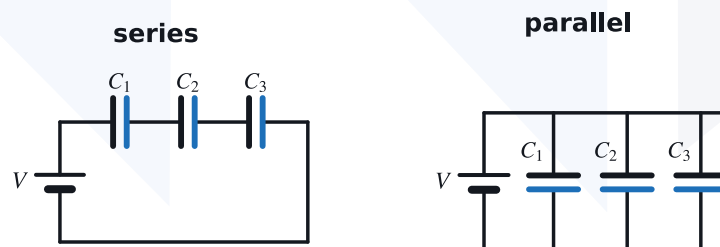


Figure 91. Capacitors in series carry the same charge Q and share the applied p.d.; in parallel they have the same p.d. V and share the charge.

EQUIVALENT CAPACITANCE IN SERIES, $\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$

We start from the two facts that fix a series chain — same charge, shared voltage — and reach the rule for the single capacitor that could replace them.

Setting up. Three capacitors C_1 , C_2 , C_3 are joined end to end across a supply of potential difference V . Let the p.d. across them be V_1 , V_2 and V_3 .

Step 1 — state why the charge is the same on all three. The battery puts $+Q$ on the first plate. Charge cannot jump across the insulating gap, so the facing plate gains $-Q$ by induction, its partner on the next capacitor gains $+Q$, and so on down the chain. Hence every capacitor carries the same charge:

$$Q_1 = Q_2 = Q_3 = Q$$

Step 2 — state why the voltages add. Going from one end of the chain to the other, the total drop in potential is the sum of the individual drops:

$$V = V_1 + V_2 + V_3$$

Step 3 — write each voltage using $C = Q/V$, rearranged as $V = Q/C$.

$$V_1 = \frac{Q}{C_1}, \quad V_2 = \frac{Q}{C_2}, \quad V_3 = \frac{Q}{C_3}$$

Step 4 — substitute these into Step 2.

$$V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

Step 5 — write the same total voltage for the single equivalent capacitor. By definition C_{eq} is the one capacitor that would take the same charge Q at the same total voltage V :

$$V = \frac{Q}{C_{eq}}$$

Step 6 — set the two expressions for V equal.

$$\frac{Q}{C_{eq}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

Step 7 — divide every term by Q . The charge is the same everywhere (Step 1), so this one division clears it from the whole equation:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Result. For capacitors in series the reciprocals add: $1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3$.

What it means. Adding another capacitor in series adds another positive term to the right-hand side, so $1/C_{eq}$ grows and C_{eq} shrinks. A series combination is therefore always smaller than the smallest capacitor in it — use it when you need the chain to withstand a high voltage, not to store more charge.

Conditions used. Step 1 needs the capacitors to be genuinely in series, with nothing else connected to the junctions between them, so that no charge can leak away sideways.

EQUIVALENT CAPACITANCE IN PARALLEL, $C_{eq} = \sum C_i$

We start from the two facts that fix a parallel group — same voltage, shared charge — and reach the rule for the single replacement capacitor.

Setting up. Three capacitors C_1 , C_2 , C_3 have all their left plates joined to one terminal and all their right plates joined to the other. Let them carry charges Q_1 , Q_2 , Q_3 .

Step 1 — state why the voltage is the same across all three. Each capacitor has its two plates connected directly to the two terminals of the supply, so each one feels the full supply voltage:

$$V_1 = V_2 = V_3 = V$$

Step 2 — state why the charges add. The total charge the supply pushes out is shared among the three capacitors:

$$Q = Q_1 + Q_2 + Q_3$$

Step 3 — write each charge using $C = Q/V$, rearranged as $Q = CV$.

$$Q_1 = C_1 V, \quad Q_2 = C_2 V, \quad Q_3 = C_3 V$$

Step 4 — substitute these into Step 2.

$$Q = C_1 V + C_2 V + C_3 V$$

Step 5 — write the same total charge for the single equivalent capacitor.

$$Q = C_{eq} V$$

Step 6 — set the two expressions for Q equal.

$$C_{eq} V = C_1 V + C_2 V + C_3 V$$

Step 7 — divide every term by V . The voltage is the same everywhere (Step 1), so it cancels right through:

$$C_{eq} = C_1 + C_2 + C_3$$

Result. For capacitors in parallel the capacitances simply add, $C_{eq} = C_1 + C_2 + C_3$.

What it means. Connecting capacitors in parallel is the same as building one capacitor with a bigger plate area, and $C = \epsilon_0 A/d$ says bigger area means bigger capacitance. The total is always larger than the largest member.

Conditions used. All the capacitors must share the same pair of nodes, and each must be able to withstand the full supply voltage.

THE EXAMINER IS LOOKING FOR

1. A clear statement, **with a reason**, of what is common: same Q in series (induction, charge cannot cross the gap), same V in parallel (both plates on the terminals).
2. The matching sum: $V = V_1 + V_2 + V_3$ in series, $Q = Q_1 + Q_2 + Q_3$ in parallel.
3. Substituting $V = Q/C$ or $Q = CV$ for each capacitor.
4. The definition of C_{eq} as the single capacitor that behaves the same way.
5. The final division by Q (series) or V (parallel).

	Series	Parallel
Same for all	charge Q	potential difference V
Shared	potential difference	charge
Rule	$1/C_{eq} = \sum 1/C_i$	$C_{eq} = \sum C_i$
Result	less than the smallest	more than the largest
Used to	withstand a higher voltage	obtain a larger capacitance

SERIES AND PARALLEL ARE OPPOSITE TO RESISTORS

For resistors the series rule is the simple sum; for capacitors it is the parallel rule. Students who learn DC circuits first regularly swap them. Check your answer: a series combination must always come out smaller than the smallest capacitor in it.

WORKED EXAMPLE 22.2

Problem. A $6 \mu\text{F}$ capacitor is joined in series with a parallel pair of $3 \mu\text{F}$ capacitors, and the combination is connected across 60 V . Find the equivalent capacitance, the charge on each capacitor and the p.d. across each.

Solution. The parallel pair first: $C_p = 3 + 3 = 6 \mu\text{F}$.

In series with the $6 \mu\text{F}$:

$$\frac{1}{C_{eq}} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \implies C_{eq} = 3 \mu\text{F}$$

Total charge: $Q = C_{eq}V = 3 \times 10^{-6} \times 60 = 180 \mu\text{C}$.

This whole charge sits on the series $6 \mu\text{F}$, so its p.d. is $V_1 = 180/6 = 30 \text{ V}$. The remaining $60 - 30 = 30 \text{ V}$ appears across the parallel pair, and each $3 \mu\text{F}$ there carries

$$Q = CV = 3 \times 10^{-6} \times 30 = 90 \mu\text{C}$$

As a check, $90 + 90 = 180 \mu\text{C}$, matching the series charge.

22.4 Energy of charged capacitor

Charging a capacitor is not free: each extra bit of charge must be pushed onto a plate that is already charged and repelling it.

ENERGY STORED IN A CHARGED CAPACITOR, $U = \frac{1}{2}CV^2$

We start from the work needed to move one more small piece of charge onto a partly charged capacitor, and we reach the total energy stored.

Setting up. The capacitor starts completely uncharged. The battery moves charge from one plate to the other a little at a time until the final charge is Q and the final p.d. is V .

Step 1 — write the p.d. at a moment when the charge so far is q . From the definition $C = q/v$:

$$v = \frac{q}{C}$$

Notice that v **grows** as q grows. Each extra bit of charge is harder to push on than the one before, so we cannot just write "work = QV ".

Step 2 — write the work needed for one extra small charge dq . Work equals charge moved times the p.d. it is moved through, and dq is small enough that v does not change during it:

$$dW = v \, dq$$

Step 3 — substitute v from Step 1.

$$dW = \frac{q}{C} \, dq$$

Step 4 — add up all these small amounts of work, from empty to full. The charge on the plates runs from 0 to Q :

$$W = \int_0^Q \frac{q}{C} \, dq$$

Step 5 — take the constant C outside the integral. The capacitance does not change while the capacitor charges:

$$W = \frac{1}{C} \int_0^Q q \, dq$$

Step 6 — integrate q .

$$W = \frac{1}{C} \left[\frac{q^2}{2} \right]_0^Q$$

Step 7 — substitute the limits, upper minus lower.

$$W = \frac{1}{C} \left(\frac{Q^2}{2} - \frac{0^2}{2} \right)$$

Step 8 — the second term is zero.

$$W = \frac{Q^2}{2C}$$

Step 9 — this work is not lost; it is stored in the field between the plates. So the energy stored is

$$U = \frac{Q^2}{2C}$$

Step 10 — get the second form. Replace one of the two Q factors using $Q = CV$:

$$U = \frac{Q \times Q}{2C} = \frac{Q \times CV}{2C}$$

Step 11 — cancel C .

$$U = \frac{1}{2}QV$$

Step 12 — get the third form. Replace Q by CV once more:

$$U = \frac{1}{2}(CV)V = \frac{1}{2}CV^2$$

Result. The energy stored in a charged capacitor is

$$U = \frac{Q^2}{2C} = \frac{1}{2}QV = \frac{1}{2}CV^2$$

What it means. The three forms are the same energy; pick the one whose two quantities you already know. The factor $\frac{1}{2}$ appears because the average p.d. during charging is only half the final value — the first charge is moved across almost no p.d., the last across the full V .

Conditions used. The capacitance C is constant during charging (no dielectric is moved, no plate separation changed), and none of the energy counted here includes the heat lost in the connecting wires and the battery.

The factor $\frac{1}{2}$ is not arbitrary. On a graph of Q against V the energy is the **area under the line**, and the line is straight through the origin, so the area is a triangle.

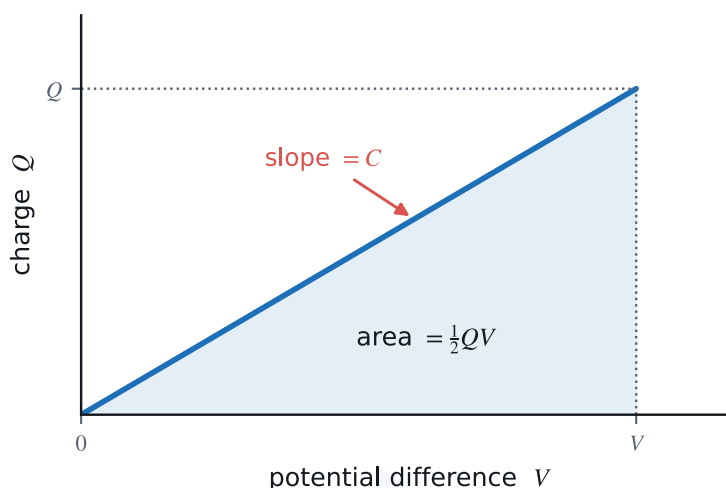


Figure 92. Charge against potential difference for a capacitor. The gradient is C and the shaded area $\frac{1}{2}QV$ is the stored energy.

Energy density

Where is the energy? In the **field**. For a parallel plate capacitor, put $C = \epsilon_0 A/d$ and $V = Ed$ into $U = \frac{1}{2}CV^2$:

$$U = \frac{1}{2} \cdot \frac{\epsilon_0 A}{d} \cdot (Ed)^2 = \frac{1}{2} \epsilon_0 E^2 (Ad)$$

Since Ad is the volume between the plates, the energy stored per unit volume is

$$u = \frac{1}{2} \epsilon_0 E^2$$

This turns out to be true of any electrostatic field, not just this one.

Energy lost in sharing charge

Connect a capacitor C_1 at V_1 to a capacitor C_2 at V_2 . Charge is conserved, so the common final potential is

$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

Comparing the energy before and after gives a loss of

$$\Delta U = \frac{C_1 C_2 (V_1 - V_2)^2}{2(C_1 + C_2)}$$

which is always positive. The "missing" energy is dissipated as heat in the connecting wires and as a small amount of electromagnetic radiation — it is never recovered.

WORKED EXAMPLE 22.3

Problem. A $10 \mu\text{F}$ capacitor is charged to 200 V and then disconnected from the supply and joined to an uncharged $10 \mu\text{F}$ capacitor. Find the common potential difference, the energy before and after, and the energy lost.

Solution. Initial charge: $Q = 10 \times 10^{-6} \times 200 = 2.0 \times 10^{-3} \text{ C}$.

Common p.d.:

$$V = \frac{Q}{C_1 + C_2} = \frac{2.0 \times 10^{-3}}{20 \times 10^{-6}} = 100 \text{ V}$$

Energy before: $U_i = \frac{1}{2}(10 \times 10^{-6})(200)^2 = 0.20 \text{ J}$.

Energy after: $U_f = \frac{1}{2}(20 \times 10^{-6})(100)^2 = 0.10 \text{ J}$.

Loss = 0.10 J, i.e. **half** the original energy. The formula agrees:

$$\Delta U = \frac{(10 \times 10^{-6})(10 \times 10^{-6})(200)^2}{2(20 \times 10^{-6})} = \frac{1.0 \times 10^{-10} \times 4 \times 10^4}{4.0 \times 10^{-5}} = 0.10 \text{ J}$$

22.5 Effect of a dielectric: Polarization and displacement

A **dielectric** is an insulator placed between the plates. It has no free electrons, but its molecules respond to the field.

- **Non-polar** molecules (H_2 , O_2 , CO_2) have no permanent dipole. The field pulls the electron cloud one way and the nucleus the other, **inducing** a dipole.
- **Polar** molecules (H_2O , HCl) already possess a dipole moment. The field simply turns them so that they line up with it, against the randomising effect of thermal motion.

Either way the slab acquires a net dipole moment per unit volume. This is called the **polarization** $\vec{P}$. Inside the bulk, neighbouring dipoles cancel, but on the two faces uncanceled **bound charges** appear — negative on the face towards the positive plate and positive on the other. Their surface density equals the polarization, $\sigma_p = P$.

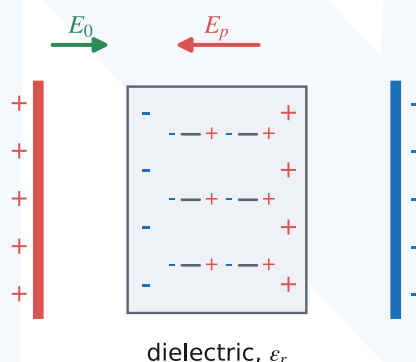


Figure 93. A dielectric slab polarized between charged plates. The bound surface charges set up a field E_p opposing E_0 , so the net field falls to E_0/ϵ_r .

The bound charges produce their own field E_p **opposing** the original field E_0 , so the net field inside the dielectric is reduced:

$$E = E_0 - E_p = \frac{E_0}{\epsilon_r}$$

The number ϵ_r (also written K) is the **dielectric constant** or relative permittivity. Since $V = Ed$ falls by the same factor while Q is unchanged, the capacitance rises:

$$\epsilon_r = \frac{E_0}{E} = \frac{V_0}{V} = \frac{C}{C_0}$$

Material	ϵ_r	Material	ϵ_r
Vacuum	1 (exactly)	Mica	6
Air	1.0006	Glass	5–10
Paper	3.7	Water	80

Electric displacement

Inside a dielectric two kinds of charge exist: **free** charge on the plates and **bound** charge on the dielectric faces. It is convenient to define a vector that responds only to the free charge, the **electric displacement**:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \epsilon_r \vec{E}$$

Its magnitude equals the free surface charge density, $D = \sigma_{free}$, and its unit is $C\ m^{-2}$. Gauss's law then takes the compact form

$$\oint \vec{D} \cdot d\vec{A} = q_{free}$$

so the bound charges never have to be counted explicitly.

Every dielectric also has a **dielectric strength** — the largest field it can withstand before it ionises and conducts. For air this is about $3 \times 10^6\ V\ m^{-1}$, which is why a capacitor is rated for a maximum working voltage as well as a capacitance.

BATTERY CONNECTED, OR BATTERY REMOVED?

The effect of inserting a dielectric depends entirely on what is held fixed.

Quantity	Battery still connected (V fixed)	Battery removed (Q fixed)
C	$\times \epsilon_r$	$\times \epsilon_r$
Q	$\times \epsilon_r$	unchanged
V	unchanged	$\div \epsilon_r$
E	unchanged	$\div \epsilon_r$
U	$\times \epsilon_r$	$\div \epsilon_r$

Decide which column applies **before** you start the arithmetic.

WORKED EXAMPLE 22.4

Problem. A parallel plate capacitor has plates of area 100 cm^2 separated by 5 mm . A slab of thickness 3 mm and dielectric constant 3 is inserted between the plates. Find the capacitance before and after insertion.

Solution. $A = 1.0 \times 10^{-2} \text{ m}^2$, $d = 5.0 \times 10^{-3} \text{ m}$.

Before:

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 1.0 \times 10^{-2}}{5.0 \times 10^{-3}} = 1.77 \times 10^{-11} \text{ F} = 17.7 \text{ pF}$$

After, with $t = 3.0 \times 10^{-3} \text{ m}$ and $\epsilon_r = 3$, the effective separation is

$$d - t + \frac{t}{\epsilon_r} = 5.0 - 3.0 + 1.0 = 3.0 \text{ mm}$$

$$C = \frac{8.85 \times 10^{-12} \times 1.0 \times 10^{-2}}{3.0 \times 10^{-3}} = 2.95 \times 10^{-11} \text{ F} = 29.5 \text{ pF}$$

The slab raises the capacitance by a factor of $5/3$, not by $\epsilon_r = 3$, because it fills only part of the gap.

✓ Chapter summary

- $C = Q/V$, unit farad (C V^{-1}); capacitance depends on geometry and medium only.
- Isolated sphere: $C = 4\pi\epsilon_0 R$.
- Parallel plate capacitor: $E = \sigma/\epsilon_0$, $V = Ed$, and $C = \epsilon_0\epsilon_r A/d$. With a partial slab, $C = \epsilon_0 A/(d - t + t/\epsilon_r)$.
- Series: same Q , $1/C_{eq} = \sum 1/C_i$, result smaller than the smallest. Parallel: same V , $C_{eq} = \sum C_i$, result larger than the largest.
- Stored energy $U = \frac{Q^2}{2C} = \frac{1}{2}QV = \frac{1}{2}CV^2$; it is the area under the Q - V line.
- Energy per unit volume of any electric field: $u = \frac{1}{2}\epsilon_0 E^2$.
- Sharing charge always wastes energy: $\Delta U = C_1 C_2 (V_1 - V_2)^2 / (C_1 + C_2)$.
- A dielectric polarizes, producing bound surface charge $\sigma_p = P$ which reduces the field to E_0/ϵ_r and multiplies C by ϵ_r . Displacement $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ responds to free charge only.

Practice questions

Group A — Multiple choice (1 mark each)

- The SI unit of capacitance, the farad, is equivalent to [1]

a C V
b C V⁻¹
c V C⁻¹
d J C⁻¹
- Three capacitors each of $3 \mu\text{F}$ are connected in series. The equivalent capacitance is [1]

a $9 \mu\text{F}$
b $3 \mu\text{F}$
c $1 \mu\text{F}$
d $0.33 \mu\text{F}$

3. The energy stored in a capacitor of capacitance C at a p.d. V is [1]
- a** CV **b** CV^2 **c** $\frac{1}{2}CV^2$ **d** $\frac{1}{2}C^2V$
4. A charged capacitor is disconnected from the battery and a dielectric of constant K is inserted. The potential difference [1]
- a** becomes K times **b** becomes $1/K$ times
c is unchanged **d** becomes zero
5. The capacitance of a parallel plate capacitor does **not** depend on [1]
- a** plate area **b** plate separation
c the dielectric **d** the charge on the plates

ANSWERS TO GROUP A

1. (b) — from $C = Q/V$, coulomb per volt.
 2. (c) — $1/C = 3/3 = 1$, so $C = 1 \mu\text{F}$.
 3. (c) — the area of the triangle under the Q - V line.
 4. (b) — Q is fixed, C becomes KC , so $V = Q/C$ falls by K .
 5. (d) — C is fixed by geometry and medium alone.

Group B — Short answer (5 marks each)

1. Define capacitance and the farad. Derive an expression for the capacitance of a parallel plate capacitor with air between the plates. [5]
 2. Derive expressions for the equivalent capacitance of three capacitors connected
(a) in series and
(b) in parallel. [5]
 3. Show that the energy stored in a capacitor is $\frac{1}{2}CV^2$, and hence obtain the energy density of an electric field. [5]
 4. Capacitors of $2 \mu\text{F}$, $3 \mu\text{F}$ and $6 \mu\text{F}$ are joined in series across a 100 V supply. Find the equivalent capacitance, the charge on each and the p.d. across each. [5]
 5. A $2 \mu\text{F}$ capacitor charged to 100 V is connected across an uncharged $3 \mu\text{F}$ capacitor. Find the common potential and the energy lost. [5]

ANSWERS TO GROUP B

4. $\frac{1}{C} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$, so $C_{eq} = 1 \mu\text{F}$. The charge is the same on each: $Q = 1 \times 10^{-6} \times 100 = 100 \mu\text{C}$. The potential differences are $V = Q/C$: $100/2 = 50 \text{ V}$, $100/3 = 33.3 \text{ V}$ and $100/6 = 16.7 \text{ V}$, which add to 100 V as required.
5. Charge is conserved: $V = \frac{C_1 V_1}{C_1 + C_2} = \frac{2 \times 100}{5} = 40 \text{ V}$. $U_i = \frac{1}{2}(2 \times 10^{-6})(100)^2 = 1.0 \times 10^{-2} \text{ J}$;
 $U_f = \frac{1}{2}(5 \times 10^{-6})(40)^2 = 4.0 \times 10^{-3} \text{ J}$. Energy lost = $6.0 \times 10^{-3} \text{ J}$, which agrees with $C_1 C_2 (V_1 - V_2)^2 / 2(C_1 + C_2)$.

Group C — Long answer (8 marks each)

1. **(a)** Derive an expression for the capacitance of a parallel plate capacitor and show that inserting a dielectric slab of thickness t and dielectric constant K changes it to $C = \epsilon_0 A / (d - t + t/K)$. [5]
(b) Explain, in terms of polarization, why a dielectric increases the capacitance. [3]
2. **(a)** Derive an expression for the energy stored in a charged capacitor. [3]
(b) A $2 \mu\text{F}$ and a $3 \mu\text{F}$ capacitor are connected in parallel, and this combination is joined in series with a $5 \mu\text{F}$ capacitor across a 50 V supply. Find the equivalent capacitance, the charge on each capacitor, and the total energy stored. [5]

ANSWER TO GROUP C QUESTION 2

(a) With charge q on the plates the p.d. is q/C , so $dW = (q/C)dq$ and $W = \int_0^Q (q/C) dq = Q^2/2C = \frac{1}{2}CV^2$.

(b) Parallel section: $C_p = 2 + 3 = 5 \mu\text{F}$. In series with $5 \mu\text{F}$: $1/C_{eq} = 1/5 + 1/5 = 2/5$, so $C_{eq} = 2.5 \mu\text{F}$.

Total charge $Q = 2.5 \times 10^{-6} \times 50 = 125 \mu\text{C}$, which is the charge on the $5 \mu\text{F}$ capacitor. Its p.d. is $125/5 = 25 \text{ V}$, so the parallel pair also has 25 V across it. Hence $Q_1 = 2 \times 25 = 50 \mu\text{C}$ and $Q_2 = 3 \times 25 = 75 \mu\text{C}$ (and $50 + 75 = 125 \mu\text{C}$, as it must).

Total energy $U = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}(2.5 \times 10^{-6})(50)^2 = 3.125 \times 10^{-3} \text{ J} = 3.125 \text{ mJ}$.

UNIT 23

DC Circuits

Electricity and Magnetism • 10 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Electric Currents; Drift velocity and its relation with current
- Ohm's law, Electrical Resistance, Resistivity, Conductivity
- Current-voltage relations: Ohmic and Non-Ohmic resistance
- Resistances in series and parallel
- Potential divider
- Electromotive force of a source, internal resistance
- Work and power in electrical circuits

A direct-current (DC) circuit is a closed path in which charge is pushed round in one steady direction by a source such as a cell, a battery or a DC power supply. This is the longest unit in the electricity area, and it is the most practical: almost every measuring instrument you meet in the school laboratory — ammeter, voltmeter, metre bridge, potentiometer — is a DC circuit designed to answer one question accurately. Here you learn what current really is at the level of the electrons, how resistance arises, how to reduce any network to a single resistance, how to analyse networks that refuse to reduce (Kirchhoff), and finally where the energy goes.

WHAT THE EXAMINER ASKS FROM THIS UNIT

Every year this unit supplies a derivation (drift velocity, series/parallel, Wheatstone balance, or $V = E - Ir$), a network numerical, and one instrument question (metre bridge, potentiometer, or converting a galvanometer). Learn the four derivations properly and practise reducing networks quickly.

23.1 Electric Currents; Drift velocity and its relation with current

An **electric current** is the rate of flow of charge across a section of a conductor. If a charge ΔQ crosses a section in time Δt ,

$$I_{av} = \frac{\Delta Q}{\Delta t}, \quad I = \frac{dQ}{dt}$$

The SI unit is the **ampere** (A): $1 \text{ A} = 1 \text{ C s}^{-1}$. Current is a scalar, even though we draw arrows for it. By convention the arrow shows the direction in which positive charge would move — in a metal the actual carriers are electrons, moving the opposite way.

A related quantity is the **current density**, the current per unit cross-sectional area:

$$J = \frac{I}{A} \quad (\text{unit: A m}^{-2})$$

Why a drift velocity exists

In a metal the outermost electrons are free. Even with no battery connected they fly about randomly at about 10^5 m s^{-1} , colliding with the vibrating ions roughly every $\tau \approx 10^{-14} \text{ s}$. Because the directions are random, the average velocity is zero and there is no current.

When a battery maintains a field E inside the wire, every electron gets a small extra acceleration $a = eE/m$ between collisions. Each collision wipes out the memory of the previous push, so on average the electrons acquire a small steady velocity opposite to E :

$$v_d = \frac{eE\tau}{m}$$

DRIFT VELOCITY

Drift velocity is the average velocity with which free electrons in a conductor move along the conductor under an applied electric field. Its magnitude is of the order of 10^{-4} m s^{-1} — less than a millimetre per second.

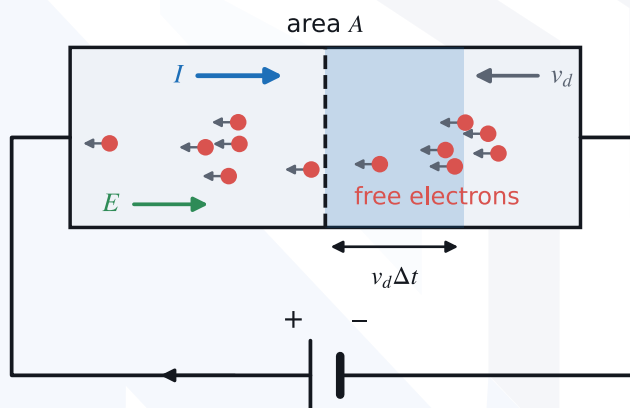


Figure 94. Free electrons drift slowly opposite to the field E . Every electron within a distance $v_d \Delta t$ of the shaded section crosses it in time Δt .

RELATION BETWEEN CURRENT AND DRIFT VELOCITY, $I = nAv_d e$

We start from the definition of current as charge per second and reach a formula linking it to the slow drift of the free electrons.

Setting up. A metal wire has cross-sectional area A . It contains n free electrons in every cubic metre, each carrying charge e . Under the applied field they all drift along the wire with the same average speed v_d . Fix your attention on one cross-section of the wire and count the charge that crosses it in a time Δt .

Step 1 — find how far an electron travels in the time Δt . Distance equals speed times time:

$$l = v_d \Delta t$$

Step 2 — identify which electrons manage to cross the section. Only those starting no further back than l can reach it in time. They are exactly the electrons inside a cylinder of length l and end area A .

Step 3 — find the volume of that cylinder. Volume = area $\times$ length:

$$\text{Volume} = A l = A v_d \Delta t$$

Step 4 — find how many electrons are inside it. Number = number per unit volume $\times$ volume:

$$N = n \times A v_d \Delta t$$

Step 5 — find the total charge those electrons carry. Charge = number $\times$ charge on each:

$$\Delta Q = Ne = n A v_d \Delta t e$$

Step 6 — use the definition of current, charge per unit time.

$$I = \frac{\Delta Q}{\Delta t} = \frac{n A v_d \Delta t e}{\Delta t}$$

Step 7 — cancel Δt from top and bottom.

$$I = n A v_d e$$

Step 8 — divide both sides by A to get the current density. Current density J is current per unit area:

$$J = \frac{I}{A} = n e v_d$$

Result. $I = n A v_d e$, and equivalently $J = n e v_d$.

What it means. The current is large not because the electrons are fast but because n is enormous — about 10^{29} per cubic metre in copper. That is why a drift velocity of a fraction of a millimetre per second still gives amperes of current. Doubling the current in a given wire only doubles v_d .

Conditions used. A steady current; a uniform conductor of constant area A ; one kind of charge carrier, all drifting with the same average speed; and n constant along the wire.

THE EXAMINER IS LOOKING FOR

1. The distance $v_d \Delta t$ and the statement that the electrons which cross are those inside a cylinder of that length.
2. Volume = $A v_d \Delta t$, then number = $n A v_d \Delta t$.
3. Charge $\Delta Q = n A v_d \Delta t e$.
4. $I = \Delta Q / \Delta t$ and the cancellation of Δt .

WORKED EXAMPLE 23.1

Problem. A copper wire of cross-section 1.0 mm^2 carries a current of 5.0 A . Copper has $n = 8.5 \times 10^{28}$ free electrons per cubic metre and $e = 1.6 \times 10^{-19} \text{ C}$. Find the drift velocity, and the time an electron takes to drift 1 m along the wire.

Solution. $A = 1.0 \text{ mm}^2 = 1.0 \times 10^{-6} \text{ m}^2$.

$$v_d = \frac{I}{nAe} = \frac{5.0}{(8.5 \times 10^{28})(1.0 \times 10^{-6})(1.6 \times 10^{-19})}$$

The denominator is $8.5 \times 10^{28} \times 10^{-6} = 8.5 \times 10^{22}$, and $8.5 \times 10^{22} \times 1.6 \times 10^{-19} = 1.36 \times 10^4$. Hence

$$v_d = \frac{5.0}{1.36 \times 10^4} = 3.7 \times 10^{-4} \text{ m s}^{-1}$$

Time for 1 m: $t = 1/(3.7 \times 10^{-4}) = 2.7 \times 10^3 \text{ s}$, about 45 minutes.

THEN WHY DOES THE BULB LIGHT INSTANTLY?

The electrons crawl, but the electric field that sets them moving is established along the whole wire at nearly the speed of light. Electrons everywhere in the circuit — including those already inside the filament — start drifting almost at once. Nothing has to travel from the switch to the bulb.

23.2 Ohm's law, Electrical Resistance, Resistivity, Conductivity

OHM'S LAW

The current flowing through a conductor is directly proportional to the potential difference across its ends, provided its physical conditions (temperature, length, cross-section, nature of material) remain constant: $V \propto I$, i.e. $V = IR$.

The constant R is the **resistance** of the conductor. Its SI unit is the **ohm** (Ω): a conductor has a resistance of one ohm if a potential difference of one volt drives a current of one ampere through it.

$$R = \frac{V}{I} \quad (1 \Omega = 1 \text{ V A}^{-1})$$

Experiment shows that for a uniform wire the resistance is proportional to its length l and inversely proportional to its area of cross-section A :

$$R = \rho \frac{l}{A}$$

ρ is the **resistivity** (specific resistance) of the material, with unit $\Omega \text{ m}$. Numerically it is the resistance of a piece of the material one metre long and one square metre in cross-section. Its reciprocal is the **conductivity**

$$\sigma = \frac{1}{\rho} \quad (\text{unit: } \Omega^{-1} \text{ m}^{-1} \text{ or S m}^{-1})$$

RESISTANCE VS RESISTIVITY

Resistance depends on the material and on the shape and size of the specimen. Resistivity depends only on the material and its temperature. Cutting a wire in half halves its resistance but leaves its resistivity unchanged.

Material	ρ at 20 °C (Ω m)	Class
Silver	1.59×10^{-8}	conductor
Copper	1.68×10^{-8}	conductor (house wiring)
Aluminium	2.65×10^{-8}	conductor (transmission lines)
Tungsten	5.60×10^{-8}	lamp filament
Nichrome	1.10×10^{-6}	heating element
Germanium	0.46	semiconductor
Silicon (pure)	≈ 640	semiconductor
Glass	$10^{10} - 10^{14}$	insulator

MICROSCOPIC FORM OF OHM'S LAW, $\rho = \frac{m}{ne^2\tau}$

We start from the drift velocity of the free electrons and reach an explanation of why a metal obeys Ohm's law, together with a formula for its resistivity.

Setting up. A wire of length l and area A carries a current I under a potential difference V . It has n free electrons per unit volume, each of mass m and charge e , and τ is the average time between two collisions of an electron with the vibrating ions.

Step 1 — write the field inside the wire. The field is uniform along the wire, so from $E = V/d$ with $d = l$:

$$E = \frac{V}{l}$$

Step 2 — write the drift velocity produced by that field. From §23.1, the electron is accelerated for an average time τ before each collision, giving

$$v_d = \frac{eE\tau}{m}$$

Step 3 — start from the current equation of §23.1.

$$I = nAv_d e$$

Step 4 — substitute the drift velocity from Step 2.

$$I = nAe \cdot \frac{eE\tau}{m}$$

Step 5 — collect the two factors of e .

$$I = \frac{ne^2\tau A}{m} E$$

Step 6 — substitute the field from Step 1.

$$I = \frac{ne^2\tau A}{m} \cdot \frac{V}{l}$$

Step 7 — divide both sides by I and multiply by the big fraction, so as to make V/I the subject.

$$\frac{V}{I} = \frac{m}{ne^2\tau} \cdot \frac{l}{A}$$

Step 8 — recognise V/I as the resistance.

$$R = \frac{m}{ne^2\tau} \cdot \frac{l}{A}$$

Step 9 — compare this, term by term, with the experimental law $R = \rho l/A$. The factor multiplying l/A must be the resistivity:

$$\rho = \frac{m}{ne^2\tau}$$

Result. The resistivity of a metal is $\rho = m/(ne^2\tau)$, and the resistance is $R = \rho l/A$.

What it means. Not one of m , n , e or τ depends on the applied voltage, so at a fixed temperature R is a constant — and $V = IR$, which is Ohm's law. The formula also explains heating: warmer ions vibrate harder, the electrons collide sooner, τ falls and ρ rises.

Conditions used. A metal with free electrons as the only carriers; constant temperature (so τ is fixed); and a uniform wire in a steady state.

For a metal over a moderate range of temperature,

$$R_t = R_0(1 + \alpha t)$$

where α is the temperature coefficient of resistance (for copper $\alpha \approx 4.0 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$). For semiconductors and electrolytes α is **negative**: resistance falls as temperature rises, because heating frees many more charge carriers.

WORKED EXAMPLE 23.2

Problem. A copper wire is 10 m long and 1.0 mm in diameter ($\rho = 1.68 \times 10^{-8} \text{ } \Omega \text{ m}$). (a) Find its resistance. (b) The wire is now stretched uniformly to twice its original length. What is its new resistance?

Solution.

(a) $A = \pi r^2 = \pi(0.5 \times 10^{-3})^2 = 7.85 \times 10^{-7} \text{ m}^2$.

$$R = \frac{\rho l}{A} = \frac{1.68 \times 10^{-8} \times 10}{7.85 \times 10^{-7}} = 0.214 \, \Omega$$

(b) Stretching does not change the volume, so $A'l' = Al$. With $l' = 2l$ we get $A' = A/2$. Hence

$$R' = \rho \frac{l'}{A'} = \rho \frac{2l}{A/2} = 4\rho \frac{l}{A} = 4R = 0.856 \, \Omega$$

In general $R \propto l^2$ for a stretched wire of fixed volume.

23.3 Current-voltage relations: Ohmic and Non-Ohmic resistance

A conductor that obeys Ohm's law is **ohmic**: its I - V graph is a straight line through the origin, and $R = V/I$ is the same at every point. A conductor whose I - V graph is not a straight line through the origin is **non-ohmic**.

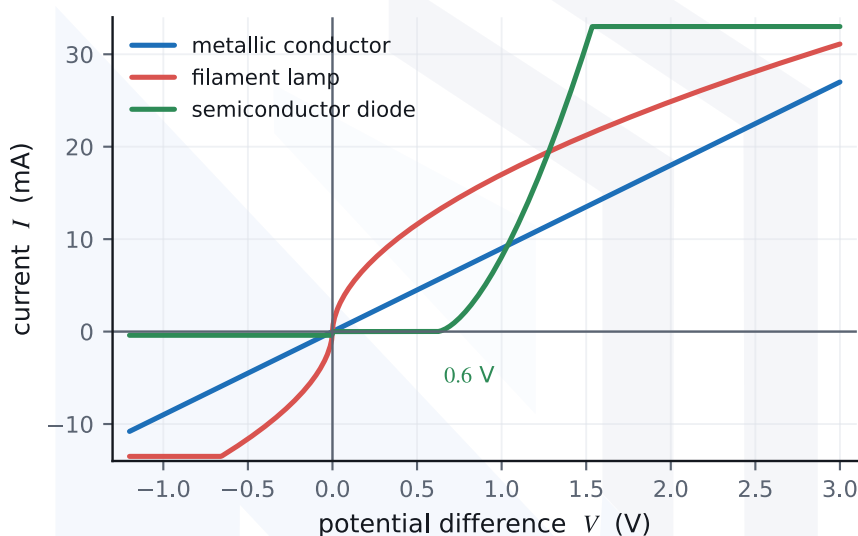


Figure 95. I - V characteristics. Only the metallic conductor is ohmic; the lamp and the diode are not.

Ohmic	Non-ohmic
I - V graph is a straight line through origin	graph is curved, or does not pass through origin
$R = V/I$ constant	$R = V/I$ changes with V
Same behaviour for either polarity	may depend on the direction of current
Metals at constant temperature, nichrome, carbon resistors	filament lamp, diode, thermistor, electrolyte, gas discharge tube, transistor

A **filament lamp** is non-ohmic only because the current heats the tungsten: as V rises the filament goes from about 300 K to about 2500 K, its resistance rises several times, and the graph bends towards the V -axis. A **junction diode** conducts almost nothing until the forward voltage reaches about 0.6 V (silicon), and blocks current completely in reverse.

SLOPE IS NOT ALWAYS RESISTANCE

For a non-ohmic device, the resistance at a point is V/I (the slope of the chord from the origin), **not** dV/dI (the slope of the tangent, which is the dynamic resistance). For an ohmic conductor the two happen to be equal — which is why students confuse them.

23.4 Resistances in series and parallel

Any combination of resistors between two terminals can be replaced by one **equivalent resistance** R_{eq} that would draw the same current from the same supply.

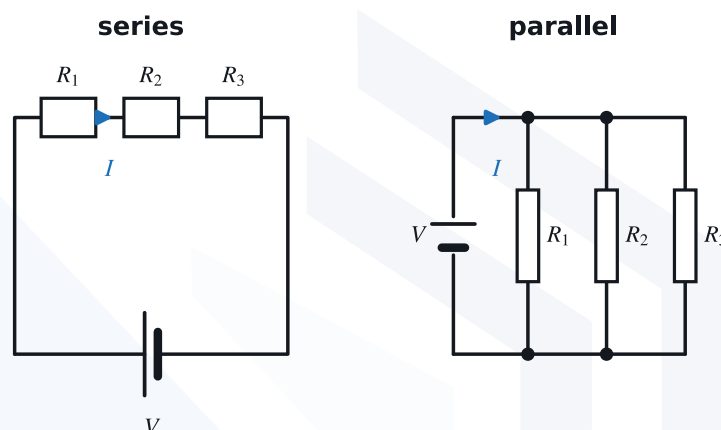


Figure 96. Left: resistors in series carry the same current and share the voltage. Right: resistors in parallel have the same voltage and share the current.

EQUIVALENT RESISTANCE IN SERIES, $R_{eq} = R_1 + R_2 + R_3$

We start from the two facts that define a series connection and reach the rule for the single resistor that could replace the three.

Setting up. Three resistors R_1 , R_2 , R_3 are joined end to end across a supply of p.d. V . The p.d. across them are V_1 , V_2 , V_3 .

Step 1 — state why the current is the same in all three. There is only one path, and charge does not pile up anywhere, so whatever flows into R_1 must flow out of R_3 :

$$I_1 = I_2 = I_3 = I$$

Step 2 — state why the voltages add. Moving from one end of the chain to the other, the total fall in potential is the sum of the separate falls:

$$V = V_1 + V_2 + V_3$$

Step 3 — write each p.d. from Ohm's law, $V = IR$.

$$V_1 = IR_1, \quad V_2 = IR_2, \quad V_3 = IR_3$$

Step 4 — substitute these into Step 2.

$$V = IR_1 + IR_2 + IR_3$$

Step 5 — write the same total p.d. for the single equivalent resistor. By definition R_{eq} carries the same current I at the same total p.d. V :

$$V = IR_{eq}$$

Step 6 — set the two expressions for V equal.

$$IR_{eq} = IR_1 + IR_2 + IR_3$$

Step 7 — divide every term by I , which is the same throughout by Step 1:

$$R_{eq} = R_1 + R_2 + R_3$$

Result. In series the resistances simply add.

What it means. Each extra resistor adds more length of "difficult path", so the total is always **larger** than the largest single resistor.

Conditions used. One single path with nothing joined at the junctions, and each resistor obeying Ohm's law at the temperature it reaches.

EQUIVALENT RESISTANCE IN PARALLEL, $\frac{1}{R_{eq}} = \sum \frac{1}{R_i}$

We start from the two facts that define a parallel connection and reach the rule for the single replacement resistor.

Setting up. Three resistors R_1 , R_2 , R_3 are connected between the same two points, across a p.d. V . They carry currents I_1 , I_2 , I_3 .

Step 1 — state why the p.d. is the same across all three. Every resistor has its two ends joined to the same two points, so each feels the same potential difference:

$$V_1 = V_2 = V_3 = V$$

Step 2 — state why the currents add. At the junction the incoming current splits into the three branches, and by Kirchhoff's first law nothing is lost:

$$I = I_1 + I_2 + I_3$$

Step 3 — write each branch current from Ohm's law, $I = V/R$.

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}$$

Step 4 — substitute these into Step 2.

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

Step 5 — write the same total current for the single equivalent resistor.

$$I = \frac{V}{R_{eq}}$$

Step 6 — set the two expressions for I equal.

$$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

Step 7 — divide every term by V , which is the same for all of them by Step 1:

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Step 8 — write the special case of just two resistors. Add the two fractions over a common denominator and turn the result upside down:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

Result. In parallel the reciprocals add; for two resistors this is "product over sum".

What it means. Each new branch gives the current another road to travel, so the combination is always **smaller** than the smallest resistor in it. This is why adding appliances in a house circuit lowers the total resistance and raises the total current drawn.

Conditions used. All the resistors share the same two nodes, the connecting wires have negligible resistance, and each resistor is ohmic.

In a parallel pair the current divides in the inverse ratio of the resistances:

$$I_1 = I \frac{R_2}{R_1 + R_2}, \quad I_2 = I \frac{R_1}{R_1 + R_2}$$

WORKED EXAMPLE 23.3

Problem. A battery of emf 12 V and negligible internal resistance is connected to a $2\ \Omega$ resistor in series with a parallel combination of $3\ \Omega$ and $6\ \Omega$. Find (a) the equivalent resistance, (b) the current drawn from the battery, and (c) the current in each parallel branch.

Solution.

(a) Parallel part: $R_p = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\ \Omega$. Total: $R_{eq} = 2 + 2 = 4\ \Omega$.

(b) $I = \frac{V}{R_{eq}} = \frac{12}{4} = 3\ \text{A}$.

(c) Voltage across the parallel section: $V_p = IR_p = 3 \times 2 = 6\ \text{V}$. So $I_3 = 6/3 = 2\ \text{A}$ and $I_6 = 6/6 = 1\ \text{A}$.

Check: $2 + 1 = 3\ \text{A}$, as required.

Kirchhoff's laws

Many networks — a bridge, or two cells feeding one resistor — cannot be reduced by series and parallel rules at all. Kirchhoff's two laws solve every such network.

KIRCHHOFF'S LAWS

First law (junction law). The algebraic sum of the currents meeting at any junction is zero: $\sum I = 0$. Currents entering the junction are taken positive and those leaving negative. This is conservation of charge — charge does not pile up at a junction.

Second law (loop law). In any closed loop, the algebraic sum of the emfs equals the algebraic sum of the IR products: $\sum E = \sum IR$. This is conservation of energy — a unit charge taken once round a loop returns to the same potential.

SIGN RULES THAT NEVER FAIL

1. Choose a direction for each unknown current and mark it. A negative answer simply means the real current is the other way.
2. Go round the loop in one fixed sense (say clockwise).
3. A resistance traversed **along** the assumed current gives $+IR$; against it, $-IR$.
4. A cell traversed from $-$ to $+$ inside it gives $+E$; from $+$ to $-$ gives $-E$.

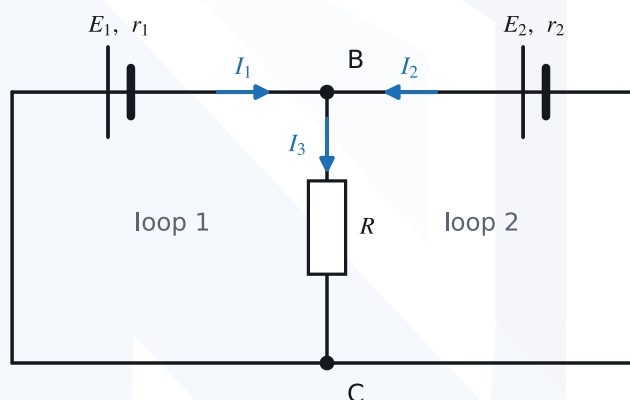


Figure 97. A two-loop network. Applying the junction law at B gives $I_3 = I_1 + I_2$; the loop law applied to loops 1 and 2 then fixes both currents.

WORKED EXAMPLE 23.4

Problem. In the network of the figure, $E_1 = 8\text{ V}$ with $r_1 = 1\ \Omega$ (left branch), $E_2 = 10\text{ V}$ with $r_2 = 4\ \Omega$ (right branch), and the middle branch BC is a $R = 2\ \Omega$ resistor. Find the current in each branch.

Solution. Let I_1 and I_2 flow towards B in the left and right branches. By the junction law at B the middle branch carries $I_3 = I_1 + I_2$ downwards.

Loop 1 (left cell and R):

$$E_1 = I_1 r_1 + I_3 R \Rightarrow 8 = I_1 + 2(I_1 + I_2) = 3I_1 + 2I_2$$

Loop 2 (right cell and R):

$$E_2 = I_2 r_2 + I_3 R \Rightarrow 10 = 4I_2 + 2(I_1 + I_2) = 2I_1 + 6I_2$$

From the first, $I_2 = (8 - 3I_1)/2$. Substituting in the second:

$$2I_1 + 3(8 - 3I_1) = 10 \Rightarrow -7I_1 = -14 \Rightarrow I_1 = 2 \text{ A}$$

Then $I_2 = (8 - 6)/2 = 1 \text{ A}$ and $I_3 = 3 \text{ A}$.

Check with energy: the cells supply $8(2) + 10(1) = 26 \text{ W}$; the resistors dissipate $2^2(1) + 1^2(4) + 3^2(2) = 4 + 4 + 18 = 26 \text{ W}$.

The Wheatstone bridge

The Wheatstone bridge is the standard way of measuring an unknown resistance accurately, because at balance the answer depends only on a ratio — no ammeter or voltmeter reading is needed.

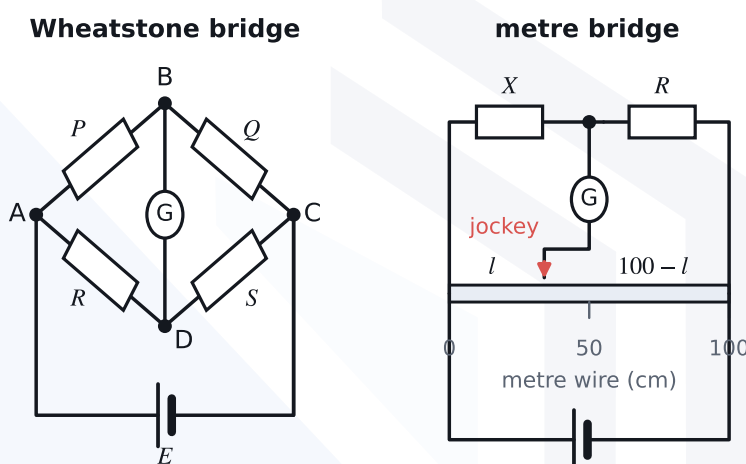


Figure 98. The Wheatstone bridge balances when $P/Q = R/S$; the metre bridge is its practical form, balanced by sliding the jockey along a uniform 1 m wire.

BALANCE CONDITION OF THE WHEATSTONE BRIDGE, $\frac{P}{Q} = \frac{R}{S}$

We start from the fact that the galvanometer reads zero and reach a simple relation between the four arm resistances.

Setting up. Four resistances P , Q , R , S form the arms AB, BC, AD and DC of a diamond. A galvanometer joins B to D, and a cell is connected across A and C. The bridge is called **balanced** when the galvanometer shows no deflection.

Step 1 — say what "no deflection" means for the potentials. No current flows through the galvanometer only if there is no potential difference driving it, so B and D must be at the same potential:

$$V_B = V_D$$

Step 2 — follow the currents. Since nothing leaks away at B into the galvanometer, the current I_1 that enters through P carries straight on through Q . In the same way the current I_2 through R carries on through S .

Step 3 — write the p.d. across the two upper arms. Ohm's law gives

$$V_A - V_B = I_1 P \quad \text{and} \quad V_A - V_D = I_2 R$$

Step 4 — these two are equal, because $V_B = V_D$ from Step 1.

$$I_1 P = I_2 R \quad (1)$$

Step 5 — write the p.d. across the two lower arms.

$$V_B - V_C = I_1 Q \quad \text{and} \quad V_D - V_C = I_2 S$$

Step 6 — these two are also equal, for the same reason.

$$I_1 Q = I_2 S \quad (2)$$

Step 7 — divide equation (1) by equation (2). Both sides are non-zero, so the division is allowed:

$$\frac{I_1 P}{I_1 Q} = \frac{I_2 R}{I_2 S}$$

Step 8 — cancel I_1 on the left and I_2 on the right.

$$\frac{P}{Q} = \frac{R}{S}$$

Step 9 — rearrange to find the unknown arm. If P , Q and R are known and S is the unknown, cross-multiply and divide:

$$S = \frac{QR}{P}$$

Result. The bridge is balanced when $P/Q = R/S$, and the unknown resistance is then $S = QR/P$.

What it means. The currents I_1 and I_2 cancelled, so the answer does not depend on the emf of the cell, on its internal resistance, or on the resistance of the galvanometer and the leads. Only a ratio of resistances is measured, which is why the method is so accurate.

Conditions used. The galvanometer must read exactly zero (true balance), all four arms must be ohmic, and the resistances must not drift as they warm up.

THE EXAMINER IS LOOKING FOR

1. A labelled diagram of the diamond with P , Q , R , S , the galvanometer between B and D, and the cell across A and C.
2. The statement $V_B = V_D$ at balance, with the reason.
3. The two current paths I_1 through P and Q , I_2 through R and S .
4. The two equations $I_1 P = I_2 R$ and $I_1 Q = I_2 S$.
5. Dividing one by the other and cancelling the currents to get $P/Q = R/S$.

WHY BALANCE METHODS BEAT DIRECT MEASUREMENT

At balance the galvanometer carries no current, so its resistance, the resistance of the connecting wires and the emf and internal resistance of the cell **do not enter the answer at all**. Only a ratio of resistances does. That is why bridge and potentiometer methods are called null methods and are far more accurate than an ammeter-voltmeter measurement.

The metre bridge

The metre bridge is the Wheatstone bridge built around a uniform resistance wire exactly one metre long. The unknown X is put in the left gap and a resistance box R in the right gap; the two parts of the wire on either side of the jockey replace the other two arms. Because the wire is uniform, its resistance is proportional to its length, so if the balance point is at l cm from the left end,

$$\frac{X}{R} = \frac{l}{100-l} \implies X = R \frac{l}{100-l}$$

WORKED EXAMPLE 23.5

Problem. In a metre bridge the unknown resistance X is in the left gap and a $5\ \Omega$ resistance box in the right gap. The balance point is found at 60 cm from the left end. (a) Find X . (b) X is a wire of length 1.0 m and diameter 0.40 mm; find its resistivity.

Solution.

(a)

$$X = R \frac{l}{100-l} = 5 \times \frac{60}{40} = 7.5\ \Omega$$

(b) $A = \pi r^2 = \pi(0.20 \times 10^{-3})^2 = 1.26 \times 10^{-7}\ \text{m}^2$.

$$\rho = \frac{XA}{l} = \frac{7.5 \times 1.26 \times 10^{-7}}{1.0} = 9.4 \times 10^{-7}\ \Omega\ \text{m}$$

KEEP THE BALANCE POINT NEAR THE MIDDLE

Take readings with l between about 40 and 60 cm by choosing a suitable R . Near the ends, a small error in l produces a large percentage error in X , and the unaccounted resistance of the end connections (end error) matters most there.

Converting a galvanometer into an ammeter and a voltmeter

A galvanometer is a sensitive current detector with resistance G that gives full-scale deflection for a very small current I_g . Series and parallel combinations turn it into a practical meter.

Ammeter. Connect a small resistance S (a **shunt**) in parallel, so that most of the current bypasses the coil. If the meter is to read up to I , then the shunt carries $I - I_g$ and, since both have the same potential difference,

$$I_g G = (I - I_g)S \implies S = \frac{I_g G}{I - I_g}$$

Voltmeter. Connect a large resistance R (a **multiplier**) in series, so that only I_g flows when the full voltage V is applied:

$$V = I_g(G + R) \implies R = \frac{V}{I_g} - G$$

	Ammeter	Voltmeter
Made by	low shunt in parallel	high multiplier in series
Connected	in series with the element	in parallel with the element
Ideal resistance	zero	infinite
Effect of a real meter	slightly reduces the current	slightly reduces the pd

WORKED EXAMPLE 23.6

Problem. A galvanometer of resistance $100 \, \Omega$ gives full-scale deflection for $1.0 \, \text{mA}$. Convert it into (a) an ammeter reading $0 - 1.0 \, \text{A}$ and (b) a voltmeter reading $0 - 10 \, \text{V}$.

Solution.

(a)

$$S = \frac{I_g G}{I - I_g} = \frac{(1.0 \times 10^{-3})(100)}{1.0 - 1.0 \times 10^{-3}} = \frac{0.100}{0.999} = 0.100 \, \Omega$$

so a shunt of about $0.1 \, \Omega$ is connected in parallel.

(b)

$$R = \frac{V}{I_g} - G = \frac{10}{1.0 \times 10^{-3}} - 100 = 10000 - 100 = 9900 \, \Omega$$

so a $9900 \, \Omega$ multiplier is connected in series.

23.5 Potential divider

Two resistors in series across a supply form a **potential divider** (voltage divider). The supply voltage divides between them in the ratio of their resistances, so the output taken across R_2 is

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

If the two resistors are the two parts of a single uniform wire or a track with a sliding contact, V_{out} can be varied continuously from 0 to V_{in} — this is how a volume control works. A **rheostat** (two terminals) can only reduce the voltage down to a limit set by the load; a potential divider (three terminals) can take the output all the way down to zero, which is why it is preferred.

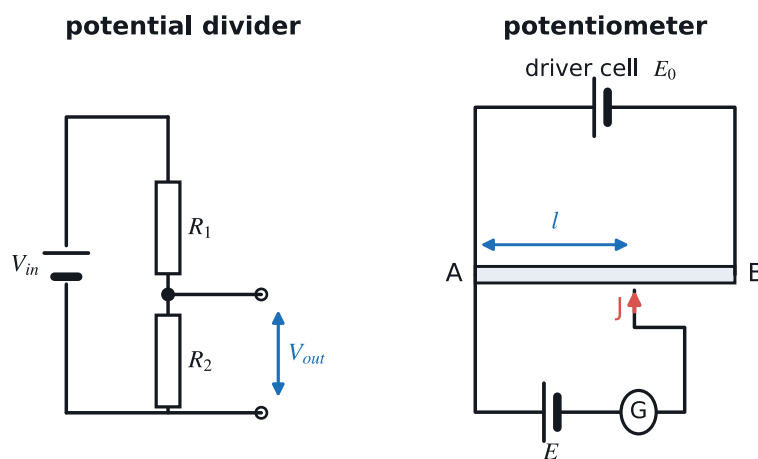


Figure 99. Left: a potential divider gives $V_{out} = V_{in}R_2/(R_1 + R_2)$. Right: the potentiometer — the jockey J is moved until the galvanometer reads zero.

The potentiometer

A potentiometer is a potential divider made from a long uniform resistance wire AB (usually 4 m or 10 m of manganin, arranged in 1 m lengths) carrying a steady current from a driver cell of emf E_0 . Because the wire is uniform, the potential falls uniformly along it. The **potential gradient** is

$$k = \frac{\text{pd across AB}}{\text{length AB}} \quad (\text{V m}^{-1})$$

and the pd across the first l metres is $V = kl$.

PRINCIPLE OF THE POTENTIOMETER

The cell being tested is connected with its positive terminal to A, through a galvanometer to the jockey. The jockey is slid until the galvanometer reads **zero**. At that balance point the cell drives no current, so the pd it would supply is exactly balanced by the pd kl across the wire, and the reading is the cell's **emf**, not its terminal pd:

$$E = kl$$

Comparison of two emfs. Balancing the two cells separately at lengths l_1 and l_2 gives $E_1 = kl_1$ and $E_2 = kl_2$, hence

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

Internal resistance of a cell. First balance the cell on open circuit at length l_1 , so $E = kl_1$. Then close a known resistance R across the cell and re-balance at l_2 , so the terminal pd is $V = kl_2$. Since $E = I(R + r)$ and $V = IR$,

$$\frac{E}{V} = \frac{R + r}{R} = \frac{l_1}{l_2} \implies r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

WORKED EXAMPLE 23.7

Problem. A potentiometer wire AB is 10.0 m long and the whole 2.0 V of the driver cell falls across it. A cell connected in the secondary circuit balances at 7.50 m from A. When a $4.0\ \Omega$ resistor is connected across this cell, the balance length becomes 6.00 m. Find (a) the potential gradient, (b) the emf of the cell and (c) its internal resistance.

Solution.

(a) $k = \frac{2.0}{10.0} = 0.20\ \text{V m}^{-1}$.

(b) $E = kl_1 = 0.20 \times 7.50 = 1.50\ \text{V}$.

(c) Terminal pd $V = kl_2 = 0.20 \times 6.00 = 1.20\ \text{V}$, so

$$r = R \left(\frac{l_1 - l_2}{l_2} \right) = 4.0 \times \frac{7.50 - 6.00}{6.00} = 4.0 \times 0.25 = 1.0\ \Omega$$

THE POTENTIOMETER WILL NOT BALANCE IF...

...the positive terminals of the driver cell and the test cell are not both connected to the same end A, or if the emf of the test cell is greater than the pd across the whole wire. In both cases the galvanometer deflects the same way at every point of the wire — that is the symptom to recognise in the lab.

23.6 Electromotive force of a source, internal resistance**EMF AND TERMINAL POTENTIAL DIFFERENCE**

The **electromotive force** E of a source is the energy given to each coulomb of charge driven round the complete circuit; it equals the pd across the terminals of the source on **open circuit**.

The **terminal potential difference** V is the pd across the terminals when the source is actually delivering current. Both are measured in volts; emf is not a force.

Every real cell has some resistance of its own — the resistance of its electrolyte and electrodes — called the **internal resistance** r . When the cell drives a current, part of the emf is used up inside the cell itself.

TERMINAL POTENTIAL DIFFERENCE OF A CELL, $V = E - Ir$

We start from the energy given to each coulomb by the cell and reach a formula for the voltage you would actually measure across its terminals.

Setting up. A cell of emf E and internal resistance r drives a steady current I through an external resistance R . The emf is the energy the cell gives to each coulomb; that energy has to be shared between the two resistances the charge meets.

Step 1 — apply conservation of energy per coulomb around the loop. The energy supplied to each coulomb equals the energy it gives up in the outside circuit plus the energy it gives up inside the cell:

$$E = (\text{p.d. across } R) + (\text{p.d. across } r)$$

Step 2 — write each p.d. from Ohm's law. The same current I passes through both, since the cell and R are in one single loop:

$$E = IR + Ir$$

Step 3 — take I out as a common factor and solve for the current.

$$E = I(R + r) \implies I = \frac{E}{R + r}$$

Step 4 — say what the voltmeter across the terminals actually reads. The terminals are the two ends of R , so the terminal p.d. is

$$V = IR$$

Step 5 — go back to Step 2 and make IR the subject. Subtract Ir from both sides:

$$IR = E - Ir$$

Step 6 — replace IR by V , using Step 4.

$$V = E - Ir$$

Result. The terminal potential difference is $V = E - Ir$, and the current in the circuit is $I = E/(R + r)$.

What it means. The term Ir is the "lost volts" — the part of the emf spent pushing the current through the cell's own resistance. It also gives a way to measure E and r : a graph of V against I is a straight line whose intercept is E and whose slope is $-r$.

Conditions used. A steady current in a single loop; E and r treated as constant (in a real cell r grows as the cell ages); and the cell is discharging. If the cell is being **charged**, the current is reversed and $V = E + Ir$ instead.

So the terminal pd is always **less** than the emf while the cell is discharging, and the graph of V against I is a straight line of intercept E and slope $-r$. If the terminals are short-circuited ($R = 0$), the current is limited only by r : $I_{\max} = E/r$. This is why a "dead" torch cell, whose r has grown large, still shows nearly 1.5 V on a voltmeter but cannot light a bulb.

Connection	Equivalent emf	Equivalent internal resistance	Use when
n identical cells in series	nE	nr	R is much larger than r
n identical cells in parallel	E	r/n	R is much smaller than r

WORKED EXAMPLE 23.8

Problem. A cell gives a terminal pd of 1.40 V when it supplies 0.40 A, and 1.20 V when it supplies 0.80 A. Find the emf and internal resistance of the cell, and the current if its terminals were short-circuited.

Solution. Using $V = E - Ir$ twice:

$$1.40 = E - 0.40r, \quad 1.20 = E - 0.80r$$

Subtracting the second from the first: $0.20 = 0.40r$, so $r = 0.50 \, \Omega$. Then $E = 1.40 + 0.40(0.50) = 1.60 \, \text{V}$.

Short-circuit current: $I = E/r = 1.60/0.50 = 3.2 \, \text{A}$.

23.7 Work and power in electrical circuits

ELECTRICAL POWER, $P = VI = I^2R = \frac{V^2}{R}$

We start from the definition of potential difference and reach the three forms of the power formula.

Setting up. A steady current I flows for a time t through a resistor R across which the potential difference is V .

Step 1 — use the definition of potential difference. Potential difference is work done per unit charge, so the work done on a charge Q falling through V is

$$W = QV$$

Step 2 — express the charge in terms of the current. Current is charge per second, so in a time t a steady current I carries

$$Q = It$$

Step 3 — substitute this into Step 1.

$$W = VIt$$

Step 4 — divide by the time to get the power. Power is work done per second:

$$P = \frac{W}{t} = \frac{VIt}{t}$$

Step 5 — cancel t .

$$P = VI$$

Step 6 — get the second form. For an ohmic resistor, Ohm's law gives $V = IR$; put that in place of V :

$$P = (IR)I$$

Step 7 — multiply out.

$$P = I^2 R$$

Step 8 — get the third form. Start again from $P = VI$ and this time replace I by V/R :

$$P = V\left(\frac{V}{R}\right)$$

Step 9 — multiply out.

$$P = \frac{V^2}{R}$$

Result.

$$P = VI = I^2 R = \frac{V^2}{R}$$

and the heat produced in a time t is $H = I^2 R t$.

What it means. All three expressions give the same number for the same resistor — choose whichever uses the two quantities you already know. In a series circuit the current is common, so $P = I^2 R$ is the quick one; in a parallel circuit the voltage is common, so $P = V^2/R$ is.

Conditions used. Steady current and steady p.d.; and Steps 6 to 9 need the element to be **ohmic**, so that $V = IR$ holds. The general form $P = VI$ works for any device, ohmic or not.

For a source of emf E delivering current I , the total power generated is $P = EI$; of this, $I^2 r$ is wasted inside the source and $VI = I^2 R$ reaches the external circuit.

JOULE'S LAW OF HEATING

In a pure resistance all the electrical work appears as heat:

$$H = I^2 R t \text{ joules}$$

The heat produced is proportional to (i) the square of the current, (ii) the resistance, and (iii) the time. In calories, $H = I^2 R t / 4.2$.

WHICH FORM OF P TO USE

Use $P = I^2 R$ when the elements are in **series** (same I): the largest resistance dissipates most power. Use $P = V^2/R$ when they are in **parallel** (same V): the smallest resistance dissipates most power. Choosing the right form makes "which bulb is brighter?" questions immediate.

THE 100 W BULB IS NOT ALWAYS THE BRIGHTER ONE

Two bulbs rated 100 W and 60 W at 220 V have resistances $484 \, \Omega$ and $807 \, \Omega$. Joined **in series**, the same current flows, so $P = I^2 R$ makes the 60 W bulb glow brighter. Power ratings only hold at the rated voltage.

Electrical energy is sold in **kilowatt-hours**: $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$. This is the "unit" on a NEA bill.

WORKED EXAMPLE 23.9

Problem. An electric heater is marked 1500 W, 220 V. (a) Find its resistance and the current it draws. (b) How long does it take to heat 2.0 kg of water from 20°C to 100°C , assuming no heat is lost? Take $c = 4200 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$. (c) If it runs 2 hours a day for 30 days, what does it cost at Rs 10 per unit?

Solution.

$$(a) R = \frac{V^2}{P} = \frac{220^2}{1500} = \frac{48400}{1500} = 32.3 \, \Omega; I = \frac{P}{V} = \frac{1500}{220} = 6.8 \text{ A}.$$

$$(b) \text{ Heat needed: } Q = mc\Delta\theta = 2.0 \times 4200 \times 80 = 6.72 \times 10^5 \text{ J}.$$

$$t = \frac{Q}{P} = \frac{6.72 \times 10^5}{1500} = 448 \text{ s} \approx 7.5 \text{ min}$$

$$(c) \text{ Energy} = 1.5 \text{ kW} \times 2 \text{ h} \times 30 = 90 \text{ kWh, so the cost is } 90 \times 10 = \text{Rs } 900.$$

✓ Chapter summary

- Current is $I = dQ/dt$; in a metal $I = nAv_d e$ and $J = I/A = nev_d$, with a drift velocity of order 10^{-4} m s^{-1} .
- Ohm's law: $V = IR$ at constant temperature. $R = \rho l/A$, $\sigma = 1/\rho$, and microscopically $\rho = m/(ne^2\tau)$.
- Ohmic conductors give a straight I - V line through the origin; lamps, diodes and thermistors are non-ohmic.
- Series: $R_{eq} = R_1 + R_2 + R_3$. Parallel: $1/R_{eq} = 1/R_1 + 1/R_2 + 1/R_3$.
- Kirchhoff: $\sum I = 0$ at a junction (charge conservation); $\sum E = \sum IR$ round a loop (energy conservation).
- Wheatstone bridge balances when $P/Q = R/S$; the metre bridge form is $X = Rl/(100 - l)$.
- Galvanometer to ammeter: shunt $S = I_g G/(I - I_g)$ in parallel. To voltmeter: multiplier $R = VI/I_g - G$ in series.
- Potential divider: $V_{out} = V_{in} R_2/(R_1 + R_2)$. Potentiometer: $E = kl$, $E_1/E_2 = l_1/l_2$, and $r = R(l_1 - l_2)/l_2$.
- For a source, $I = E/(R + r)$ and $V = E - Ir$; power $P = VI = I^2 R = V^2/R$ and heat $H = I^2 R t$.

Practice questions

Group A — Multiple choice (1 mark each)

- The drift velocity of free electrons in a copper wire carrying a normal current is of the order of [1]
 a 10^8 m s^{-1} b 10^5 m s^{-1} c 10^{-4} m s^{-1} d $10^{-10} \text{ m s}^{-1}$
- A wire of resistance R is stretched uniformly until its length is doubled. Its new resistance is [1]
 a $R/2$ b $2R$ c $4R$ d $R/4$
- Three resistors of 6Ω each are connected in parallel. The equivalent resistance is [1]
 a 18Ω b 6Ω c 3Ω d 2Ω
- Kirchhoff's junction law is a statement of the conservation of [1]
 a energy b charge c momentum d mass
- A galvanometer is converted into an ammeter by connecting [1]
 a a high resistance in series b a low resistance in series
 c a high resistance in parallel d a low resistance in parallel
- A 100 W and a 60 W bulb, both rated 220 V , are joined in series across 220 V . Then [1]
 a the 100 W bulb is brighter b the 60 W bulb is brighter
 c both are equally bright d neither glows

ANSWERS TO GROUP A

- (c) — from $v_d = I/nAe$, a few tenths of a millimetre per second.
- (c) — volume is constant, so A halves and $R \propto l^2$ gives $4R$.
- (d) — $1/R_{eq} = 3/6$, so $R_{eq} = 2 \Omega$.
- (b) — charge cannot accumulate at a junction.
- (d) — a low shunt in parallel carries the excess current.
- (b) — same current in series, and $P = I^2 R$ with $R_{60} > R_{100}$.

Group B — Short answer (5 marks each)

- Define drift velocity and derive the relation $I = nAv_d e$ for a metallic conductor. [5]
- Distinguish between ohmic and non-ohmic conductors with one I - V graph and two examples of each. [5]
- Derive expressions for the equivalent resistance of three resistors connected
 (a) in series and
 (b) in parallel. [5]
- A battery of emf 10 V and negligible internal resistance is connected to a 2Ω resistor in series with a parallel combination of 4Ω and 12Ω . Find the equivalent resistance, the current drawn, and the current in each parallel branch. [5]
- A cell of emf 2.0 V and internal resistance 0.5Ω is connected across a 4.5Ω resistor. Find the current, the terminal potential difference, and the power dissipated in the external resistor. [5]

6. In a metre bridge experiment the unknown resistance X is in the left gap and a $6\ \Omega$ resistance box in the right gap. The balance point is at 40 cm from the left end. Find X , and state where the balance point would be if the two were interchanged. [5]

ANSWERS TO GROUP B

4. Parallel part: $4 \times 12/16 = 3\ \Omega$; total $= 2 + 3 = 5\ \Omega$. Current drawn $I = 10/5 = 2\text{ A}$. Pd across the parallel part $= 2 \times 3 = 6\text{ V}$, so $I_4 = 6/4 = 1.5\text{ A}$ and $I_{12} = 6/12 = 0.5\text{ A}$ (sum $= 2\text{ A}$, as it must be).

5. $I = E/(R + r) = 2.0/5.0 = 0.40\text{ A}$; $V = IR = 0.40 \times 4.5 = 1.8\text{ V}$ (or $E - Ir = 2.0 - 0.2$); $P = I^2 R = 0.16 \times 4.5 = 0.72\text{ W}$.

6. $X = R l/(100 - l) = 6 \times 40/60 = 4\ \Omega$. On interchanging, the balance length measured from the left end becomes $100 - 40 = 60\text{ cm}$, since now $R/X = l'/(100 - l')$ gives $6/4 = 60/40$.

1., 2., 3. are bookwork: see §23.1, §23.3 and §23.4 respectively. Marks are awarded for the labelled diagram, the statement of the principle, and each algebraic step.

Group C — Long answer (8 marks each)

- (a) State Kirchhoff's laws of electrical networks and name the conservation principle behind each. [2]

(b) Draw a Wheatstone bridge and derive its balance condition. [3]

(c) In a metre bridge, an unknown resistance in the left gap balances against $3\ \Omega$ in the right gap at 75 cm from the left end. Find the unknown resistance. [3]
- (a) Define emf and terminal potential difference and derive $V = E - Ir$. [3]

(b) Describe how a potentiometer is used to measure the internal resistance of a cell, deriving the formula used. [3]

(c) A cell balances at 250 cm of a potentiometer wire; with a $5\ \Omega$ resistor across it the balance length falls to 200 cm . Find its internal resistance. [2]
- (a) Derive $P = I^2 R$ from the definition of potential difference, and state Joule's law of heating. [3]

(b) A battery of emf 12 V and internal resistance $1\ \Omega$ is connected to a $3\ \Omega$ resistor in series with a parallel combination of $6\ \Omega$ and $12\ \Omega$. Find the current drawn, the terminal potential difference, and the power dissipated in the $6\ \Omega$ resistor. [5]

ANSWERS TO GROUP C

1.(c) $X = R l/(100 - l) = 3 \times 75/25 = 9\ \Omega$.

2.(c) $r = R(l_1 - l_2)/l_2 = 5 \times (250 - 200)/200 = 5 \times 0.25 = 1.25\ \Omega$.

3.(b) Parallel part $= 6 \times 12/18 = 4\ \Omega$; total $= 3 + 4 + 1 = 8\ \Omega$. Current $I = 12/8 = 1.5\text{ A}$. Terminal pd $V = E - Ir = 12 - 1.5 = 10.5\text{ V}$. Pd across the parallel part $= 1.5 \times 4 = 6\text{ V}$, so the $6\ \Omega$ resistor carries 1.0 A and dissipates $P = I^2 R = 1.0^2 \times 6 = 6\text{ W}$.

1.(a), 1.(b), 2.(a), 2.(b), 3.(a) are bookwork: see the boxed statements in §23.4, the Wheatstone derivation, §23.6 and §23.7.

UNIT 24

Nuclear physics

Modern Physics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Nucleus: Discovery of nucleus
- Nuclear density, Mass number, Atomic number
- Atomic mass, Isotopes
- Einstein's mass-energy relation
- Mass defect, packing fraction, BE per nucleon
- Creation and annihilation
- Nuclear fission and fusion; energy released

Almost the whole mass of an atom sits in a nucleus about 10^5 times smaller than the atom itself. This unit is about that tiny object: how it was discovered, how we describe it, and why rearranging its nucleons releases a million times more energy per kilogram than burning coal. The single equation behind all of it is Einstein's $E = mc^2$ — in nuclear physics, mass and energy are two ways of writing the same thing.

WHAT THE EXAMINER ASKS FROM THIS UNIT

Two things appear every year: a numerical on mass defect, binding energy or energy released in fission/fusion (always via $1 \text{ u} = 931.5 \text{ MeV}$), and a question on the binding-energy-per-nucleon curve. Learn the curve, its peak at $A \approx 56$, and what its two slopes mean.

24.1 Nucleus: Discovery of nucleus

In 1911 the accepted picture was J. J. Thomson's "plum pudding" atom: positive charge spread uniformly through the whole atom with electrons embedded in it. If that were true, a fast alpha particle fired at a thin foil would pass through almost undeflected, because the spread-out charge could never exert a large force.

Geiger and Marsden, working under Rutherford, fired alpha particles (helium nuclei, energy about 5 MeV) from a radium source at a gold foil only $\approx 10^{-7} \text{ m}$ thick, and counted the flashes they produced on a movable zinc-sulphide screen.

Observation	Conclusion
Most alpha particles passed through with no deflection	the atom is mostly empty space
A small fraction were deflected through large angles	there is a concentrated, positively charged core that repels them
About 1 in 8000 was turned back through more than 90°	that core is very massive and very small

Rutherford's conclusion (1911) was the **nuclear model**: all the positive charge and nearly all the mass of the atom are concentrated in a nucleus of radius about 10^{-14} m, while the electrons move around it at a distance of about 10^{-10} m. In his own words, the result was "as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you."

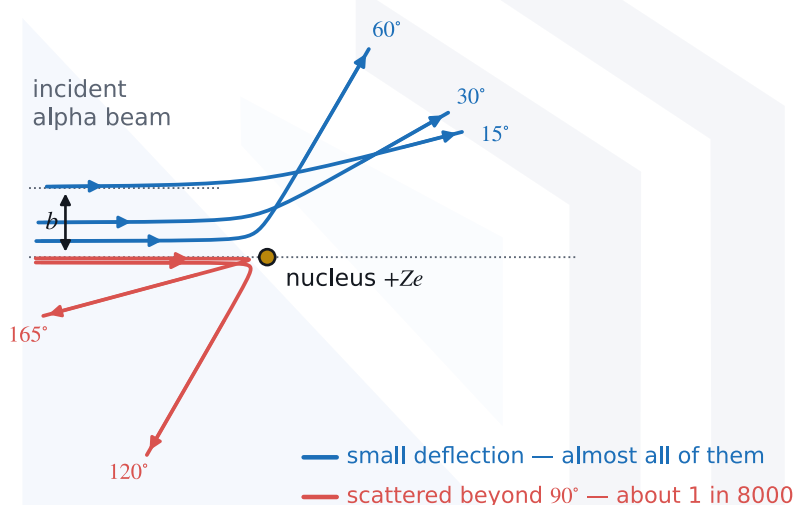


Figure 100. Rutherford scattering. The deflection depends on the impact parameter b : only an alpha particle aimed almost straight at the nucleus is turned through a large angle, while distant ones pass practically undeviated.

How big is the nucleus? An alpha particle fired straight at it never touches it: the electrostatic repulsion stops it first. The point where it stops gives an upper limit on the nuclear radius.

DISTANCE OF CLOSEST APPROACH

An alpha particle of kinetic energy E is fired **head-on** at a nucleus of atomic number Z . The alpha particle carries charge $+2e$, the nucleus carries $+Ze$, so the two repel. As the alpha particle comes closer it slows down, and at the closest point it has stopped completely for an instant. Call that separation r_0 .

Step 1 — the energy far away. At a large distance the potential energy of the two charges is zero, so all the energy of the alpha particle is kinetic:

$$E_{\text{far}} = E$$

Step 2 — the energy at the closest point. Here the speed is zero, so the kinetic energy is zero. All the energy is now electrostatic potential energy of two charges $2e$ and Ze a distance r_0 apart:

$$E_{\text{near}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{(2e)(Ze)}{r_0} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{r_0}$$

Step 3 — energy is conserved, so the two expressions are equal:

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{r_0}$$

Step 4 — multiply both sides by r_0 :

$$E r_0 = \frac{2Ze^2}{4\pi\epsilon_0}$$

Step 5 — divide both sides by E to make r_0 the subject:

$$r_0 = \frac{2Ze^2}{4\pi\epsilon_0 E}$$

r_0 is the **distance of closest approach**. Because the alpha particle is turned back before touching, the true nuclear radius must be smaller than r_0 — so this calculation puts an upper limit on the size of the nucleus. Later work completed the picture: Rutherford identified the **proton** in 1919, and Chadwick discovered the neutral **neutron** in 1932. Protons and neutrons together are called **nucleons**, and they are held together by the short-range, strongly attractive **nuclear force**, which acts only over about 10^{-15} m but is far stronger than the electrostatic repulsion between protons at that range.

24.2 Nuclear density, Mass number, Atomic number

ATOMIC NUMBER AND MASS NUMBER

The **atomic number** Z is the number of protons in the nucleus (and so the number of electrons in the neutral atom). The **mass number** A is the total number of nucleons, protons plus neutrons. The number of neutrons is $N = A - Z$.

A nuclide is written with the mass number as a left superscript and the atomic number as a left subscript, for example $^{235}_{92}\text{U}$ — uranium with $Z = 92$, $A = 235$, hence $N = 143$ neutrons.

Experiments in which fast electrons are scattered from nuclei show that the nuclear radius depends only on the number of nucleons:

$$R = R_0 A^{1/3}, \quad R_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fm}$$

WHY NUCLEAR DENSITY IS THE SAME FOR EVERY NUCLEUS

The volume of a nucleus is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^3 A$$

so the volume is directly proportional to A . The mass of the nucleus is approximately $A m_n$, where $m_n \approx 1.67 \times 10^{-27}$ kg is the mass of one nucleon. Hence

$$\rho = \frac{Am_n}{\frac{4}{3}\pi R_0^3 A} = \frac{3m_n}{4\pi R_0^3}$$

A cancels: **nuclear density is independent of the mass number**, the same for hydrogen and for uranium. Numerically $\rho \approx 2.3 \times 10^{17} \text{ kg m}^{-3}$, about 10^{14} times the density of water. This constancy tells us that nucleons are packed like marbles in a bag — the nuclear force saturates, each nucleon binding only to its nearest neighbours.

WORKED EXAMPLE 24.1

Problem. Find the radius and the density of a nucleus with mass number $A = 125$. Take $R_0 = 1.2 \text{ fm}$ and the mass of a nucleon as $1.67 \times 10^{-27} \text{ kg}$.

Solution. Since $125^{1/3} = 5$,

$$R = R_0 A^{1/3} = 1.2 \times 10^{-15} \times 5 = 6.0 \times 10^{-15} \text{ m}$$

Volume:

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi (6.0 \times 10^{-15})^3 = 9.05 \times 10^{-43} \text{ m}^3$$

Mass = $125 \times 1.67 \times 10^{-27} = 2.09 \times 10^{-25} \text{ kg}$. Hence

$$\rho = \frac{2.09 \times 10^{-25}}{9.05 \times 10^{-43}} = 2.3 \times 10^{17} \text{ kg m}^{-3}$$

24.3 Atomic mass, Isotopes

Nuclear masses are far too small for the kilogram, so we use the **unified atomic mass unit** (u), also called the amu.

ATOMIC MASS UNIT

One atomic mass unit is exactly one-twelfth of the mass of one neutral atom of carbon-12:

$$1 \text{ u} = 1.6605 \times 10^{-27} \text{ kg}$$

Particle	Mass (u)	Mass (kg)	Charge
Proton	1.007276	1.6726×10^{-27}	$+e$
Neutron	1.008665	1.6749×10^{-27}	0
Electron	0.000549	9.109×10^{-31}	$-e$
^1H atom	1.007825	—	neutral

The **atomic mass** of an element is the mass of one of its atoms in u. It is close to, but never exactly equal to, the mass number A — partly because protons and neutrons are not exactly 1 u , and mainly because some mass is "missing" as binding energy (§24.5).

ISOTOPES

Isotopes are atoms of the same element that have the **same atomic number Z but different mass numbers A** — that is, the same number of protons but different numbers of neutrons. They are chemically identical but physically different.

Term	Same	Examples
Isotopes	Z	^1H , ^2H , ^3H ; ^{35}Cl , ^{37}Cl ; ^{235}U , ^{238}U
Isobars	A	^{40}Ar , ^{40}K , ^{40}Ca
Isotones	$N = A - Z$	^{36}S , ^{37}Cl , ^{38}Ar (each $N = 20$)

The atomic mass quoted in a data book is the **weighted average** over the naturally occurring isotopes, which is why chlorine is listed as 35.45 and not as a whole number.

WORKED EXAMPLE 24.2

Problem. Natural chlorine is 75.77% ^{35}Cl (atomic mass 34.96885 u) and 24.23% ^{37}Cl (atomic mass 36.96590 u). Find its average atomic mass.

Solution.

$$\bar{M} = \frac{75.77 \times 34.96885 + 24.23 \times 36.96590}{100}$$

The two products are 2649.59 and 895.68 , so

$$\bar{M} = \frac{3545.27}{100} = 35.45\text{ u}$$

24.4 Einstein's mass-energy relation

Einstein's special theory of relativity (1905) showed that mass and energy are equivalent: a body of mass m possesses a rest energy

$$E = mc^2$$

with $c = 3.0 \times 10^8\text{ m s}^{-1}$. If a system loses mass Δm , it releases energy $\Delta E = \Delta m c^2$. Because c^2 is enormous, a tiny mass change gives a huge energy: this is why nuclear reactions are millions of times more energetic than chemical ones, in which the mass change is far too small to detect.

Where $E = mc^2$ comes from

Relativity says that mass is not a fixed property of a body. A body whose mass is m_0 when it is at rest — its **rest mass** — behaves, when it moves at speed v , as though its mass were

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

At everyday speeds v/c is so small that m and m_0 agree to many decimal places, which is why nobody noticed for two hundred years. As v approaches c the square root approaches zero and m grows without limit. The mass-energy relation follows from this one formula together with the ordinary work-energy theorem.

EINSTEIN'S MASS-ENERGY RELATION $E = mc^2$

Step 1 — work done becomes kinetic energy. A force F acting through a small distance dx does work $F dx$, and all of that work goes into the kinetic energy of the body:

$$dE_k = F dx$$

Step 2 — write the force as a rate of change of momentum. Newton's second law in its general form is $F = \frac{d(mv)}{dt}$, so

$$dE_k = \frac{d(mv)}{dt} dx$$

Step 3 — rearrange the differentials. dx and dt are just numbers, so we may group them as dx/dt , which is the speed v :

$$dE_k = d(mv) \cdot \frac{dx}{dt} = v d(mv)$$

Step 4 — expand the product $d(mv)$. Both m and v are changing now, so the product rule gives $d(mv) = m dv + v dm$:

$$dE_k = v(m dv + v dm) = mv dv + v^2 dm \quad (1)$$

Step 5 — go back to the mass formula and clear the square root. Squaring $m = m_0/\sqrt{1 - v^2/c^2}$ gives

$$m^2 \left(1 - \frac{v^2}{c^2}\right) = m_0^2$$

Step 6 — multiply every term by c^2 so there are no fractions left:

$$m^2 c^2 - m^2 v^2 = m_0^2 c^2$$

Step 7 — differentiate both sides. The right-hand side is a constant, so its differential is zero. Differentiating $m^2 c^2$ gives $2mc^2 dm$, and differentiating $m^2 v^2$ by the product rule gives $2mv^2 dm + 2m^2 v dv$. Hence

$$2mc^2 dm - 2mv^2 dm - 2m^2 v dv = 0$$

Step 8 — divide every term by $2m$:

$$c^2 dm - v^2 dm - mv dv = 0$$

Step 9 — move the last two terms across:

$$c^2 dm = mv dv + v^2 dm \quad (2)$$

Step 10 — compare (1) and (2). Their right-hand sides are word for word the same, so their left-hand sides must be equal:

$$dE_k = c^2 dm$$

Step 11 — add up all the small bits. Integrate from rest, where the mass is m_0 and the kinetic energy is zero, up to speed v , where the mass is m and the kinetic energy is E_k :

$$\int_0^{E_k} dE_k = c^2 \int_{m_0}^m dm$$

$$E_k = c^2(m - m_0) = (m - m_0)c^2$$

Step 12 — read what this says. Giving the body kinetic energy E_k has increased its mass by exactly E_k/c^2 . Energy has mass. Turning the statement round, the body already owned a hidden **rest energy** m_0c^2 before it ever moved, and its total energy is

$$E = E_k + m_0c^2 = (m - m_0)c^2 + m_0c^2 = mc^2$$

THE CONVERSION EVERY NUCLEAR NUMERICAL NEEDS

$$1 \text{ u} = 931.5 \text{ MeV}/c^2$$

So a mass defect expressed in u is turned into energy simply by multiplying by 931.5 to get MeV. Also $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ and $1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$.

WORKED EXAMPLE 24.3

Problem. (a) Show that 1 u is equivalent to about 931 MeV. (b) How much energy would be released if 1.0 g of matter were completely converted into energy?

Solution.

(a) $E = mc^2 = (1.6605 \times 10^{-27})(2.998 \times 10^8)^2 = 1.4924 \times 10^{-10} \text{ J}$. Converting to MeV,

$$E = \frac{1.4924 \times 10^{-10}}{1.602 \times 10^{-13}} = 931.5 \text{ MeV}$$

(b) $E = mc^2 = (1.0 \times 10^{-3})(3.0 \times 10^8)^2 = 9.0 \times 10^{13} \text{ J}$.

For comparison, burning 1 g of coal gives about 3×10^4 J — roughly three billion times less.

24.5 Mass defect, packing fraction, BE per nucleon

Measure the mass of a helium nucleus and compare it with the masses of the two protons and two neutrons it contains: the nucleus is **lighter**. The missing mass is called the **mass defect**.

MASS DEFECT AND BINDING ENERGY

The **mass defect** is the difference between the total mass of the free nucleons and the actual mass of the nucleus:

$$\Delta m = [Zm_p + (A - Z)m_n] - M$$

The **binding energy** is the energy equivalent of the mass defect, $B = \Delta m c^2$. It is the energy that must be supplied to break the nucleus completely into free nucleons — and the energy released when they come together.

USE ATOMIC MASSES CONSISTENTLY

If you use the mass of the ^1H **atom** (1.007825 u) instead of the proton, you must also use the **atomic** mass of the nuclide, not the nuclear mass. The Z electron masses then cancel out. Mixing the two conventions is the commonest source of wrong binding energies in exams.

The **packing fraction** measures the mass defect per nucleon in a scaled form:

$$f = \frac{M - A}{A}$$

where M is the atomic mass in u and A the mass number; it is usually quoted after multiplying by 10^4 . A **negative** packing fraction means the nuclide is lighter than A units and so is stable; nuclides near $A \approx 56$ have the most negative values. For ^{16}O , $M = 15.9949$ u, so $f = (15.9949 - 16)/16 = -3.2 \times 10^{-4}$.

The real measure of stability, however, is the **binding energy per nucleon**, B/A . Plotting it against A gives the most important graph in nuclear physics.

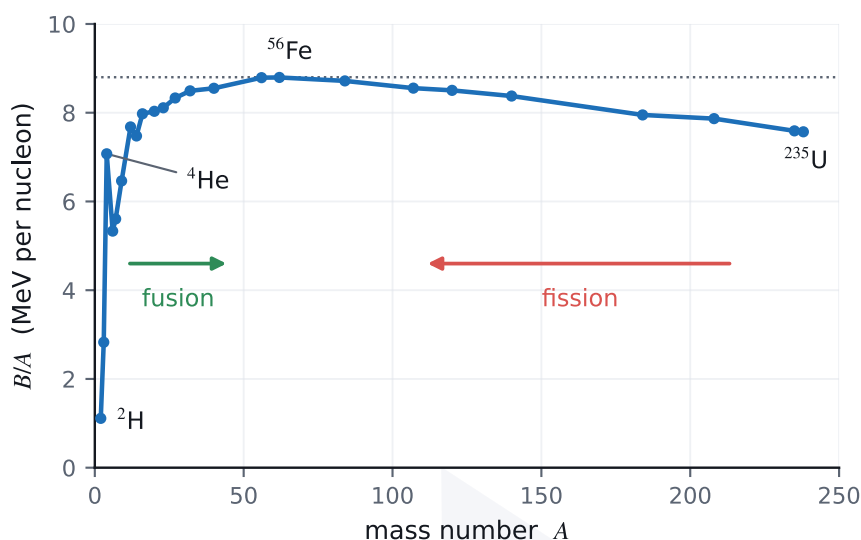


Figure 101. Binding energy per nucleon against mass number. The peak is at $A \approx 56$ (iron), about 8.8 MeV per nucleon.

Read the curve carefully — four facts are examined again and again:

1. B/A rises steeply for light nuclei, with sharp peaks at ${}^4\text{He}$, ${}^{12}\text{C}$ and ${}^{16}\text{O}$ (nuclei built of whole alpha particles are unusually stable).
2. It is nearly flat at about 8.5 to 8.8 MeV for $30 < A < 120$; the maximum, 8.79 MeV, is at ${}^{56}\text{Fe}$. These nuclei are the most stable.
3. It falls slowly for heavy nuclei (to 7.6 MeV for ${}^{235}\text{U}$) because the long-range electrostatic repulsion between many protons grows faster than the short-range nuclear attraction.
4. Therefore energy is released **either** by joining very light nuclei (**fusion**) **or** by splitting very heavy ones (**fission**) — both move the products towards the peak, so the total binding energy increases.

WORKED EXAMPLE 24.4

Problem. Calculate the mass defect, the binding energy, and the binding energy per nucleon of the helium nucleus ${}^4\text{He}$. Given: mass of ${}^1\text{H}$ atom = 1.007825 u, mass of neutron = 1.008665 u, atomic mass of ${}^4\text{He}$ = 4.002603 u.

Solution. Helium has $Z=2$, $A=4$, so $N=2$.

$$\Delta m = 2(1.007825) + 2(1.008665) - 4.002603$$

$$\Delta m = 2.015650 + 2.017330 - 4.002603 = 0.030377 \text{ u}$$

Binding energy:

$$B = 0.030377 \times 931.5 = 28.3 \text{ MeV}$$

Binding energy per nucleon:

$$\frac{B}{A} = \frac{28.3}{4} = 7.07 \text{ MeV per nucleon}$$

This is well above its neighbours on the curve, which is why the alpha particle is so stable and is emitted as a unit in radioactive decay.

24.6 Creation and annihilation

Because mass and energy are interchangeable, energy can turn into matter and matter back into energy — but only in particle–antiparticle pairs, so that charge is conserved.

Pair production (creation). A high-energy photon passing close to a heavy nucleus can vanish, creating an electron and a positron:

$$\gamma \rightarrow e^- + e^+$$

The photon must supply at least the rest energy of both particles. Each has rest energy $m_e c^2 = 0.511 \text{ MeV}$, so the **threshold energy** is

$$E_{\min} = 2m_e c^2 = 1.02 \text{ MeV}$$

Any extra photon energy appears as kinetic energy of the pair. The nearby nucleus is essential: it recoils and takes up momentum, because a lone photon cannot produce a pair and conserve energy and momentum at the same time.

Annihilation. The reverse process: when a positron meets an electron, both disappear and their entire mass becomes radiation. Two photons are produced, travelling in opposite directions,

$$e^- + e^+ \rightarrow \gamma + \gamma$$

each of energy 0.511 MeV when the particles were nearly at rest. Two photons are needed, not one, so that the total momentum stays zero. This process is the basis of the **PET scan** used in hospitals.

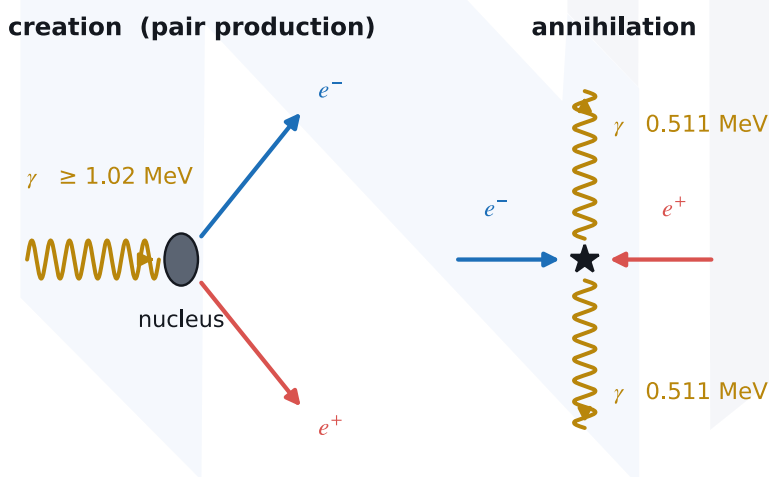


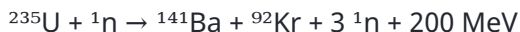
Figure 102. Creation (pair production) needs a photon of at least 1.02 MeV and a nearby nucleus; annihilation turns an electron–positron pair into two 0.511 MeV photons moving oppositely.

24.7 Nuclear fission and fusion; energy released

NUCLEAR FISSION

Nuclear fission is the splitting of a heavy nucleus into two nuclei of comparable size, together with two or three neutrons and a large release of energy, when it absorbs a slow (thermal) neutron.

A typical fission of uranium-235 is



The 200 MeV is not a fact to be memorised — it can be read straight off the binding-energy-per-nucleon curve.

WHERE THE 200 MEV OF A FISSION COMES FROM

Step 1 — count the nucleons. The ^{235}U nucleus absorbs one neutron, so the nucleus that actually splits has $A = 235 + 1 = 236$ nucleons. Nucleons are neither created nor destroyed, so the products between them also hold 236 nucleons.

Step 2 — total binding energy before the split. From the curve, uranium sits at about 7.6 MeV per nucleon. Total binding energy is (binding energy per nucleon) $\times$ (number of nucleons):

$$B_{\text{before}} = 236 \times 7.6 = 1794 \text{ MeV}$$

Step 3 — total binding energy after the split. The fragments have mass numbers near 141 and 92, which lie close to the top of the curve at about 8.5 MeV per nucleon. Counting the same 236 nucleons,

$$B_{\text{after}} = 236 \times 8.5 = 2006 \text{ MeV}$$

Step 4 — subtract. The products are held together more tightly than the uranium was. Binding energy that the system did not have before must have been supplied from somewhere — it comes out of the rest mass, and appears as the energy released:

$$Q = B_{\text{after}} - B_{\text{before}} = 2006 - 1794 = 212 \text{ MeV}$$

which is the familiar "about 200 MeV per fission". The same sum is quicker per nucleon: each of the 236 nucleons gains $8.5 - 7.6 = 0.9$ MeV, and $236 \times 0.9 \approx 212$ MeV.

About 80 % of that energy appears immediately as kinetic energy of the two fragments, which is why the fuel gets hot; the rest is carried by the neutrons, by prompt gamma rays, and by the radiation from the fragments as they later decay.

Chain reaction and critical mass

Each fission releases two or three neutrons, and each of those can split another nucleus, which releases more neutrons still.

CHAIN REACTION, MULTIPLICATION FACTOR AND CRITICAL MASS

A **chain reaction** is a self-sustaining series of fissions in which neutrons produced by one fission go on to cause further fissions. Its state is measured by the **neutron multiplication factor** k :

$$k = \frac{\text{number of fissions in one generation}}{\text{number of fissions in the generation before}}$$

- $k < 1$ — **subcritical**: each generation is smaller and the reaction dies out.
- $k = 1$ — **critical**: the rate is steady. This is how a power reactor is run.
- $k > 1$ — **supercritical**: the rate grows generation after generation. This is an atomic bomb.

The **critical mass** is the smallest mass of fissile material for which k can reach 1. Below it too many neutrons leak out through the surface before they meet a nucleus. The reason is geometric: volume grows as r^3 but surface area only as r^2 , so a bigger lump loses a smaller fraction of its neutrons. For pure ^{235}U as a bare sphere the critical mass is about 52 kg, and much less if a neutron reflector surrounds it.

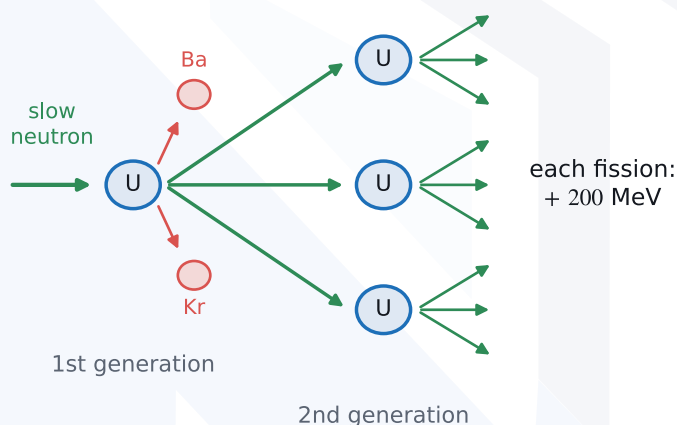


Figure 103. A chain reaction. Each fission of a ^{235}U nucleus releases about 200 MeV and two or three neutrons, which can trigger further fissions.

Parts of a nuclear reactor

A reactor is simply a chain reaction held permanently at $k = 1$ with its heat taken away and used.

Part	Material	Function
Fuel rods	uranium enriched to 2–3 % ^{235}U	supply the nuclei that fission
Moderator	graphite, heavy water (D_2O) or ordinary water	slows the fast neutrons (about 2 MeV) down to thermal speeds (about 0.025 eV), which ^{235}U captures far more readily
Control rods	boron or cadmium	absorb surplus neutrons; pushed in to slow the chain, drawn out to speed it up, holding k at exactly 1
Coolant	water, CO_2 or liquid sodium	carries heat from the core to the heat exchanger
Heat ex-changer	—	boils water to steam, which drives a turbine and generator
Shielding	thick concrete and lead	absorbs escaping neutrons and gamma rays and protects the operators

THE MODERATOR AND THE CONTROL RODS DO OPPOSITE JOBS

Both are "extra" materials in the core, so students mix them up. The **moderator slows neutrons down** so that more fissions happen; the **control rods absorb neutrons** so that fewer happen. Removing the moderator stops a thermal reactor just as surely as pushing the control rods all the way in.

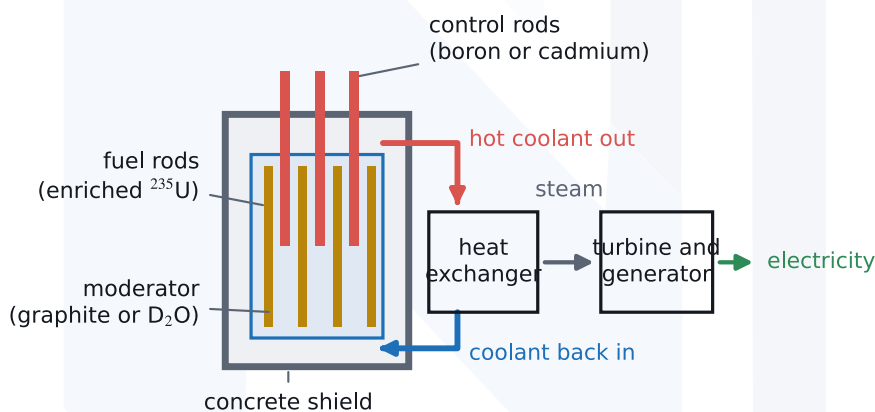


Figure 104. Schematic of a thermal nuclear reactor. Control rods are raised or lowered to hold the multiplication factor at $k = 1$; the coolant carries the heat of the fission fragments away to the heat exchanger.

In a bomb the chain is deliberately left uncontrolled: two subcritical pieces of almost pure ^{235}U or ^{239}Pu are slammed together to exceed the critical mass, and k jumps far above 1.

NUCLEAR FUSION

Nuclear fusion is the joining of two very light nuclei into a single heavier nucleus, with a release of energy. It requires extremely high temperature ($\approx 10^8 \text{ K}$) so that the nuclei approach fast enough to overcome their electrostatic repulsion; hence it is called a **thermonuclear** reaction.

The reaction used in fusion research is deuterium–tritium:



In the Sun, four hydrogen nuclei are effectively converted into one helium nucleus through the proton-proton cycle, releasing 26.7 MeV and consuming about 6×10^{11} kg of hydrogen every second.

	Fission	Fusion
Process	heavy nucleus splits	light nuclei join
Trigger	absorption of a slow neutron	very high temperature and pressure
Energy per event	$\approx 200 \text{ MeV}$	$\approx 17.6 \text{ MeV (D-T)}$
Energy per nucleon	$\approx 0.85 \text{ MeV}$	$\approx 3.5 \text{ MeV}$
Products	radioactive fragments	mostly non-radioactive
Controlled use	nuclear power reactors	not yet commercially achieved

WORKED EXAMPLE 24.5

Problem. (a) Calculate the energy released in the fusion reaction ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He} + {}^1\text{n}$, given the masses ${}^2\text{H} = 2.014102 \text{ u}$, ${}^3\text{H} = 3.016049 \text{ u}$, ${}^4\text{He} = 4.002603 \text{ u}$, ${}^1\text{n} = 1.008665 \text{ u}$. (b) If each fission of ${}^{235}\text{U}$ releases 200 MeV, how much energy is obtainable from 1.0 kg of ${}^{235}\text{U}$? ($N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$.)

Solution.

(a) Total mass before = $2.014102 + 3.016049 = 5.030151 \text{ u}$. Total mass after = $4.002603 + 1.008665 = 5.011268 \text{ u}$.

$$\Delta m = 5.030151 - 5.011268 = 0.018883 \text{ u}$$

$$Q = 0.018883 \times 931.5 = 17.6 \text{ MeV}$$

(b) Number of nuclei in 1.0 kg = 1000 g:

$$N = \frac{1000}{235} \times 6.022 \times 10^{23} = 2.56 \times 10^{24}$$

$$E = 2.56 \times 10^{24} \times 200 \text{ MeV} = 5.13 \times 10^{26} \text{ MeV}$$

$$E = 5.13 \times 10^{26} \times 1.6 \times 10^{-13} = 8.2 \times 10^{13} \text{ J}$$

That single kilogram would run a 100 MW power station for $8.2 \times 10^{13} / 10^8 = 8.2 \times 10^5 \text{ s}$, about 9.5 days.

✓ Chapter summary

- Rutherford's alpha-scattering experiment showed the atom is mostly empty with a tiny, massive, positively charged nucleus of radius about 10^{-14} m.
- Z = number of protons, A = number of nucleons, $N = A - Z$; $R = R_0 A^{1/3}$ with $R_0 = 1.2$ fm, so nuclear density is constant at about 2.3×10^{17} kg m $^{-3}$.
- $1 \text{ u} = 1.6605 \times 10^{-27}$ kg = 931.5 MeV. Isotopes have the same Z and different A .
- Mass varies with speed, $m = m_0 / \sqrt{1 - v^2/c^2}$; from this and $dE_k = v \, d(mv)$ follows $E_k = (m - m_0)c^2$ and hence $E = mc^2$. A mass defect in u times 931.5 gives the energy in MeV.
- Mass defect $\Delta m = [Zm_p + (A - Z)m_n] - M$; binding energy $B = \Delta mc^2$; packing fraction $f = (M - A)/A$.
- The B/A curve peaks at about 8.8 MeV near $A = 56$; light nuclei release energy by fusion, heavy nuclei by fission.
- Pair production needs $E \geq 1.02$ MeV and a nearby nucleus; annihilation of e^- and e^+ gives two 0.511 MeV photons.
- Fission of ^{235}U releases about 200 MeV (read off the B/A curve as $236 \times (8.5 - 7.6)$) and 2–3 neutrons, which sustain a chain reaction above the critical mass (≈ 52 kg for bare ^{235}U). A reactor holds the multiplication factor at $k = 1$ using a moderator to slow neutrons and control rods to absorb them.
- D-T fusion releases 17.6 MeV but needs about 10^8 K.

Practice questions

Group A — Multiple choice (1 mark each)

- Rutherford's alpha-scattering experiment established that [1]
 - the electron is negatively charged
 - the nucleus is small, massive and positive
 - the atom is a uniform sphere of charge
 - neutrons exist in the nucleus
- The density of a nucleus [1]
 - increases with A
 - decreases with A
 - is independent of A
 - is equal to that of water
- The energy equivalent of 1 u is about [1]
 - 0.511 MeV
 - 1.02 MeV
 - 931.5 MeV
 - 200 MeV
- The binding energy per nucleon is maximum for nuclei with mass number near [1]
 - 4
 - 56
 - 120
 - 235
- The minimum photon energy needed for pair production is [1]
 - 0.255 MeV
 - 0.511 MeV
 - 1.02 MeV
 - 1.44 MeV
- Two nuclides with the same mass number but different atomic numbers are called [1]

- a** isotopes **b** isobars **c** isotones **d** isomers

7. The moderator in a nuclear reactor is used to [1]

- a** absorb surplus neutrons **b** slow down fast neutrons
c cool the core **d** shield the operators

8. A chain reaction is just self-sustaining when the multiplication factor is [1]

- a** $k < 1$ **b** $k = 0$ **c** $k = 1$ **d** $k > 1$

ANSWERS TO GROUP A

- (b) — large-angle scattering can only come from a concentrated positive charge.
- (c) — mass $\propto A$ and volume $\propto A$, so A cancels.
- (c) — $(1.6605 \times 10^{-27})c^2 = 1.49 \times 10^{-10} \text{ J} = 931.5 \text{ MeV}$.
- (b) — the peak of the B/A curve is ^{56}Fe at 8.79 MeV.
- (c) — twice the electron rest energy, $2 \times 0.511 \text{ MeV}$.
- (b) — same A means isobars, e.g. ^{40}Ar and ^{40}Ca .
- (b) — slow (thermal) neutrons are captured by ^{235}U far more readily; the control rods, not the moderator, do the absorbing.
- (c) — $k = 1$ means each generation is the same size as the one before.

Group B — Short answer (5 marks each)

- Describe Rutherford's alpha-particle scattering experiment. State three observations and the conclusion drawn from each. [5]
- Show that the density of a nucleus is independent of its mass number, and calculate its value. Take $R_0 = 1.2 \text{ fm}$ and $m_n = 1.67 \times 10^{-27} \text{ kg}$. [5]
- Define mass defect and binding energy. Calculate the binding energy per nucleon of ^7Li , given atomic masses $^7\text{Li} = 7.016004 \text{ u}$, $^1\text{H} = 1.007825 \text{ u}$, $^1_0\text{n} = 1.008665 \text{ u}$. [5]
- An alpha particle of energy 5.0 MeV is fired head-on at a gold nucleus ($Z = 79$). Find its distance of closest approach. Take $1/4\pi\epsilon_0 = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$. [5]
- What is meant by creation and annihilation of matter? Why does pair production need a minimum energy of 1.02 MeV, and why must it occur near a nucleus? [5]
- Sketch the graph of binding energy per nucleon against mass number and use it to explain why both fission and fusion release energy. [5]
- Starting from the relativistic variation of mass with speed, derive Einstein's mass-energy relation $E = mc^2$. [5]
- With the help of a labelled diagram, describe the main parts of a nuclear reactor and state the function of each. [5]

ANSWERS TO GROUP B

2. $\rho = 3m_n/4\pi R_0^3$; numerically
 $3(1.67 \times 10^{-27})/[4\pi(1.2 \times 10^{-15})^3] = 5.01 \times 10^{-27}/7.24 \times 10^{-45} = 2.3 \times 10^{17} \text{ kg m}^{-3}$.

3. $\Delta m = 3(1.007825) + 4(1.008665) - 7.016004 = 3.023475 + 4.034660 - 7.016004 = 0.042131 \text{ u}$,
 $B = 0.042131 \times 931.5 = 39.2 \text{ MeV}$, so $B/A = 39.2/7 = 5.6 \text{ MeV per nucleon}$.

4. $E = 5.0 \text{ MeV} = 5.0 \times 1.6 \times 10^{-13} = 8.0 \times 10^{-13} \text{ J}$.

$$r_0 = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{E} = \frac{9 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{8.0 \times 10^{-13}}$$

The numerator is $9 \times 10^9 \times 158 \times 2.56 \times 10^{-38} = 3.64 \times 10^{-26}$, so $r_0 = 4.6 \times 10^{-14} \text{ m}$.

1., 5., 6. are bookwork: see §24.1, §24.6 and §24.5. For 6, marks are given for the labelled axes, the peak at $A \approx 56$ with $B/A \approx 8.8 \text{ MeV}$, and the statement that both processes move the products towards the peak, so binding energy increases and energy is released.

7. See the derivation in §24.4. One mark each for: $dE_k = v d(mv)$; expanding it to $mv dv + v^2 dm$; differentiating $m^2 c^2 - m^2 v^2 = m_0^2 c^2$ to reach $c^2 dm = mv dv + v^2 dm$; integrating $dE_k = c^2 dm$ to get $E_k = (m - m_0)c^2$; and concluding $E = mc^2$.

8. See §24.7. Marks for the labelled diagram plus fuel rods, moderator, control rods, coolant and shielding with one correct function each.

Group C — Long answer (8 marks each)

- (a) Define mass defect, binding energy and binding energy per nucleon. [3]

(b) Draw the binding-energy-per-nucleon curve and describe its main features. [2]

(c) Calculate the energy released in the reaction ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He} + {}^1_0\text{n}$, given ${}^2\text{H} = 2.014102 \text{ u}$, ${}^3\text{H} = 3.016049 \text{ u}$, ${}^4\text{He} = 4.002603 \text{ u}$ and ${}^1_0\text{n} = 1.008665 \text{ u}$. [3]
- (a) What is nuclear fission? Explain chain reaction and critical mass, and state the function of a moderator and of control rods in a reactor. [4]

(b) Each fission of ${}^{235}\text{U}$ releases 200 MeV. Calculate the energy released by the complete fission of 0.50 kg of ${}^{235}\text{U}$, and the mass that disappears in the process. [4]

ANSWERS TO GROUP C

1.(c) $\Delta m = (2.014102 + 3.016049) - (4.002603 + 1.008665) = 5.030151 - 5.011268 = 0.018883 \text{ u}$, so
 $Q = 0.018883 \times 931.5 = 17.6 \text{ MeV}$.

2.(b) $N = (500/235) \times 6.022 \times 10^{23} = 1.28 \times 10^{24}$ nuclei. $E = 1.28 \times 10^{24} \times 200 \times 1.6 \times 10^{-13} = 4.1 \times 10^{13} \text{ J}$.
 The mass lost follows from $E = \Delta mc^2$: $\Delta m = 4.1 \times 10^{13} / (9 \times 10^{16}) = 4.6 \times 10^{-4} \text{ kg}$, i.e. about 0.46 g — less than one part in a thousand of the fuel.

1.(a), 1.(b), 2.(a) are bookwork: see §24.5 and §24.7.

UNIT 25

Solids

Modern Physics • 3 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Energy bands in solids (qualitative ideas)
- Difference between metals, insulators and semi-conductors using band theory
- Intrinsic and extrinsic semi-conductors

In a single isolated atom an electron may occupy only certain sharp energy levels. Push 10^{29} atoms together into a crystal and those sharp levels smear out into continuous **bands** of allowed energy separated by **forbidden gaps**. One number — the width of the gap above the highest filled band — decides whether the solid is a copper wire, a slab of mica, or the silicon chip inside a mobile phone. This short unit builds that picture and uses it to explain doping.

WHAT THE EXAMINER WANTS

Four things, almost every year: a labelled band diagram distinguishing conductor, semiconductor and insulator; the reason a semiconductor's resistance falls as it is heated; the covalent-bond (lattice) diagrams of n-type and p-type silicon; and a clear comparison of the two. Learn the diagrams as diagrams — they are worth full marks on their own.

25.1 Energy bands in solids (qualitative ideas)

An electron in an isolated atom has discrete energies $E_1, E_2, E_3, \dots$ given by quantum theory. The **Pauli exclusion principle** allows at most two electrons (opposite spin) in any one state.

Now bring N identical atoms close together to build a crystal. When the atoms are far apart, each of the N atoms has its own identical $3s$ level, so that energy is N -fold repeated. As the separation shrinks, the outer electron clouds overlap; the electron of one atom now feels the nuclei of its neighbours. Pauli's principle forbids N electrons from sharing one identical state, so the single level **splits into N very closely spaced levels**.

How closely spaced? Take one cubic metre of solid, so $N \approx 10^{29}$, and let the whole set of split levels spread over about 1 eV. The average step from one level to the next is then

$$\Delta E \approx \frac{1 \text{ eV}}{10^{29}} = 10^{-29} \text{ eV}$$

No instrument can resolve a step that small, and no experiment can tell it apart from zero. The set of levels therefore behaves as one continuous **energy band**. Inner-shell electrons are buried deep inside the atom, their orbitals barely overlap, and their levels hardly split at all — only the outermost (valence) levels broaden into wide bands.

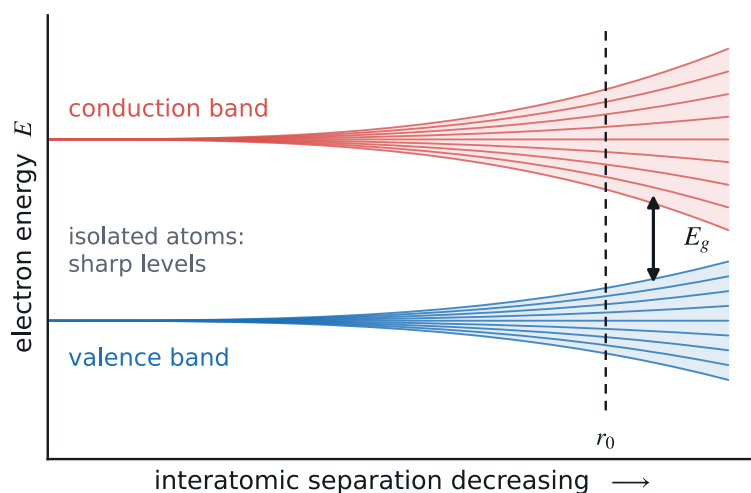


Figure 105. A single sharp atomic level splits into a band of N closely packed levels as the atoms are brought together. At the real lattice spacing r_0 the two bands are separated by the forbidden gap E_g .

Two bands matter:

VALENCE BAND AND CONDUCTION BAND

The **valence band** is the highest energy band that is completely or partly filled by the valence electrons of the atoms at 0 K . The **conduction band** is the next allowed band above it; an electron in the conduction band is no longer tied to one atom and can drift through the whole crystal when a field is applied.

The energy range between the top of the valence band and the bottom of the conduction band contains no allowed states. It is the **forbidden energy gap** E_g .

A current flows only if electrons can gain small amounts of energy from the applied field, i.e. only if there are **empty states immediately above occupied states**. A completely full band carries no net current however strong the field, because every electron that moves one way is matched by one moving the other. This single sentence explains the whole of the next section.

25.2 Difference between metals, insulators and semi-conductors using band theory

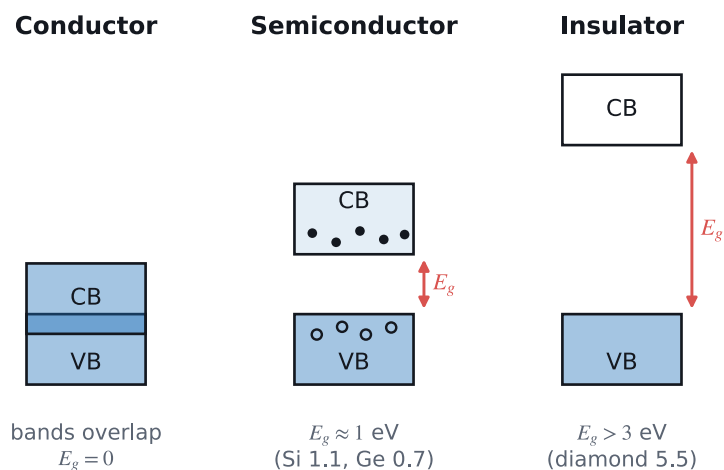


Figure 106. Band structure at room temperature, all three drawn to the same energy scale. Shading shows how full a band is; filled dots are electrons and open circles are holes. Every solid has both bands — only the gap between them changes.

Conductors (metals). The valence band is only partly filled (sodium), or a filled valence band **overlaps** the empty conduction band (magnesium, zinc). Either way there is no gap: empty states sit immediately above occupied ones, a huge number of free electrons ($\sim 10^{28} \text{ m}^{-3}$) is available, and the resistivity is tiny.

Insulators. The valence band is completely full, the conduction band completely empty, and E_g is large — 5.47 eV for diamond. At 300 K the average thermal energy of a particle is only about $kT \approx 0.026 \text{ eV}$, so practically no electron can be lifted across.

Semiconductors. The band picture is the same as an insulator's, but E_g is small: 1.1 eV for silicon, 0.7 eV for germanium. At 0 K a semiconductor is a perfect insulator. At room temperature a small fraction of the bonds break, electrons jump the gap into the conduction band and leave holes behind in the valence band, and conduction begins.

Property	Conductor	Semiconductor	Insulator
Forbidden gap E_g	zero (bands overlap)	$\leq 3 \text{ eV}$ (Ge 0.7, Si 1.1)	$> 3 \text{ eV}$ (diamond 5.47)
Resistivity at 300 K	$10^{-8} - 10^{-6} \Omega \text{ m}$	$10^{-5} - 10^6 \Omega \text{ m}$	$10^8 - 10^{16} \Omega \text{ m}$
Carrier density	$\sim 10^{28} \text{ m}^{-3}$	$10^{16} - 10^{19} \text{ m}^{-3}$	negligible
Temperature coefficient of resistance	positive	negative	negative
Behaviour at 0 K	still conducts	perfect insulator	perfect insulator
Examples	Cu, Ag, Al	Si, Ge, GaAs	diamond, mica, glass

The **negative temperature coefficient** is the giveaway property of a semiconductor. Heating a metal makes the lattice ions vibrate more, electrons are scattered more often, and resistance rises. Heating a semiconductor does that too, but it also breaks far more bonds. In a pure semiconductor the Fermi level (the energy at which a state is half likely to be occupied) sits in the middle of the gap, so on the Boltzmann scale an electron only has to climb $E_g/2$ to reach the conduction band. The number of carri-

ers therefore grows like $e^{-E_g/2kT}$, and this exponential growth easily overwhelms the extra scattering, so the resistivity falls.

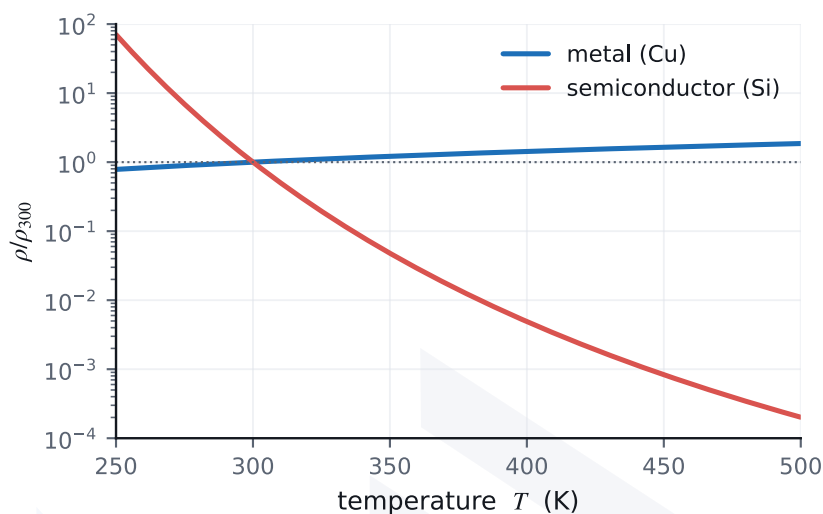


Figure 107. Resistivity against temperature, plotted relative to its value at 300 K. The metal rises gently and linearly; the semiconductor ($E_g = 1.1$ eV) falls by orders of magnitude.

WORKED EXAMPLE 25.1

Problem. The forbidden gap of silicon is 1.1 eV. Find the longest wavelength of light that can lift an electron from its valence band into its conduction band. Take $h = 6.63 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m s $^{-1}$, $e = 1.6 \times 10^{-19}$ C.

Solution. One photon carries energy hc/λ . A long wavelength means a small energy, so the longest usable wavelength is the one whose photon energy is exactly E_g :

$$\frac{hc}{\lambda_{\max}} = E_g$$

Make $\lambda_{\max}$ the subject by cross-multiplying:

$$\lambda_{\max} = \frac{hc}{E_g}$$

First change E_g from electron-volts to joules by multiplying by the electronic charge:

$$E_g = 1.1 \times 1.6 \times 10^{-19} = 1.76 \times 10^{-19} \text{ J}$$

Next work out the numerator hc :

$$hc = 6.63 \times 10^{-34} \times 3 \times 10^8 = 1.989 \times 10^{-25} \text{ J m}$$

Now divide:

$$\lambda_{\max} = \frac{1.989 \times 10^{-25}}{1.76 \times 10^{-19}} = 1.13 \times 10^{-6} \text{ m}$$

That is 1130 nm, in the infrared. Visible light (400–700 nm) has a shorter wavelength and therefore more energy than this, which is why a silicon solar cell works in sunlight.

25.3 Intrinsic and extrinsic semi-conductors

Intrinsic semiconductors

A **pure** semiconductor is called intrinsic. Silicon and germanium are tetravalent: each atom shares its four valence electrons in covalent bonds with four neighbours, so at 0 K every bond is complete and the crystal does not conduct.

At room temperature thermal energy breaks a few bonds. Each broken bond frees one electron into the conduction band and leaves behind a vacancy in the valence band — a **hole**. A neighbouring bound electron can hop into the vacancy, moving the vacancy the other way, so the hole behaves like a mobile particle of charge $+e$.

Because carriers are created only in pairs,

$$n_e = n_h = n_i$$

where n_i is the **intrinsic carrier concentration**: $1.5 \times 10^{16} \text{ m}^{-3}$ for Si and $2.4 \times 10^{19} \text{ m}^{-3}$ for Ge at 300 K. Compare this with roughly 5×10^{28} silicon atoms per cubic metre: only about one atom in 10^{12} contributes a carrier.

CONDUCTIVITY OF A SEMICONDUCTOR

Step 1 — put the sample in a field. Take a bar of cross-section A and apply an electric field E along it. By the definition of **mobility**, the average (drift) speed a carrier picks up is mobility $\times$ field:

$$v_e = \mu_e E \quad \text{and} \quad v_h = \mu_h E$$

Step 2 — count the electrons that cross a line in time t . Every electron within a distance $v_e t$ of a chosen cross-section will reach it in time t . That slab has volume $A v_e t$, so the number of electrons in it is

$$N_e = n_e \times (A v_e t)$$

Step 3 — turn that into charge, then into current. Each electron carries charge e , so the charge crossing is $q = e n_e A v_e t$. Current is charge per unit time, so divide by t :

$$I_e = \frac{e n_e A v_e t}{t} = e n_e A v_e$$

Step 4 — divide by the area to get current density.

$$J_e = \frac{I_e}{A} = e n_e v_e$$

Step 5 — put the drift speed of Step 1 back in.

$$J_e = e n_e (\mu_e E) = e n_e \mu_e E$$

Step 6 — do the same for holes. Holes drift the opposite way but carry the opposite (positive) charge, so their current points the same way as the electron current and the two simply add:

$$J_h = e n_h \mu_h E, \quad J = J_e + J_h$$

Step 7 — add and take out the common factors e and E .

$$J = e n_e \mu_e E + e n_h \mu_h E = e(n_e \mu_e + n_h \mu_h)E$$

Step 8 — compare with Ohm's law. In point form Ohm's law reads $J = \sigma E$. Comparing the two expressions for J , the bracket must be the conductivity:

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

Step 9 — specialise to a pure sample. Putting $n_e = n_h = n_i$ and taking n_i out as a common factor,

$$\sigma_i = e n_i \mu_e + e n_i \mu_h = e n_i (\mu_e + \mu_h), \quad \rho_i = \frac{1}{\sigma_i}$$

Here μ_e and μ_h are the electron and hole **mobilities** (drift velocity per unit field, in $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$). Electrons move through the lattice more easily than holes, so $\mu_e > \mu_h$ always.

WORKED EXAMPLE 25.2

Problem. For intrinsic germanium at 300 K, $n_i = 2.4 \times 10^{19} \text{ m}^{-3}$, $\mu_e = 0.39 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$ and $\mu_h = 0.19 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$. Find its conductivity and resistivity.

Solution. The sample is pure, so $n_e = n_h = n_i$ and Step 9 above applies:

$$\sigma = e n_i (\mu_e + \mu_h)$$

Add the two mobilities first:

$$\mu_e + \mu_h = 0.39 + 0.19 = 0.58 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$$

Now multiply e by n_i . The powers of ten cancel ($10^{-19} \times 10^{19} = 10^0 = 1$):

$$e n_i = 1.6 \times 10^{-19} \times 2.4 \times 10^{19} = 3.84 \text{ C m}^{-3}$$

Multiply the two results:

$$\sigma = 3.84 \times 0.58 = 2.23 \text{ } \Omega^{-1} \text{m}^{-1}$$

Finally invert to get resistivity:

$$\rho = \frac{1}{\sigma} = \frac{1}{2.23} = 0.45 \text{ } \Omega \text{ m}$$

This lies neatly between copper ($1.7 \times 10^{-8} \text{ } \Omega \text{ m}$) and glass.

Extrinsic semiconductors

Intrinsic material is far too resistive to be useful. Adding a controlled trace of impurity — typically one atom in 10^6 to 10^8 — is called **doping**, and the result is an **extrinsic** semiconductor.

n-type. Dope silicon with a **pentavalent** atom (P, As, Sb, Bi). The impurity takes the place of a silicon atom in the lattice. Four of its five valence electrons pair up with electrons of the four silicon neighbours and make four ordinary covalent bonds. The fifth electron has no partner to bond with. It is held only weakly by its parent atom, and about 0.045 eV in silicon (about 0.01 eV in germanium) is enough to set it free — far less than the 1.1 eV needed to break a bond. Since $kT \approx 0.026$ eV at 300 K, practically every impurity atom has already given up its fifth electron at room temperature. The impurity **donates** an electron, so it is called a **donor**, and electrons become the **majority carriers**.

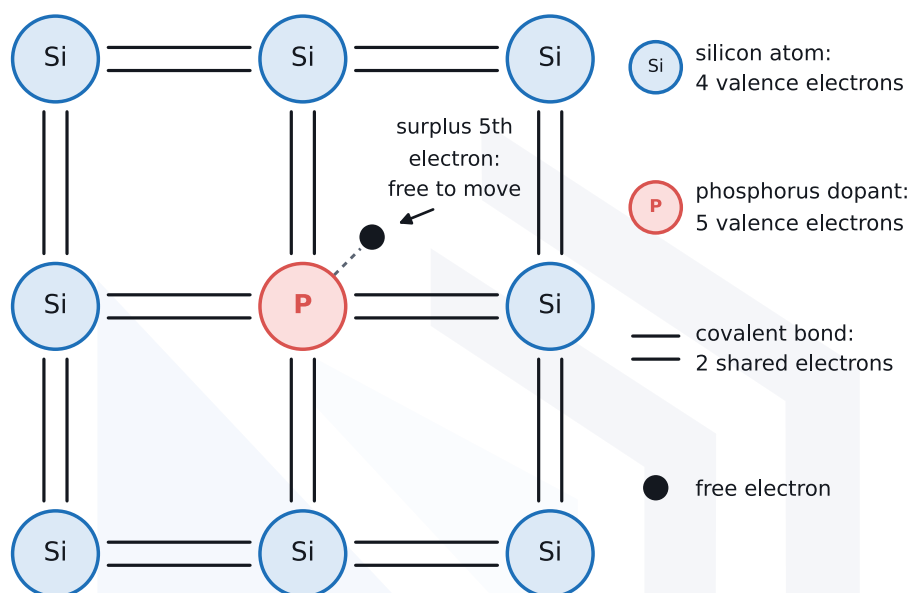


Figure 108. n-type silicon. One silicon atom has been replaced by a pentavalent phosphorus atom. Four of the five valence electrons of P complete the four covalent bonds with its silicon neighbours; the fifth has no bond to join and is free to wander through the crystal.

p-type. Dope silicon with a **trivalent** atom (B, Al, Ga, In). This atom brings only three valence electrons, so it can complete only three of the four bonds with its silicon neighbours. The fourth bond is left with a single electron instead of a pair. That empty half-bond is a **hole**. A bound electron from a neighbouring bond needs only about 0.05 eV to jump across and fill it, which moves the hole to where that electron came from. Because the impurity **accepts** an electron it is called an **acceptor**, and holes become the majority carriers.

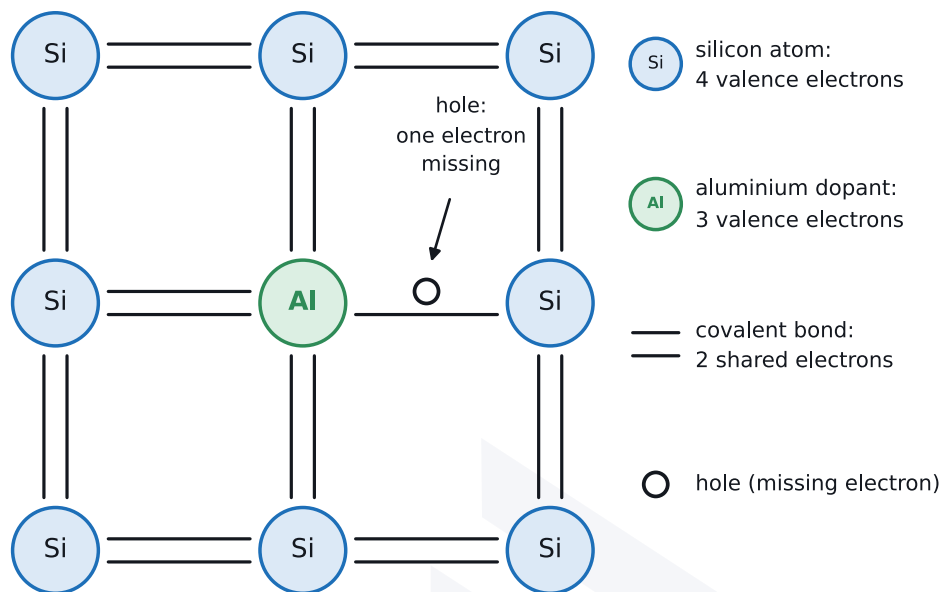


Figure 109. p-type silicon. One silicon atom has been replaced by a trivalent aluminium atom. Al can complete only three covalent bonds; the fourth bond is one electron short, and that vacancy is a hole which moves like a positive charge.

In band language the two dopants create new allowed levels inside the forbidden gap. A donor level sits only 0.045 eV below the bottom of the conduction band, so its electron needs almost no help to get in. An acceptor level sits only about 0.05 eV above the top of the valence band, so a valence electron can hop up into it and leave a hole behind.

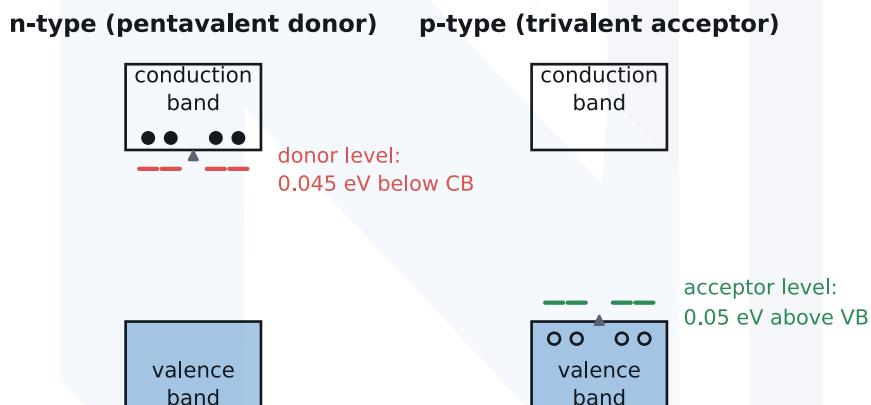


Figure 110. Donor levels sit just below the conduction band in n-type material; acceptor levels just above the valence band in p-type material. Filled circles are electrons, open circles are holes. The grey arrow shows the tiny jump an electron makes at room temperature.

LAW OF MASS ACTION, $n_e n_h = n_i^2$

Step 1 — what makes carriers. Heat breaks bonds and creates electron-hole pairs. How fast pairs appear (the generation rate g) depends only on the temperature and the gap E_g — it does not care how much impurity we added.

Step 2 — what destroys carriers. A free electron disappears only when it meets a hole and drops into it. The chance of a meeting is proportional to how many electrons there are and to how many holes there are, so the recombination rate is

$$r = C n_e n_h$$

where C is a constant fixed by the material and the temperature.

Step 3 — steady state. At a fixed temperature the carrier numbers are not changing, so pairs must be destroyed exactly as fast as they are made:

$$g = C n_e n_h$$

Step 4 — make the product the subject. Divide both sides by C :

$$n_e n_h = \frac{g}{C}$$

The right-hand side contains only g and C , and both depend on temperature alone. So the product $n_e n_h$ is the same number for every sample of that material at that temperature, however it is doped.

Step 5 — evaluate that number using the pure sample. In pure material at the same temperature, $n_e = n_h = n_i$, so

$$n_e n_h = n_i \times n_i = n_i^2$$

Step 6 — combine. Since the product is the same for all doping levels and equals n_i^2 for the pure sample, it must equal n_i^2 always:

$$n_e n_h = n_i^2$$

So doping to raise one carrier density automatically suppresses the other: push n_e up by a factor of a million and n_h falls by the same factor.

Feature	n-type	p-type
Dopant valency	pentavalent (P, As, Sb)	trivalent (B, Al, Ga, In)
Bonds formed by dopant	4 complete bonds + 1 spare electron	3 complete bonds + 1 hole
Impurity level created	donor, just below CB	acceptor, just above VB
Majority carriers	electrons	holes
Minority carriers	holes	electrons
Carrier relation	$n_e \gg n_h, n_e \approx N_d$	$n_h \gg n_e, n_h \approx N_a$
Net charge on crystal	zero	zero

A DOPED CRYSTAL IS STILL ELECTRICALLY NEUTRAL

"n-type is negatively charged" is wrong, and it costs marks every year. A donor atom releases one electron but is left as a fixed positive ion, so the totals still balance. The labels n and p name the **majority carrier**, not a charge on the block.

A HOLE IS NOT A POSITRON

A hole is the absence of an electron in a bond — a bookkeeping device for the motion of the many remaining valence electrons. It exists only inside the crystal, whereas a positron is a real particle that can travel through vacuum.

WORKED EXAMPLE 25.3

Problem. Pure silicon has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$ at 300 K. It is doped with 10^{22} donor atoms per m^3 , all ionised. Find the electron and hole concentrations, and say which are the minority carriers.

Solution. Every ionised donor hands one electron to the conduction band, and $N_d = 10^{22}$ is far bigger than $n_i = 1.5 \times 10^{16}$, so the thermally generated electrons are negligible in comparison:

$$n_e \approx N_d = 1 \times 10^{22} \text{ m}^{-3}$$

Now use the law of mass action, $n_e n_h = n_i^2$. Make n_h the subject by dividing both sides by n_e :

$$n_h = \frac{n_i^2}{n_e}$$

Square n_i first — square the number and double the power of ten:

$$n_i^2 = (1.5 \times 10^{16})^2 = 2.25 \times 10^{32} \text{ m}^{-6}$$

Then divide, subtracting the powers of ten ($32 - 22 = 10$):

$$n_h = \frac{2.25 \times 10^{32}}{1 \times 10^{22}} = 2.25 \times 10^{10} \text{ m}^{-3}$$

Holes are the minority carriers. Note that one impurity atom per 5×10^6 silicon atoms has raised the electron density by a factor of about 10^6 — that is the whole point of doping.

✓ Chapter summary

- Bringing N atoms together splits each sharp atomic level into N levels so close ($\sim 10^{-29}$ eV apart) that they form a continuous **energy band**; between bands lie **forbidden gaps**.
- The **valence band** is the highest band filled at 0 K; the **conduction band** is the empty band above it; their separation is E_g .
- A completely filled band carries no current. Conduction needs empty states just above occupied ones.
- Conductor: bands overlap, $E_g = 0$. Semiconductor: $E_g \leq 3$ eV (Ge 0.7 eV, Si 1.1 eV). Insulator: $E_g > 3$ eV (diamond 5.47 eV).
- Semiconductors have a **negative** temperature coefficient of resistance because carrier number grows like $e^{-E_g/2kT}$.
- Intrinsic: $n_e = n_h = n_i$, and $\sigma = e(n_e\mu_e + n_h\mu_h)$ reduces to $\sigma_i = e n_i(\mu_e + \mu_h)$, with $\rho = 1/\sigma$.
- Extrinsic: a pentavalent dopant makes four bonds and leaves one spare electron (n-type, donor level below CB, electrons majority); a trivalent dopant makes three bonds and leaves one hole (p-type, acceptor level above VB, holes majority). Both crystals remain electrically neutral.
- Law of mass action: $n_e n_h = n_i^2$ at a given temperature.

Practice questions

Group A — Multiple choice (1 mark each)

- In an intrinsic semiconductor at room temperature [1]
 - $n_e > n_h$
 - $n_e < n_h$
 - $n_e = n_h$
 - $n_h = 0$
- The forbidden energy gap of germanium is about [1]
 - 0.7 eV
 - 1.1 eV
 - 3.0 eV
 - 5.5 eV
- On heating a semiconductor its resistance [1]
 - increases
 - decreases
 - stays constant
 - first rises then falls
- Silicon doped with indium becomes [1]
 - n-type
 - p-type
 - intrinsic
 - an insulator
- A pure semiconductor at 0 K behaves as [1]
 - a conductor
 - a superconductor
 - a perfect insulator
 - an n-type semiconductor
- In n-type silicon the phosphorus atom forms [1]
 - 5 covalent bonds
 - 4 covalent bonds and has 1 spare electron
 - 3 covalent bonds and 1 hole
 - no bonds

ANSWERS TO GROUP A

1. (c) — carriers are produced only in electron-hole pairs.
2. (a) — Ge 0.7 eV, Si 1.1 eV.
3. (b) — the exponential rise in carrier number beats the extra scattering.
4. (b) — indium is trivalent, so it supplies acceptor levels.
5. (c) — every covalent bond is intact, so both bands are full or empty.
6. (b) — silicon offers only four neighbours, so the fifth electron of P has no bond to join.

Group B — Short answer (5 marks each)

1. Explain qualitatively how the discrete energy levels of isolated atoms become energy bands in a solid. [5]
2. Using band diagrams, distinguish between a conductor, a semiconductor and an insulator. [5]
3. Why does the resistance of a metal increase but that of a semiconductor decrease with rise in temperature? [5]
4. The energy gap of a certain semiconductor is 0.72 eV. Calculate the maximum wavelength of radiation it can absorb. ($h = 6.63 \times 10^{-34}$ J s) [5]
5. Intrinsic silicon has $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. After doping, the hole concentration becomes $4.5 \times 10^{21} \text{ m}^{-3}$. Find the electron concentration and name the type of the semiconductor. [5]
6. Draw the covalent-bond diagram of silicon doped with a pentavalent atom and use it to explain why the material is called n-type although the crystal as a whole is neutral. [5]

ANSWERS TO GROUP B

1. Outline: sharp levels in an isolated atom $\rightarrow$ Pauli exclusion forbids N identical states $\rightarrow$ overlapping of outer orbitals on bringing N atoms together splits each level into N sub-levels $\rightarrow$ with $N \sim 10^{29} \text{ m}^{-3}$ the spacing is $\sim 10^{-29} \text{ eV}$, so the set is effectively a continuous band, separated from the next by a forbidden gap.

3. Outline: in a metal the carrier number is fixed, so more violent lattice vibrations scatter electrons more and ρ rises. In a semiconductor the carrier number grows like $e^{-E_g/2kT}$; this exponential increase dominates the extra scattering, so ρ falls (negative temperature coefficient).

4. First convert the gap to joules:

$$E_g = 0.72 \times 1.6 \times 10^{-19} = 1.152 \times 10^{-19} \text{ J}$$

Then $\lambda_{\max} = hc/E_g$ with $hc = 1.989 \times 10^{-25} \text{ J m}$:

$$\lambda_{\max} = \frac{1.989 \times 10^{-25}}{1.152 \times 10^{-19}} = 1.73 \times 10^{-6} \text{ m} = 1730 \text{ nm}$$

5. By the law of mass action, $n_e = n_i^2/n_h$. Square n_i first:

$$n_i^2 = (1.5 \times 10^{16})^2 = 2.25 \times 10^{32} \text{ m}^{-6}$$

Then divide:

$$n_e = \frac{2.25 \times 10^{32}}{4.5 \times 10^{21}} = 5 \times 10^{10} \text{ m}^{-3}$$

Holes greatly outnumber electrons, so the material is **p-type**.

6. Outline: draw the lattice of the n-type figure — P on a silicon site, four double-line covalent bonds to four Si neighbours, one extra electron drawn outside any bond. That spare electron is easily freed (0.045 eV), so electrons are the majority carriers and the material is called n-type. But the phosphorus atom that lost the electron is left behind as a fixed positive ion inside the lattice, so the positive and negative charges still balance exactly and the block is neutral.

Group C — Long answer (8 marks each)

1. **(a)** Define valence band, conduction band and forbidden energy gap. [3]
(b) Draw band diagrams for a conductor, a semiconductor and an insulator and use them to explain the enormous difference in their resistivities. [5]
2. **(a)** Distinguish between intrinsic and extrinsic semiconductors. [2]
(b) Draw the covalent-bond lattice of n-type and p-type silicon, showing the free electron and the hole, and name the majority carriers in each. [4]
(c) A germanium sample has $n_i = 2.4 \times 10^{19} \text{ m}^{-3}$, $\mu_e = 0.39$ and $\mu_h = 0.19 \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$. Find the resistivity of the pure sample. [2]

ANSWER TO GROUP C QUESTION 2(C)

Add the mobilities: $\mu_e + \mu_h = 0.39 + 0.19 = 0.58$. Then

$$\sigma = e n_i (\mu_e + \mu_h) = 1.6 \times 10^{-19} \times 2.4 \times 10^{19} \times 0.58$$

$$\sigma = 3.84 \times 0.58 = 2.23 \text{ } \Omega^{-1} \text{m}^{-1}$$

$$\rho = \frac{1}{\sigma} = \frac{1}{2.23} = 0.45 \text{ } \Omega \text{ m}$$

UNIT 26

Recent Trends in physics

Modern Physics • 6 teaching hours

CDC SYLLABUS — THIS UNIT MUST COVER

- Particle physics: Particles and antiparticles, Quarks (baryons and mesons), leptons (neutrinos)
- Universe: Big Bang and Hubble law, expansion of the Universe, Dark matter, Black Hole, gravitational wave

The last hundred years have pushed physics to both ends of the scale. Downwards, the atom has been opened up to reveal quarks and leptons — particles with no known internal structure. Upwards, the same equations describe a universe 13.8 billion years old, still expanding, and mostly made of something nobody has yet held in a laboratory. This unit tours both frontiers. Much of it is descriptive, but the numbers matter too: Hubble's law, the size of a black hole and the charge of a quark are all things you can be asked to calculate.

HOW THIS UNIT IS EXAMINED

Most marks come from **definitions, classifications and evidence**: the quark content of a proton, why neutrinos are hard to detect, three pieces of evidence for the Big Bang. Learn the tables and the dates. But be ready for short numerical parts as well — adding up quark charges, using $v = H_0 d$, and putting a mass into $R_s = 2GM/c^2$ are all fair game.

26.1 Particle physics

Particles and antiparticles

In 1928 **P. A. M. Dirac** combined quantum mechanics with special relativity. His equation had negative-energy solutions which he interpreted as a new particle: an electron with positive charge. In 1932 **Carl Anderson** found it in cosmic-ray tracks in a cloud chamber and named it the **positron**.

ANTIPARTICLE

For every particle there exists an antiparticle having the **same mass, the same spin and the same lifetime**, but **opposite** electric charge, magnetic moment and other internal quantum numbers (lepton number, baryon number, strangeness).

Antiparticles are written with a bar or the opposite charge sign: antiproton $\bar{p}$, antineutrino $\bar{\nu}$, antielectron e^+ . The antiproton was made in 1955 at the Bevatron. A few particles — the photon, the neutral pion — are their own antiparticles.

Two processes connect matter and energy directly:

- **Pair production.** A photon passing close to a heavy nucleus converts into a particle-antiparticle pair: $\gamma \rightarrow e^- + e^+$. It needs at least $2m_e c^2 = 1.02 \text{ MeV}$; the nucleus is needed to conserve momentum.
- **Annihilation.** A particle meeting its antiparticle disappears, the whole rest energy becoming radiation: $e^- + e^+ \rightarrow 2\gamma$, with each photon carrying 0.511 MeV — the principle behind the hospital PET scan.

WORKED EXAMPLE 26.1

Problem. An electron and a positron, both at rest, annihilate to give two identical photons. Find the energy, frequency and wavelength of each photon. Take $m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.63 \times 10^{-34} \text{ J s}$.

Solution. The total rest energy shared between two photons is $2m_e c^2$, so each photon gets $m_e c^2$:

$$E = m_e c^2 = 9.1 \times 10^{-31} \times (3 \times 10^8)^2 = 8.19 \times 10^{-14} \text{ J}$$

$$f = \frac{E}{h} = \frac{8.19 \times 10^{-14}}{6.63 \times 10^{-34}} = 1.24 \times 10^{20} \text{ Hz}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1.24 \times 10^{20}} = 2.4 \times 10^{-12} \text{ m}$$

Both are gamma rays, emitted back to back so that momentum stays zero.

Classification of particles

Every particle is either a **fermion** (matter, half-integer spin, obeys the Pauli exclusion principle) or a **boson** (force carrier, integer spin). Fermions divide into **quarks**, which feel the strong force, and **leptons**, which do not.

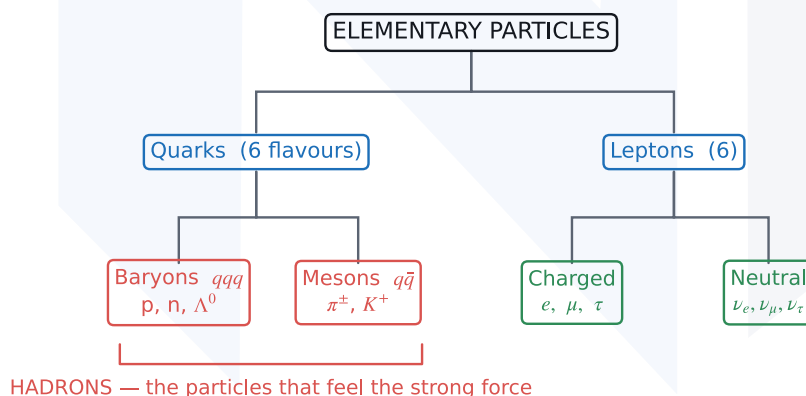


Figure 111. Classification of the fundamental particles. Hadrons are not elementary: they are bound states of quarks.

Force	Relative strength	Exchange particle	Range
Strong	1	gluon	$\sim 10^{-15}$ m
Electromagnetic	10^{-2}	photon	infinite
Weak	10^{-6}	$W^{\pm}, Z^0$	$\sim 10^{-18}$ m
Gravitational	10^{-39}	graviton (not yet found)	infinite

Quarks: baryons and mesons

In 1964 **Murray Gell-Mann** and **George Zweig** independently proposed that the hundreds of known hadrons are not elementary but are built from a few constituents, which Gell-Mann named **quarks**. Six flavours are known; the last, the top quark, was found at Fermilab in 1995.

Generation	Flavour	Symbol	Charge
First	up	u	$+\frac{2}{3}e$
First	down	d	$-\frac{1}{3}e$
Second	charm	c	$+\frac{2}{3}e$
Second	strange	s	$-\frac{1}{3}e$
Third	top	t	$+\frac{2}{3}e$
Third	bottom	b	$-\frac{1}{3}e$

Quarks alone carry fractional charge. Each has baryon number $+\frac{1}{3}$ and a property called **colour** — the "charge" of the strong force. Hadrons come in exactly two combinations:

- **Baryons** — three quarks, qqq , baryon number $+1$. Proton $= uud$, giving charge $\frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$. Neutron $= udd$, giving $\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$. The $\Lambda^0 = uds$.
- **Mesons** — a quark and an antiquark, $q\bar{q}$, baryon number 0 . The $\pi^+ = u\bar{d}$ and the $K^+ = u\bar{s}$. All mesons are unstable.

WORKED EXAMPLE 26.2

Problem. Using the quark charges in the table, work out the electric charge and the baryon number of (a) the combination uds and (b) the combination $u\bar{s}$. In each case say whether it is a baryon or a meson and name it.

Solution. Every quark has baryon number $+\frac{1}{3}$; every antiquark has $-\frac{1}{3}$. An antiquark also carries the **opposite** charge to its quark.

(a) uds is three quarks, so it is a **baryon**. Adding the charges from the table,

$$Q = +\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = \frac{2-1-1}{3} = 0$$

$$B = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = +1$$

A neutral baryon: this is the Λ^0 .

(b) $u\bar{s}$ is one quark with one antiquark, so it is a **meson**. The s quark has charge $-\frac{1}{3}$, so $\bar{s}$ has $+\frac{1}{3}$:

$$Q = +\frac{2}{3} + \frac{1}{3} = \frac{2+1}{3} = +1$$

$$B = +\frac{1}{3} - \frac{1}{3} = 0$$

A meson of charge $+1$: this is the K^+ .

QUARK CONFINEMENT

No free quark has ever been seen. The strong force between two quarks does not weaken with distance, so pulling them apart costs ever more energy until it is cheaper to create a new quark-antiquark pair. You always end up with more hadrons, never a loose quark.

Leptons and neutrinos

Leptons are elementary fermions that do not feel the strong force. There are six, in three generations, each charged lepton paired with its own neutrino.

Generation	Charged lepton	Rest energy	Neutrino
First	electron e^-	0.511 MeV	ν_e
Second	muon μ^-	105.7 MeV	ν_μ
Third	tau τ^-	1777 MeV	ν_τ

The **neutrino** was proposed by **Wolfgang Pauli in 1930** to rescue the conservation of energy and momentum in beta decay: the emitted electrons had a continuous range of energies instead of a single value, so an unseen third particle must be carrying the balance away. Its beta-decay role is

$$n \rightarrow p + e^- + \bar{\nu}_e$$

Neutrinos are electrically neutral, have almost no mass and interact **only** through the weak force and gravity. About 6.5×10^{14} solar neutrinos cross each square metre of your body every second, and almost none interact; detecting them needs thousands of tonnes of material deep underground. **Cowan and Reines** finally did so in 1956.

Neutrinos were long assumed massless. Then **Super-Kamiokande** (1998) and **SNO** (2001–02) showed that neutrinos change flavour in flight — **neutrino oscillation** — which is possible only if they have a non-zero mass (Nobel Prize 2015). The tightest direct laboratory limit, from the KATRIN experiment in 2025, is $m_\nu < 0.45 \text{ eV}/c^2$ — less than a millionth of the electron's mass.

The six quarks, six leptons, the force-carrying bosons and the **Higgs boson** (found at CERN's Large Hadron Collider in July 2012, mass about $125 \text{ GeV}/c^2$) form the **Standard Model** of particle physics.

26.2 Universe

The Big Bang and Hubble's law

Observing distant galaxies in 1929, **Edwin Hubble** found that almost all of them are moving away from us, and that the recession speed v is proportional to the distance d :

$$v = H_0 d$$

H_0 is the **Hubble constant**. The speed comes from the **redshift** of the galaxy's spectral lines; for $v \ll c$,

$$z = \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$

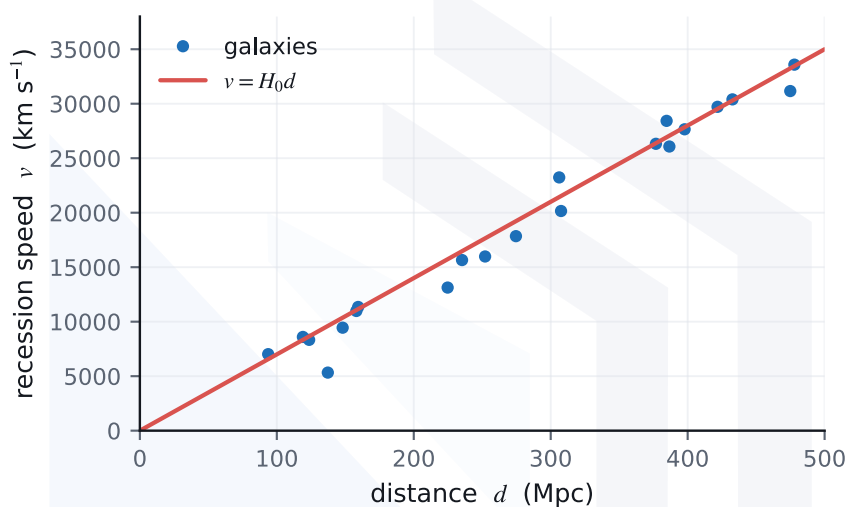


Figure 112. Hubble diagram. Each point is a galaxy; the slope of the line is $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (1 Mpc = $3.086 \times 10^{22} \text{ m}$).

Run the expansion backwards and all the matter was once packed into an extremely hot, dense state. The **Big Bang** model says the universe began in such a state about **13.8 billion years ago** and has expanded and cooled ever since. Three independent observations support it:

Evidence	What is observed
Hubble expansion	every distant galaxy recedes, with $v \propto d$
Cosmic microwave background	a 2.725 K blackbody glow from every direction, found by Penzias and Wilson in 1965 — the cooled radiation of the hot early universe
Light-element abundance	about 75% hydrogen and 25% helium by mass everywhere, exactly the amount predicted to form in the first few minutes

WORKED EXAMPLE 26.3

Problem. Taking $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $1 \text{ Mpc} = 3.086 \times 10^{22} \text{ m}$, find (a) the recession speed of a galaxy 150 Mpc away and (b) the Hubble time $1/H_0$ in years.

Solution.

(a) $v = H_0 d = 70 \times 150 = 1.05 \times 10^4 \text{ km s}^{-1}$, i.e. 3.5% of the speed of light.

(b) In SI units,

$$H_0 = \frac{70 \times 10^3 \text{ m s}^{-1}}{3.086 \times 10^{22} \text{ m}} = 2.27 \times 10^{-18} \text{ s}^{-1}$$

$$\frac{1}{H_0} = 4.41 \times 10^{17} \text{ s} = \frac{4.41 \times 10^{17}}{3.15 \times 10^7} = 1.4 \times 10^{10} \text{ years}$$

About 14 billion years — close to the measured age of 13.8 billion years.

THE UNIVERSE HAS NO CENTRE

Galaxies are not flying outwards through space from an explosion at some point. **Space itself stretches**, carrying the galaxies with it, so every observer in every galaxy sees the same law $v = H_0 d$. Think of dots on an inflating balloon, not of shrapnel.

Expansion of the universe and dark energy

In 1998 two teams found distant Type Ia supernovae to be **fainter than expected**: the expansion is not slowing under gravity but **accelerating**. The agent driving it is called **dark energy**: an unknown energy of empty space that pushes outwards.

The microwave background fixes the present composition of the universe (Planck mission, 2018 values):

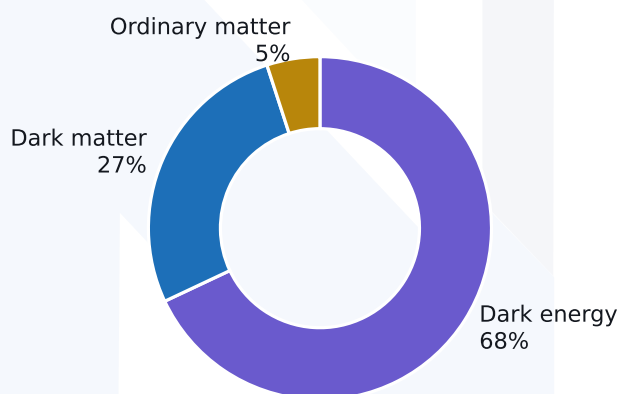


Figure 113. Energy content of the present universe (Planck 2018 values). Everything ever seen with a telescope is the 5% slice.

One open problem: the distance-ladder value $H_0 = 73.0 \pm 1.0$ (SH0ES, 2022) and the microwave-background value $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck, 2018) disagree by far more than their error bars. This **Hubble tension** is still unresolved.

Dark matter

Stars in the outer parts of a spiral galaxy orbit far too fast. To see why that is a problem, work out what speed they should have.

WHY THE ROTATION CURVE OUGHT TO FALL AWAY

Step 1 — what holds a star in its orbit? A star of mass m moving in a circle of radius r about the centre of its galaxy needs a centripetal force mv^2/r , and gravity must supply it. If $M(r)$ is the total mass lying inside radius r , then

$$\frac{GM(r)m}{r^2} = \frac{mv^2}{r}$$

Step 2 — cancel m , which appears on both sides (the answer cannot depend on which star we pick):

$$\frac{GM(r)}{r^2} = \frac{v^2}{r}$$

Step 3 — multiply both sides by r :

$$\frac{GM(r)}{r} = v^2$$

Step 4 — take the square root:

$$v = \sqrt{\frac{GM(r)}{r}}$$

Step 5 — apply this outside the bright disc. Out there almost no visible stars are left, so if visible matter were all there is, $M(r)$ would stop growing and settle down to a constant M . Putting a constant into Step 4,

$$v = \sqrt{GM} \cdot \frac{1}{\sqrt{r}} \implies v \propto \frac{1}{\sqrt{r}}$$

The speed should **fall off**, in exactly the way Neptune crawls round the Sun more slowly than Mercury.

Step 6 — compare with what is measured. Real rotation curves stay flat: $v \approx$ constant, far beyond the edge of the light. Put $v =$ constant back into Step 3 and make $M(r)$ the subject:

$$M(r) = \frac{v^2 r}{G} \implies M(r) \propto r$$

Mass must go on piling up in proportion to r long after the light has stopped. A spiral galaxy needs roughly ten times more of this invisible mass than of the matter we can see.

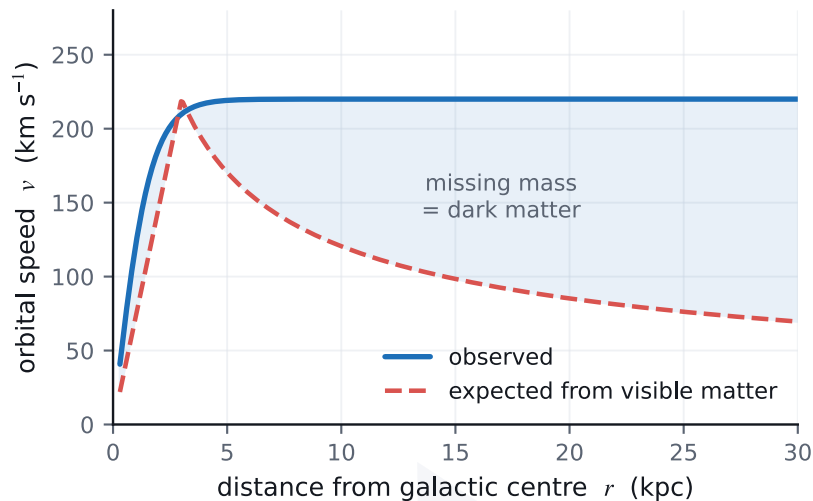


Figure 114. Rotation curve of a spiral galaxy. The flat measured curve needs far more mass than the visible matter can supply.

DARK MATTER

Dark matter is matter that exerts gravitational attraction but neither emits, absorbs nor reflects electromagnetic radiation. It is detected only through its gravity — flat galaxy rotation curves, the speeds of galaxies in clusters (noticed by Zwicky in 1933), gravitational lensing, and the ripple pattern of the microwave background.

It cannot be ordinary matter that merely fails to shine, because the light-element abundances cap how much ordinary matter exists. Leading candidates are weakly interacting massive particles (WIMPs), axions and sterile neutrinos — none yet found, despite decades of searching in underground laboratories.

Black holes

Squeeze a body small enough and its escape velocity reaches the speed of light. Beyond that point nothing at all can leave it.

THE SCHWARZSCHILD RADIUS

Step 1 — the condition for escaping. To get away from the surface of a body of mass M and radius R , a projectile of mass m must have just enough kinetic energy to pay off its gravitational potential energy. At the surface the potential energy is $-GMm/R$; infinitely far away it is zero. So the smallest launch speed v_e satisfies

$$\frac{1}{2}mv_e^2 = \frac{GMm}{R}$$

Step 2 — cancel m from both sides; the escape speed does not depend on what you throw:

$$\frac{1}{2}v_e^2 = \frac{GM}{R}$$

Step 3 — multiply both sides by 2:

$$v_e^2 = \frac{2GM}{R}$$

Step 4 — take the square root:

$$v_e = \sqrt{\frac{2GM}{R}}$$

Step 5 — now shrink the body. Keep M fixed and make R smaller and smaller. v_e grows. Call R_s the radius at which v_e has just reached c :

$$c^2 = \frac{2GM}{R_s}$$

Step 6 — multiply both sides by R_s :

$$c^2 R_s = 2GM$$

Step 7 — divide both sides by c^2 :

$$R_s = \frac{2GM}{c^2}$$

This R_s is the **Schwarzschild radius**.

THE NEWTONIAN SHORTCUT GIVES THE RIGHT ANSWER FOR THE WRONG REASON

Light does not slow down as it climbs out of a gravitational field, so the escape-velocity argument above is not strictly valid for light. Karl Schwarzschild obtained the very same formula in 1916 from Einstein's field equations, which is why NEB accepts the short derivation. Quote it — but remember the real reason: inside R_s , every path leads inwards.

BLACK HOLE

A black hole is a region where matter has collapsed to such a density that the escape velocity exceeds the speed of light. Its boundary, at $r = R_s$, is the **event horizon**: a one-way surface. Nothing that crosses it can return, so the object is seen only through its gravitational effect on its surroundings.

A star heavier than about 20 solar masses leaves a black hole when its core collapses. Far larger **super-massive** black holes, millions to billions of solar masses, sit at the centres of galaxies. Evidence includes X-ray emission from gas spiralling into Cygnus X-1; the orbits of stars whipping around an invisible $4.3 \times 10^6 M_\odot$ object at our galactic centre (Nobel Prize 2020); and direct images from the **Event Horizon Telescope** — M87* on 10 April 2019 ($6.5 \times 10^9 M_\odot$, 55 million light-years away) and **Sagittarius A*** on 12 May 2022, 27,000 light-years away.

WORKED EXAMPLE 26.4

Problem. Calculate the Schwarzschild radius of (a) the Sun ($M = 2 \times 10^{30}$ kg) and (b) the Earth ($M = 6 \times 10^{24}$ kg). Take $G = 6.67 \times 10^{-11}$ SI units.

Solution.

(a)

$$R_s = \frac{2GM}{c^2} = \frac{2 \times 6.67 \times 10^{-11} \times 2 \times 10^{30}}{(3 \times 10^8)^2} = \frac{2.67 \times 10^{20}}{9 \times 10^{16}} = 2.96 \times 10^3 \text{ m}$$

so about 3 km: the Sun would have to be squeezed from 7×10^5 km to the size of a small town.

(b)

$$R_s = \frac{2 \times 6.67 \times 10^{-11} \times 6 \times 10^{24}}{9 \times 10^{16}} = \frac{8.0 \times 10^{14}}{9 \times 10^{16}} = 8.9 \times 10^{-3} \text{ m}$$

— the Earth squeezed to the size of a marble.

Gravitational waves

General relativity treats gravity as curvature of spacetime. When a massive object accelerates asymmetrically — two black holes spiralling together — that curvature ripples outwards at the speed of light. These **gravitational waves** are transverse: they alternately stretch and squeeze space in two perpendicular directions.

Einstein predicted them in 1916 and believed they would never be measured, because the **strain** $h = \Delta L/L$ they produce on Earth is only about 10^{-21} .

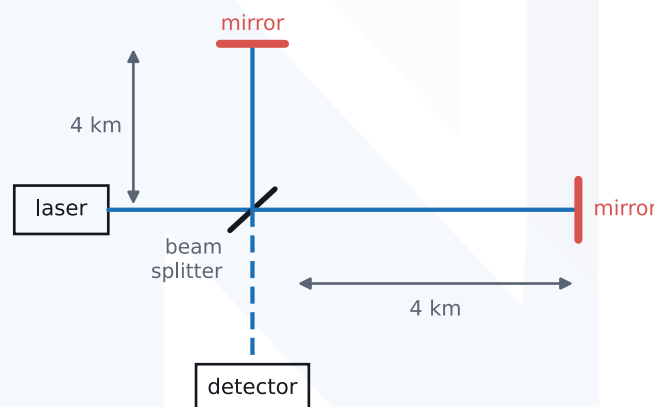


Figure 115. A LIGO detector: a Michelson interferometer with 4 km arms. A passing wave lengthens one arm and shortens the other, shifting the interference pattern at the photodetector.

LIGO solved the problem with two Michelson interferometers, at Hanford and Livingston in the USA, with 4 km arms. A laser beam is split, sent along both arms, reflected and recombined; if one arm changes length relative to the other, the interference pattern shifts. Two detectors far apart are used so that a local disturbance — a truck, an earthquake — is not mistaken for a wave from space.

WORKED EXAMPLE 26.5

Problem. A gravitational wave of strain $h = 1.0 \times 10^{-21}$ passes through a LIGO arm of length $L = 4.0$ km. (a) By how much does the arm change in length? (b) Compare that with the diameter of a proton, 1.7×10^{-15} m.

Solution.

(a) The strain is defined as $h = \Delta L/L$. Multiplying both sides by L ,

$$\Delta L = hL = (1.0 \times 10^{-21}) \times (4.0 \times 10^3 \text{ m})$$

$$\Delta L = 4.0 \times 10^{-18} \text{ m}$$

(b)

$$\frac{\Delta L}{d_{\text{proton}}} = \frac{4.0 \times 10^{-18}}{1.7 \times 10^{-15}} = 2.4 \times 10^{-3}$$

The mirrors shift by about one four-hundredth of the width of a proton — and that has to be measured over a 4 km baseline. This is why the detection took a century after the prediction.

The first detection, **GW150914**, was made on **14 September 2015** and announced in February 2016. Two black holes of about 36 and 29 solar masses merged into one of 62 solar masses, roughly $3 M_{\odot}$ vanishing as pure gravitational-wave energy in a fraction of a second. The 2017 Nobel Prize followed. In August 2017 the neutron-star merger GW170817 was seen in gravitational waves and gamma rays together, opening multi-messenger astronomy. With Virgo and KAGRA now joining the network, the cumulative LIGO–Virgo–KAGRA catalogue (GWTC-5.0, 2026) lists **391 confirmed events** — the count grows with every observing run, so quote it with its date.

WORKED EXAMPLE 26.6

Problem. In GW150914 about $3.0 M_{\odot}$ of mass was converted into gravitational-wave energy, where $M_{\odot} = 2 \times 10^{30} \text{ kg}$. How much energy was released?

Solution. By Einstein's mass–energy relation,

$$E = \Delta m c^2 = (3.0 \times 2 \times 10^{30}) \times (3 \times 10^8)^2$$

$$E = 6 \times 10^{30} \times 9 \times 10^{16} = 5.4 \times 10^{47} \text{ J}$$

Briefly, the merger radiated more power than every star in the observable universe combined.

✓ Chapter summary

- Every particle has an antiparticle of equal mass and opposite charge. Pair production needs ≥ 1.02 MeV; annihilation of e^-e^+ gives two 0.511 MeV photons.
- Quarks come in six flavours: u, c, t with charge $+\frac{2}{3}e$ and d, s, b with $-\frac{1}{3}e$. Baryons are qqq ($p = uud, n = udd$); mesons are $q\bar{q}$ ($\pi^+ = u\bar{d}$). Free quarks are never seen.
- The six leptons are e, μ, τ and their neutrinos. Neutrinos are neutral, feel only the weak force, and oscillate between flavours — proof of tiny mass.
- Hubble's law $v = H_0 d$ with $H_0 \approx 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$; the Hubble time $1/H_0 \approx 14$ billion years.
- Big Bang evidence: the expansion, the 2.725 K cosmic microwave background, and the 75% H / 25% He abundance. Age = 13.8 billion years.
- Space itself expands, with no centre, and ever faster: the universe is 68% dark energy, 27% dark matter, 5% ordinary matter.
- Dark matter is inferred from flat rotation curves, cluster dynamics and lensing; it has never been detected directly. Newtonian orbits give $v = \sqrt{GM(r)/r}$, so a flat curve means $M(r) \propto r$.
- Black hole: escape velocity exceeds c ; event horizon at $R_s = 2GM/c^2$. Imaged by the Event Horizon Telescope (M87* 2019, Sgr A* 2022).
- Gravitational waves are ripples of spacetime travelling at c , of strain $h = \Delta L/L \approx 10^{-21}$, detected by km-scale Michelson interferometers; first detection GW150914, 14 September 2015; 391 events in the catalogue by 2026.

Practice questions

Group A — Multiple choice (1 mark each)

- The quark composition of a neutron is [1]

a uud	b udd	c $u\bar{d}$	d uds
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- Which of the following is **not** a lepton? [1]

a electron	b muon	c pion	d neutrino
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- The cosmic microwave background has a temperature of about [1]

a 0 K	b 2.7 K	c 27 K	d 273 K
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- The Schwarzschild radius of a body of mass M is [1]

a GM/c^2	b $2GM/c^2$	c $GM/2c^2$	d $2GM/c$
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- Flat rotation curves of spiral galaxies are evidence for [1]

a dark energy	b dark matter	c black holes	d antimatter
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ANSWERS TO GROUP A

- (b) — udd gives $\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$.
- (c) — the pion is a meson, made of a quark and an antiquark.

3. (b) — a 2.725 K blackbody, found by Penzias and Wilson in 1965.
4. (b) — from $\sqrt{2GM/R} = c$.
5. (b) — the extra unseen mass keeps the outer stars moving fast.

Group B — Short answer (5 marks each)

1. What is an antiparticle? Describe pair production and annihilation, giving the threshold energy for electron–positron pair production. [5]
2. Distinguish between baryons and mesons, and show that the quark contents uud and $u\bar{d}$ give the correct charges of the proton and the π^+ . [5]
3. Why was the neutrino proposed, and why is it so difficult to detect? What does neutrino oscillation tell us about its mass? [5]
4. State Hubble's law. A galaxy is 200 Mpc away. Taking $H_0 = 70 \text{ km s}^{-1}\text{Mpc}^{-1}$, find its recession speed and the fractional redshift z of its spectral lines. [5]
5. What is dark matter? Give two pieces of observational evidence for it. [5]
6. Show that a star orbiting at radius r inside a galaxy of enclosed mass $M(r)$ has speed $v = \sqrt{GM(r)/r}$, and explain why a **flat** rotation curve requires $M(r) \propto r$. [5]
7. A gravitational wave of strain 1.0×10^{-21} crosses a detector whose arms are 4.0 km long. Calculate the change in length of one arm. [5]

ANSWERS TO GROUP B

2. Outline: baryons are three quarks with baryon number +1; mesons are a quark and an anti-quark with baryon number 0, and all mesons are unstable. Charges: $uud = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$ for the proton; $u\bar{d} = \frac{2}{3} + \frac{1}{3} = +1$ for the π^+ .

3. Outline: beta-decay electrons emerge with a continuous energy spectrum, so Pauli (1930) proposed an unseen neutral particle carrying off the missing energy and momentum. The neutrino is chargeless and interacts only weakly, so its chance of interacting in a detector is minute. Oscillation between flavours is possible only if at least two neutrino states have non-zero mass.

4. $v = H_0 d = 70 \times 200 = 1.4 \times 10^4 \text{ km s}^{-1}$.

$$z = \frac{v}{c} = \frac{1.4 \times 10^4}{3 \times 10^5} = 0.047$$

so the lines are shifted about 4.7% towards the red.

6. Gravity supplies the centripetal force, $GM(r)m/r^2 = mv^2/r$. Cancelling m and multiplying by r gives $v^2 = GM(r)/r$, so $v = \sqrt{GM(r)/r}$. If v is constant then $M(r) = v^2 r/G$, which is proportional to r : mass keeps accumulating outside the visible disc.

7. $\Delta L = hL = 1.0 \times 10^{-21} \times 4.0 \times 10^3 = 4.0 \times 10^{-18} \text{ m}$, roughly one four-hundredth of a proton diameter.

1. Outline: an antiparticle has the same mass and spin as its particle but opposite charge and opposite quantum numbers (§26.1). In **pair production** a photon passing close to a nucleus turns into a particle–antiparticle pair; for e^-e^+ the photon must carry at least $2m_e c^2 = 1.02 \text{ MeV}$, and the nearby nucleus is needed to take up recoil momentum. In **annihilation** an electron and a positron

meet and vanish, and two photons of 0.511 MeV go out in opposite directions so that momentum is conserved.

5. Outline: dark matter is matter that has mass and therefore gravitates, but neither emits nor absorbs light, so it is seen only through its gravity (§26.2). Evidence: (i) **flat galactic rotation curves** — outer stars orbit far faster than the visible mass allows, requiring $M(r) \propto r$ out to large radii; (ii) **gravitational lensing** by clusters, which bends background light by far more than the luminous mass could; (iii) the **speeds of galaxies inside clusters** (Zwicky, Coma cluster), which are too high for the cluster to stay bound on visible mass alone.

Group C — Long answer (8 marks each)

1. **(a)** State Hubble's law and explain how it leads to the Big Bang model. [3]
(b) Give three independent pieces of evidence supporting the Big Bang. [3]
(c) Show that the Hubble time $1/H_0$ for $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is about 14 billion years, given $1 \text{ Mpc} = 3.086 \times 10^{22} \text{ m}$. [2]
2. **(a)** Define a black hole and derive the expression for its Schwarzschild radius. [3]
(b) What are gravitational waves? Describe with a diagram how LIGO detects them, and state one major observation. [5]

ANSWERS TO GROUP C

1.(a) Outline: Hubble's law is $v = H_0 d$ — every distant galaxy recedes with a speed proportional to its distance. Running the motion backwards, every galaxy was at the same place at a time $t \approx 1/H_0$ ago, which is the Big Bang. The recession is not galaxies flying through space but space itself expanding, so there is no centre.

1.(b) Any three of: the redshift-distance relation itself; the cosmic microwave background at 2.7 K, the cooled relic of the hot early universe; the observed abundance of light elements (about 75% hydrogen, 25% helium by mass) predicted by Big-Bang nucleosynthesis; and the fact that distant galaxies look younger than nearby ones.

1.(c) $H_0 = \frac{70 \times 10^3}{3.086 \times 10^{22}} = 2.27 \times 10^{-18} \text{ s}^{-1}$, so

$$\frac{1}{H_0} = \frac{1}{2.27 \times 10^{-18}} = 4.41 \times 10^{17} \text{ s}$$

Dividing by $3.156 \times 10^7 \text{ s per year}$ gives $1.40 \times 10^{10} \text{ years}$, about 14 billion years — close to the accepted age of 13.8 billion years.

2.(b) Outline: gravitational waves are ripples in spacetime radiated by accelerating masses, stretching space in one direction while squeezing it in the perpendicular direction (§26.2). Marks for the labelled Michelson interferometer diagram (laser, beam splitter, two 4 km arms with mirrors, detector), for the statement that a passing wave lengthens one arm and shortens the other so the recombined beams fall out of step and the fringe pattern shifts, and for one named observation such as GW150914, the merger of two black holes of about 36 and 29 solar masses detected on 14 September 2015.

2.(a) A black hole is a region whose escape velocity exceeds the speed of light, so no radiation can leave it; its boundary at $r = R_s$ is the event horizon. For the derivation set out step by step, see

§26.2. The marks go to: equating kinetic energy with the magnitude of the gravitational potential energy, $\frac{1}{2}mv_e^2 = GMm/R$; cancelling m and rearranging to $v_e = \sqrt{2GM/R}$; and putting $v_e = c$ to obtain

$$R_s = \frac{2GM}{c^2}$$

